DIAGNOSTIC MANUAL



VW 2.0 BEF / CBS

LPG / NG

Engine Systems Technologies B.V.

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Abbreviations

- A/D - Analog to Digital converter.
- AL - Adaptive Learn.
- BEF - Engine code for Volkswagen 2.0 liter 4 cylinder propane engine.
- CAN - Controller Area Network.
- CBS - Engine code for Volkswagen 2.0 liter 4 cylinder natural gas engine.
- BP - Barometric Pressure.
- CL - Closed Loop.
- Connectors
  - CAM - Camshaft position sensor.
  - ChC - Chassis Connector.
  - CP - Coil Pack.
  - CRK - CRanKshaft position sensor.
  - ECU - Electronic Control Unit.
  - EGO - Exhaust Gas Oxygen sensor.
  - ETB - Electronic Throttle Body.
  - ECT - Engine Coolant Temperature.
  - FCV - Fuel Control Valve.
  - LO - Lock Off.
  - GEN - Generator.
  - IAT - Intake Air Temperature sensor.
  - KNK - KNocK sensor.
  - MAP - Manifold Absolute Pressure sensor.
  - OILP - OIL Pressure switch.
  - STM - STarter Motor.
- Colors
  - BK - Black.
  - BL - Blue.
  - BN - Brown.
  - GN - Green.
  - GY - Gray.
  - RD - Red.
  - VT - Violet.
  - WT White.
  - YL - Yellow.
- COP - Computer Operation Properly.
- DST - Diagnostic Scan Tool.
- DTC - Diagnostic Trouble Code.
- FMI - Failure Mode Indicator.
- FPP - Foot Pedal Position sensor.
- GCP - Global Control Platform.
- GND - GrouND.
- IVS - Idle Valid Switch.
- LPG - Liquefied Petroleum Gas.
- MIL - Malfunction Indicator Lamp.
- NG - Natural Gas.
- OEM - Original Equipment Manufacturer.
- OBD - On-Board Diagnostics.
- RAM - Random Access Memory.
- Rev - Revolutions (engine speed).
- RTI - Real Time Interrupt.
- Rx - Receive.
- SAE - Society of Automotive Engineering.

- SPN - Suspect Parameter Number.
- TPS - Throttle Position Sensor.
- TSC - Torque/Speed Control.
- Tx - Transmit.

Diagnostic trouble code documentation

This is the diagnostic manual for the VW 2.0 4 cylinder BEF (LPG) / CBS (NG) engine. It is divided in 4 parts.

The first part contains the diagnostic trouble codes (DTC). Every DTC is explained with a general description, an electric diagram and a flow chart to identify the cause of the problem.

The second part contains the connector information. Every connector is explained with a description of the location, a detailed drawing of the connector and a table with pin number / function / wire color.

The third part is a reference list of the SAE CAN J1939 SPN / FMI numbers, this helps to identify the DTC by CAN.

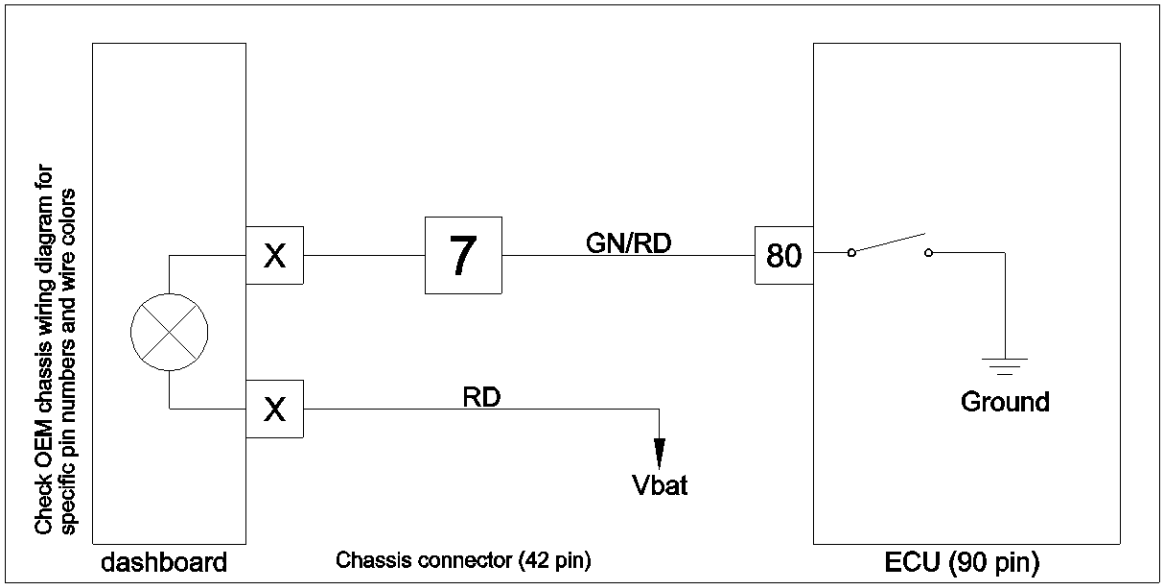
The fourth part is an electric diagram of the engine harness.

To this manual belongs a customer specific part with detailed information for each DTC.

OBD System Check

General

The Global Control Platform (GCP) - Electronic Control Unit (ECU) is equipped with On-Board Diagnostics (OBD). The system has a dash mounted Malfunction Indicator Lamp (MIL). The MIL serves as a notification of an emissions related problem. The MIL also has the ability to flash Diagnostic Trouble Codes (DTC's) in what is referred to as the blink code mode. It will display DTC's that have been stored due to a possible system malfunction. The following DTC flow charts in this manual will instruct the technician to perform the OBD system check. This simply means to verify the operation of the MIL. The lamp should illuminate when the key is in the ON position, and the engine is not running. This feature verifies that the lamp is in proper working order. If the lamp does not illuminate with the vehicle key ON and engine OFF, repair it as soon as possible. Once the engine is in start or run mode, the lamp should go OFF. If the lamp stays ON while the engine is in the start or run mode, a current diagnostic trouble code may be set or a problem may exist with the electrical wiring of the MIL. The ECU completes the circuit to ground to turn the lamp ON.

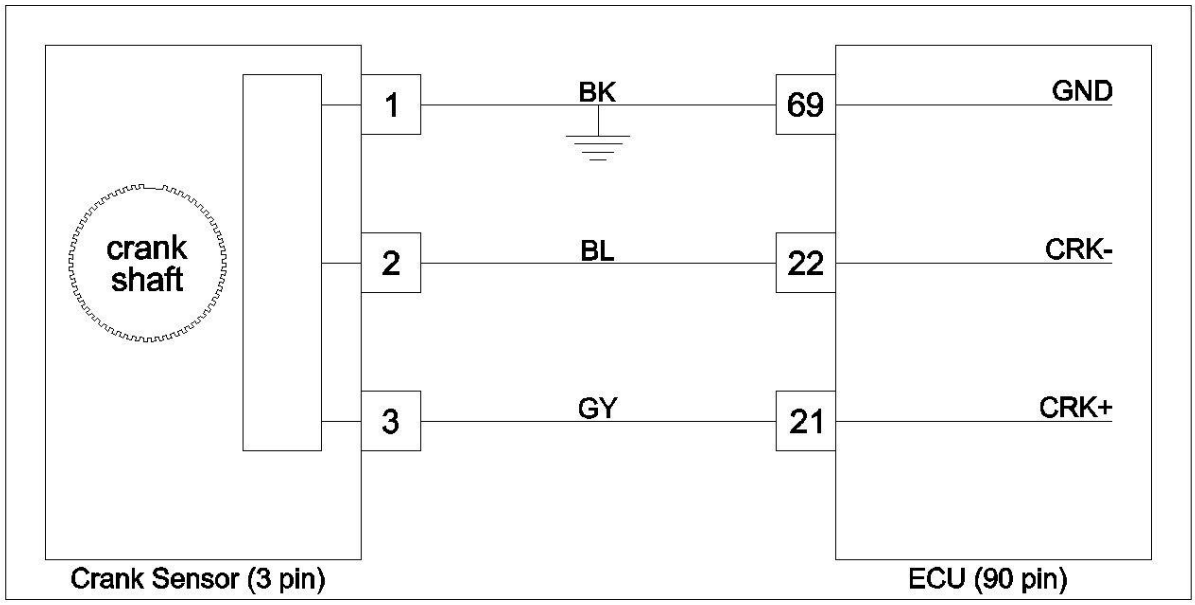


DTC 16 – Never crank synced at start

General

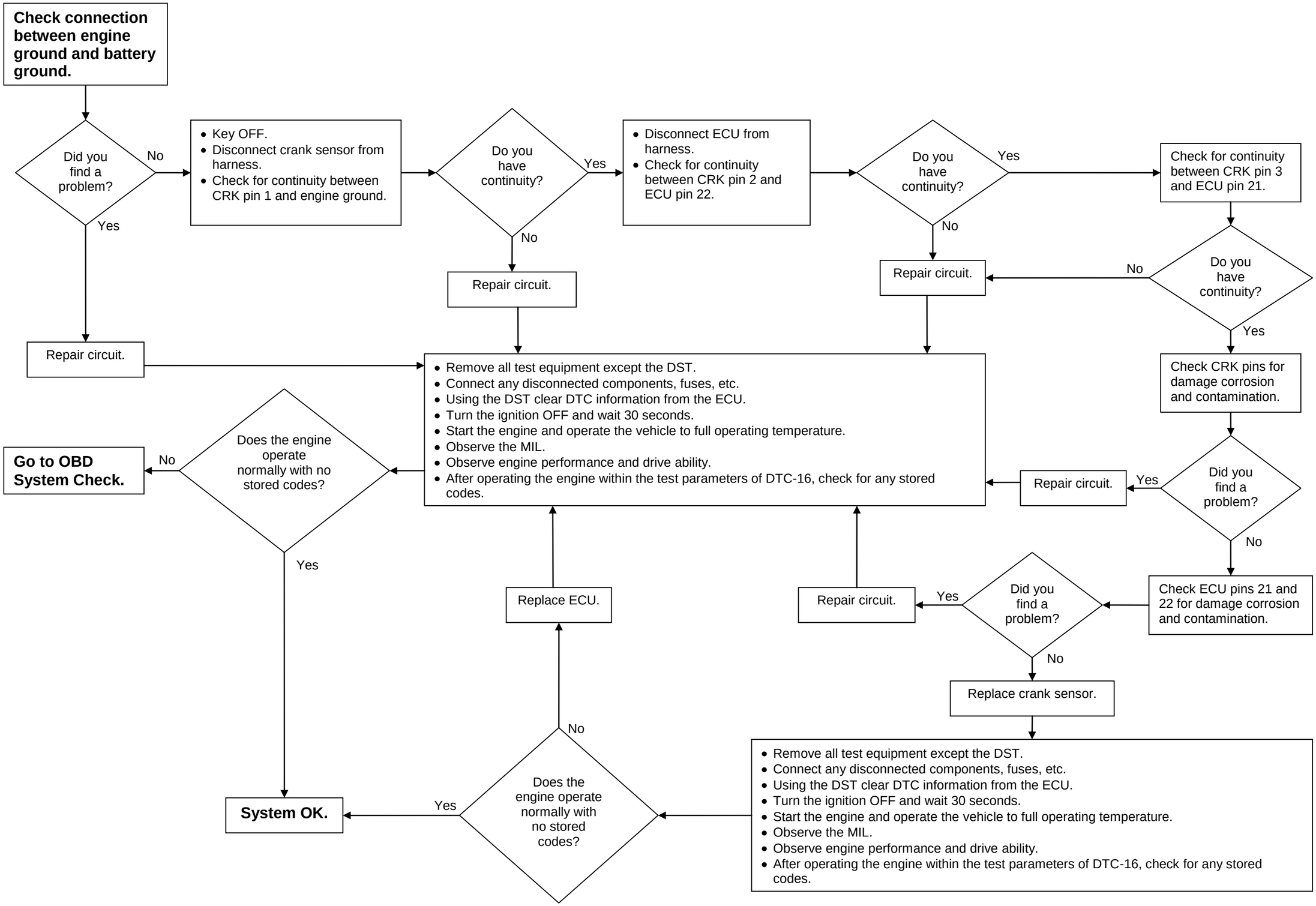
The crankshaft position (CRK) sensor is a 3 pin plug which is located on the cold side of the engine next to the oil filter. A pulse wheel located on the crankshaft is used to measure engine speed and its signal is used to synchronize the ignition and fuel systems. This fault will set if the ECU detects cranking revolutions without synchronization of the CAM and CRK sensors.

For detailed information, please see customer specific DTC settings.



All information concerning connectors, pin numbers and wire colors can be found in the connector documentation.

DTC 16 – Never crank synced at start



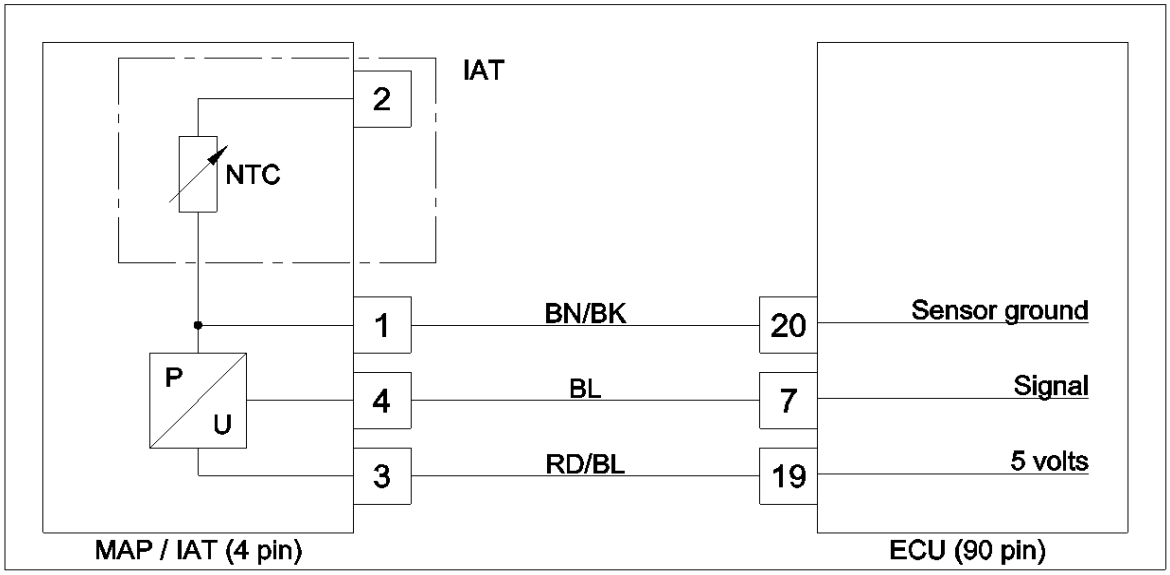
All measurements are done at the connector unless otherwise stated.

DTC 107 – MAP low voltage

General

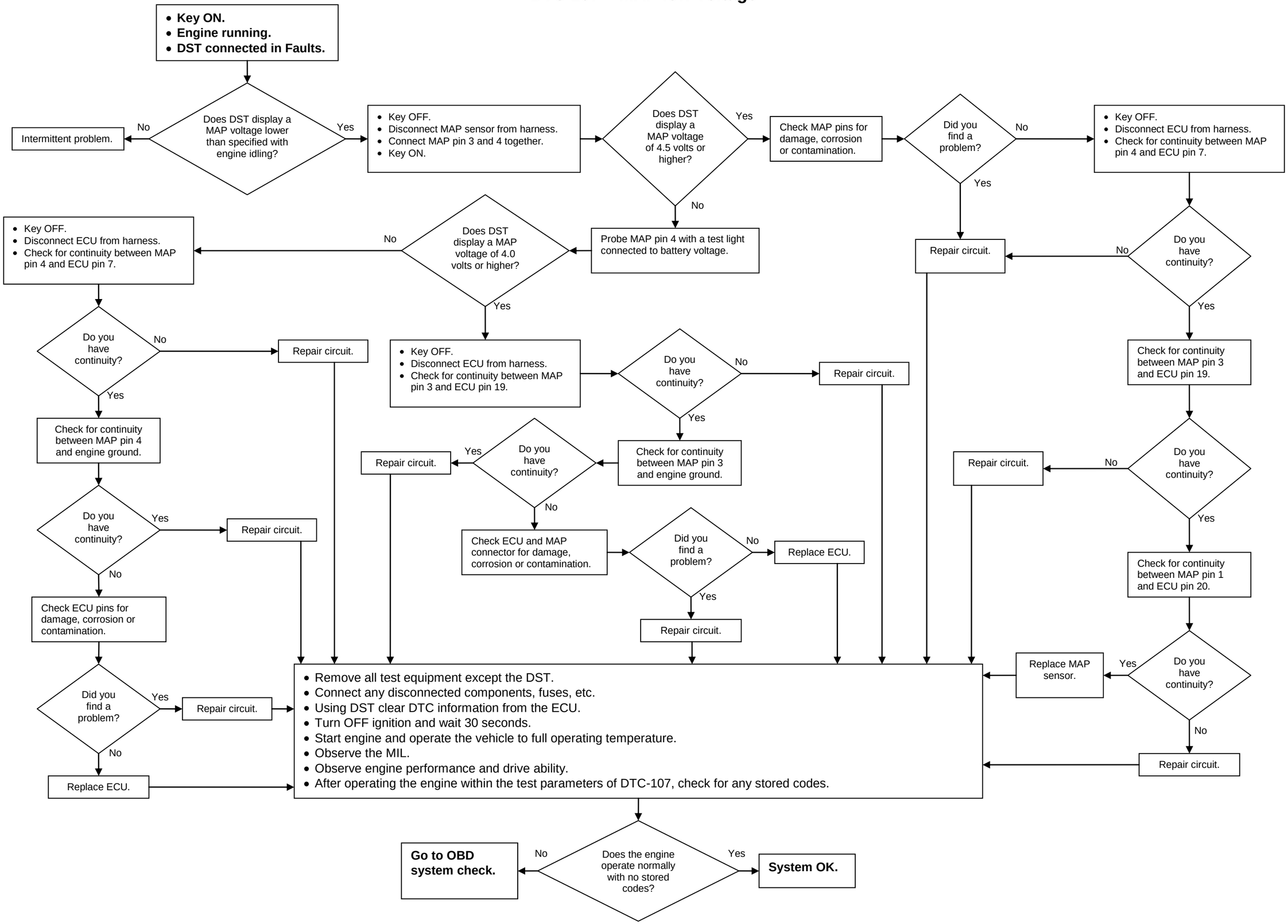
The Manifold Absolute Pressure (MAP) sensor is a pressure transducer connected to the intake manifold. It is used to measure the pressure of air in the intake manifold. This reading is used in conjunction with other inputs to estimate the airflow rate to the engine, which determines the fuel flow rate. This fault will set if the sensor voltage is too low.

For detailed information, please see customer specific DTC settings.



All information concerning connectors, pin numbers and wire colors can be found in the connector documentation.

DTC 107 – MAP low voltage



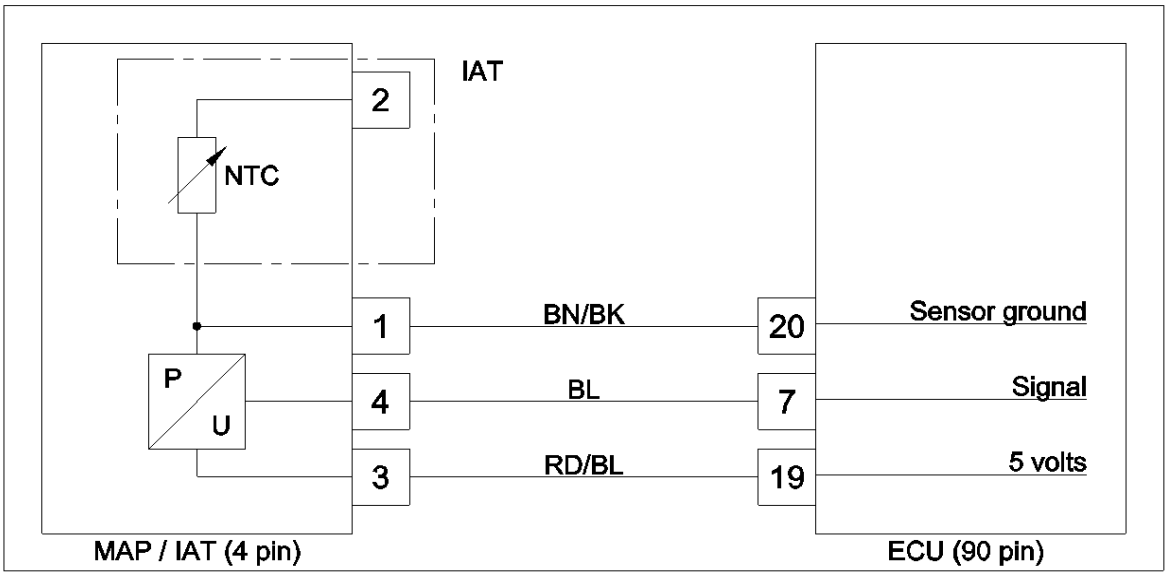
All measurements are done at the connector unless otherwise stated.

DTC 108 – MAP high pressure

General

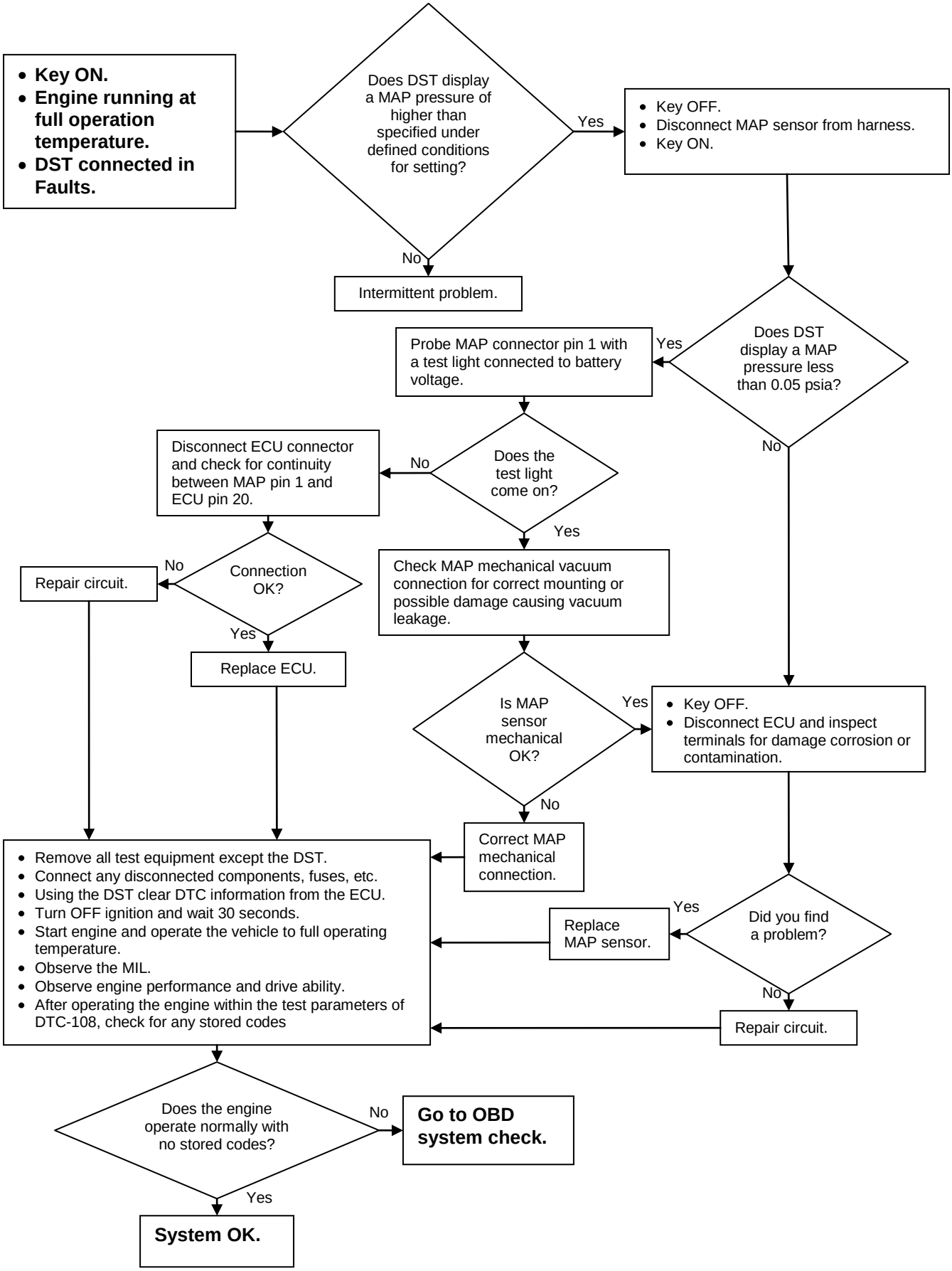
The Manifold Absolute Pressure (MAP) sensor is a pressure transducer connected to the intake manifold. It is used to measure the pressure of air in the intake manifold. This reading is used in conjunction with other inputs to estimate the airflow rate to the engine, which determines the fuel flow rate. This fault will set if the MAP pressure is too high.

For detailed information, please see customer specific DTC settings.



All information concerning connectors, pin numbers and wire colors can be found in the connector documentation.

DTC 108 – MAP high pressure



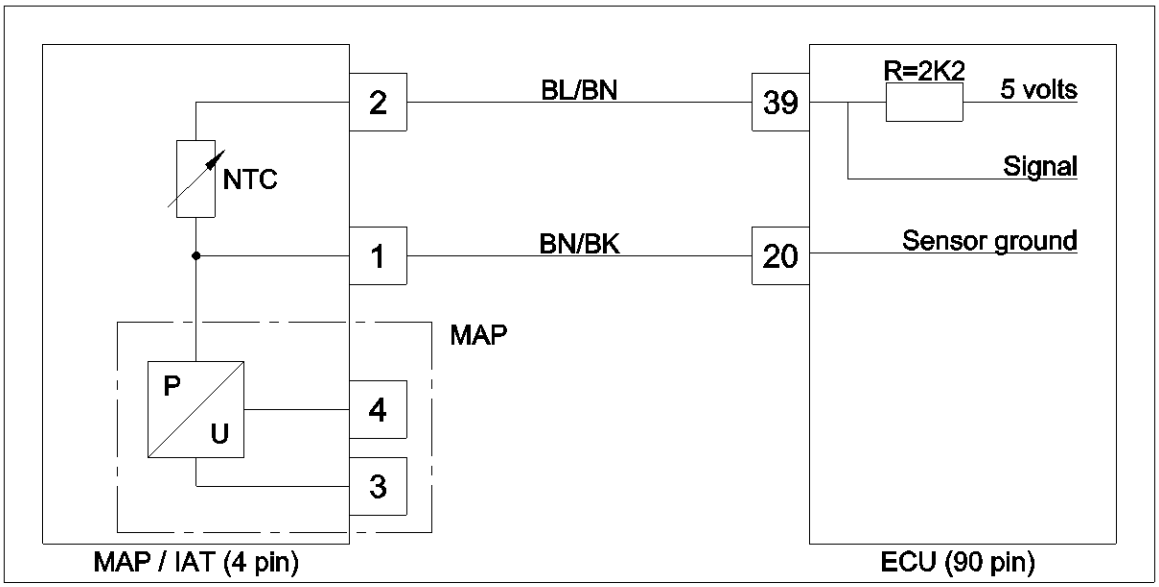
All measurements are done at the connector unless otherwise stated.

DTC 111 – IAT higher than expected 1

General

The Intake Air Temperature (IAT) is used to monitor incoming air temperature to help calculate the engines airflow requirement. The ECU provides a voltage divider circuit so that when the air is cold, the signal reads higher voltage, and lower voltage when warm. The IAT sensor is combined with the MAP sensor. This fault will set if the IAT is too high.

For detailed information, please see customer specific DTC settings.



All information concerning connectors, pin numbers and wire colors can be found in the connector documentation.

DTC 111 – IAT higher than expected 1

Trouble shooting

- This fault will set when the inlet air temperature is much higher than normal. The most common cause of high inlet air temperature is a problem with the inlet air system. Ensure that the air inlet is not obstructed, modified or damaged.

- Inspect the air inlet system for cracks or breaks that may allow unwanted under hood air into the air inlet system.

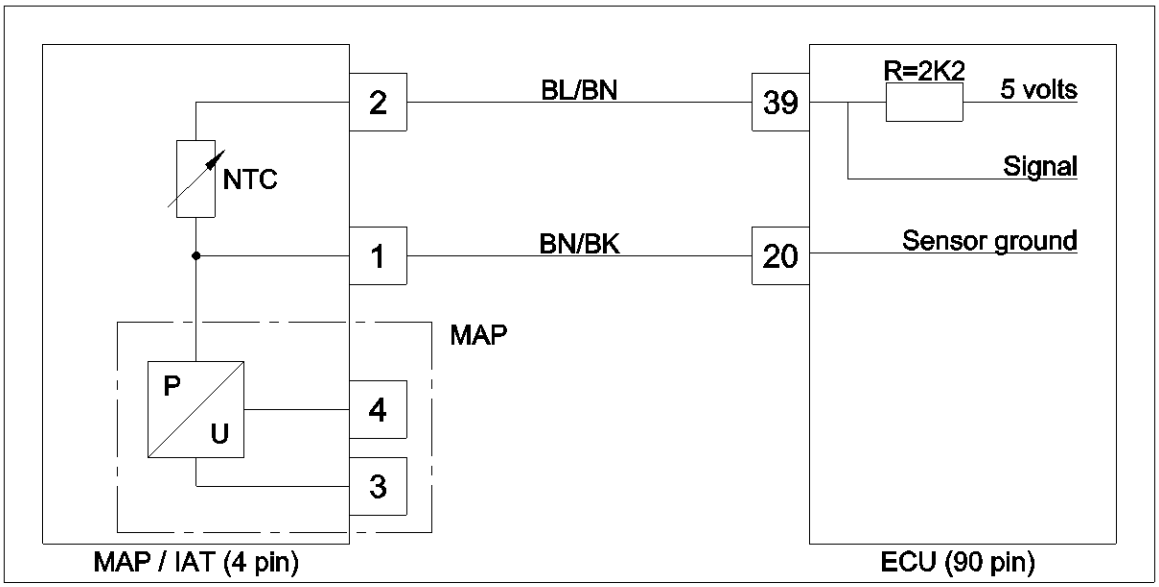
- If none of the above can be found, follow the diagnostic steps for DTC 112-IAT low voltage.

DTC 112 – IAT low voltage

General

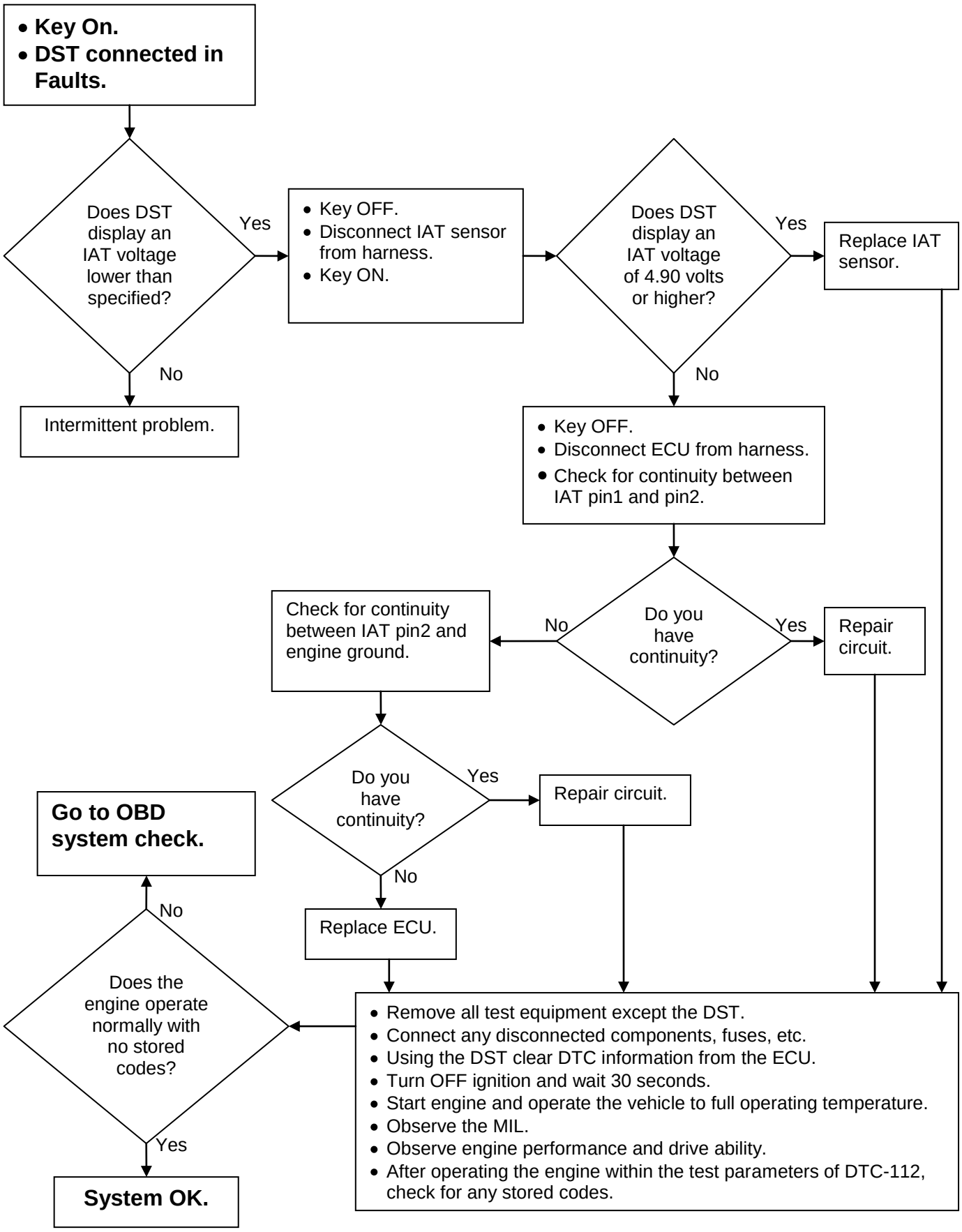
The Intake Air Temperature (IAT) is used to monitor incoming air temperature to help calculate the engines airflow requirement. The ECU provides a voltage divider circuit so that when the air is cold, the signal reads higher voltage, and lower voltage when warm. The IAT sensor is combined with the MAP sensor. This fault will set if the sensor voltage is too low.

For detailed information, please see customer specific DTC settings.



All information concerning connectors, pin numbers and wire colors can be found in the connector documentation.

DTC 112 – IAT low voltage



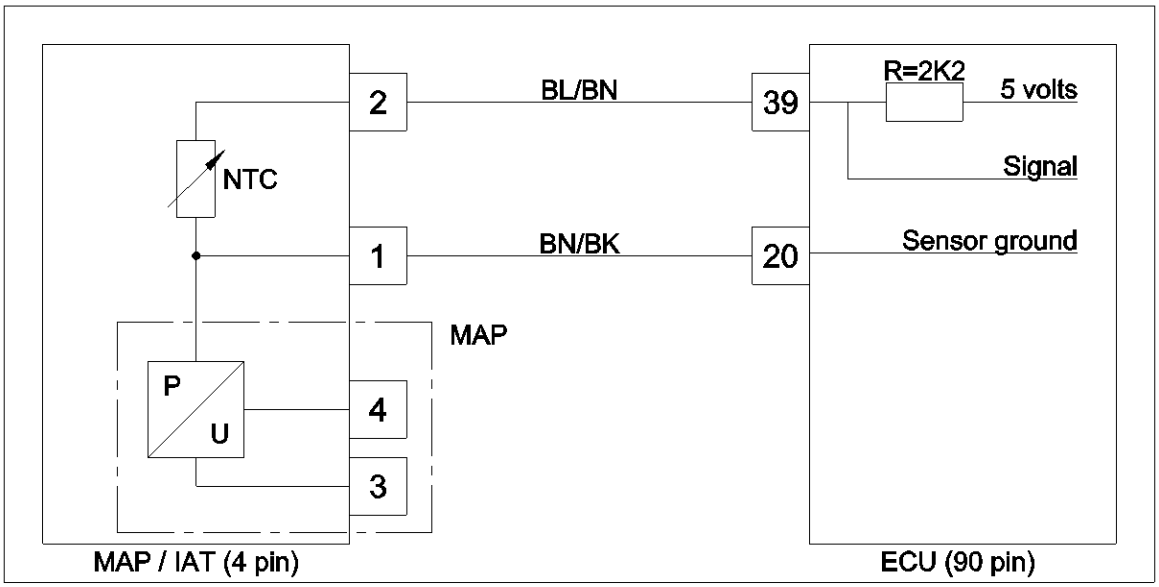
All measurements are done at the connector unless otherwise stated.

DTC 113 – IAT high voltage

General

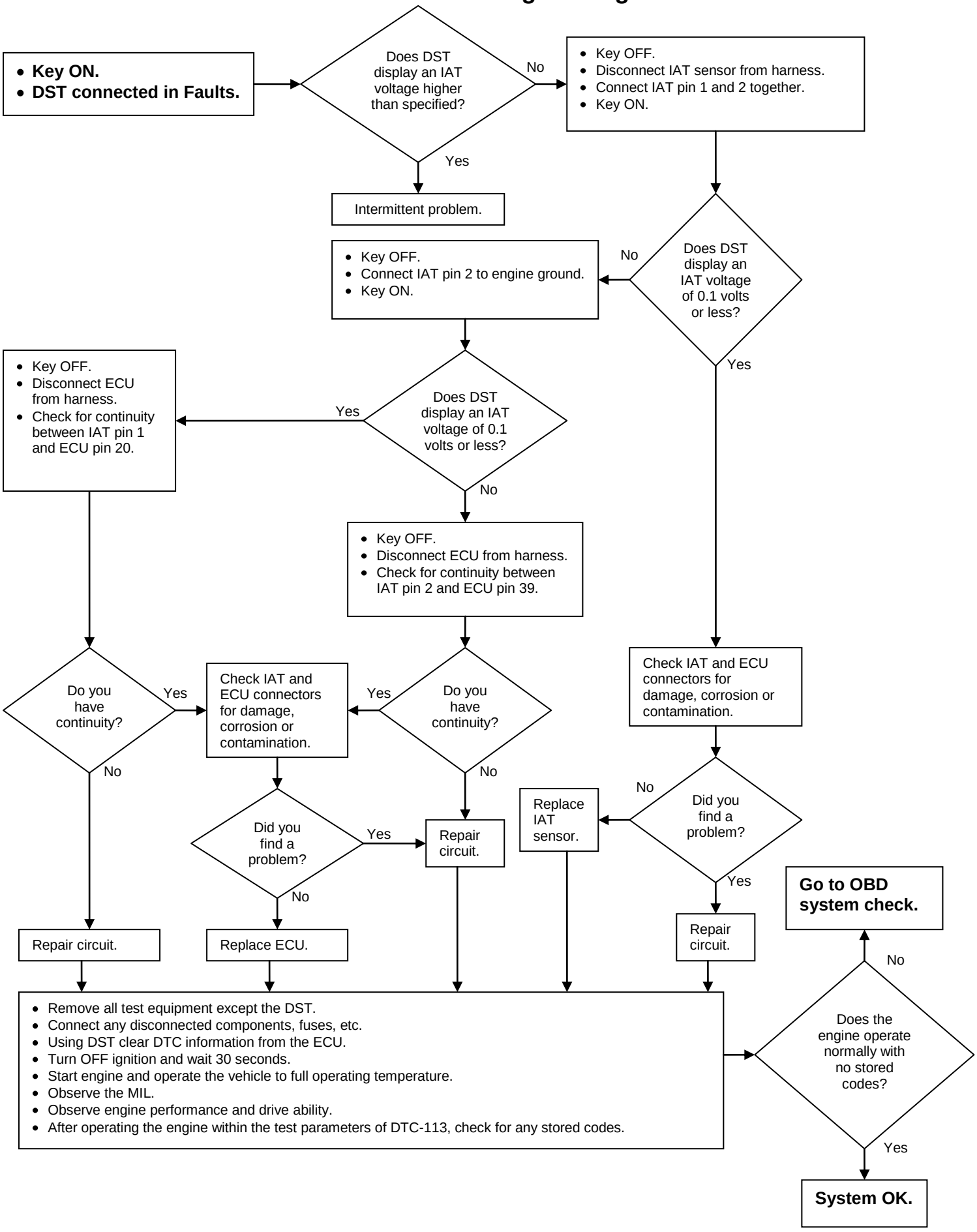
The Intake Air Temperature (IAT) is used to monitor incoming air temperature to help calculate the engines airflow requirement. The ECU provides a voltage divider circuit so that when the air is cold, the signal reads higher voltage, and lower voltage when warm. The IAT sensor is combined with the MAP sensor. This fault will set if the sensor voltage is too high.

For detailed information, please see customer specific DTC settings.



All information concerning connectors, pin numbers and wire colors can be found in the connector documentation.

DTC 113 – IAT high voltage



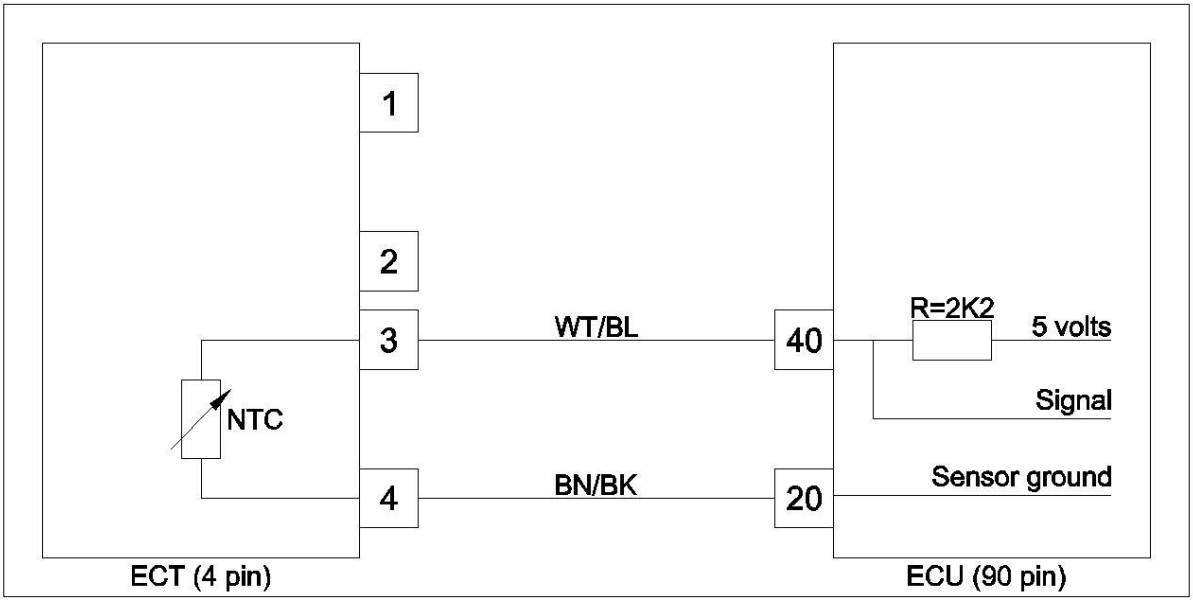
All measurements are done at the connector unless otherwise stated.

DTC 116 – ECT higher than expected 1

General

The Engine Coolant Temperature (ECT) sensor is a temperature sensitive resistor located in the engine coolant passage. It is used for the engine airflow calculation, fuel cold enrichment and to enable other temperature dependant features. The ECU provides a voltage divider circuit so that when the coolant is cold, the signal reads higher voltage, and lower when warm. This fault will set if the ECT is too high.

For detailed information, please see customer specific DTC settings.



All information concerning connectors, pin numbers and wire colors can be found in the connector documentation.

DTC 116 – ECT higher than expected 1

Trouble shooting

- This fault will set when the engine coolant temperature is much higher than normal. The most common cause of high engine coolant temperature is a problem with the engine cooling system. Ensure that the coolant level is OK, and the radiator not obstructed, blocked or damaged.

- Inspect the engine coolant system for cracks or breaks that may allow the cooling system to work inaccurate.

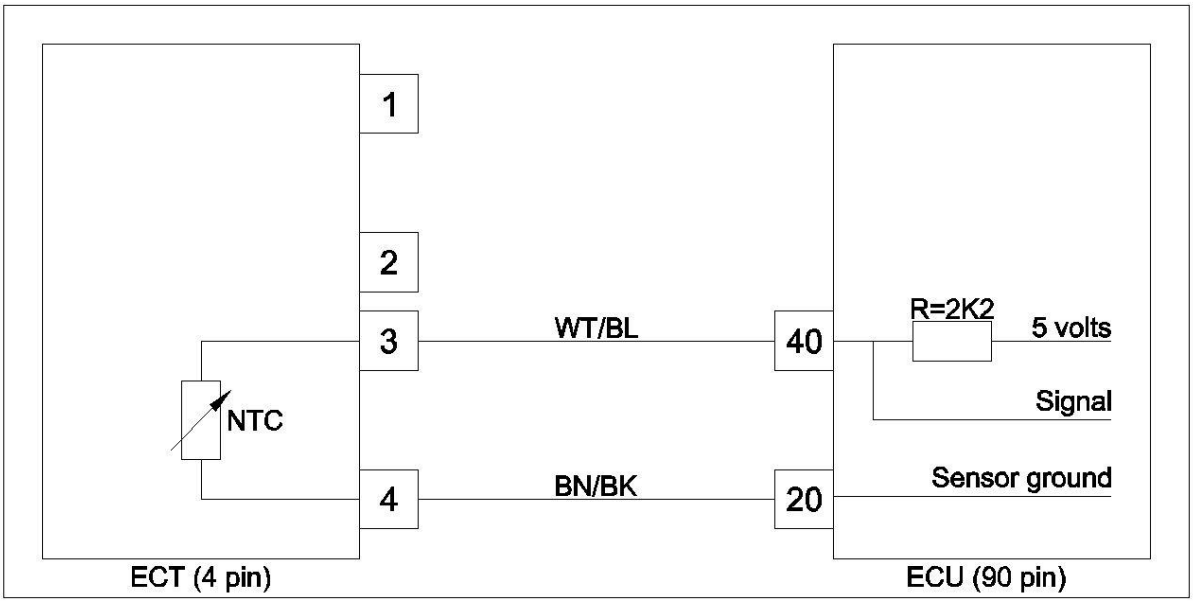
- If none of the above can be found, follow the diagnostic steps for DTC 117-ECT low voltage.

DTC 117 – ECT low voltage

General

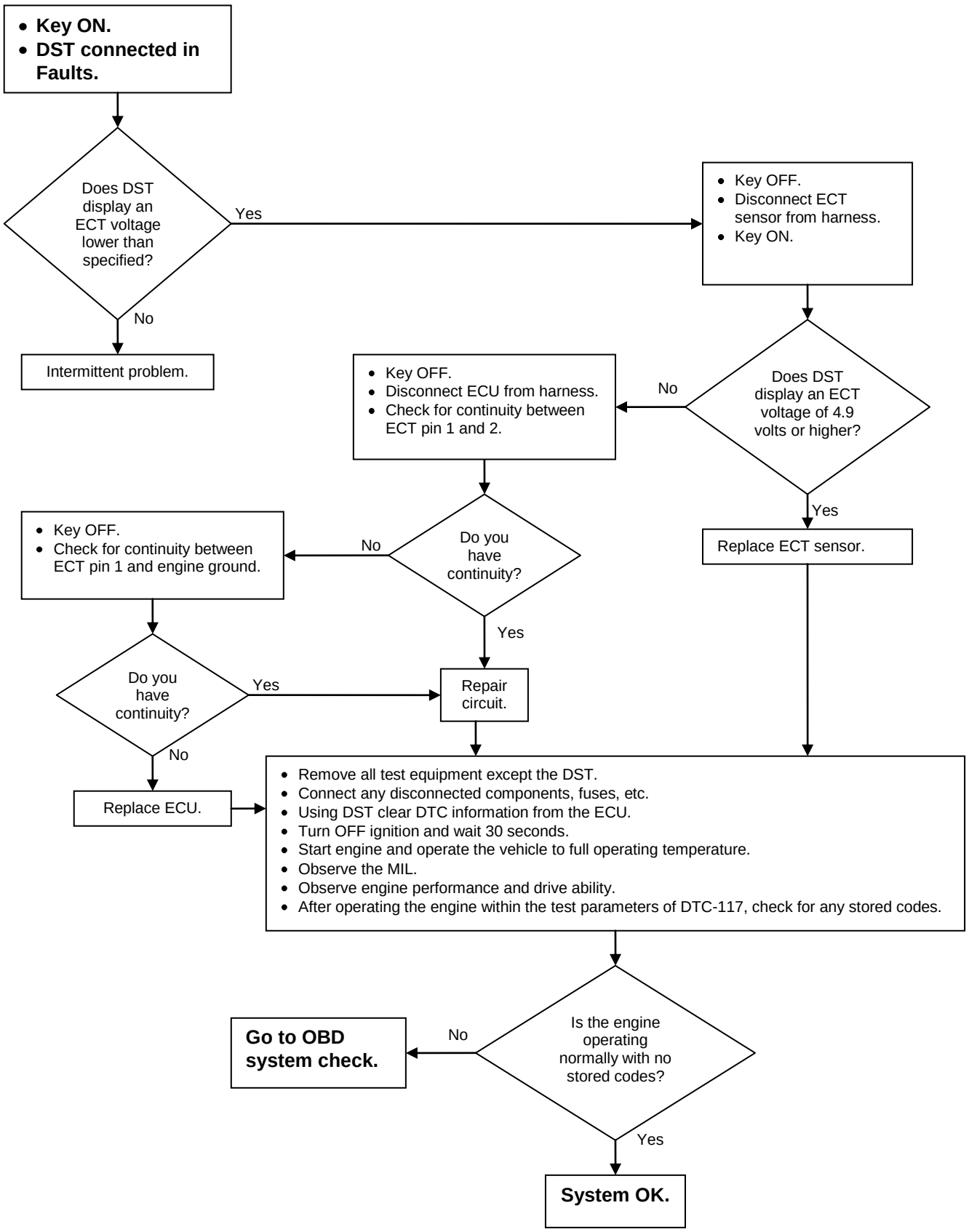
The Engine Coolant Temperature (ECT) sensor is a temperature sensitive resistor located in the engine coolant passage. It is used for the engine airflow calculation, fuel cold enrichment and to enable other temperature dependant features. The ECU provides a voltage divider circuit so that when the coolant is cold, the signal reads higher voltage, and lower when warm. This fault will set if the sensor voltage is too low.

For detailed information, please see customer specific DTC settings.



All information concerning connectors, pin numbers and wire colors can be found in the connector documentation.

DTC 117 – ECT low voltage



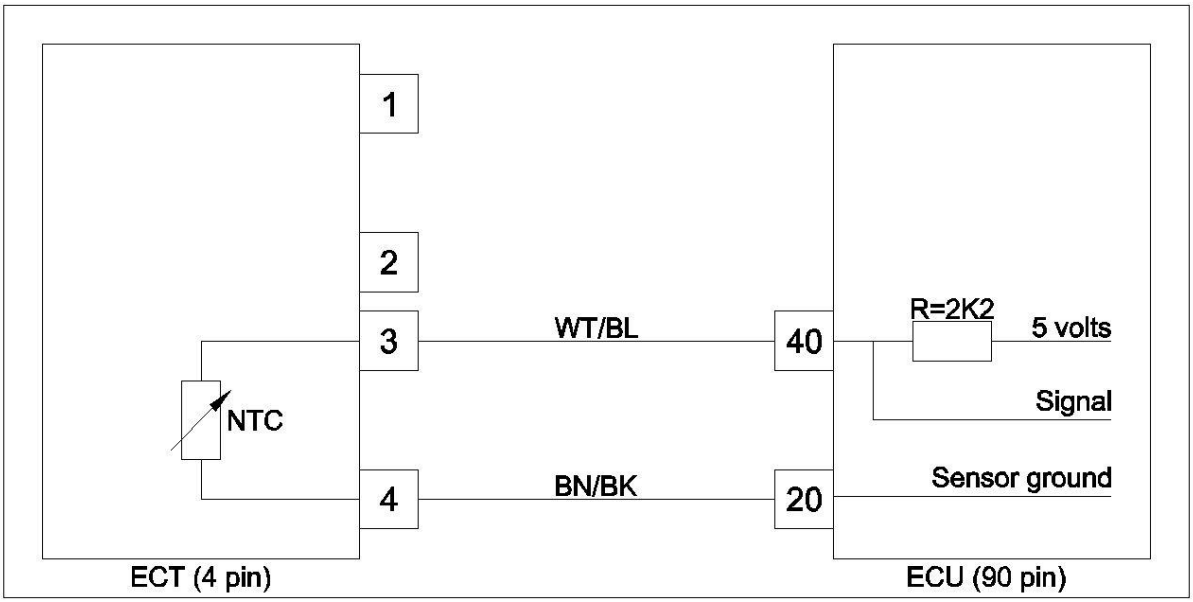
All measurements are done at the connector unless otherwise stated.

DTC 118 – ECT high voltage

General

The Engine Coolant Temperature (ECT) sensor is a temperature sensitive resistor located in the engine coolant passage. It is used for the engine airflow calculation, fuel cold enrichment and to enable other temperature dependant features. The ECU provides a voltage divider circuit so that when the coolant is cold, the signal reads higher voltage, and lower when warm. This fault will set if the sensor voltage is too high.

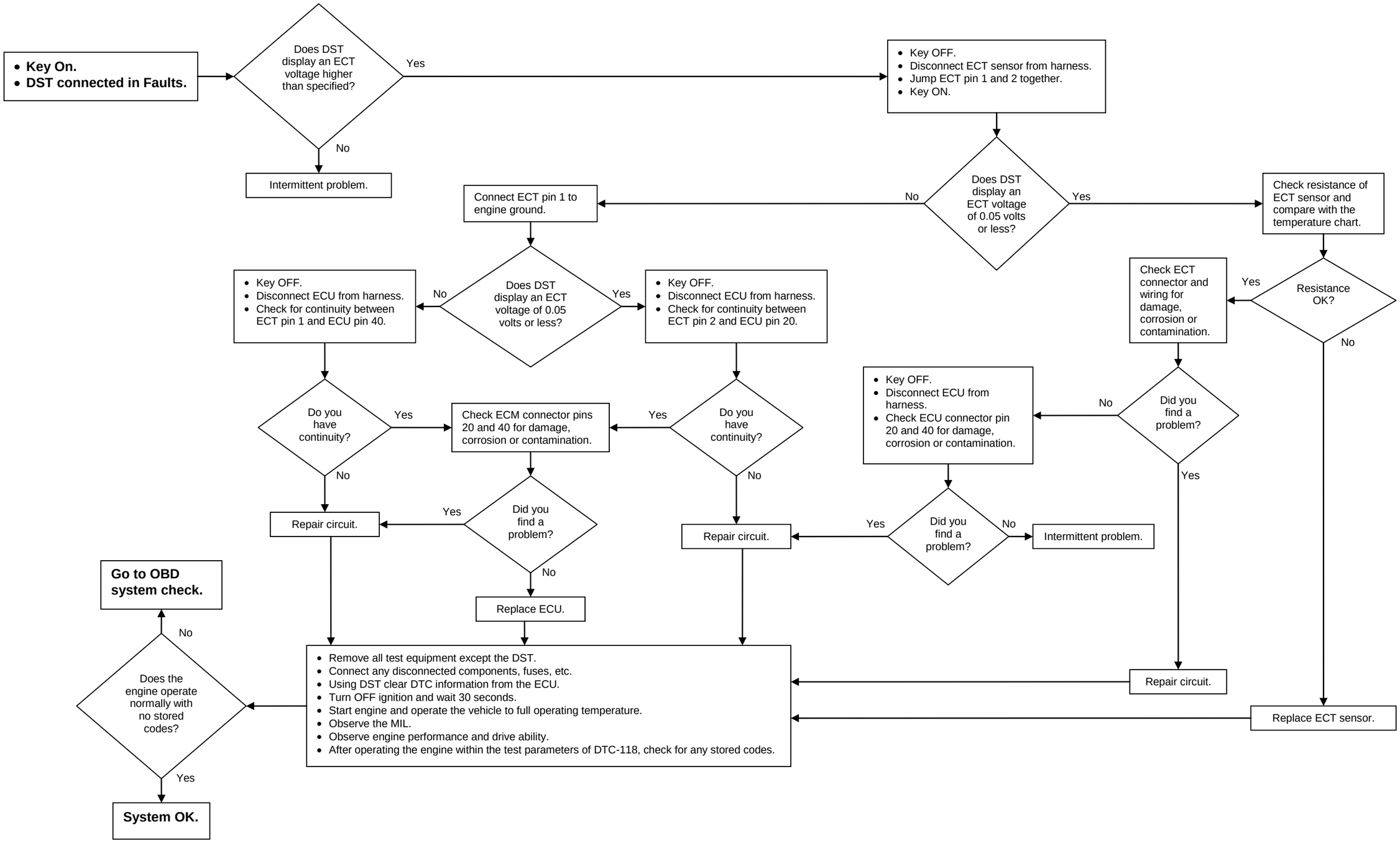
For detailed information, please see customer specific DTC settings.



All information concerning connectors, pin numbers and wire colors can be found in the connector documentation.

Temp (C)	Temp (F)	Resistance (Ω)
120	248	116
110	230	150
100	212	190
90	194	249
82	180	310
71	160	420
60	140	604
49	120	850
38	100	1250
20	68	2504
10	50	3750
0	32	5881
-7	20	7800
-10	14	9261
-30	-22	24486
-40	-40	40000

DTC 118 – ECT high voltage



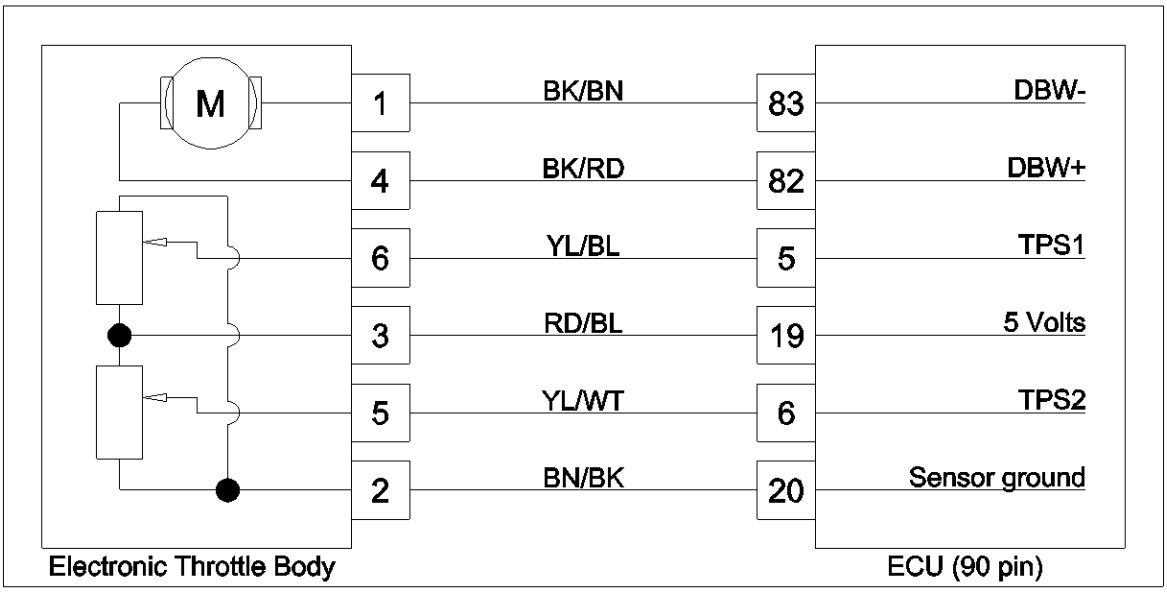
All measurements are done at the connector unless otherwise stated.

DTC 121 – TPS1 lower than TPS2

General

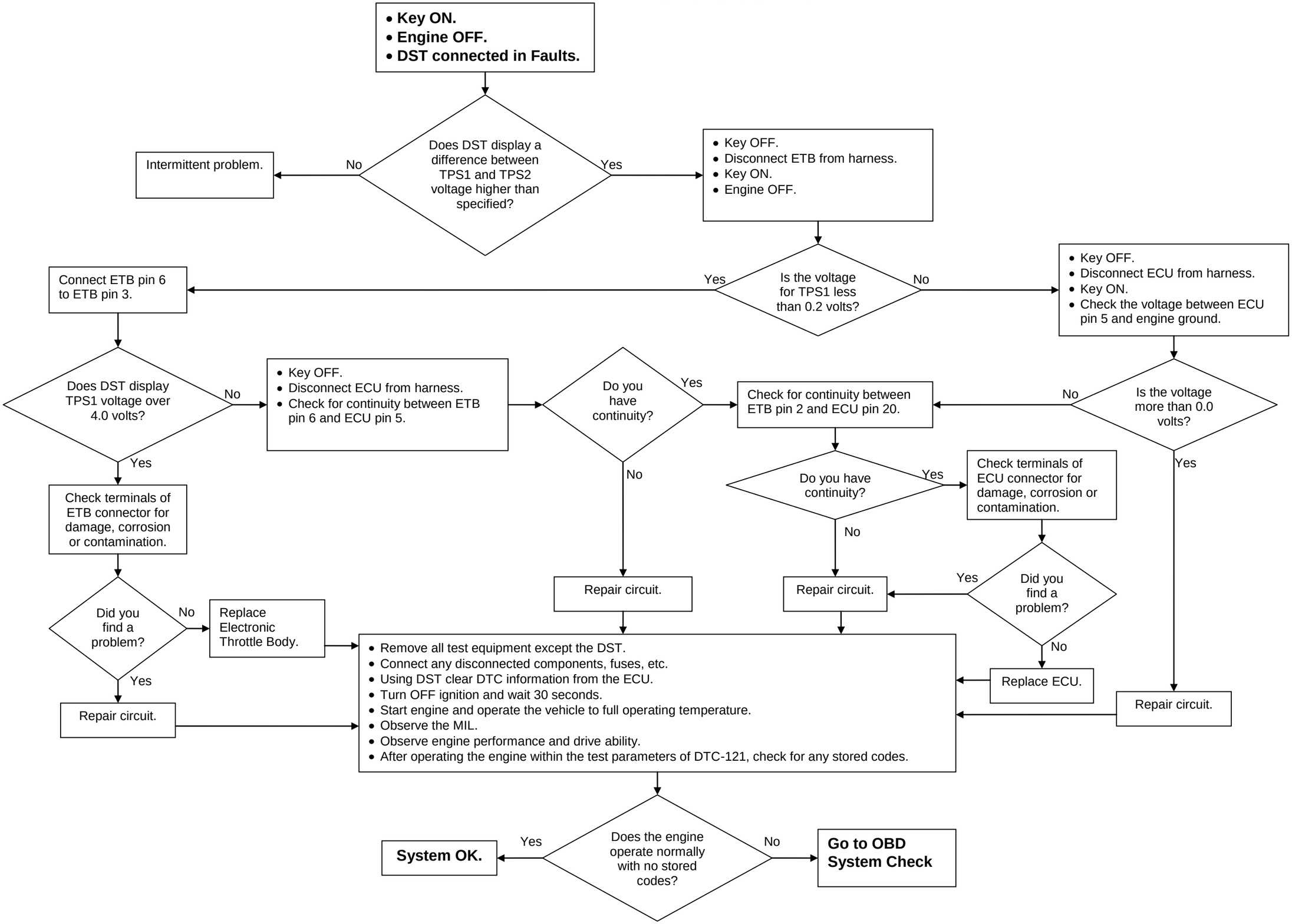
There are two Throttle Position Sensors (TPS) located within the throttle which use variable resistors to determine signal voltage based on throttle plate position. TPS1 will read low voltage when closed and TPS2 will read high voltage when closed. The TPS1 and TPS2 percentages are calculated from these voltages. Although the voltages are different, the calculated values for the throttle position percentages should be very close to the same. The TPS values are used by the ECU to determine if the throttle is opening as commanded. This fault will set if the difference between TPS1 and TPS2 is higher than specified.

For detailed information, please see customer specific DTC settings.



All information concerning connectors, pin numbers and wire colors can be found in the connector documentation.

DTC 121 – TPS1 lower than TPS2



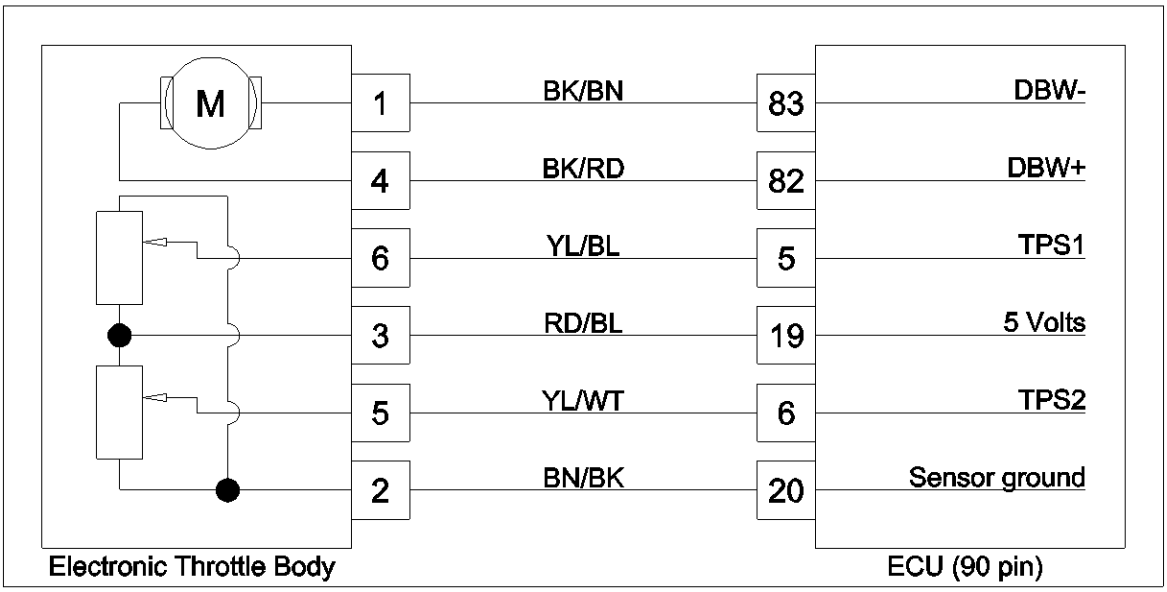
All measurements are done at the connector unless otherwise stated.

DTC 122 – TPS1 low voltage

General

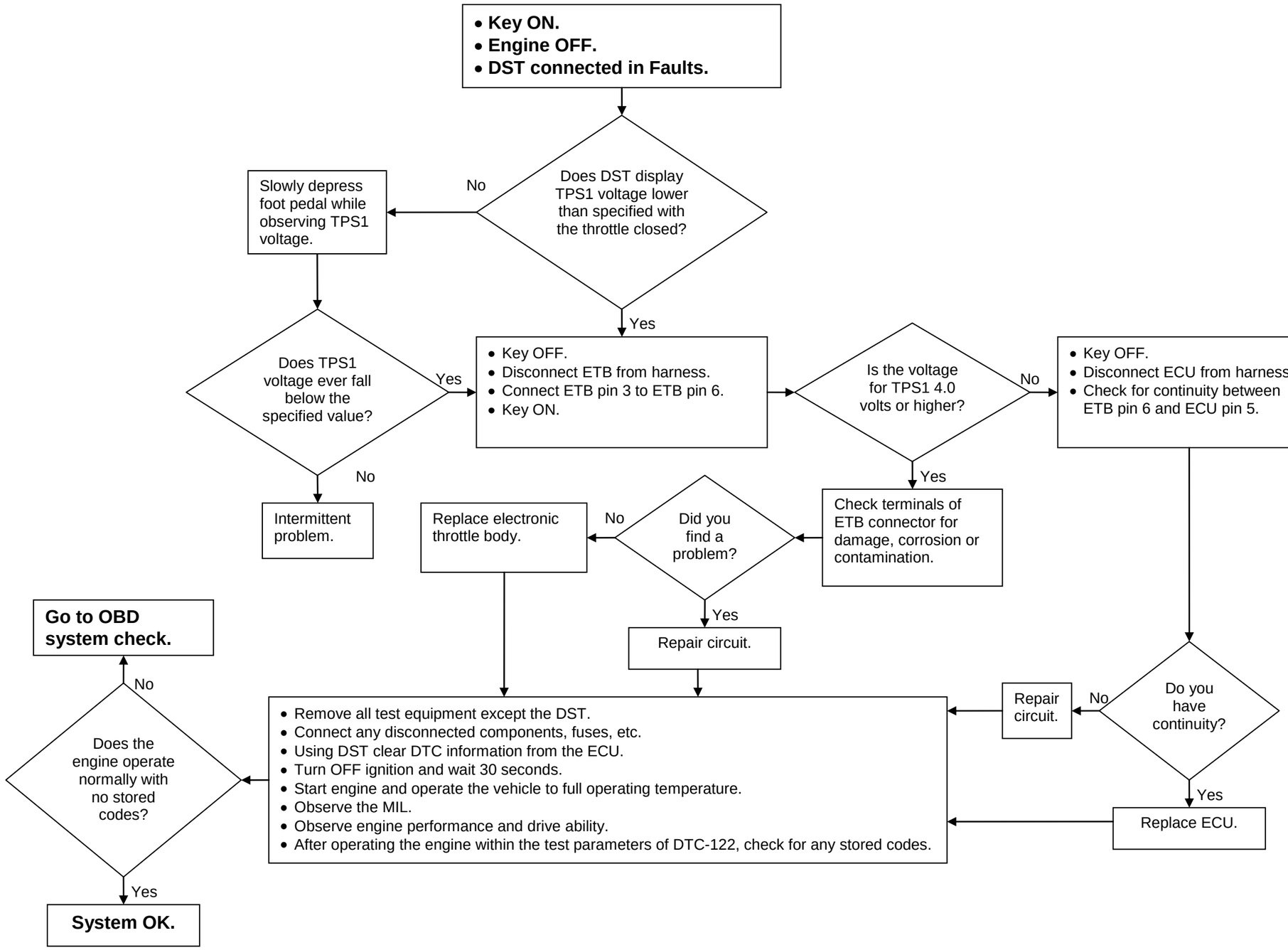
There are two Throttle Position Sensors (TPS) located within the throttle which use variable resistors to determine signal voltage based on throttle plate position. TPS1 will read low voltage when closed and TPS2 will read high voltage when closed. The TPS1 and TPS2 percentages are calculated from these voltages. Although the voltages are different, the calculated values for the throttle position percentages should be very close to the same. The TPS values are used by the ECU to determine if the throttle is opening as commanded. This fault will set if the sensor voltage is too low.

For detailed information, please see customer specific DTC settings.



All information concerning connectors, pin numbers and wire colors can be found in the connector documentation.

DTC 122 – TPS1 low voltage



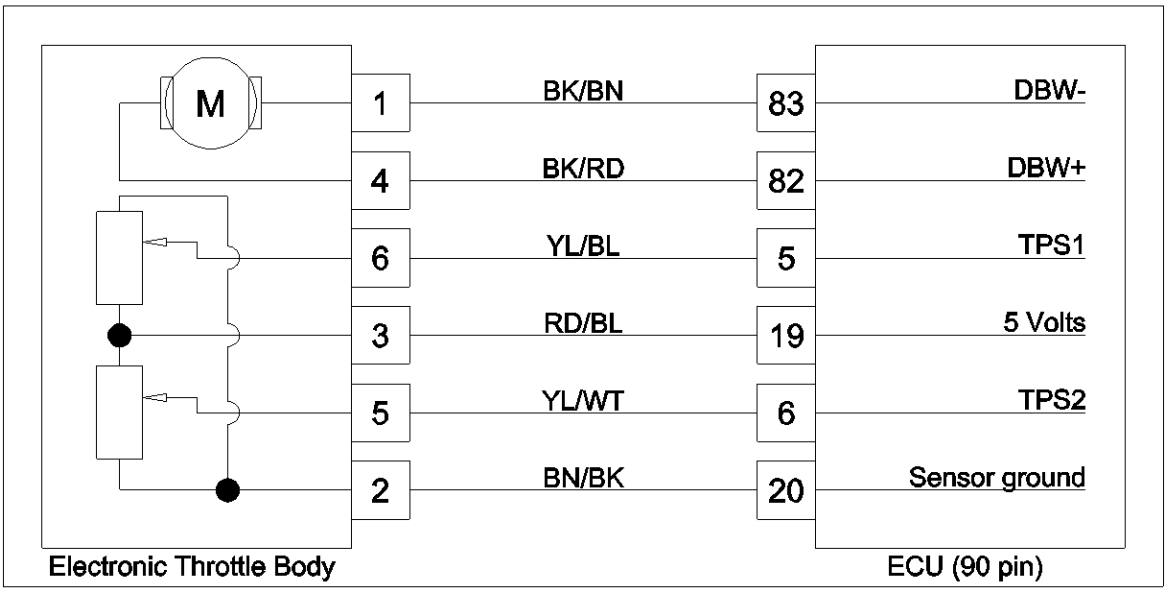
All measurements are done at the connector unless otherwise stated.

DTC 123 – TPS1 high voltage

General

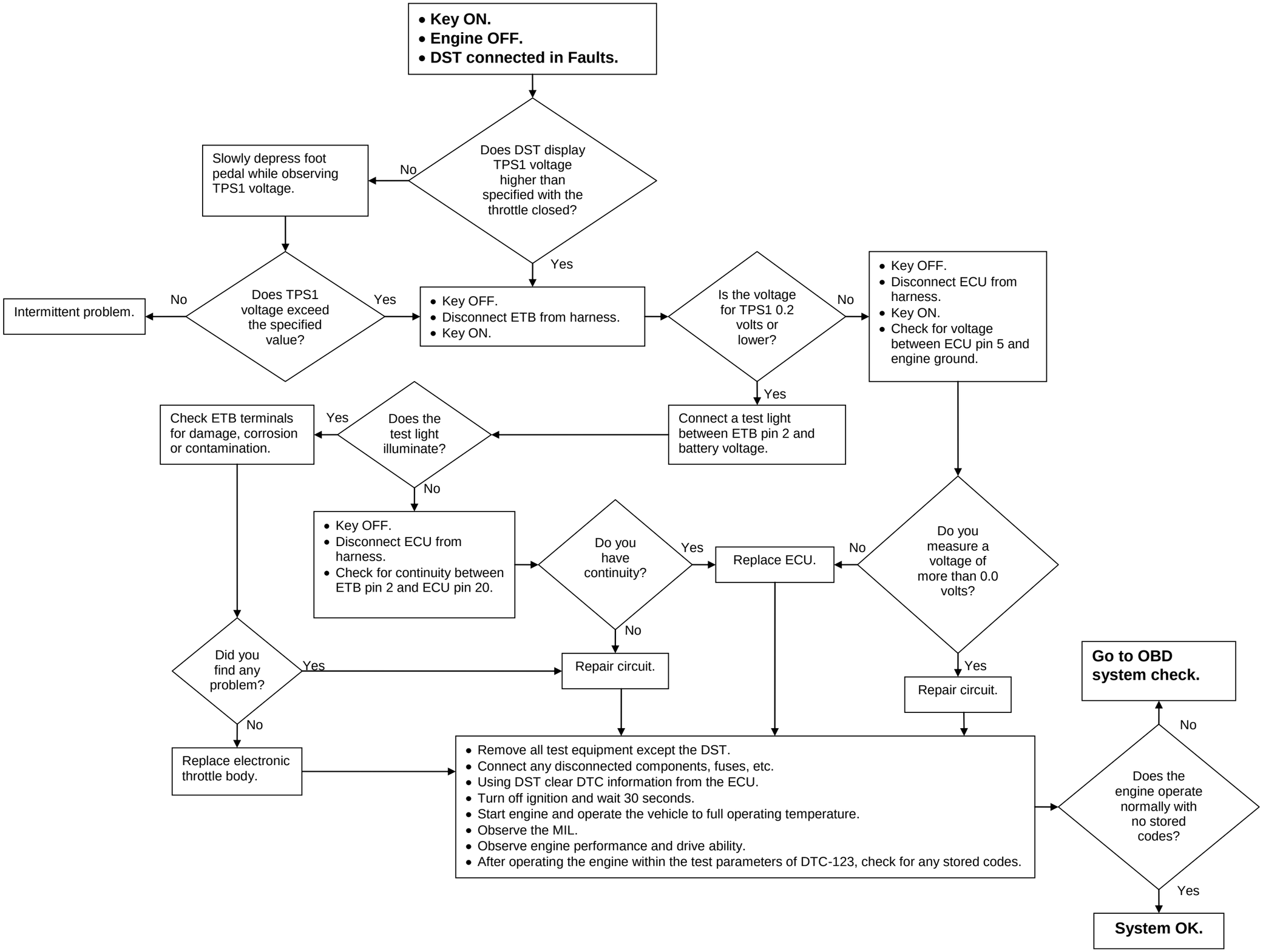
There are two Throttle Position Sensors (TPS) located within the throttle which use variable resistors to determine signal voltage based on throttle plate position. TPS1 will read low voltage when closed and TPS2 will read high voltage when closed. The TPS1 and TPS2 percentages are calculated from these voltages. Although the voltages are different, the calculated values for the throttle position percentages should be very close to the same. The TPS values are used by the ECU to determine if the throttle is opening as commanded. This fault will set if the sensor voltage is too high.

For detailed information, please see customer specific DTC settings.



All information concerning connectors, pin numbers and wire colors can be found in the connector documentation.

DTC 123 – TPS1 high voltage



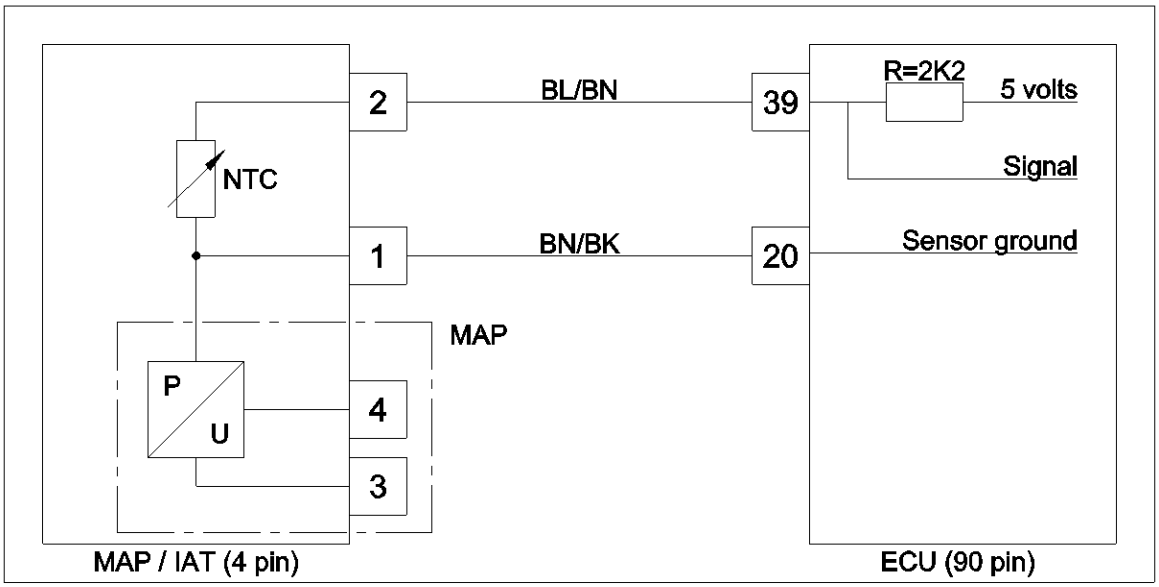
All measurements are done at the connector unless otherwise stated.

DTC 127 – IAT higher than expected 2

General

The Intake Air Temperature (IAT) is used to monitor incoming air temperature to help calculate the engines airflow requirement. The ECU provides a voltage divider circuit so that when the air is cold, the signal reads higher voltage, and lower voltage when warm. The IAT sensor is combined with the MAP sensor. This fault will set if the IAT is too high.

For detailed information, please see customer specific DTC settings.



All information concerning connectors, pin numbers and wire colors can be found in the connector documentation.

DTC 127 – IAT higher than expected 2

Trouble shooting

- This fault will set when the inlet air temperature is much higher than normal. The most common cause of high inlet air temperature is a problem with the inlet air system. Ensure that the air inlet is not obstructed, modified or damaged.

- Inspect the air inlet system for cracks or breaks that may allow unwanted under hood air into the air inlet system.

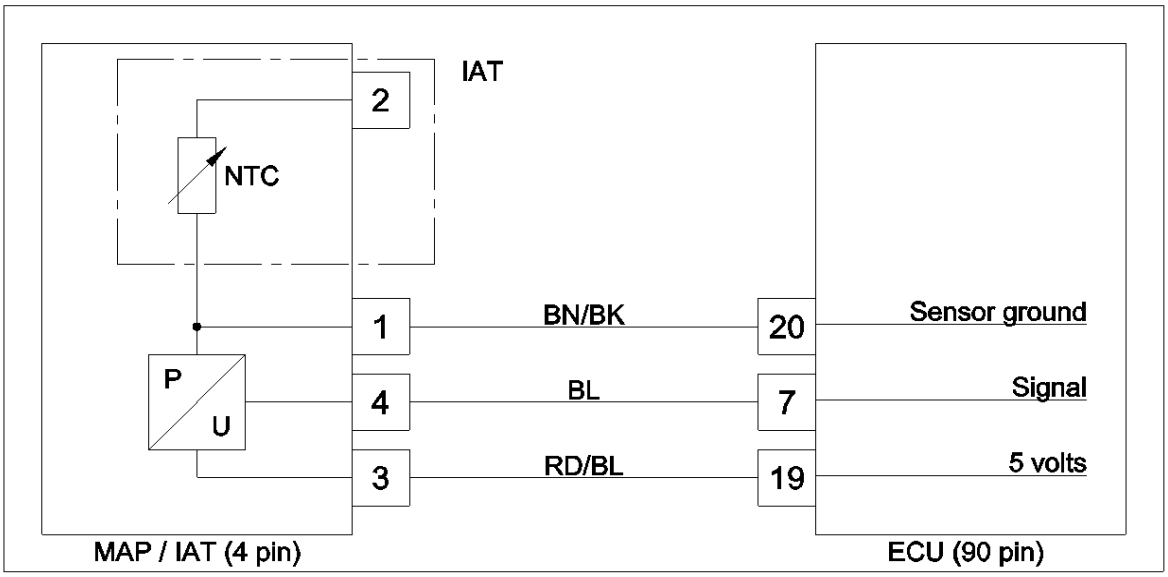
- If none of the above can be found, follow the diagnostic steps for DTC 112-IAT low voltage.

DTC 129 – BP low pressure

General

The Barometric Pressure (BP) is estimated from the MAP sensor. The barometric pressure value is used for fuel and airflow calculations. This fault will set if the BP is too low.

For detailed information, please see customer specific DTC settings.



All information concerning connectors, pin numbers and wire colors can be found in the connector documentation.

DTC 129 – BP low pressure

- **Key ON.**
- **Engine OFF.**
- **DST connected in Faults**

Does the DST display Barometric Pressure lower than specified?

Yes:
- Key OFF.
- Disconnect MAP sensor from harness.
- Connect MAP pin 3 to MAP pin 4.
- Key ON.

No: Intermittent problem.

Does the DST display a Barometric Pressure of 16.00 psia or higher?

Yes: Check ECU and MAP connector pins for damage, corrosion or contamination.

Did you find a problem?

Yes: Repair circuit.

No:
- Key OFF.
- Disconnect ECU from harness.
- Check for continuity between MAP pin 4 and ECU pin 7.

Do you have continuity?

Yes: Check for continuity between MAP pin 3 and ECU pin 19.

Do you have continuity?

Yes: Check for continuity between MAP pin 1 and ECU pin 20.

Do you have continuity?

Yes: Replace MAP sensor.

No: Repair circuit.

No: Repair circuit.

No:
- Remove connection between MAP pin 3 and MAP pin 4.
- Connect a test light between MAP pin 4 and battery voltage.

Does the DST display a Barometric Pressure of 16.00 psia or higher?

No:
- Disconnect ECU from harness.
- Check for continuity between MAP pin 4 and ECU pin 7.

Do you have continuity?

Yes: Check for continuity between MAP pin 4 and engine ground.

Do you have continuity?

Yes: Did you find a problem?

Yes: Repair circuit.

No: Replace ECU.

No: Check MAP and ECU connector pins for damage, corrosion or contamination.

Yes:
- Key OFF.
- Disconnect ECU from harness.
- Check for continuity between MAP pin 3 and ECU pin 19.

Do you have continuity?

Yes: Check for continuity between MAP pin 3 and engine ground.

Do you have continuity?

Yes: Repair circuit.

No: Check MAP and ECU connector pins for damage, corrosion or contamination.

Did you find a problem?

Yes: Repair circuit.

No: Replace ECU.

- Remove all test equipment except the DST.
- Connect any disconnected components, fuses, etc.
- Using DST clear DTC information from the ECU.
- Turn OFF ignition and wait 30 seconds.
- Start engine and operate the vehicle to full operating temperature.
- Observe the MIL.
- Observe engine performance and drive ability.
- After operating the engine within the test parameters of DTC-129, check for any stored codes.

Does the engine operate normally with no stored codes?

Yes: **System OK.**

No: **Go to OBD system check.**

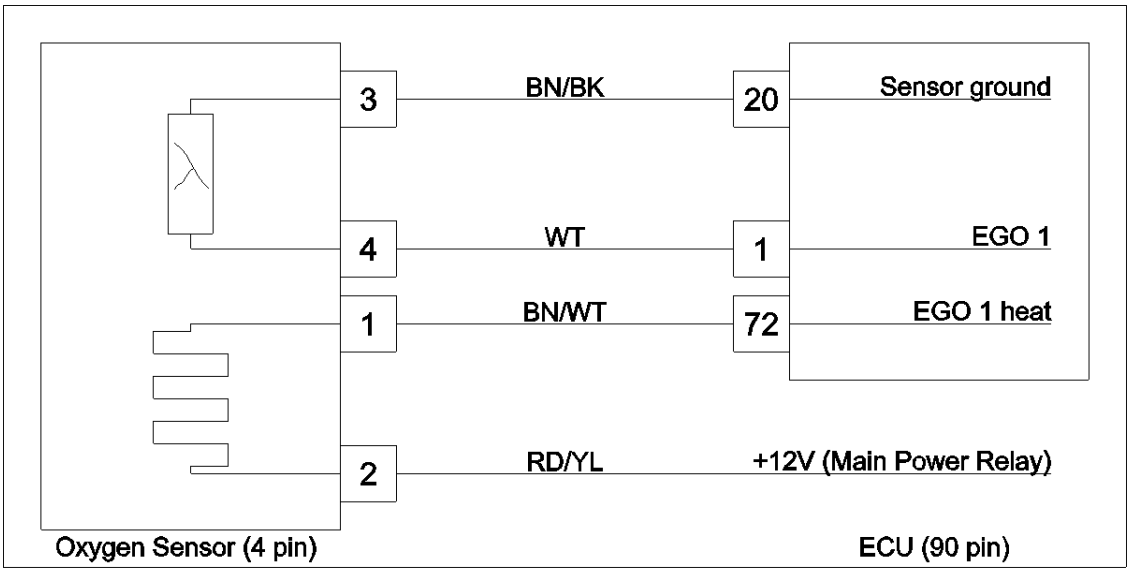
All measurements are done at the connector unless otherwise stated.

DTC 134 – EGO open / lazy pre - cat 1

General

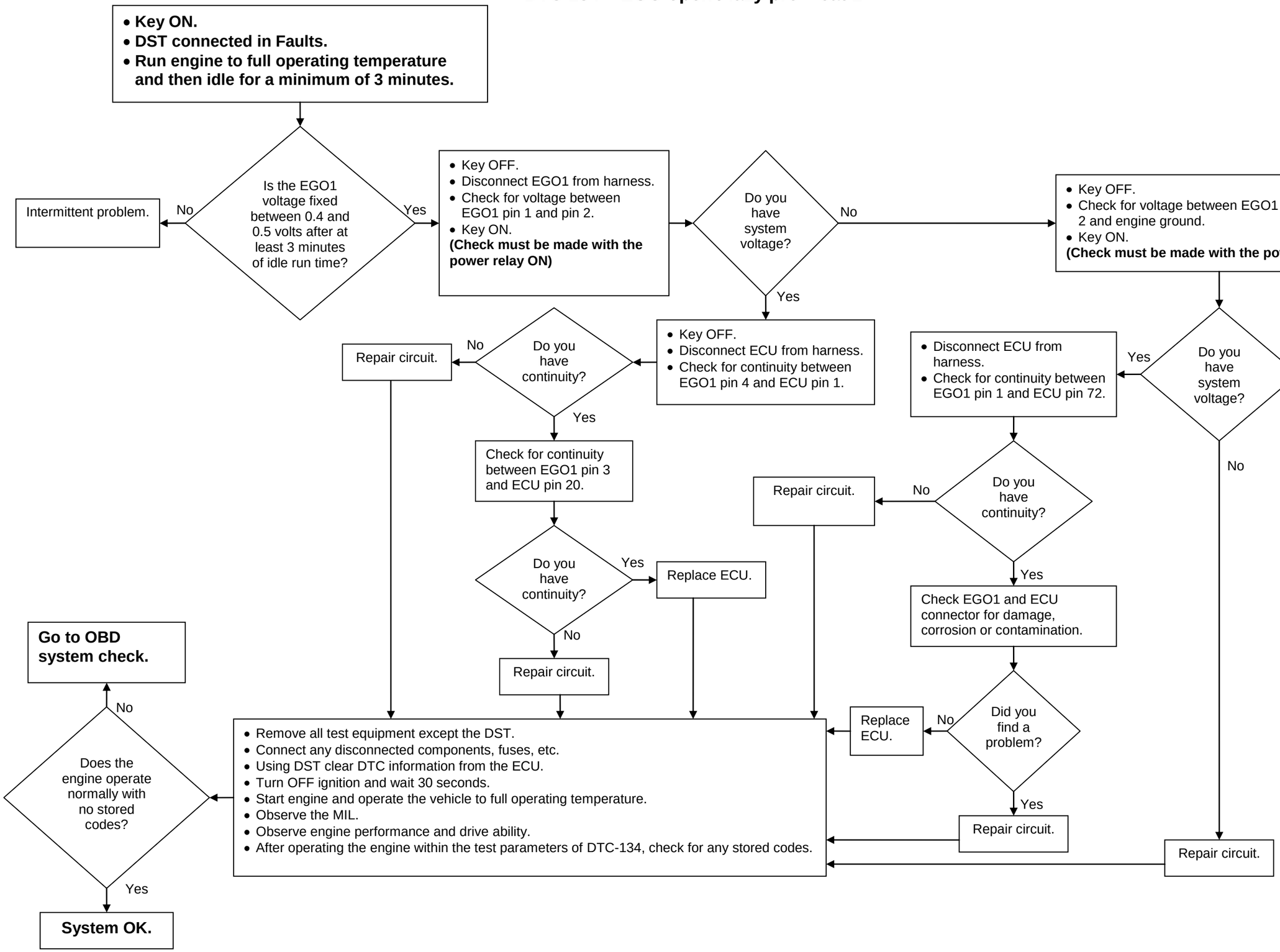
The pre-catalyst Exhaust Gas Oxygen (EGO) sensor is a switching-type sensor around stoichiometry that measures the oxygen content present in the exhaust to determine if the fuel flow to the engine is correct. If there is a deviation between the expected reading and the actual reading, fuel flow is precisely adjusted using the closed loop multiplier and then learned with the adaptive multiplier. This fault will set if the sensor element is cold, non responsive or inactive.

For detailed information, please see customer specific DTC settings.



All information concerning connectors, pin numbers and wire colors can be found in the connector documentation.

DTC 134 – EGO open / lazy pre – cat 1



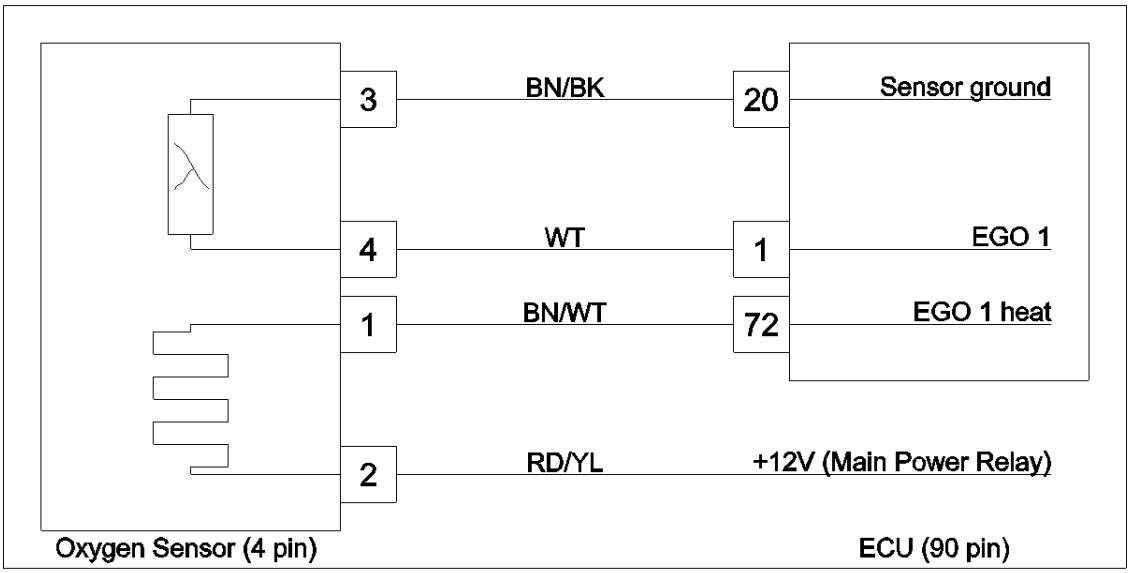
All measurements are done at the connector unless otherwise stated.

DTC 154 – EGO open / lazy post – cat 1

General

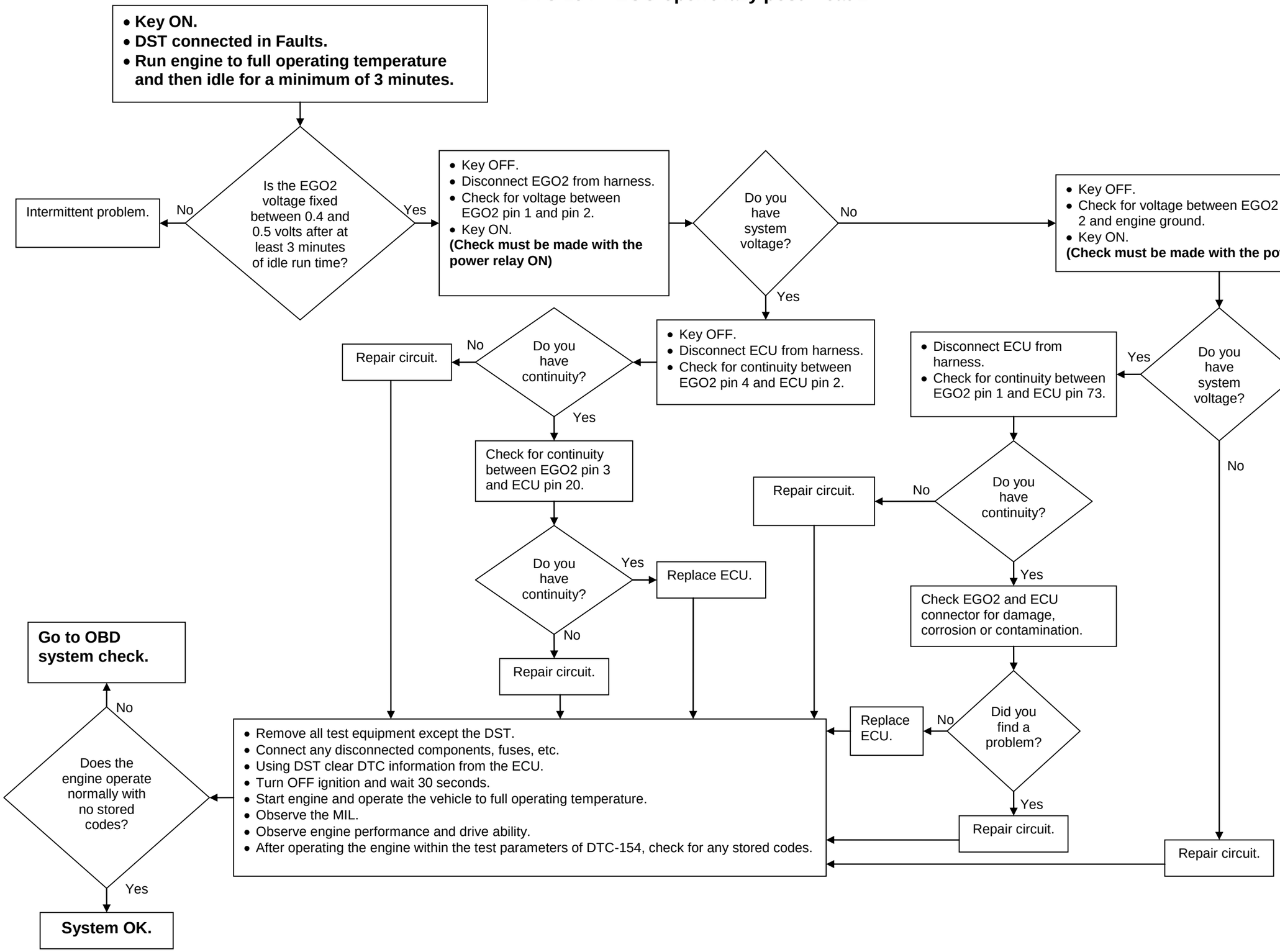
The post-catalyst Exhaust Gas Oxygen (EGO) sensor is a switching-type sensor around stoichiometry that measures the oxygen content downstream of the catalyst to trim the commanded equivalence ratio target to maximize the catalyst conversion efficiency. This fault will set if the sensor element is cold, non responsive or inactive.

For detailed information, please see customer specific DTC settings.



All information concerning connectors, pin numbers and wire colors can be found in the connector documentation.

DTC 154 – EGO open / lazy post – cat 1



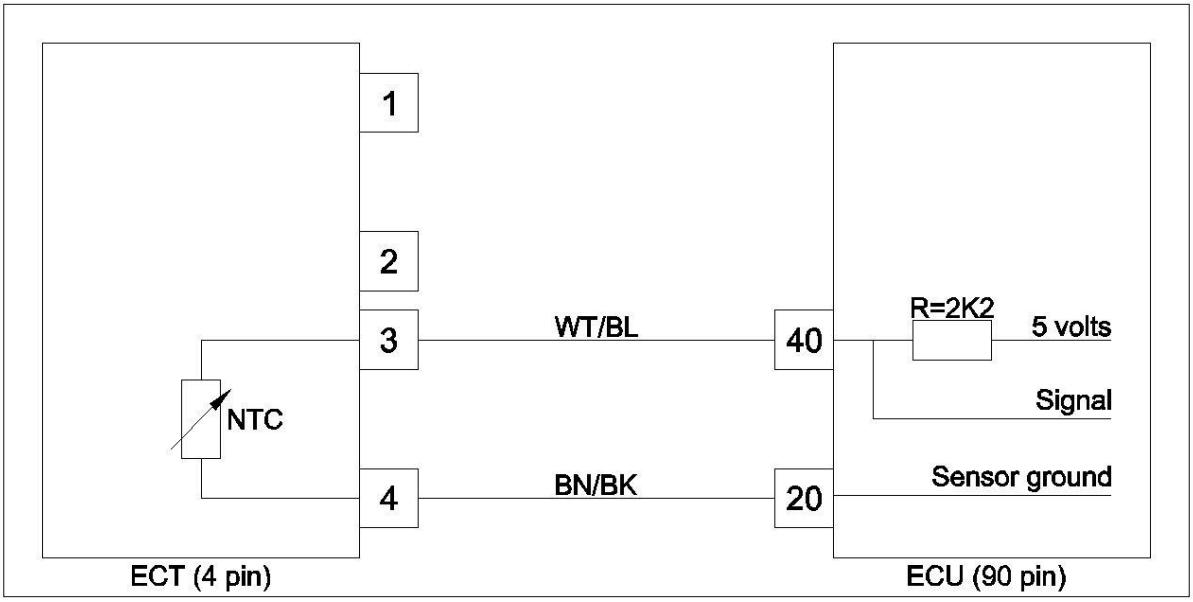
All measurements are done at the connector unless otherwise stated.

DTC 217 – ECT higher than expected 2

General

The Engine Coolant Temperature (ECT) sensor is a temperature sensitive resistor located in the engine coolant passage. It is used for the engine airflow calculation, fuel cold enrichment and to enable other temperature dependant features. The ECU provides a voltage divider circuit so that when the coolant is cold, the signal reads higher voltage, and lower when warm. This fault will set if the ECT is too high.

For detailed information, please see customer specific DTC settings.



All information concerning connectors, pin numbers and wire colors can be found in the connector documentation.

DTC 217 – ECT higher than expected 2

Trouble shooting

- This fault will set when the engine coolant temperature is much higher than normal. The most common cause of high engine coolant temperature is a problem with the engine cooling system. Ensure that the coolant level is OK, and the radiator not obstructed, blocked or damaged.

- Inspect the engine coolant system for cracks or breaks that may allow the cooling system to work inaccurate.

- If none of the above can be found, follow the diagnostic steps for DTC 117-ECT low voltage.

DTC 219 – Max govern speed override

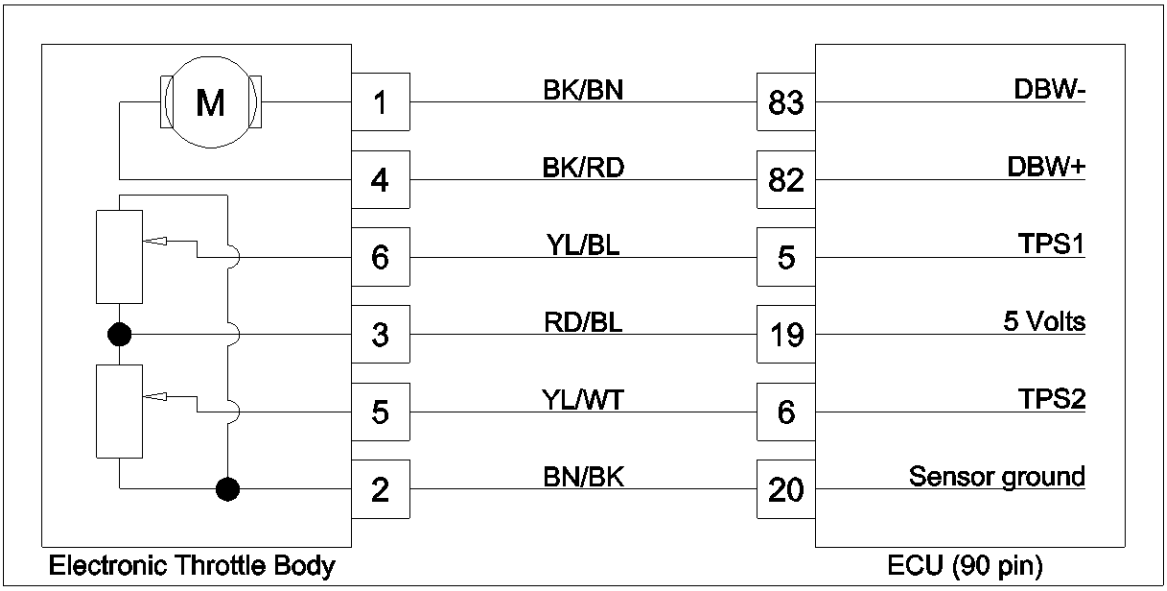
General

In case of an engine over speed, the governor (throttle valve) will be closed. This fault will set if the engine speed exceeds the specified speed.

For detailed information, please see customer specific DTC settings.

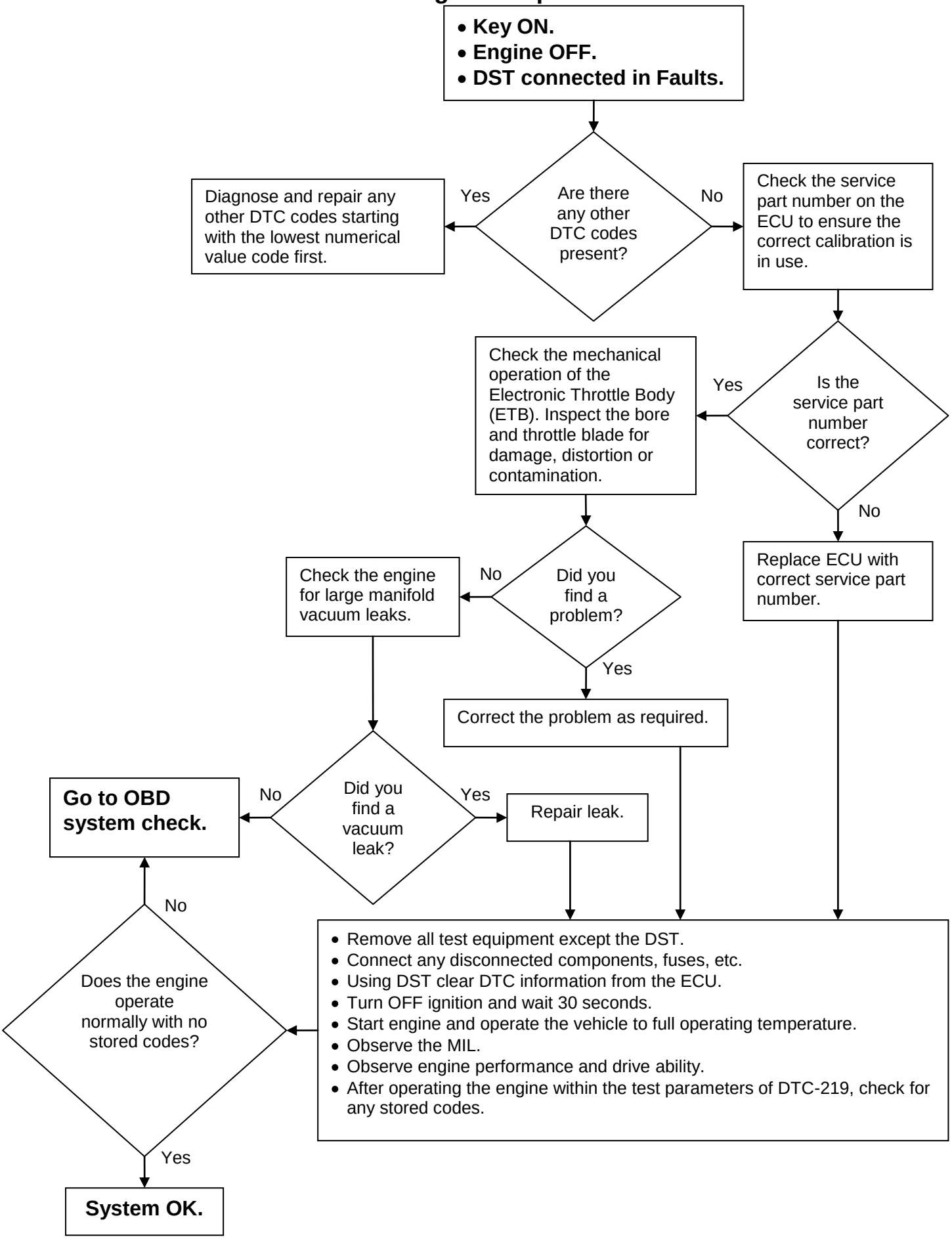
Diagnostic Aid

Check for other stored DTC codes before using the following diagnostic chart for this code set. It is quite likely that this DTC will be accompanied by other DTC sets. Always diagnose and repair any existing codes starting with the lowest numerical code first. Repairing the lower numerical value code will most likely correct this DTC.



All information concerning connectors, pin numbers and wire colors can be found in the connector documentation.

DTC 219 – Max govern speed override



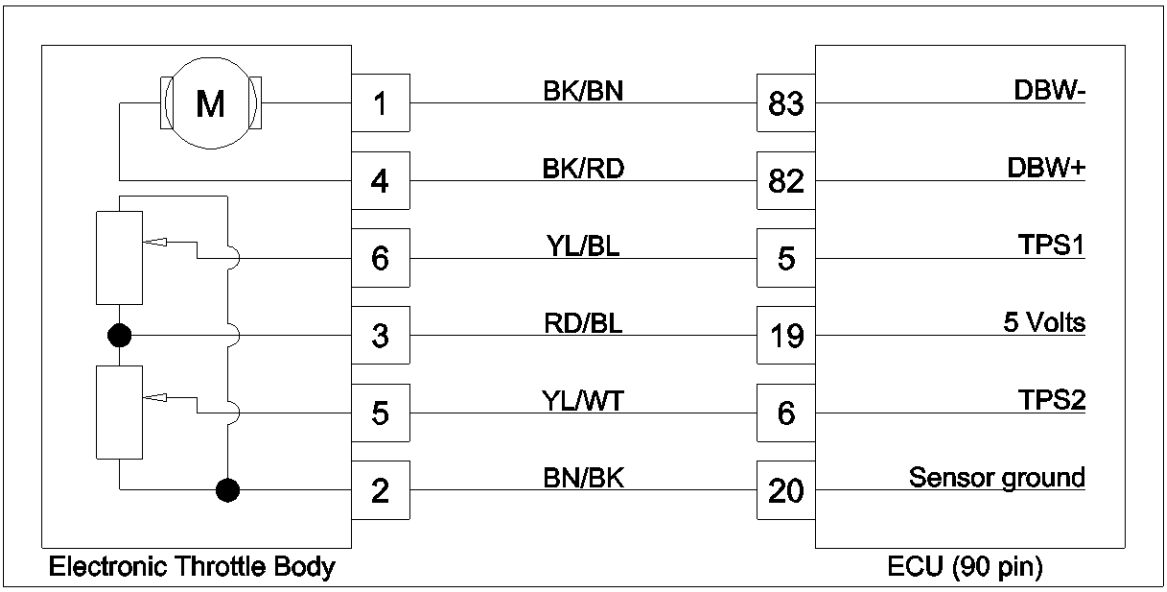
All measurements are done at the connector unless otherwise stated.

DTC 221 – TPS1 higher than TPS2

General

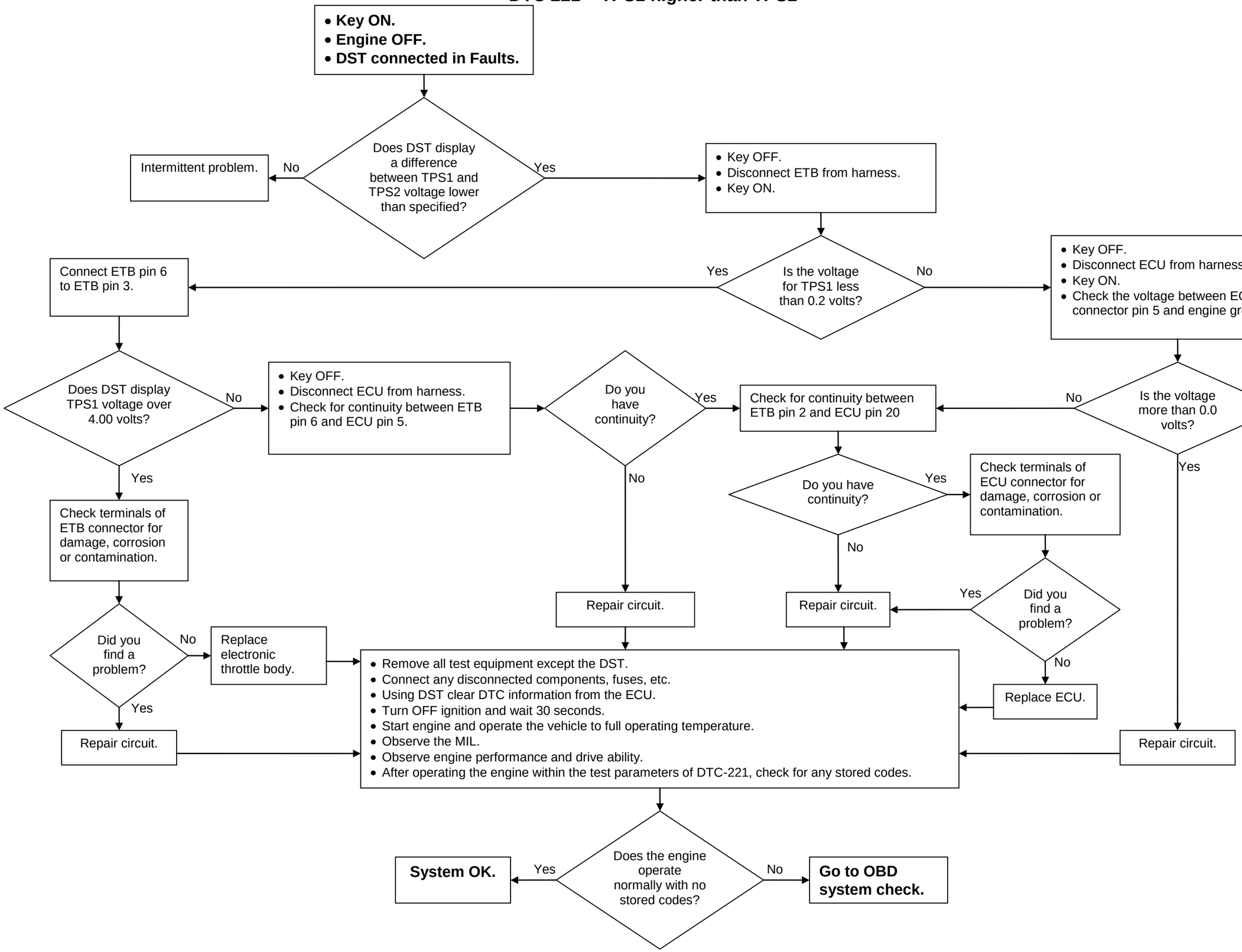
There are two Throttle Position Sensors (TPS) located within the throttle which use variable resistors to determine signal voltage based on throttle plate position. TPS1 will read low voltage when closed and TPS2 will read high voltage when closed. The TPS1 and TPS2 percentages are calculated from these voltages. Although the voltages are different, the calculated values for the throttle position percentages should be very close to the same. The TPS values are used by the ECU to determine if the throttle is opening as commanded. This fault will set if the difference between TPS1 and TPS2 is higher than specified.

For detailed information, please see customer specific DTC settings.



All information concerning connectors, pin numbers and wire colors can be found in the connector documentation.

DTC 221 – TPS1 higher than TPS2



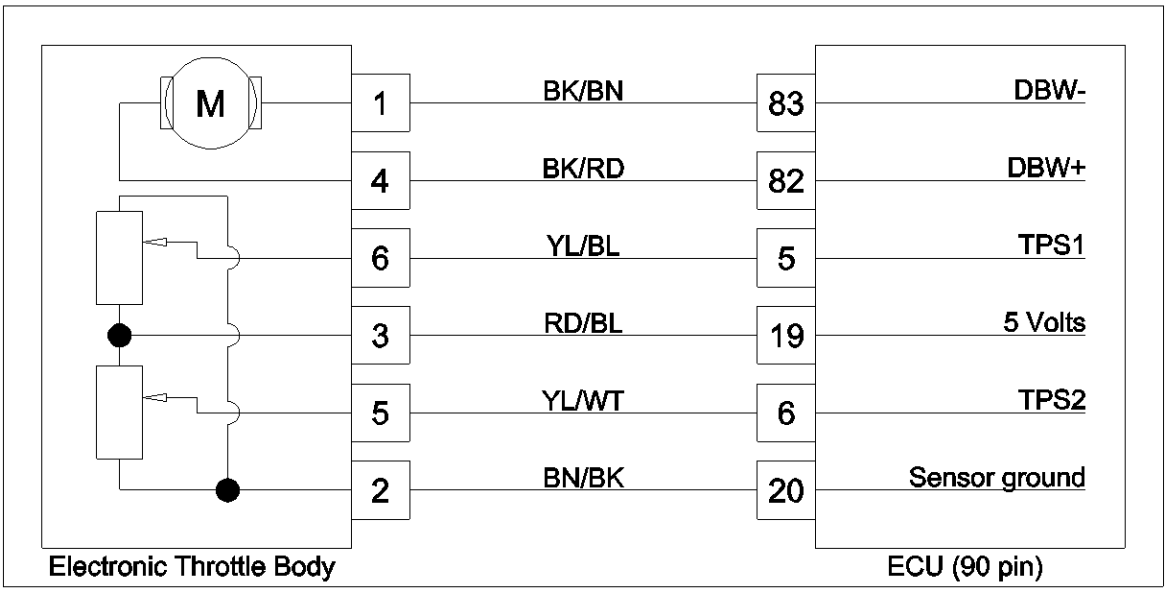
All measurements are done at the connector unless otherwise stated.

DTC 222 – TPS2 low voltage

General

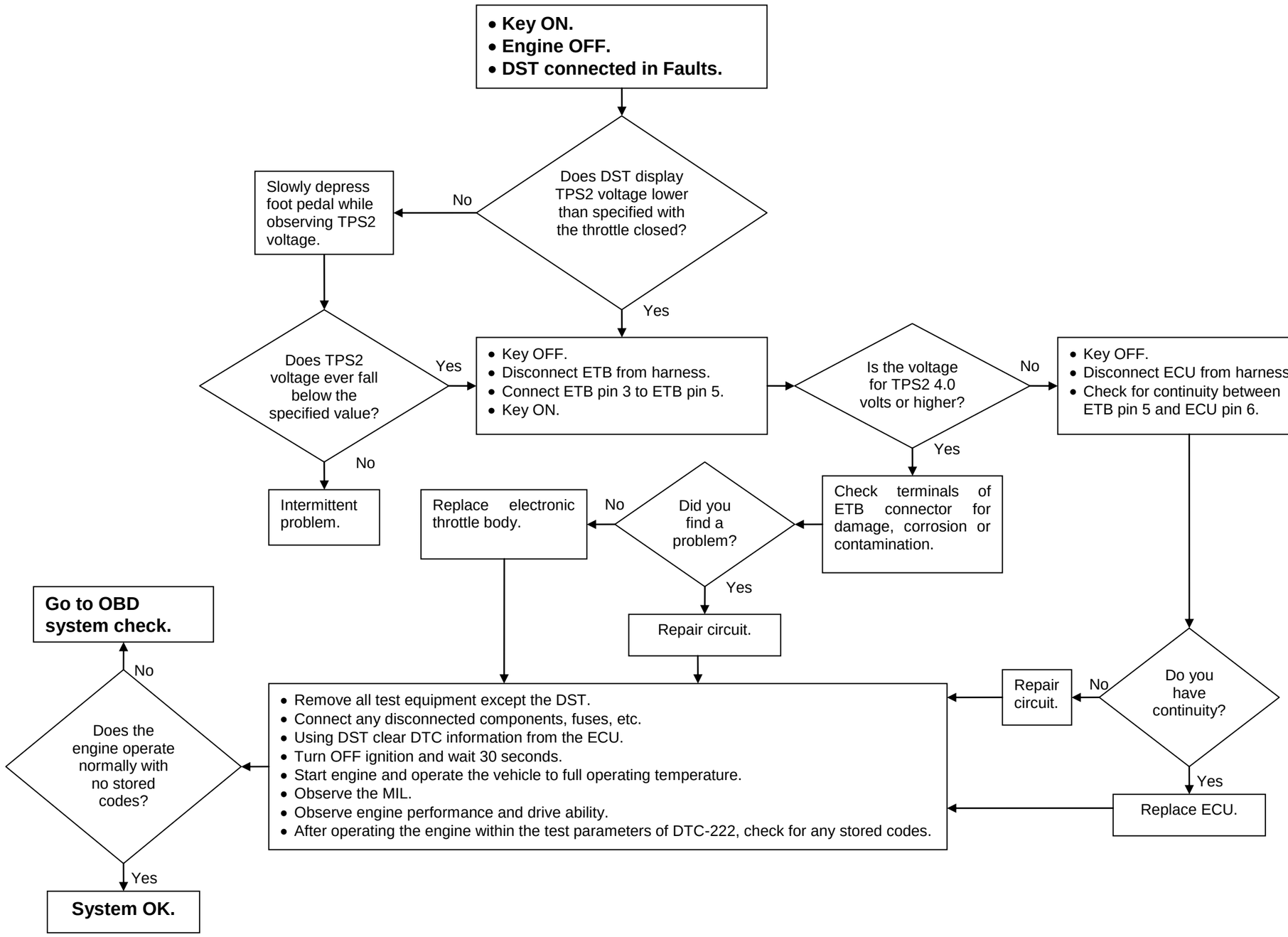
There are two Throttle Position Sensors (TPS) located within the throttle which use variable resistors to determine signal voltage based on throttle plate position. TPS1 will read low voltage when closed and TPS2 will read high voltage when closed. The TPS1 and TPS2 percentages are calculated from these voltages. Although the voltages are different, the calculated values for the throttle position percentages should be very close to the same. The TPS values are used by the ECU to determine if the throttle is opening as commanded. This fault will set if the sensor voltage is too low.

For detailed information, please see customer specific DTC settings.



All information concerning connectors, pin numbers and wire colors can be found in the connector documentation.

DTC 222 – TPS2 low voltage



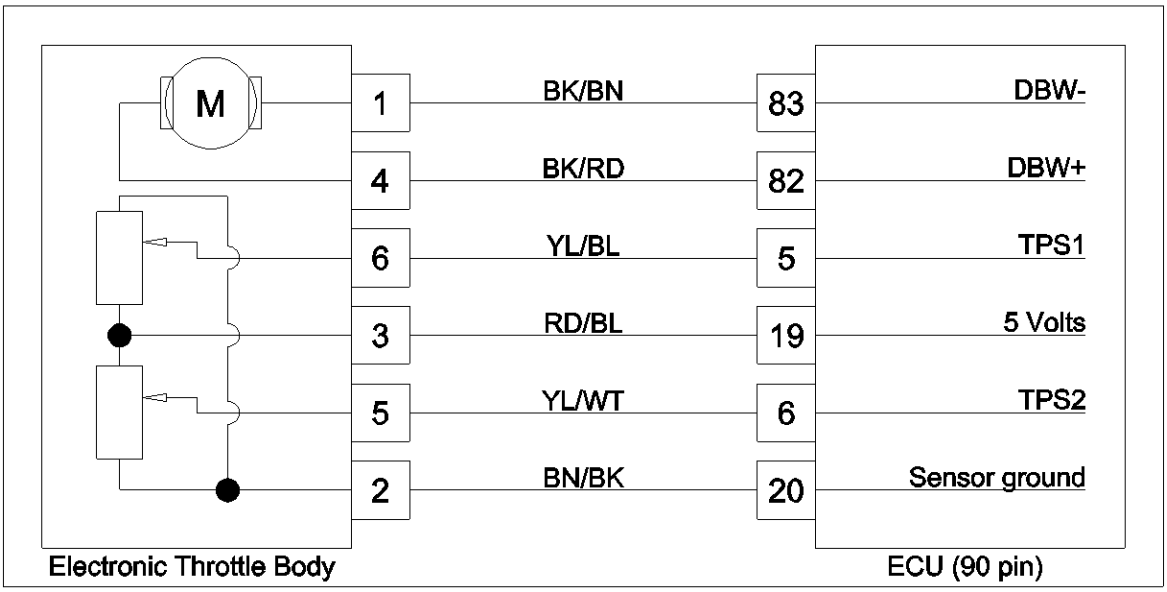
All measurements are done at the connector unless otherwise stated.

DTC 223 – TPS2 high voltage

General

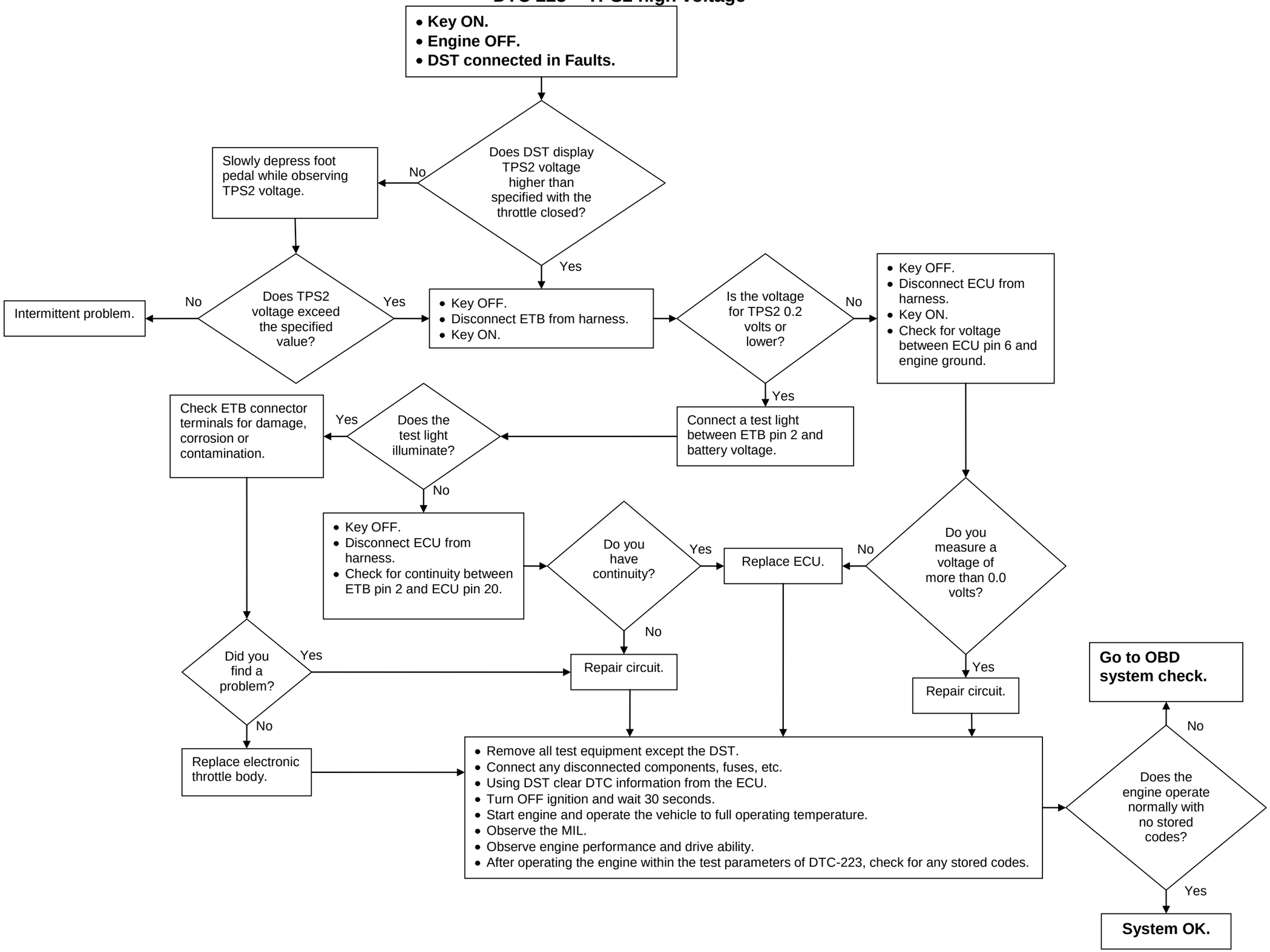
There are two Throttle Position Sensors (TPS) located within the throttle which use variable resistors to determine signal voltage based on throttle plate position. TPS1 will read low voltage when closed and TPS2 will read high voltage when closed. The TPS1 and TPS2 percentages are calculated from these voltages. Although the voltages are different, the calculated values for the throttle position percentages should be very close to the same. The TPS values are used by the ECU to determine if the throttle is opening as commanded. This fault will set if the sensor voltage is too high.

For detailed information, please see customer specific DTC settings.



All information concerning connectors, pin numbers and wire colors can be found in the connector documentation.

DTC 223 – TPS2 high voltage



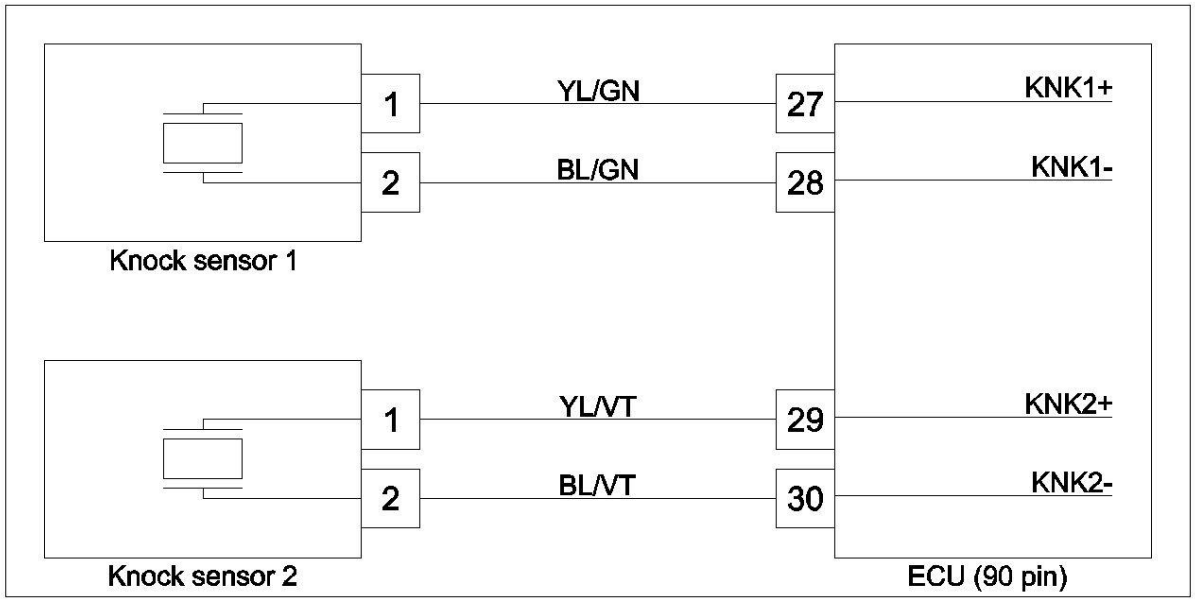
All measurements are done at the connector unless otherwise stated.

DTC 326 – Knock 1 excessive signal

General

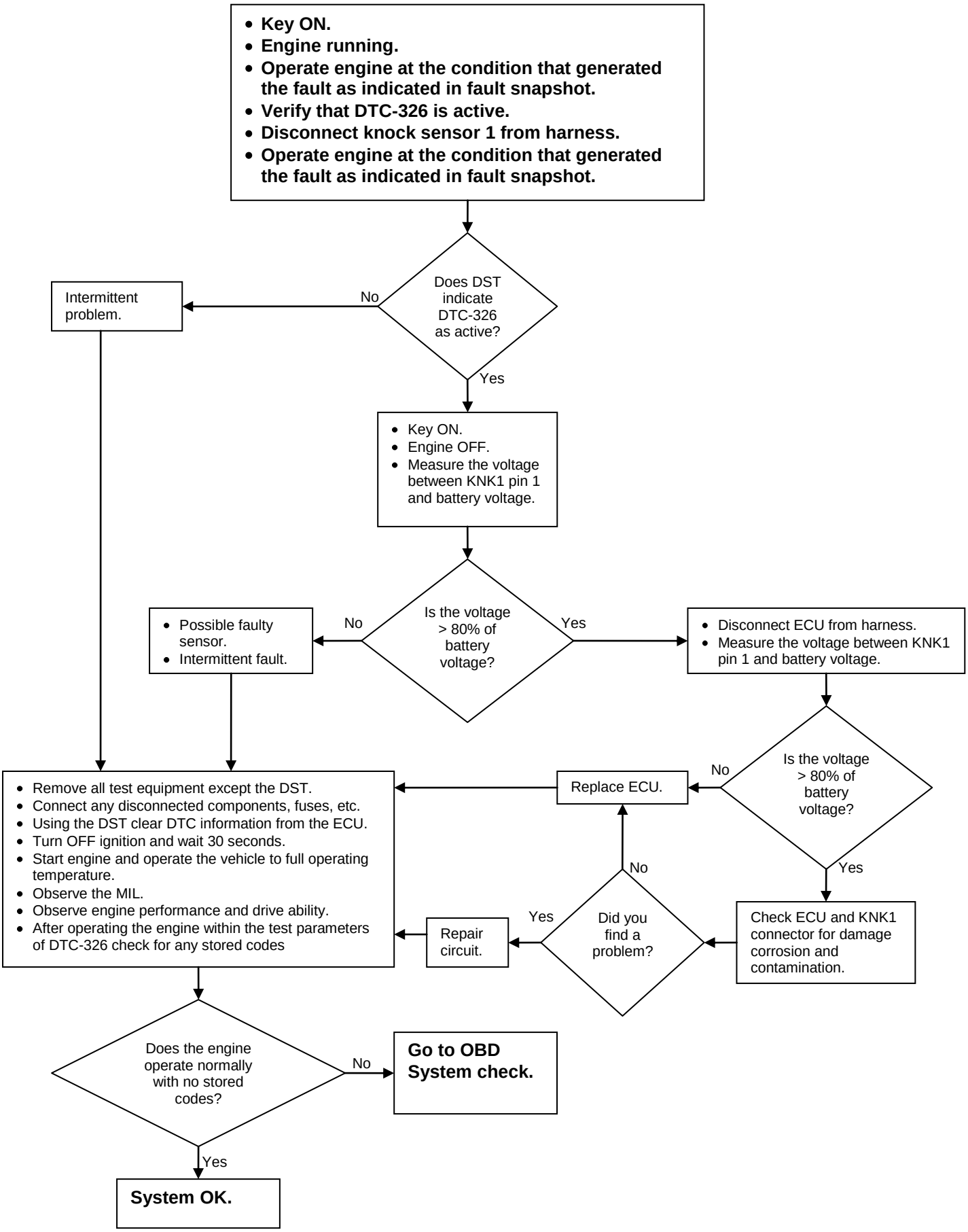
The Knock (KNK) sensor is used to detect detonation through mechanical vibration in the engine block and/or cylinder heads and provide feedback for the ignition system to retard spark to reduce knock intensity. In most applications the knock sensor is used to protect the engine from damage that can be caused from detonation or knock based on fixed spark advance. In other applications, the knock sensor is used to optimize spark advance and “learn” between spark tables based on fuel quality. This fault will set if the sensor voltage is too high.

For detailed information, please see customer specific DTC settings.



All information concerning connectors, pin numbers and wire colors can be found in the connector documentation.

DTC 326 – Knock 1 excessive signal



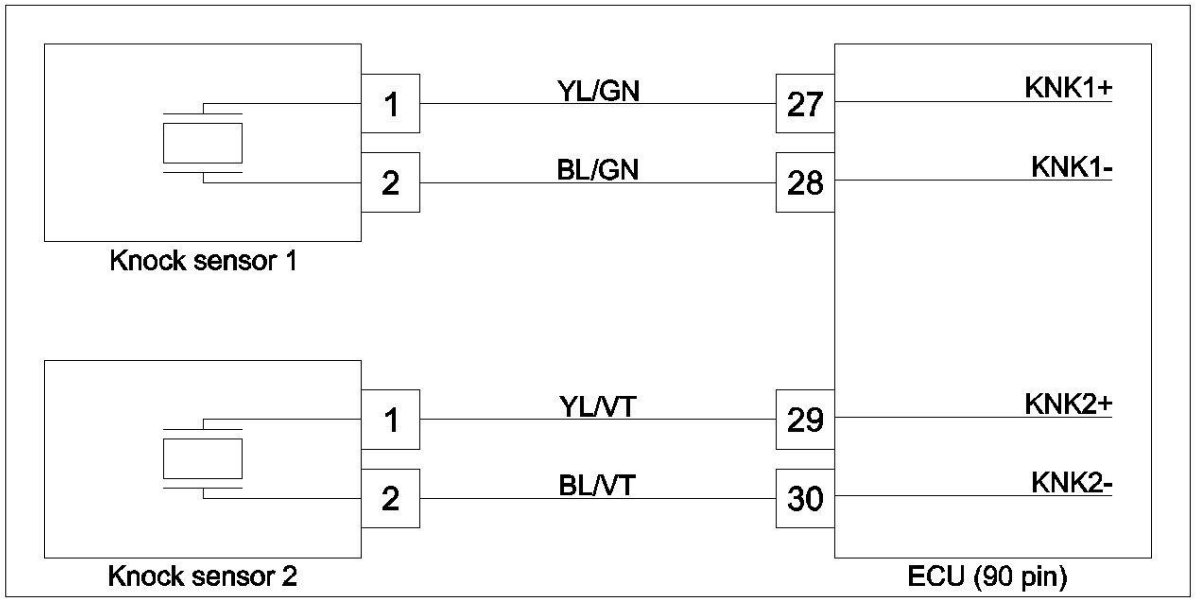
All measurements are done at the connector unless otherwise stated.

DTC 327 – Knock 1 sensor open

General

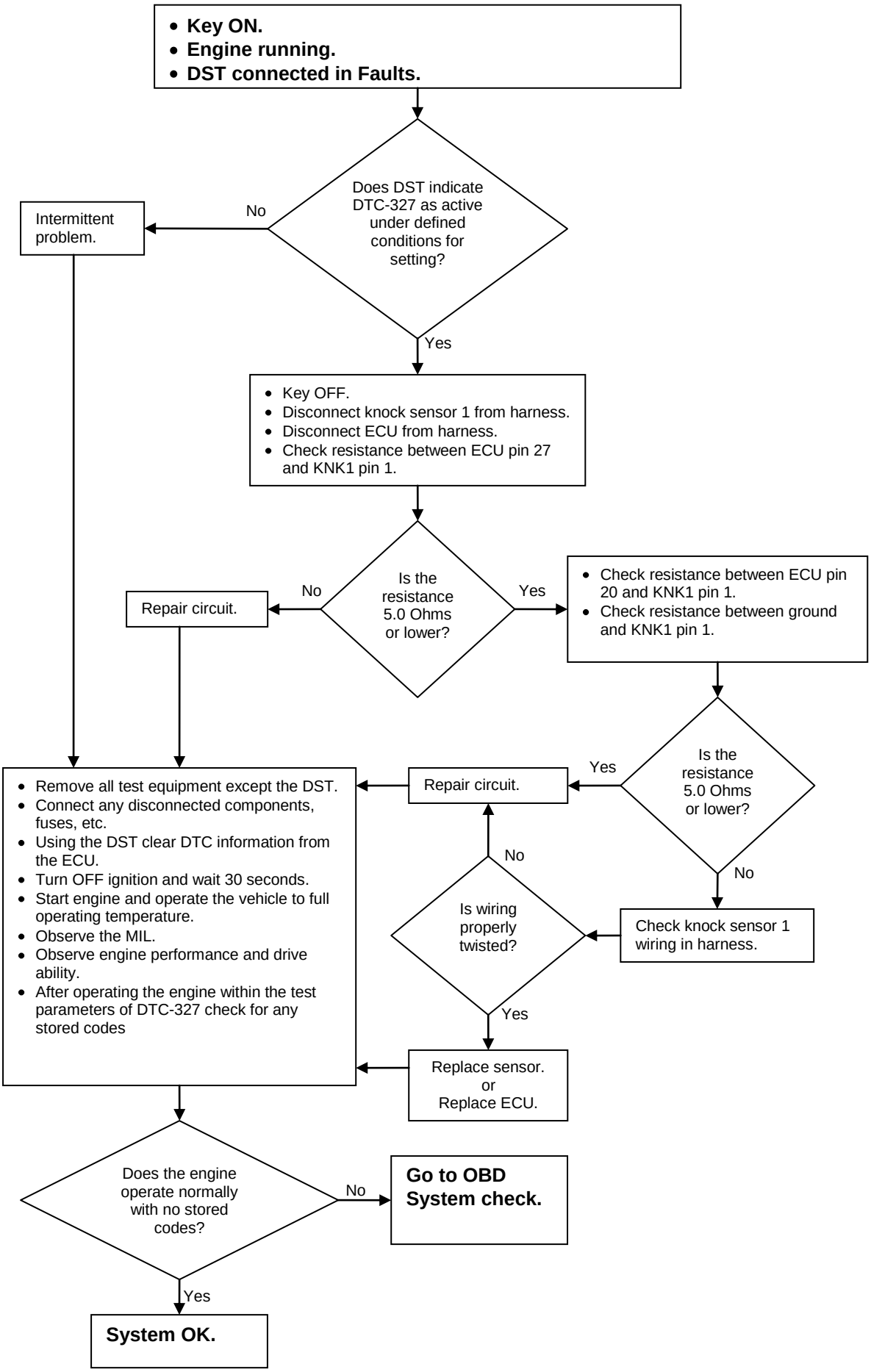
The Knock (KNK) sensor is used to detect detonation through mechanical vibration in the engine block and/or cylinder heads and provide feedback for the ignition system to retard spark to reduce knock intensity. In most applications the knock sensor is used to protect the engine from damage that can be caused from detonation or knock based on fixed spark advance. In other applications, the knock sensor is used to optimize spark advance and “learn” between spark tables based on fuel quality. This fault will set if the sensor voltage is too low.

For detailed information, please see customer specific DTC settings.



All information concerning connectors, pin numbers and wire colors can be found in the connector documentation.

DTC 327 – Knock 1 sensor open



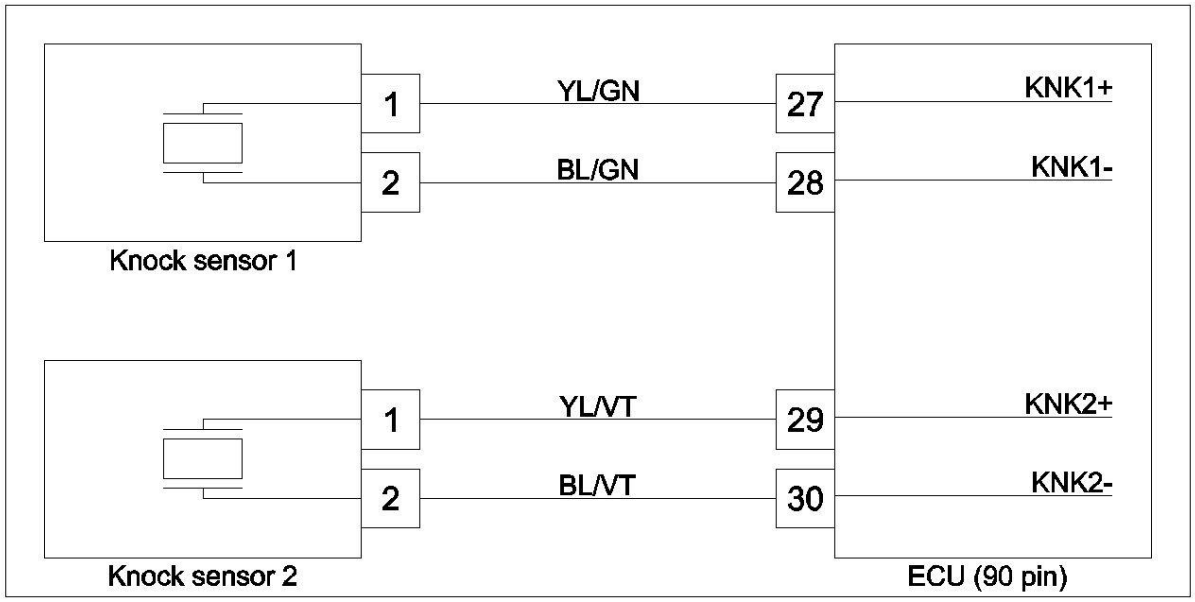
All measurements are done at the connector unless otherwise stated.

DTC 331 – Knock 2 excessive signal

General

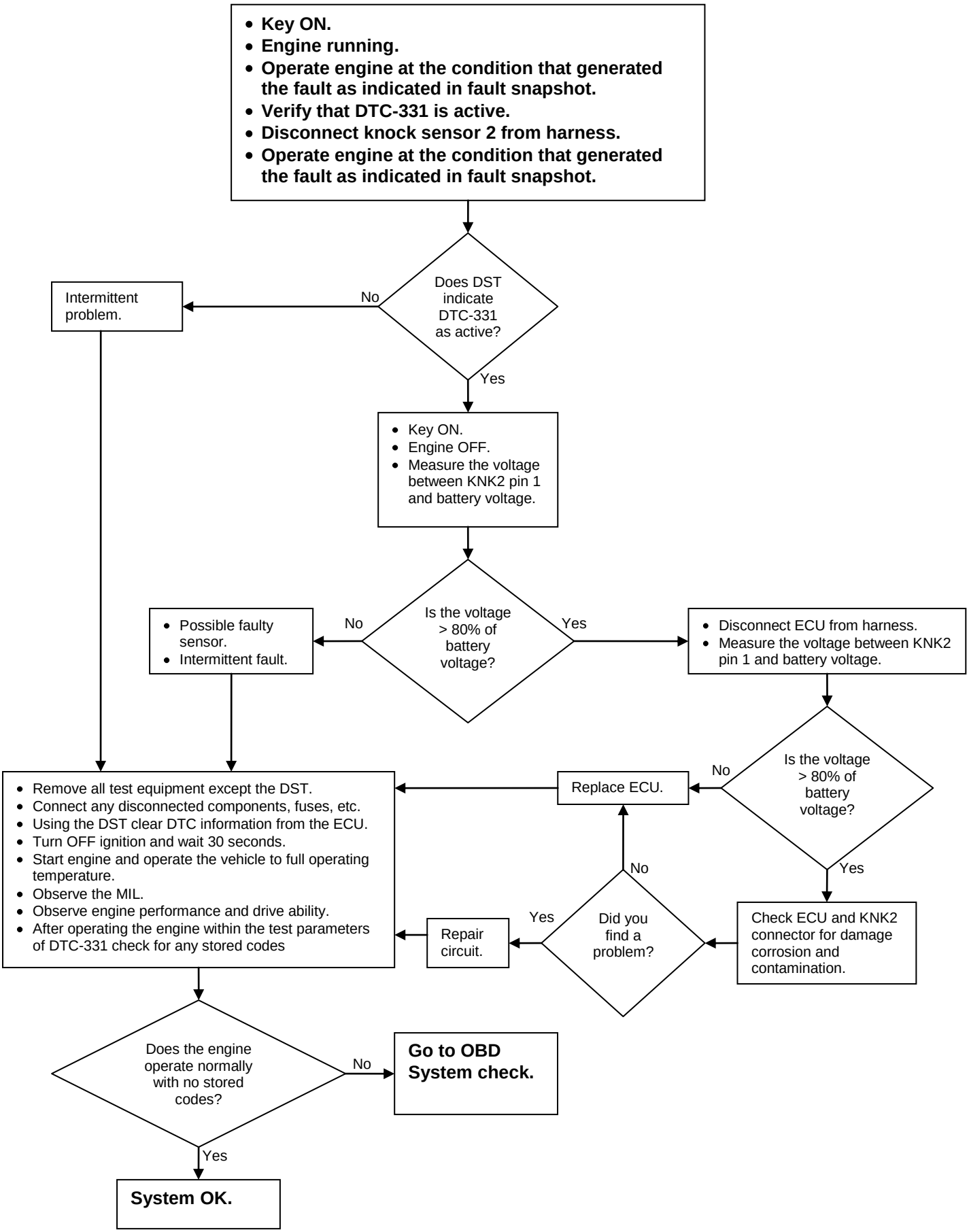
The Knock (KNK) sensor is used to detect detonation through mechanical vibration in the engine block and/or cylinder heads and provide feedback for the ignition system to retard spark to reduce knock intensity. In most applications the knock sensor is used to protect the engine from damage that can be caused from detonation or knock based on fixed spark advance. In other applications, the knock sensor is used to optimize spark advance and “learn” between spark tables based on fuel quality. This fault will set if the sensor voltage is too high.

For detailed information, please see customer specific DTC settings.



All information concerning connectors, pin numbers and wire colors can be found in the connector documentation.

DTC 331 – Knock 2 excessive signal



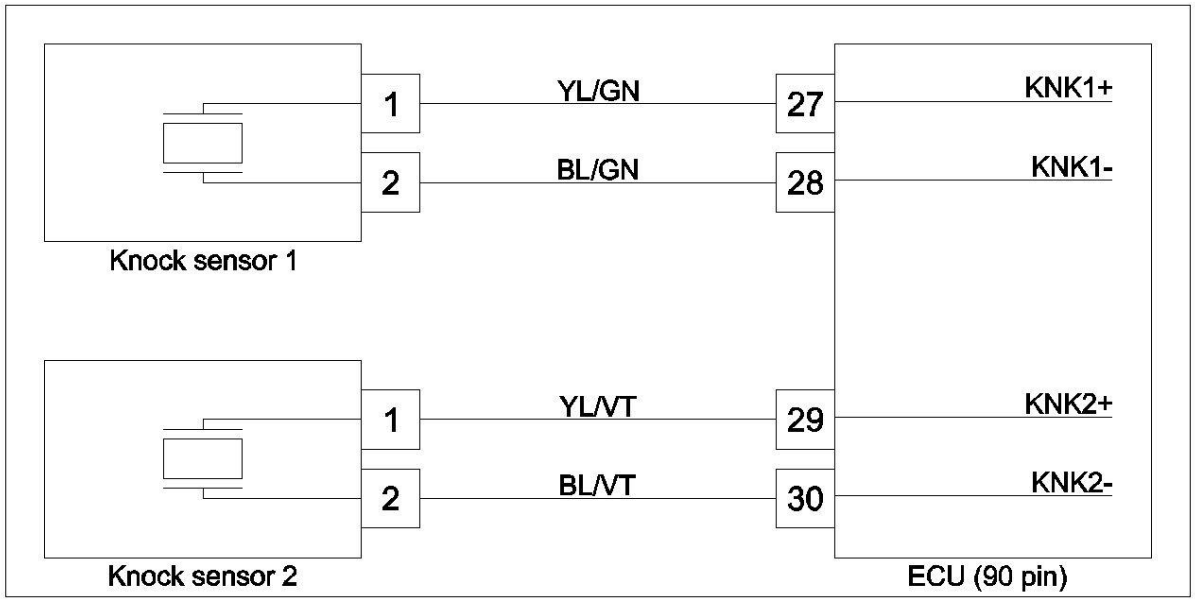
All measurements are done at the connector unless otherwise stated.

DTC 332 – Knock 2 sensor open

General

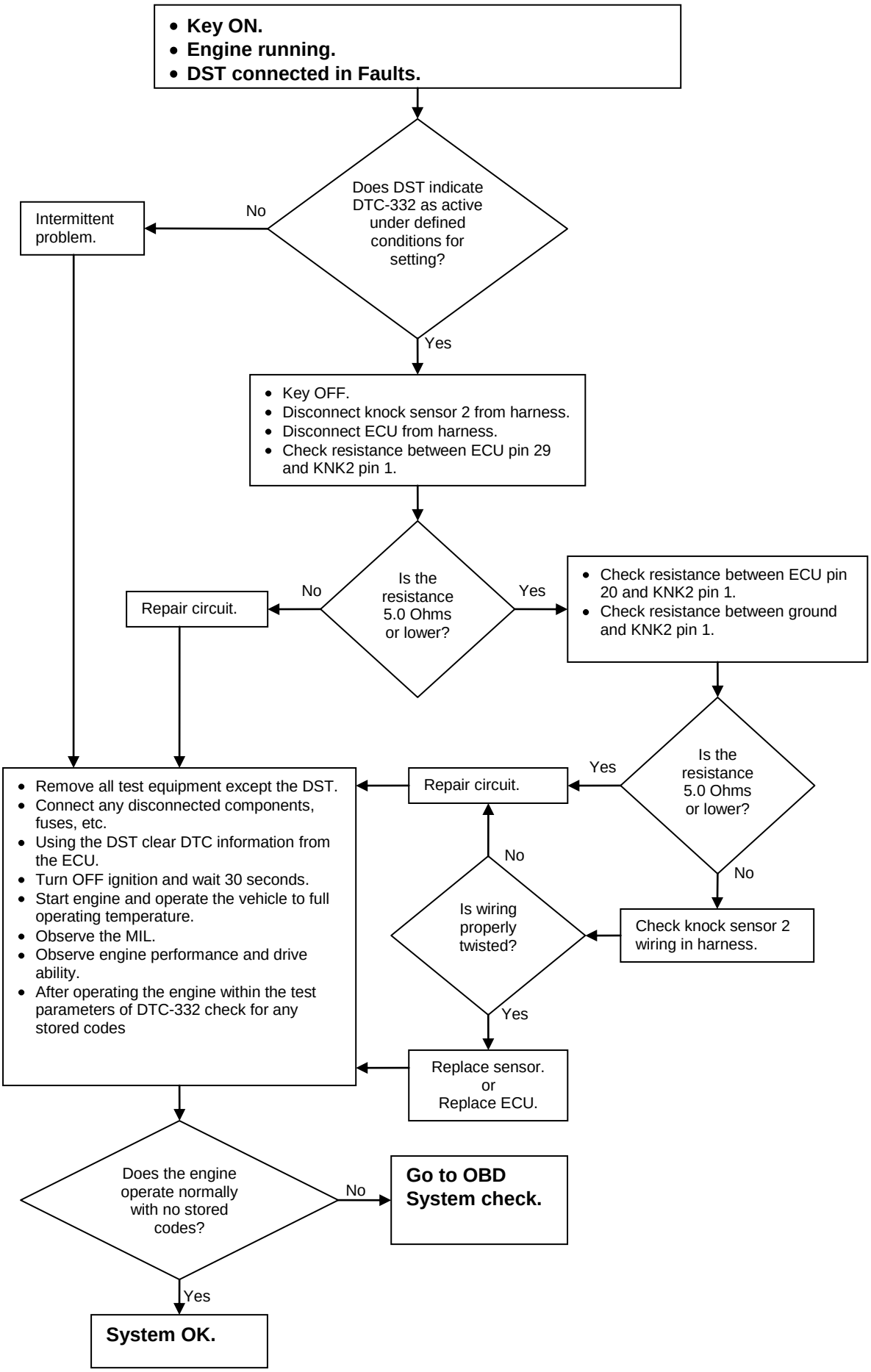
The Knock (KNK) sensor is used to detect detonation through mechanical vibration in the engine block and/or cylinder heads and provide feedback for the ignition system to retard spark to reduce knock intensity. In most applications the knock sensor is used to protect the engine from damage that can be caused from detonation or knock based on fixed spark advance. In other applications, the knock sensor is used to optimize spark advance and “learn” between spark tables based on fuel quality. This fault will set if the sensor voltage is too low.

For detailed information, please see customer specific DTC settings.



All information concerning connectors, pin numbers and wire colors can be found in the connector documentation.

DTC 332 – Knock 2 sensor open



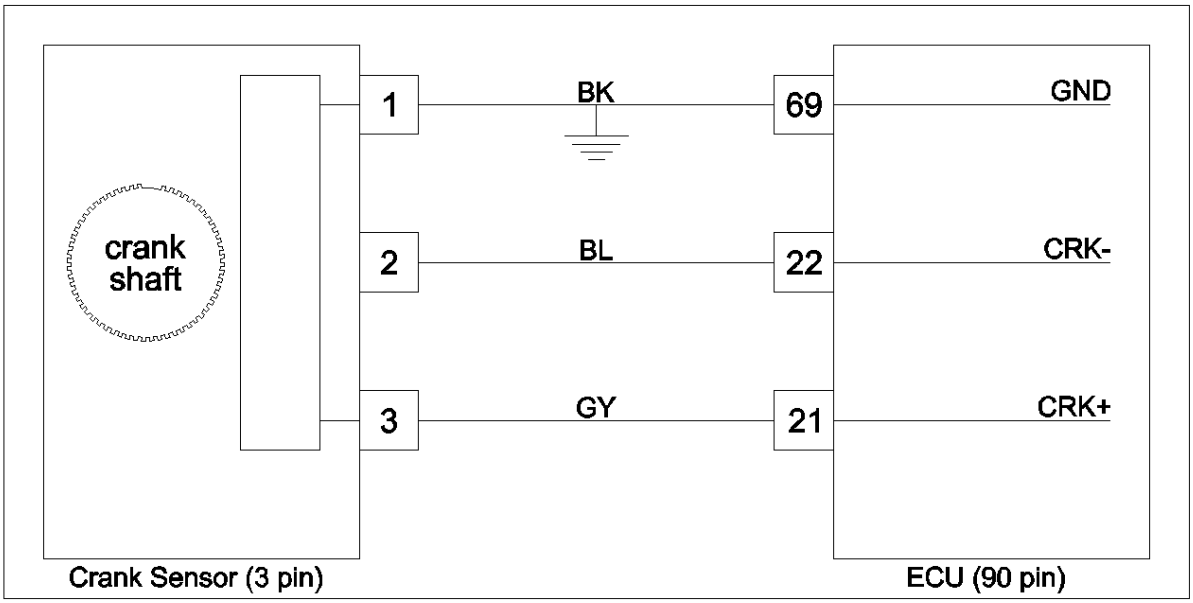
All measurements are done at the connector unless otherwise stated.

DTC 336 – Crank sync noise

General

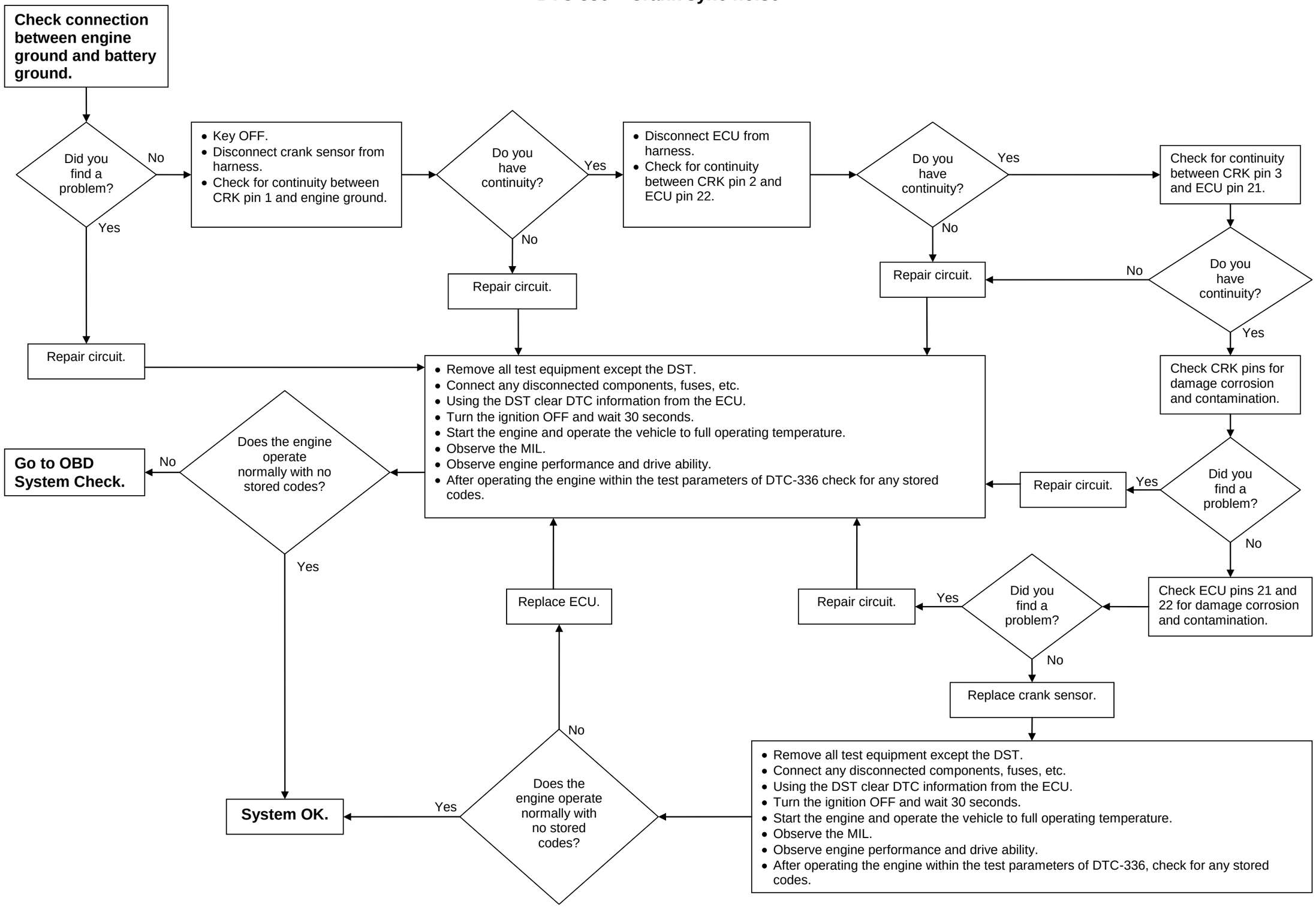
The crankshaft position (CRK) sensor is a 3 pin plug which is located on the cold side of the engine next to the oil filter. A pulse wheel located on the crankshaft is used to measure engine speed and its signal is used to synchronize the ignition and fuel systems. The ECU must see a valid crankshaft position signal while cranking. This fault will set if the crank signal is invalid.

For detailed information, please see customer specific DTC settings.



All information concerning connectors, pin numbers and wire colors can be found in the connector documentation.

DTC 336 – Crank sync noise



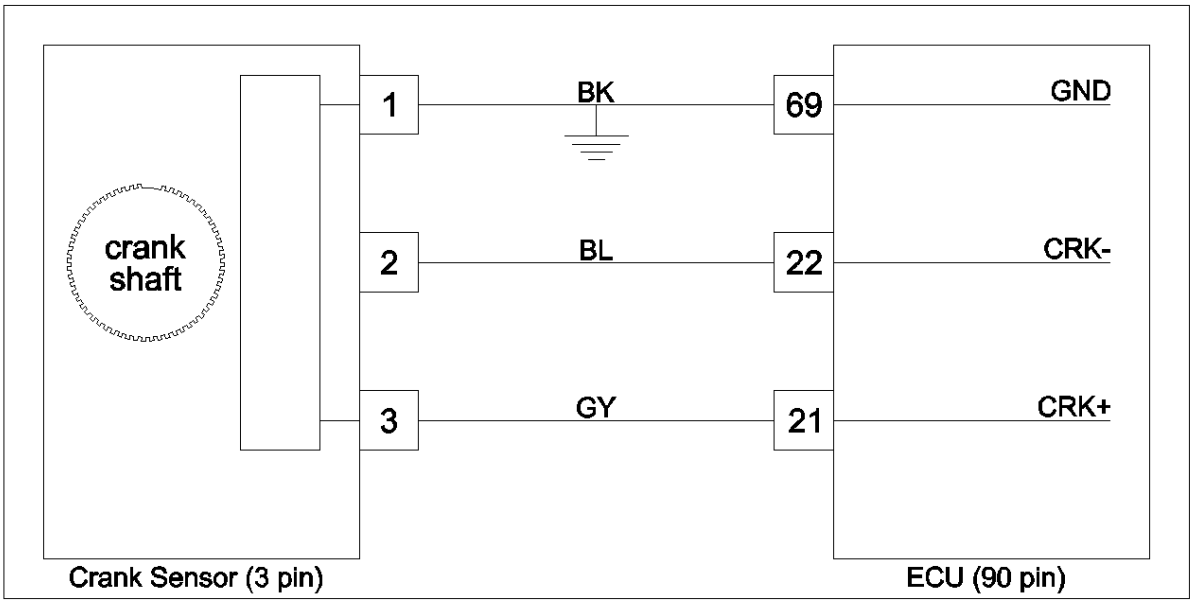
All measurements are done at the connector unless otherwise stated.

DTC 337 – Crank loss

General

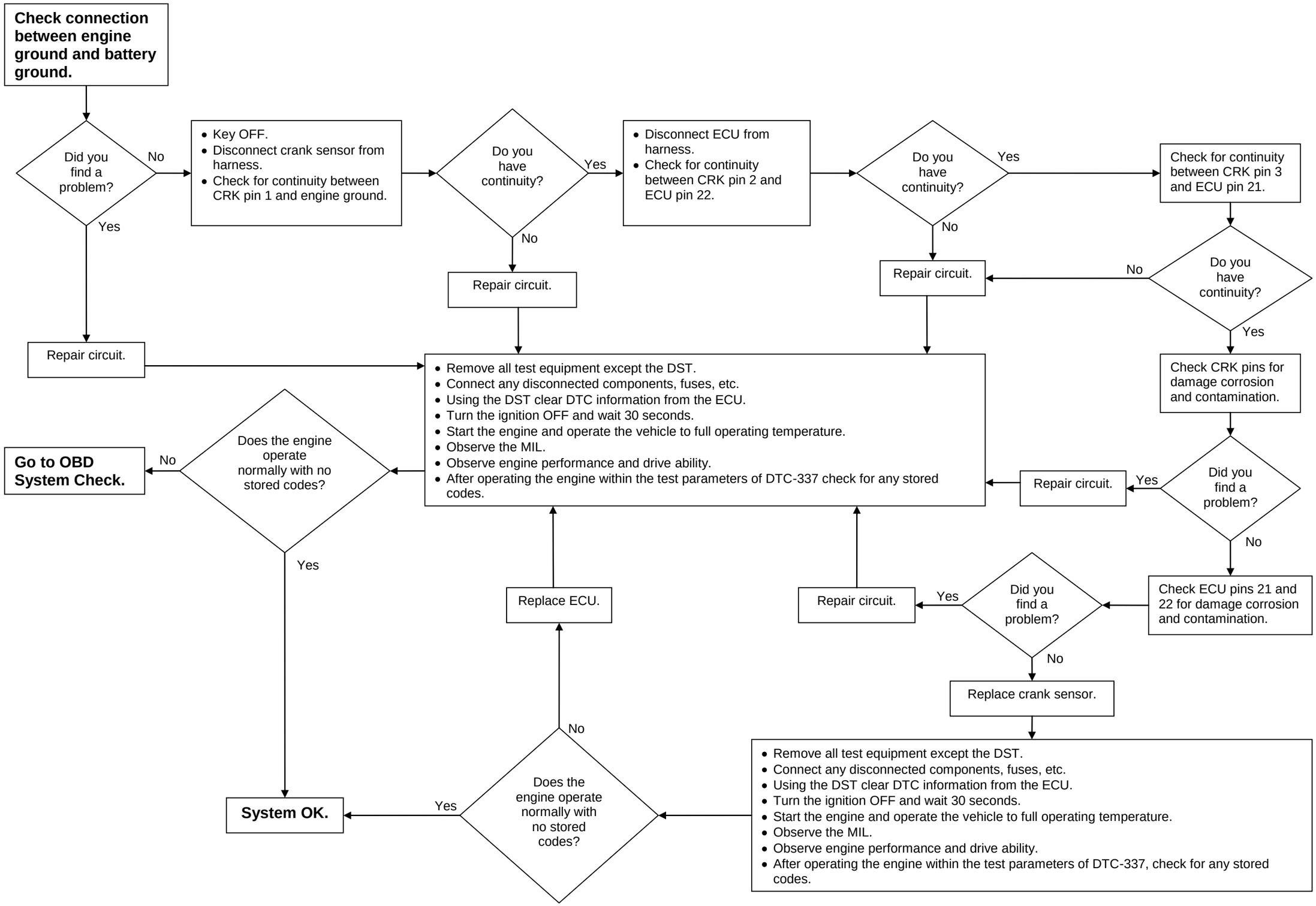
The crankshaft position (CRK) sensor is a 3 pin plug which is located on the cold side of the engine next to the oil filter. A pulse wheel located on the crankshaft is used to measure engine speed and its signal is used to synchronize the ignition and fuel systems. The ECU must see a valid crankshaft position signal while cranking. This fault will set if the crank signal is invalid.

For detailed information, please see customer specific DTC settings.



All information concerning connectors, pin numbers and wire colors can be found in the connector documentation.

DTC 337 – Crank loss



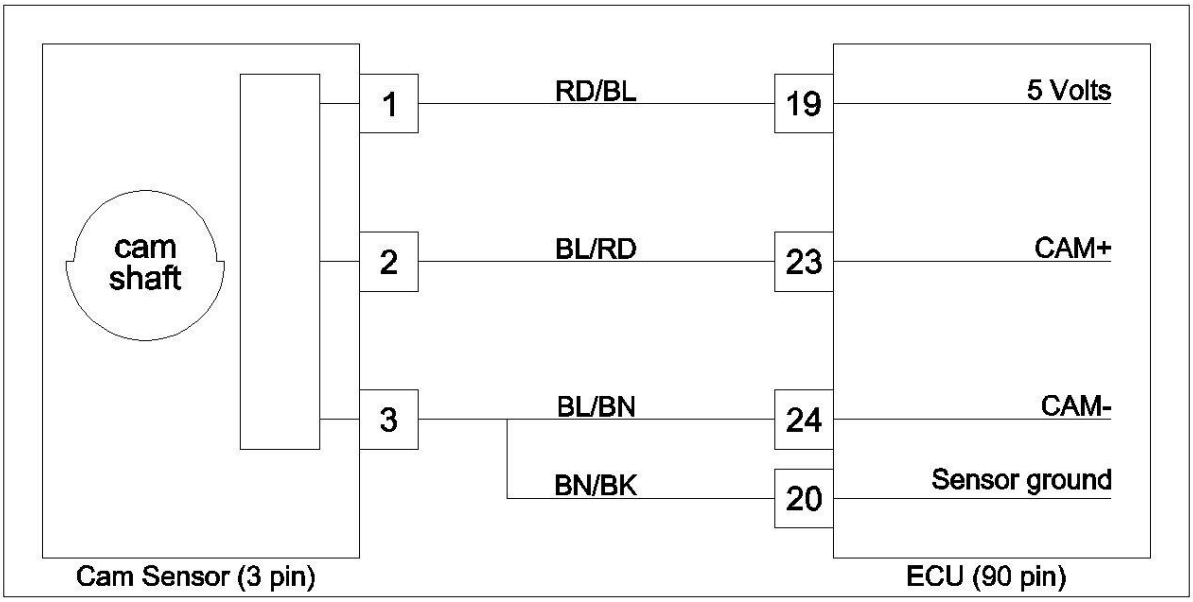
All measurements are done at the connector unless otherwise stated.

DTC 341 – Cam sync noise

General

The intake camshaft position (CAM) sensor is a 3 pin plug which is located on the cylinder head on the rear side of the engine above the flywheel housing. A pulse wheel located on the intake cam shaft is used to measure the intake cam shaft position. The intake cam solenoid is a 2 pin plug which is also located on the cylinder head on the rear side of the engine above the flywheel housing. To get an optimal volumetric efficiency the intake cam shaft position can be controlled by the ECU with the intake cam solenoid. This fault will set if the CAM signal is invalid.

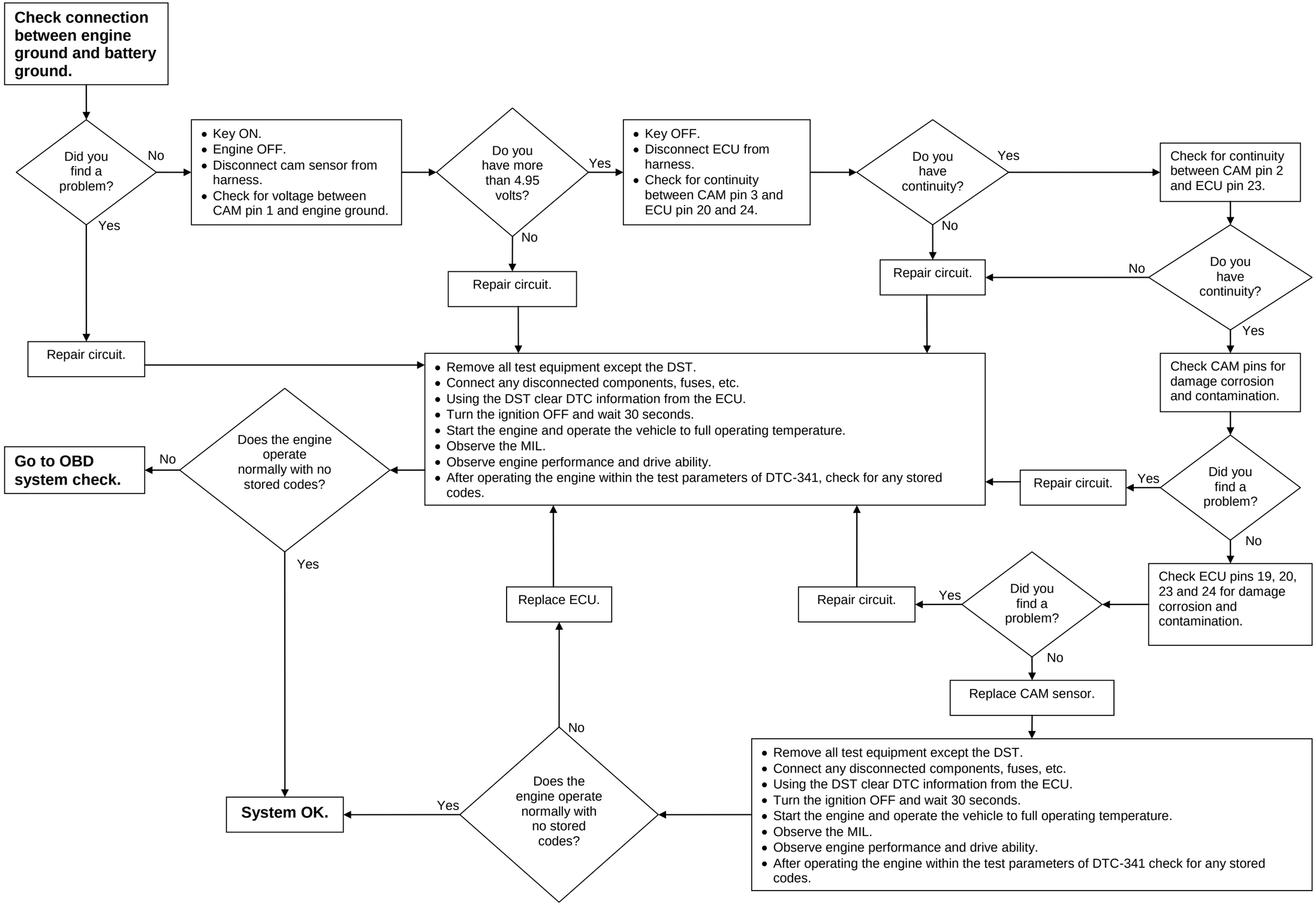
For detailed information, please see customer specific DTC settings.



All information concerning connectors, pin numbers and wire colors can be found in the connector documentation.

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DTC 341 – Cam sync noise



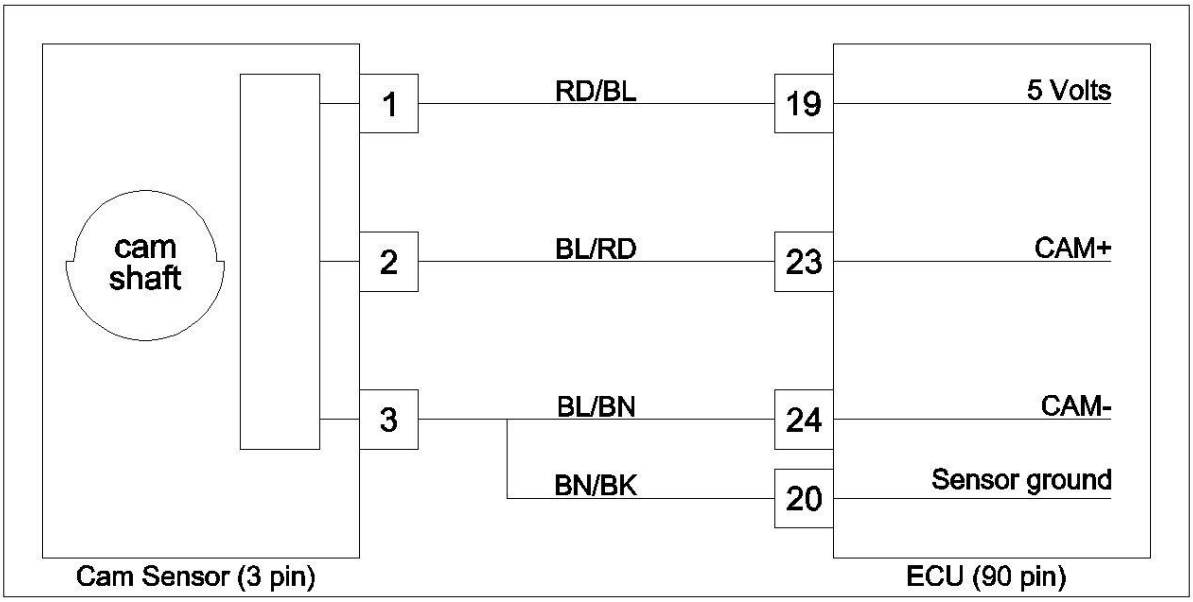
All measurements are done at the connector unless otherwise stated.

DTC 342 – Cam loss

General

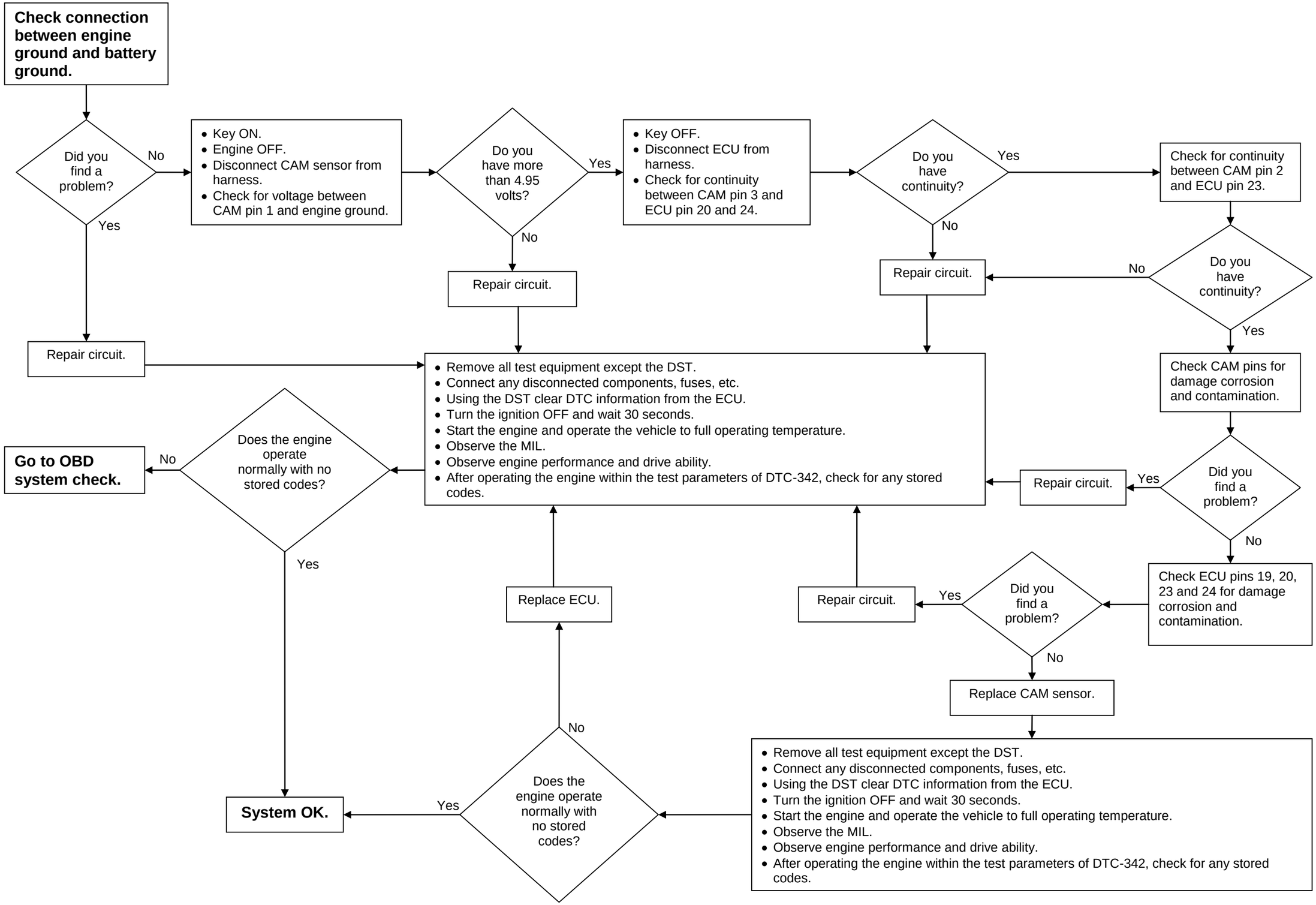
The intake camshaft (CAM) position sensor is a 3 pin plug which is located on the cylinder head on the rear side of the engine above the flywheel housing. A pulse wheel located on the intake cam shaft is used to measure the intake cam shaft position. The intake cam solenoid is a 2 pin plug which is also located on the cylinder head on the rear side of the engine above the flywheel housing. To get an optimal volumetric efficiency the intake cam shaft position can be controlled by the ECU with the intake cam solenoid. This fault will set if the CAM signal is invalid.

For detailed information, please see customer specific DTC settings.



All information concerning connectors, pin numbers and wire colors can be found in the connector documentation.

DTC 342 – Cam loss



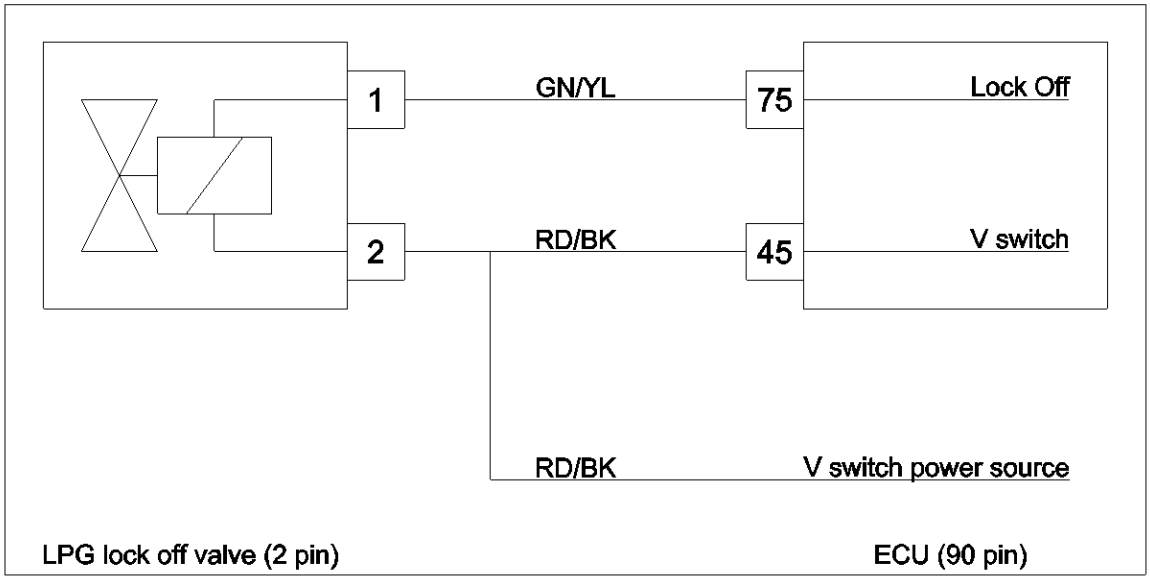
All measurements are done at the connector unless otherwise stated.

DTC 359 – Fuel run-out longer than expected

General

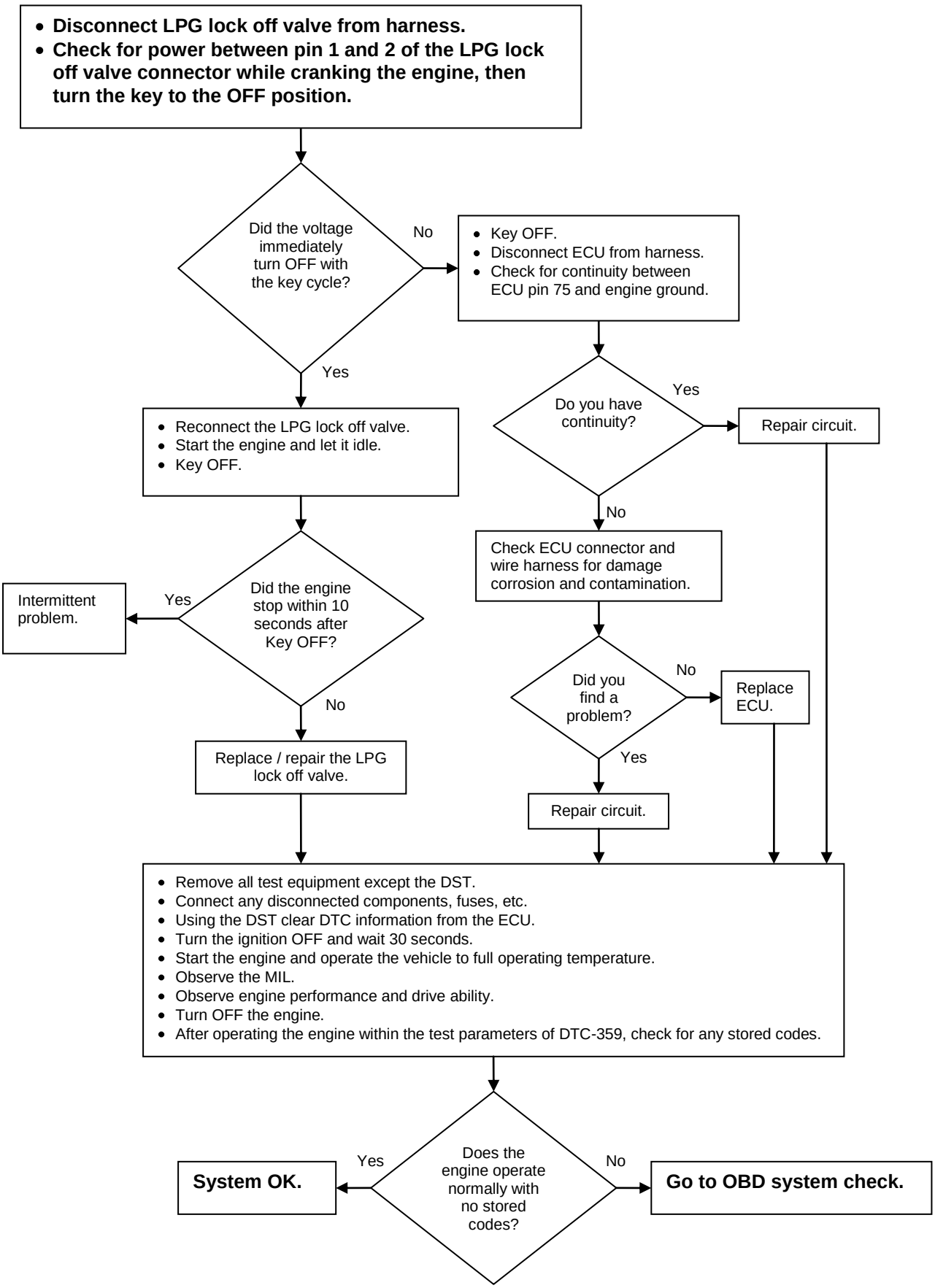
The LPG Lock Off (LO) valve is supplied with system battery power from the Vswitch power source. The ECU provides a connection to ground to open the lock off valve. This fault will set if the engine shut down takes longer than specified and indicates a possible problem with the LPG lock off valve.

For detailed information, please see customer specific DTC settings.



All information concerning connectors, pin numbers and wire colors can be found in the connector documentation.

DTC 359 – Fuel run-out longer than expected



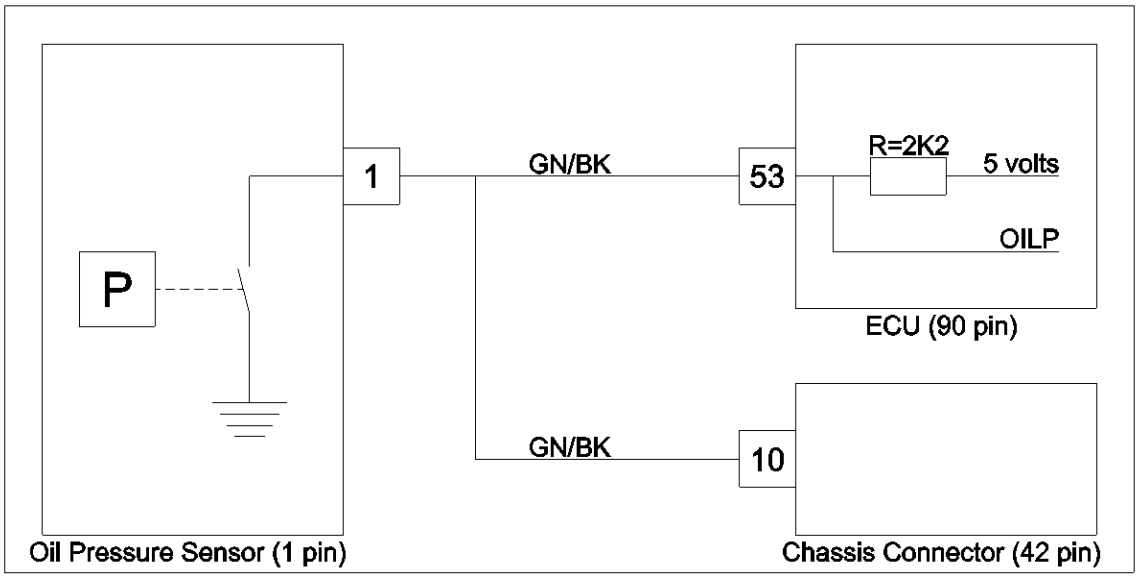
All measurements are done at the connector unless otherwise stated.

DTC 524 – Oil pressure low

General

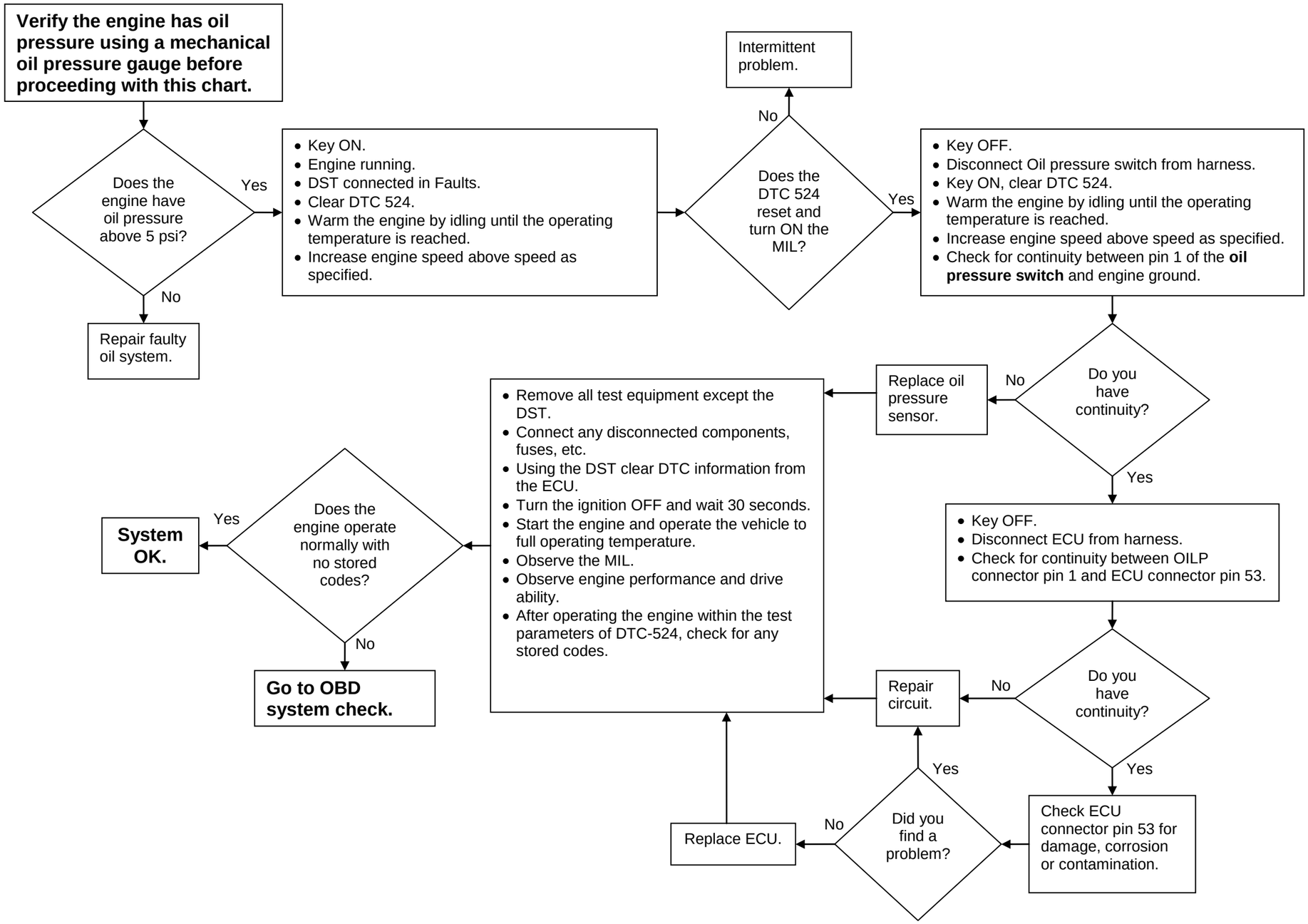
The Oil Pressure (OILP) Switch is used to communicate a low oil pressure condition to the ECU. Engine damage can occur if the engine is operated with low oil pressure. The oil pressure switch connects the ECU to engine ground when the oil pressure is correct. The ECU uses an analog voltage input with an internal 5 volt reference. This fault will set if the oil pressure is too low.

For detailed information, please see customer specific DTC settings.



All information concerning connectors, pin numbers and wire colors can be found in the connector documentation.

DTC 524 – Oil pressure low



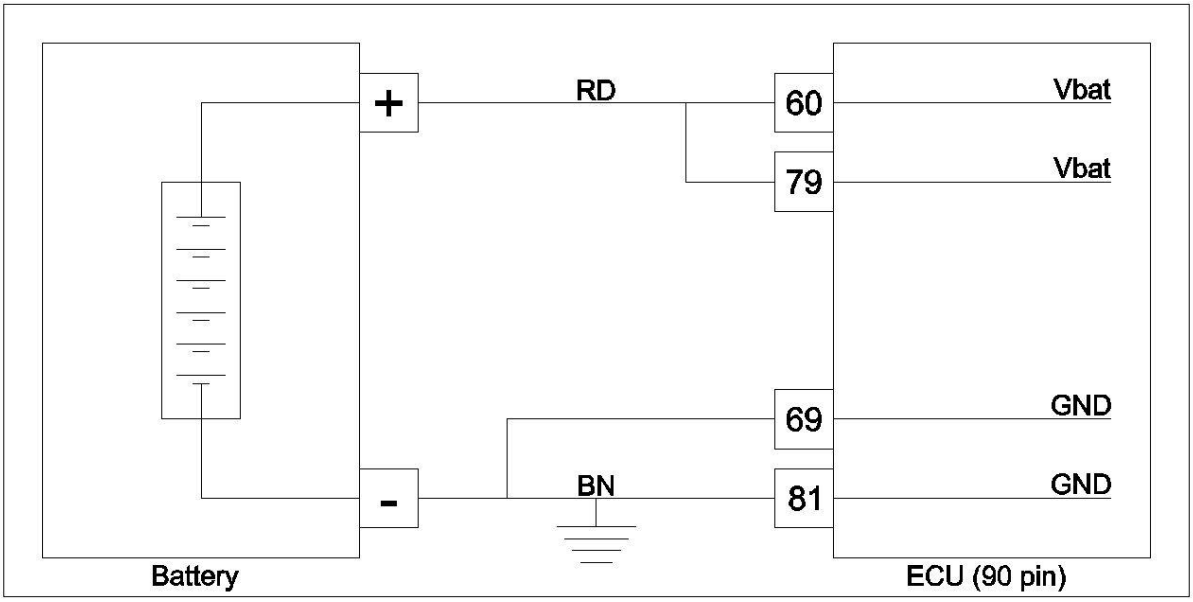
All measurements are done at the connector unless otherwise stated.

DTC 562 – System voltage low

General

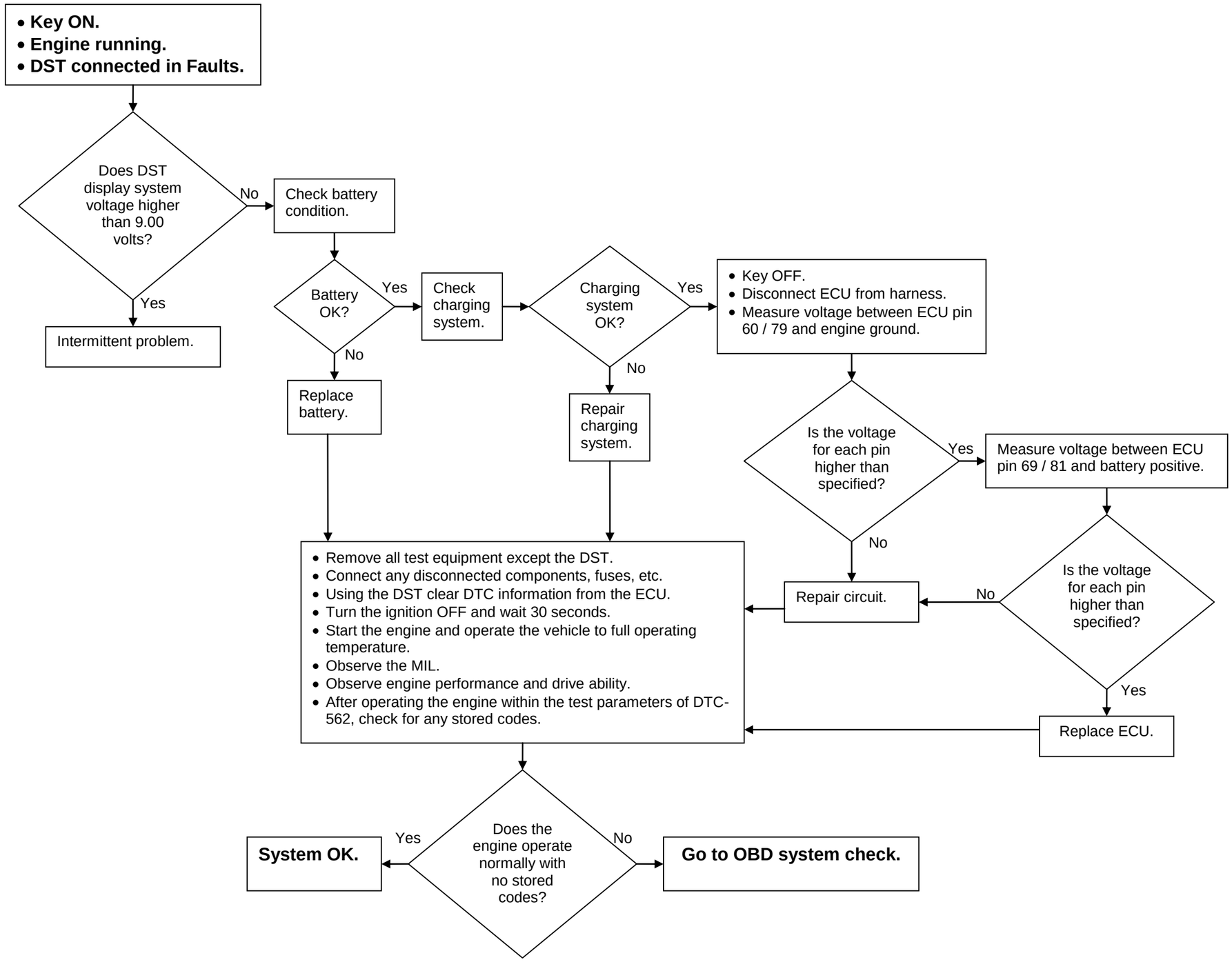
The battery voltage powers the ECU and must be measured correctly to properly operate injector drivers, solenoid valves and ignition coils. This fault will set if the system voltage is too low.

For detailed information, please see customer specific DTC settings.



All information concerning connectors, pin numbers and wire colors can be found in the connector documentation.

DTC 562 – System voltage low



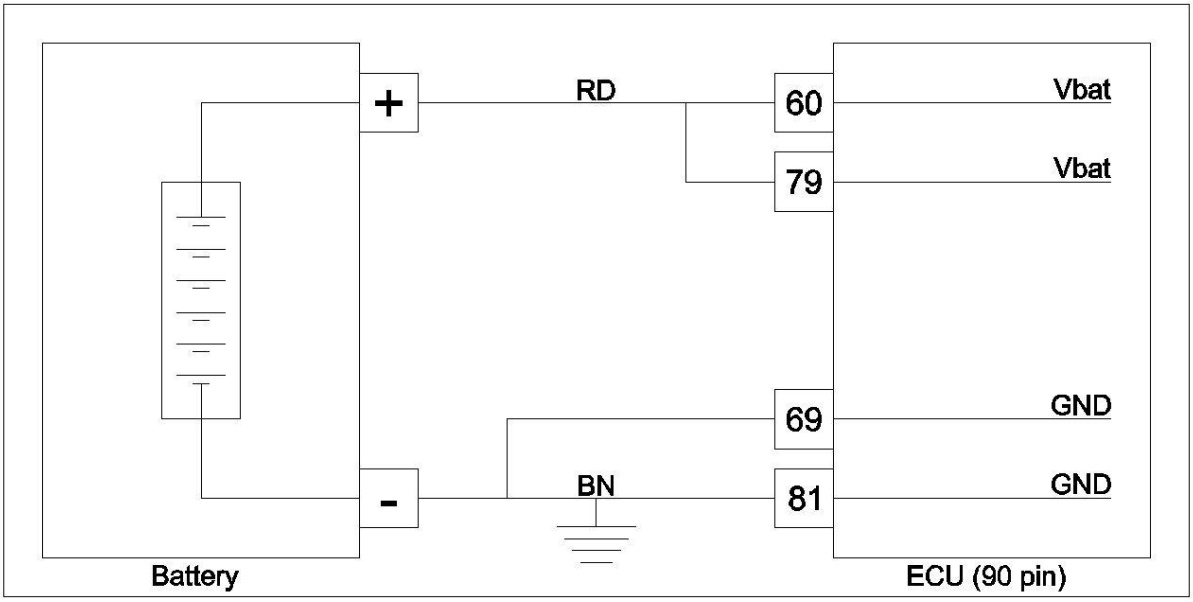
All measurements are done at the connector unless otherwise stated.

DTC 563 – System voltage high

General

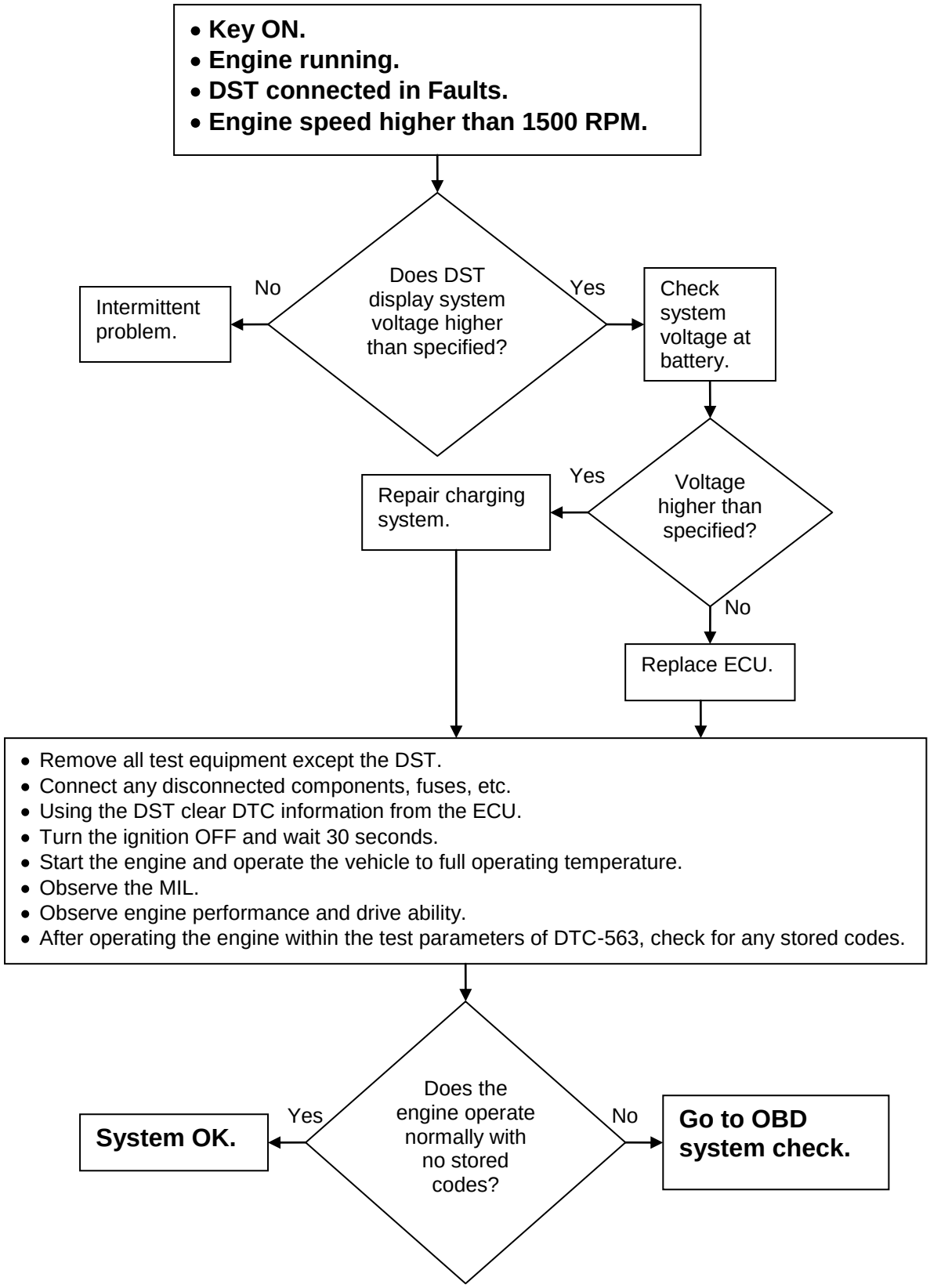
The battery voltage powers the ECU and must be measured correctly to properly operate injector drivers, solenoid valves and ignition coils. This fault will set if the system voltage is too high.

For detailed information, please see customer specific DTC settings.



All information concerning connectors, pin numbers and wire colors can be found in the connector documentation.

DTC 563 – System voltage high



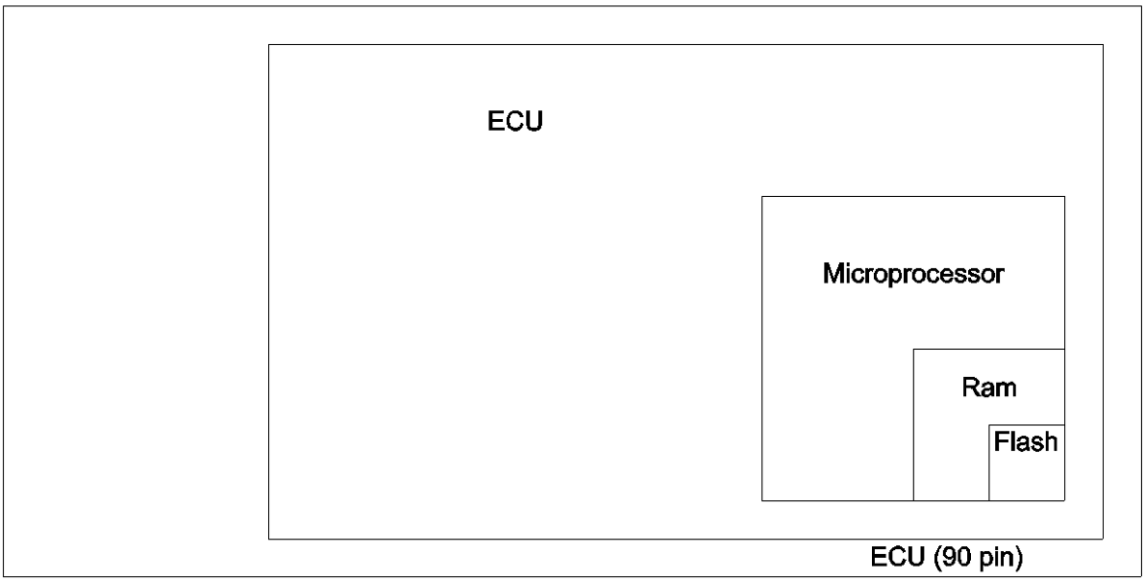
All measurements are done at the connector unless otherwise stated.

DTC 601 – Flash checksum invalid

General

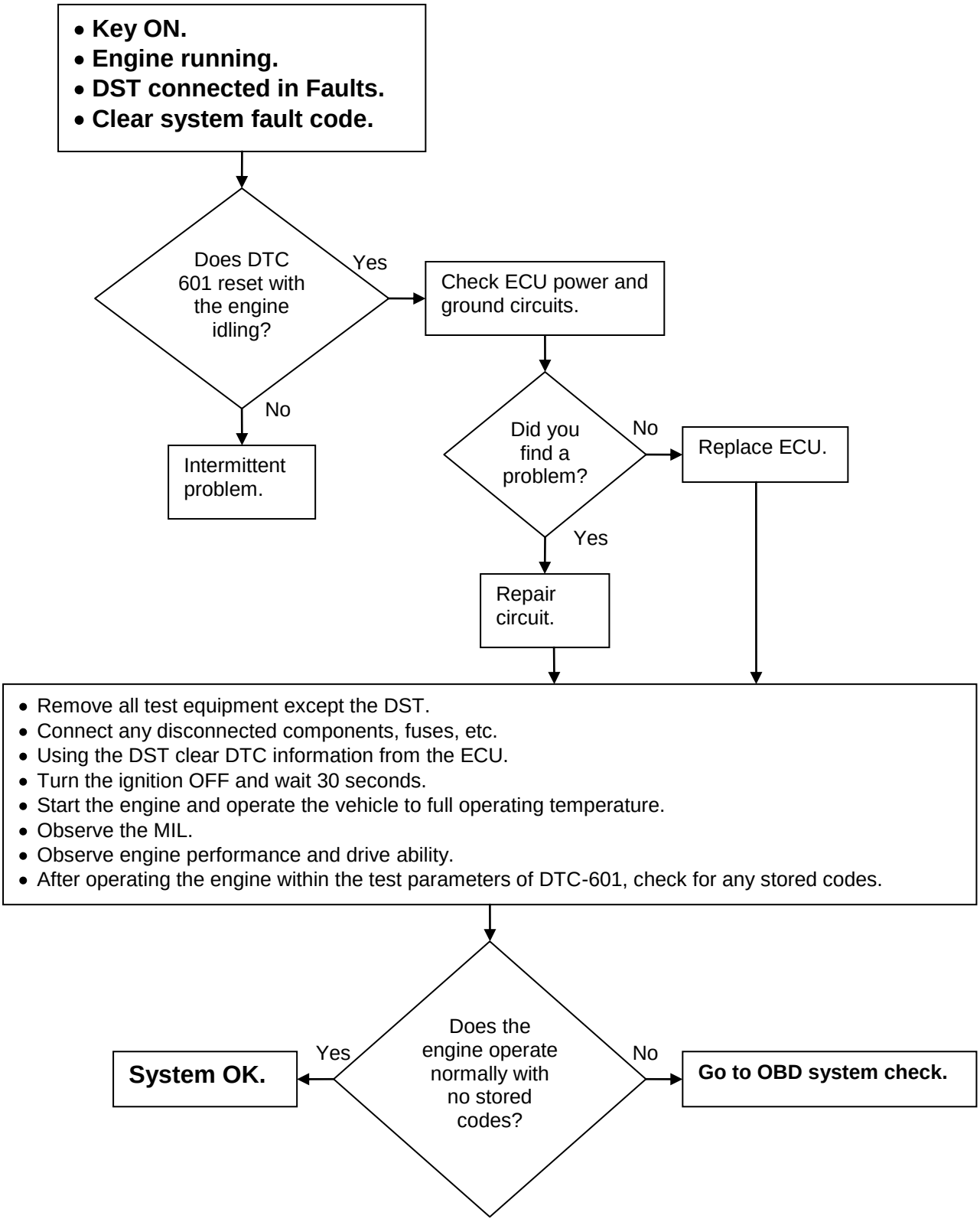
The ECU has several internal checks that must be satisfied each time an instruction is executed. Several different things can happen within the microprocessor that will cause this fault.
The ECU will attempt to reset itself in the event this fault is set.

For detailed information, please see customer specific DTC settings.



All information concerning connectors, pin numbers and wire colors can be found in the connector documentation.

DTC 601 – Flash checksum invalid



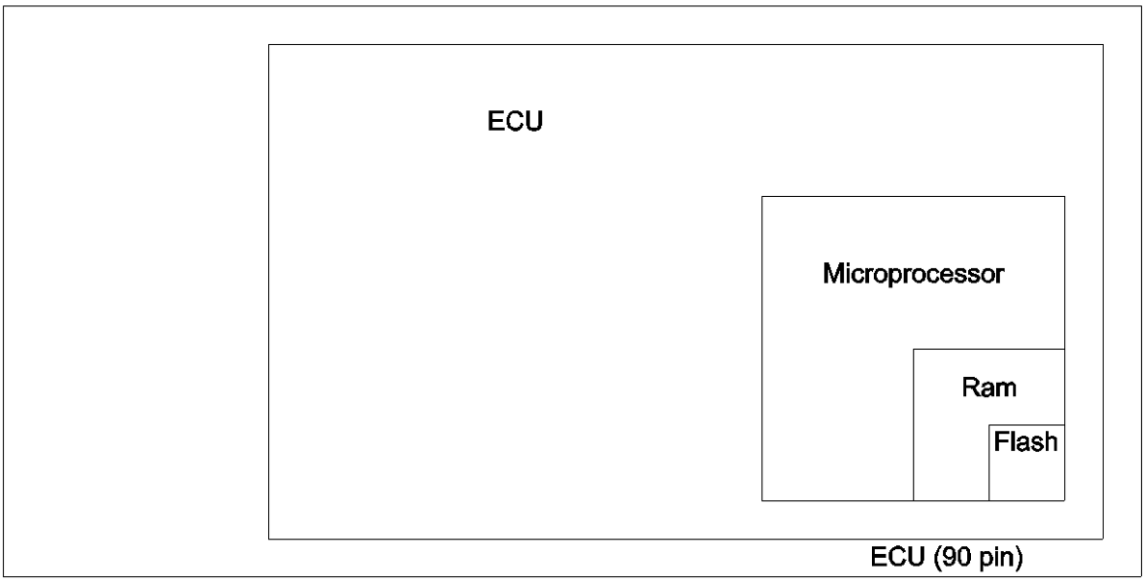
All measurements are done at the connector unless otherwise stated.

DTC 604 – RAM failure

General

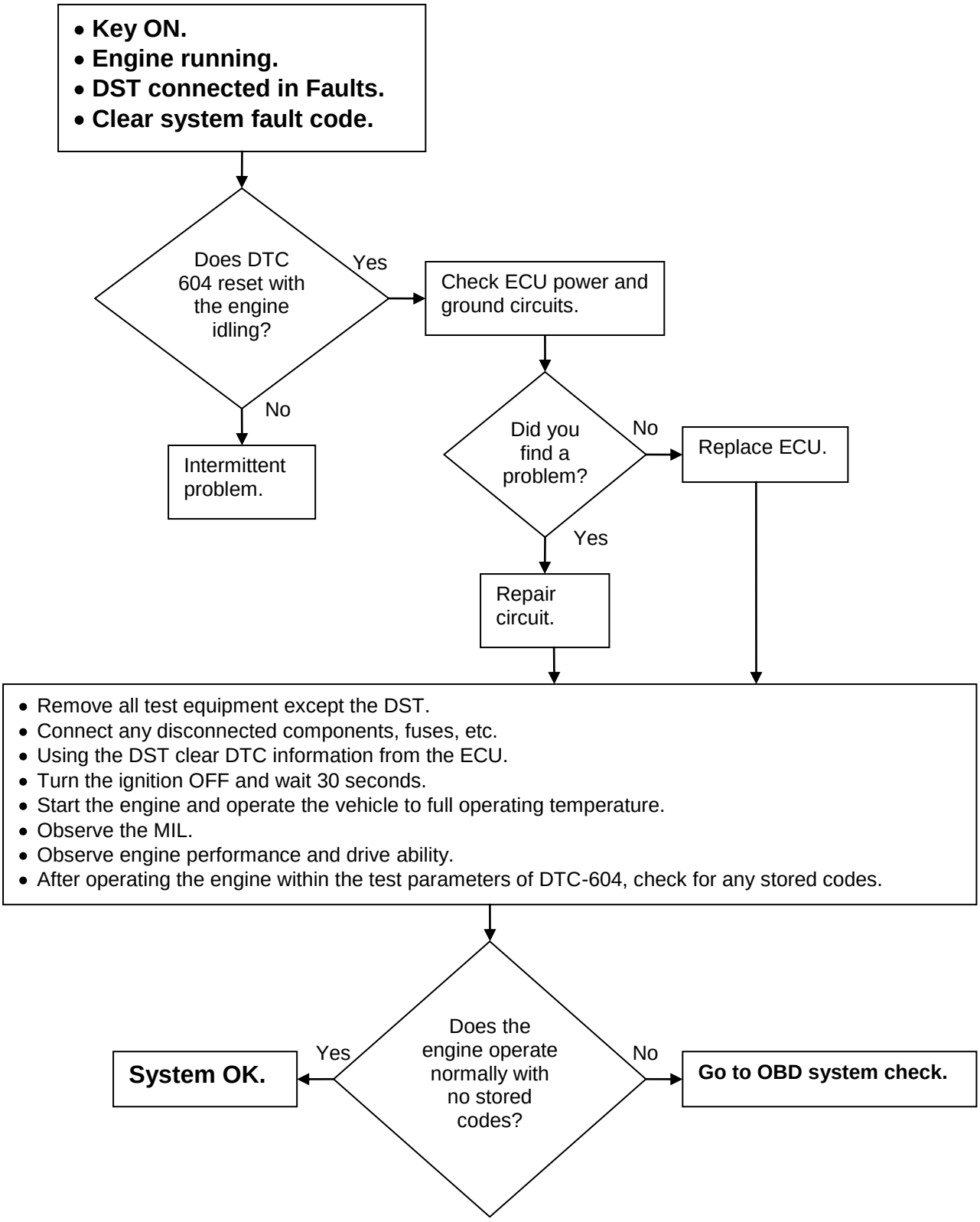
The ECU has several internal checks that must be satisfied each time an instruction is executed. Several different things can happen within the microprocessor that will cause this fault.
The ECU will attempt to reset itself in the event this fault is set.

For detailed information, please see customer specific DTC settings.



All information concerning connectors, pin numbers and wire colors can be found in the connector documentation.

DTC 604 – RAM failure



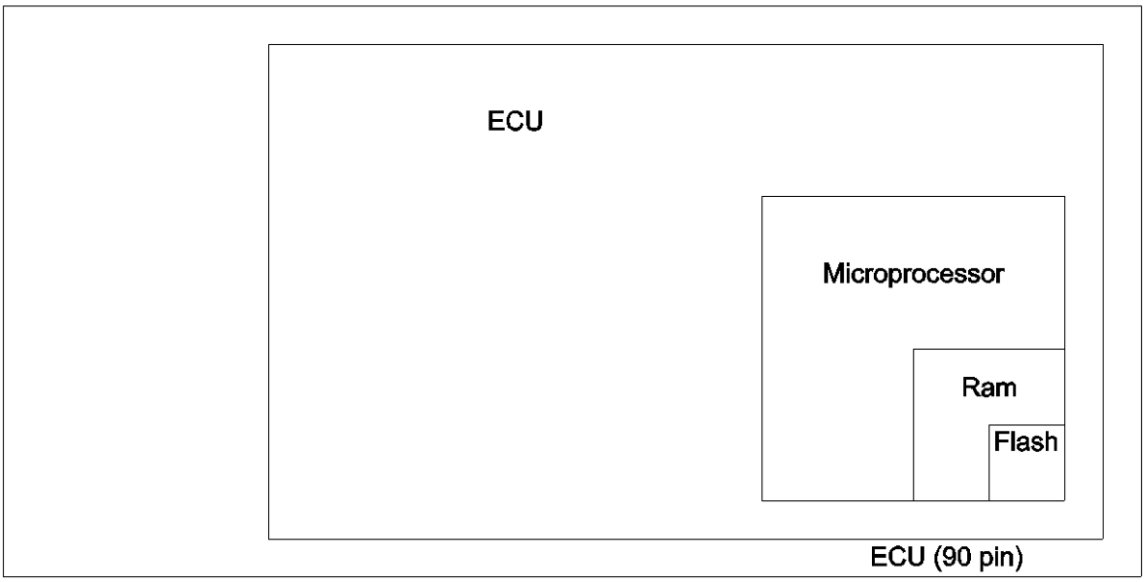
All measurements are done at the connector unless otherwise stated.

DTC 606 – COP failure

General

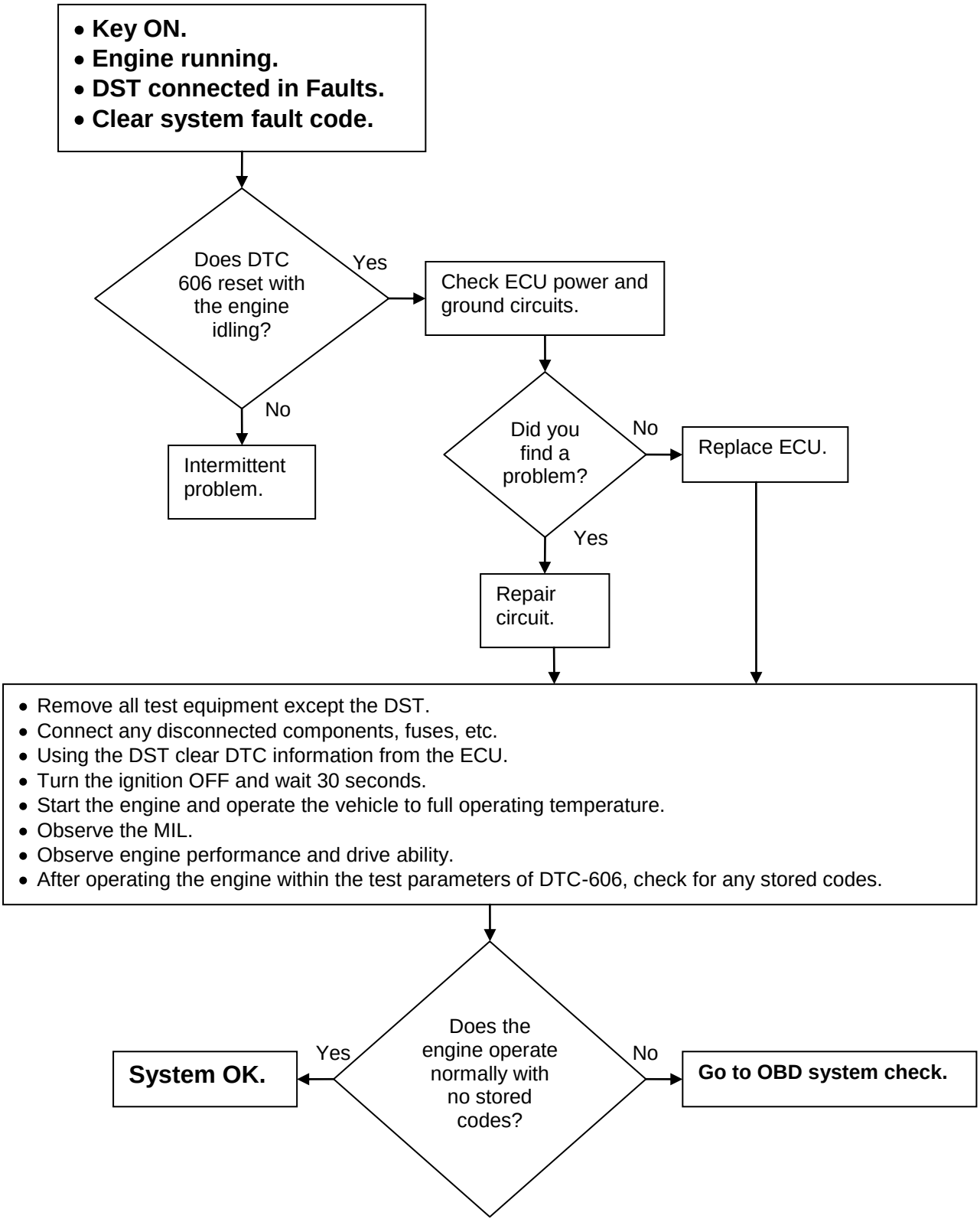
The ECU has several internal checks that must be satisfied each time an instruction is executed. Several different things can happen within the microprocessor that will cause this fault.
The ECU will attempt to reset itself in the event this fault is set.

For detailed information, please see customer specific DTC settings.



All information concerning connectors, pin numbers and wire colors can be found in the connector documentation.

DTC 606 – COP failure



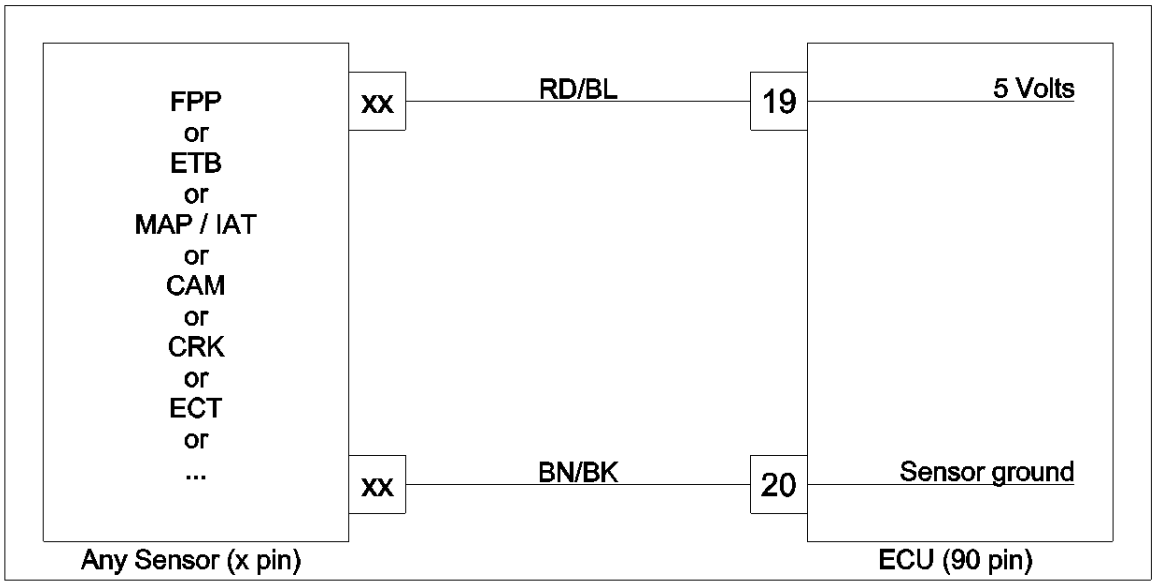
All measurements are done at the connector unless otherwise stated.

DTC 642 – 5VE1 low voltage

General

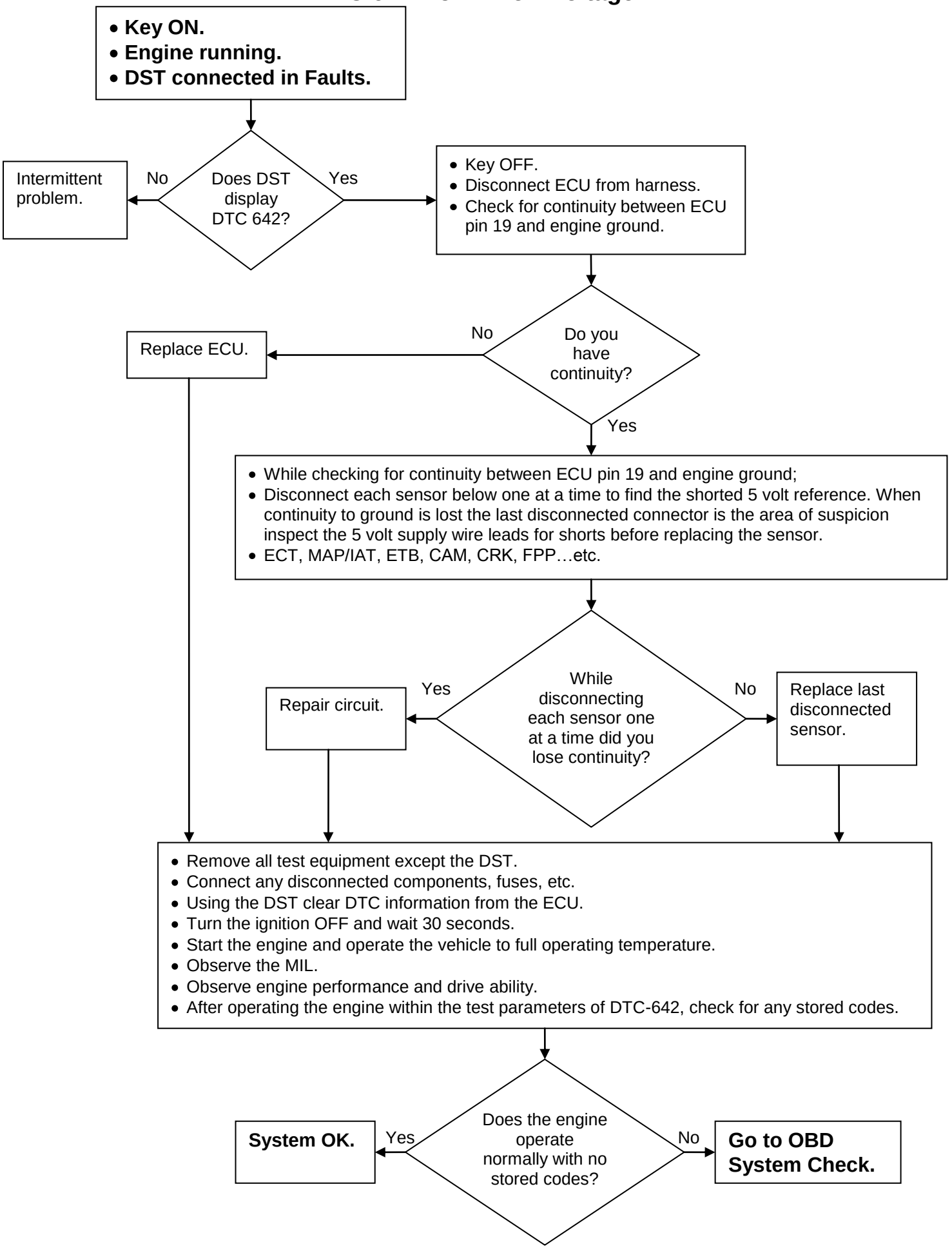
The External 5 volt supply powers many of the sensors and other components of the fuel system. The accuracy of the 5 volt supply is very important to the accuracy of the powered sensors and fuel control by the ECU. The ECU is able to determine if they are overloaded, shorted, or otherwise out of specification by monitoring the 5 volt supply. This fault will set if the 5 volt supply is too low.

For detailed information, please see customer specific DTC settings.



All information concerning connectors, pin numbers and wire colors can be found in the connector documentation.

DTC 642 – 5VE1 low voltage



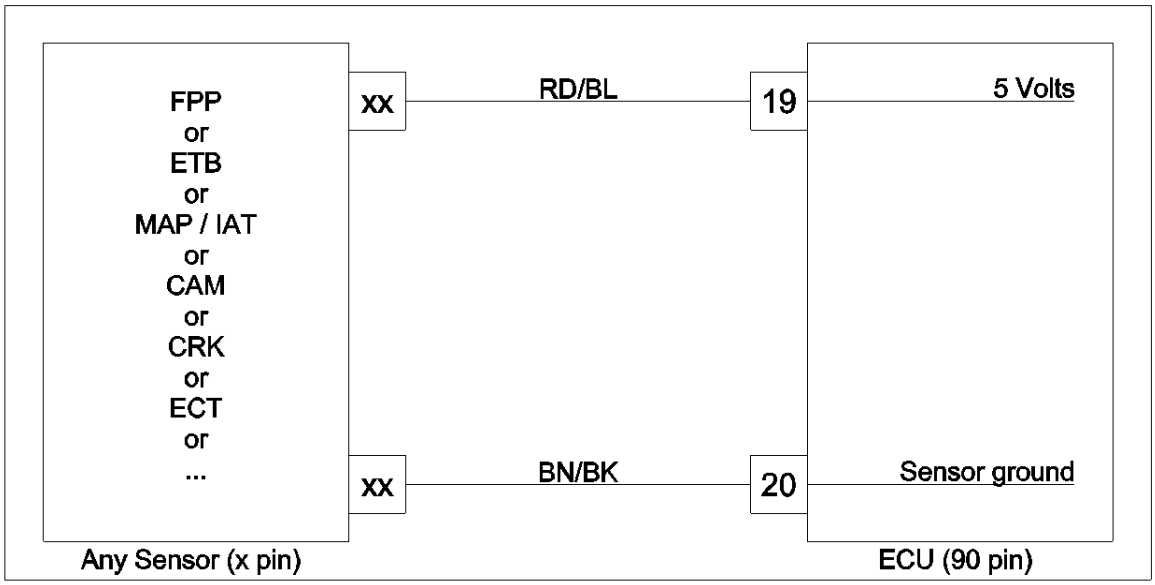
All measurements are done at the connector unless otherwise stated.

DTC 643 – 5VE1 high voltage

General

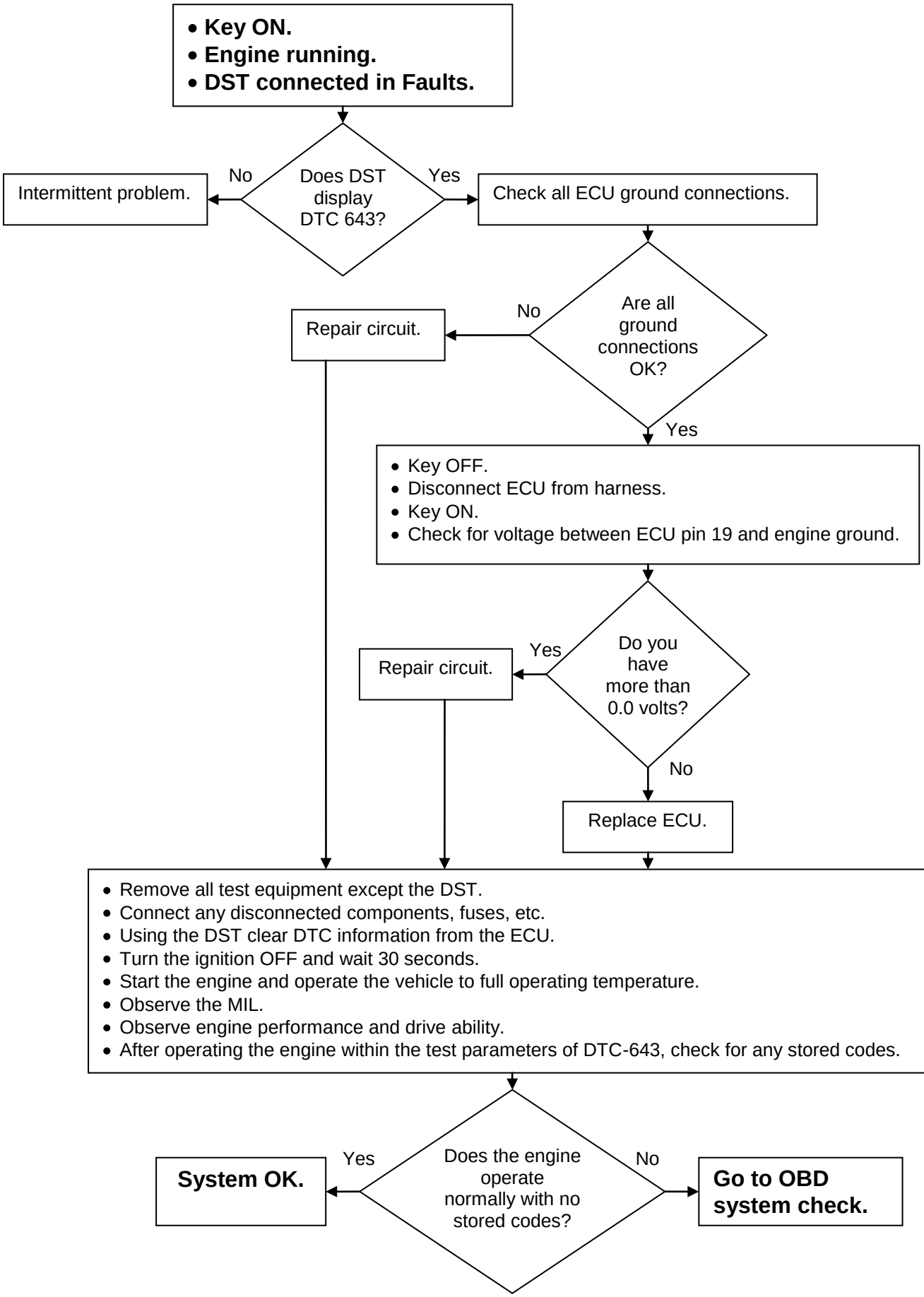
The External 5 volt supply powers many of the sensors and other components of the fuel system. The accuracy of the 5 volt supply is very important to the accuracy of the powered sensors and fuel control by the ECU. The ECU is able to determine if they are overloaded, shorted, or otherwise out of specification by monitoring the 5 volt supply. This fault will set if the 5 volt supply is too high.

For detailed information, please see customer specific DTC settings.



All information concerning connectors, pin numbers and wire colors can be found in the connector documentation.

DTC 643 – 5VE1 high voltage



All measurements are done at the connector unless otherwise stated.

DTC 1111 – Fuel rev limit

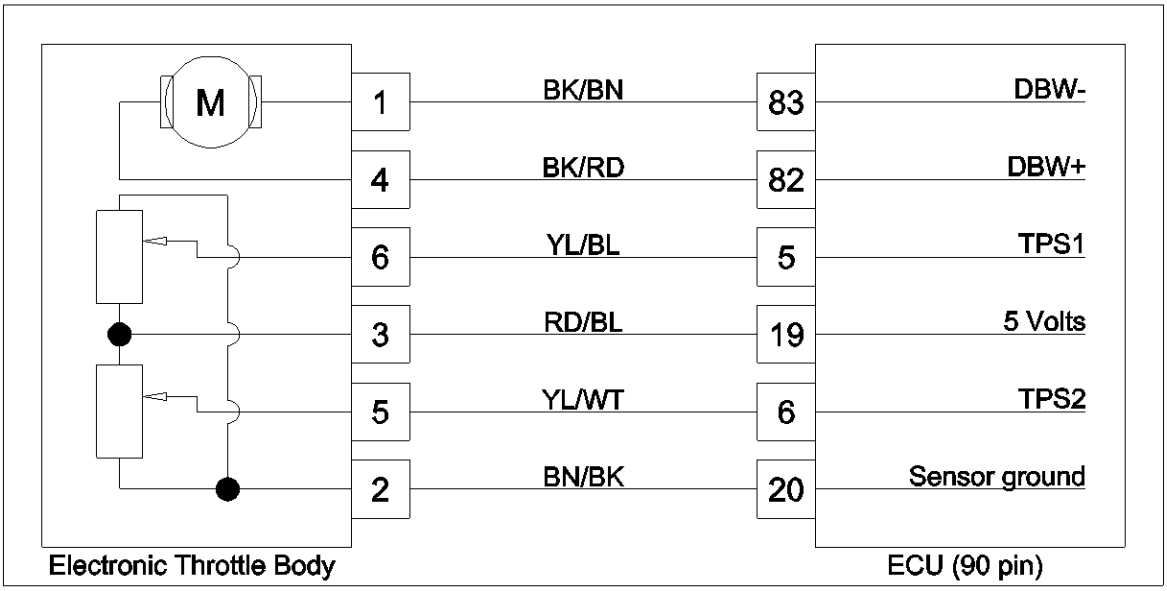
General

In case of an engine over speed, the fuel supply will be cut. This fault will set if the engine speed exceeds the specified speed.

For detailed information, please see customer specific DTC settings.

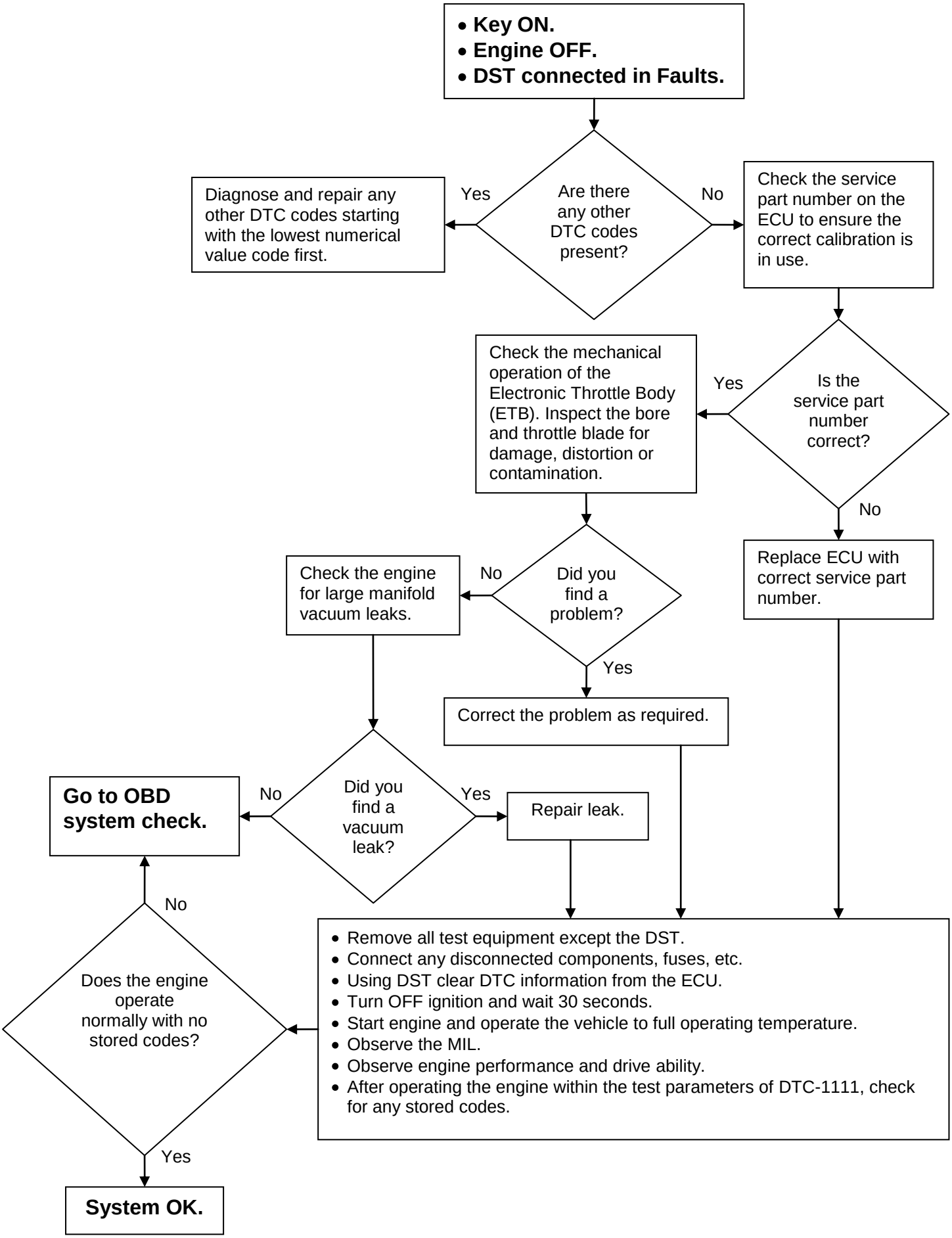
Diagnostic Aid

Check for other stored DTC codes before using the following diagnostic chart for this code set. It is quite likely that this DTC will be accompanied by other DTC sets. Always diagnose and repair any existing codes starting with the lowest numerical code first. Repairing the lower numerical value code will most likely correct this DTC.



All information concerning connectors, pin numbers and wire colors can be found in the connector documentation.

DTC 1111 – Fuel rev limit



All measurements are done at the connector unless otherwise stated.

DTC 1112 – Spark rev limit

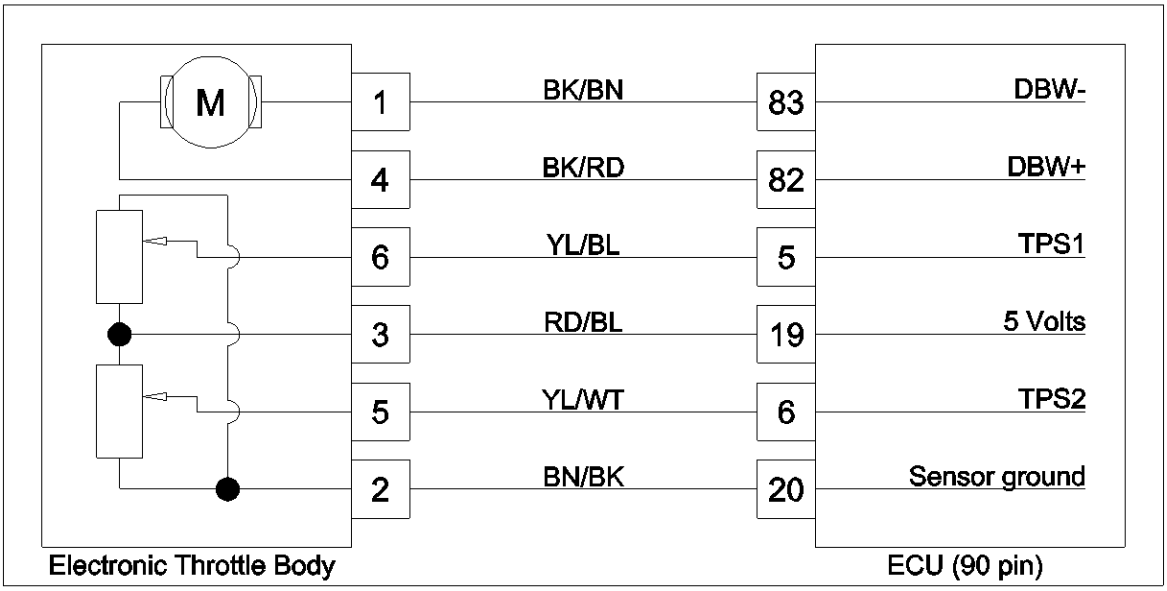
General

In case of an engine over speed, the ignition will be cut. This fault will set if the engine speed exceeds the specified speed.

For detailed information, please see customer specific DTC settings.

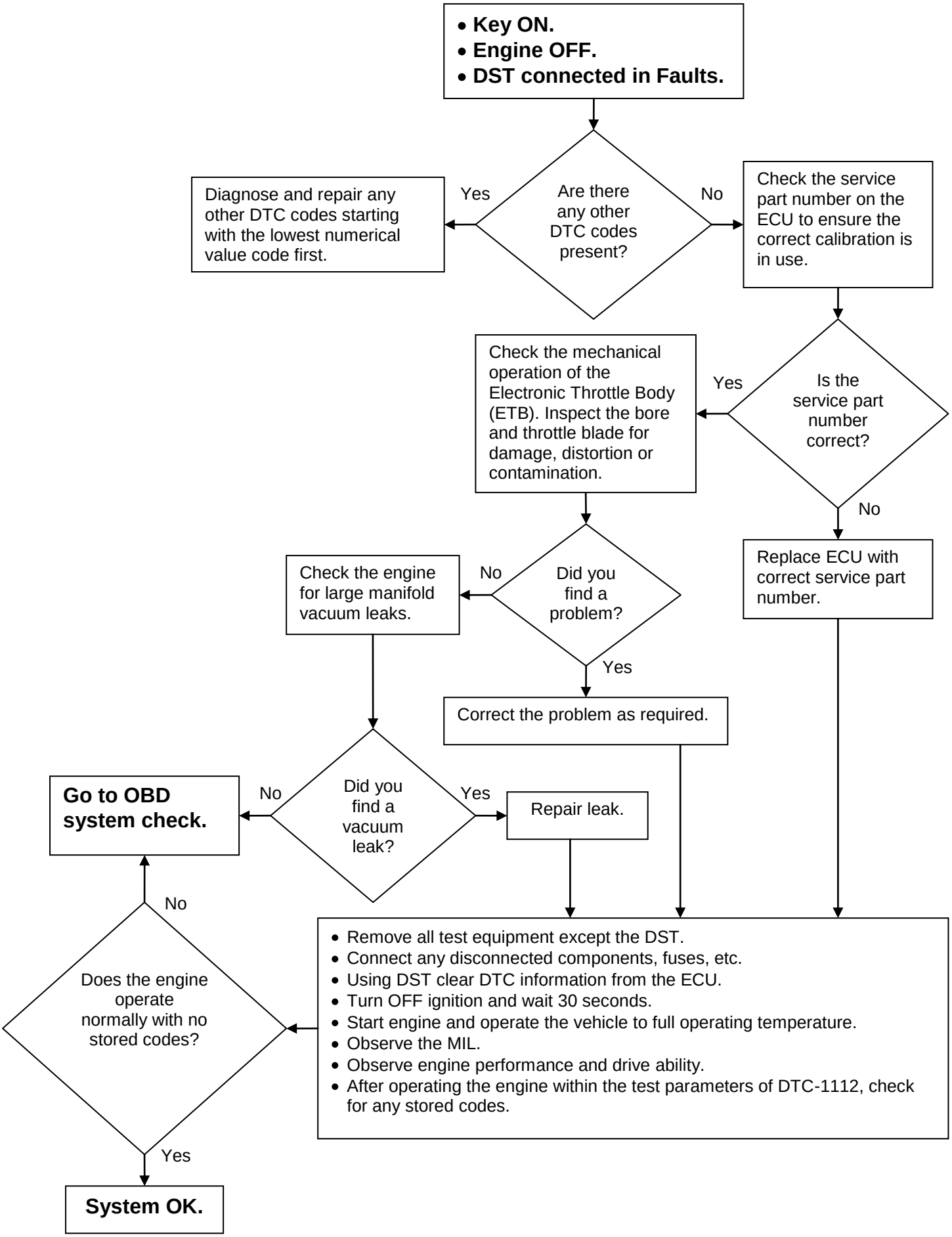
Diagnostic Aid

Check for other stored DTC codes before using the following diagnostic chart for this code set. It is quite likely that this DTC will be accompanied by other DTC sets. Always diagnose and repair any existing codes starting with the lowest numerical code first. Repairing the lower numerical value code will most likely correct this DTC.



All information concerning connectors, pin numbers and wire colors can be found in the connector documentation.

DTC 1112 – Spark rev limit



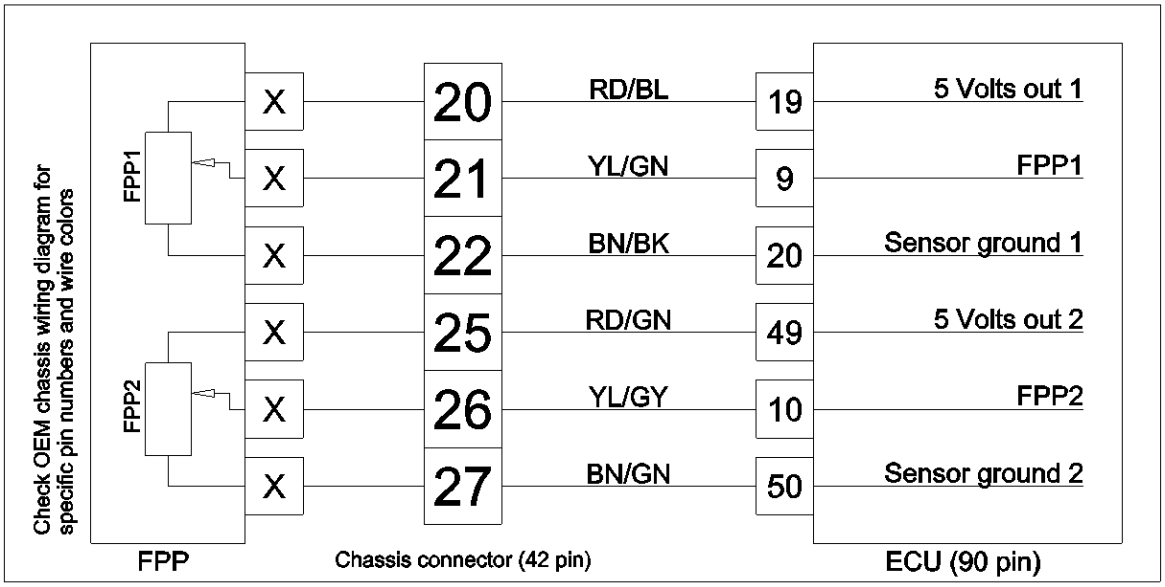
All measurements are done at the connector unless otherwise stated.

DTC 1121 – FPP1/2 simultaneous voltages out of range

General

The Foot Pedal Position (FPP) sensor is an electronic device that is coupled to a mechanically driven input which is commanded by the vehicle/engine operator. General sensor configurations consist of a single potentiometer with an Idle Valid Switch (IVS), or two potentiometers. The FPP sensor outputs are proportional to the commanded input. The ECU uses the FPP sensor inputs to control the throttle and adjust the engine's load in order to achieve the requested power. Since the FPP sensor inputs directly affect the engine's power output, redundant sensors are generally used to ensure safe, reliable operation. This fault is only applicable for dual potentiometer sensors and indicates that FPP1 and FPP2 percentages do not correlate with each other.

For detailed information, please see customer specific DTC settings.



All information concerning connectors, pin numbers and wire colors can be found in the connector documentation.

For troubleshooting, refer to DTC 2122, 2123, 2127 and 2128 procedures.

DTC 1151 – CL high LPG

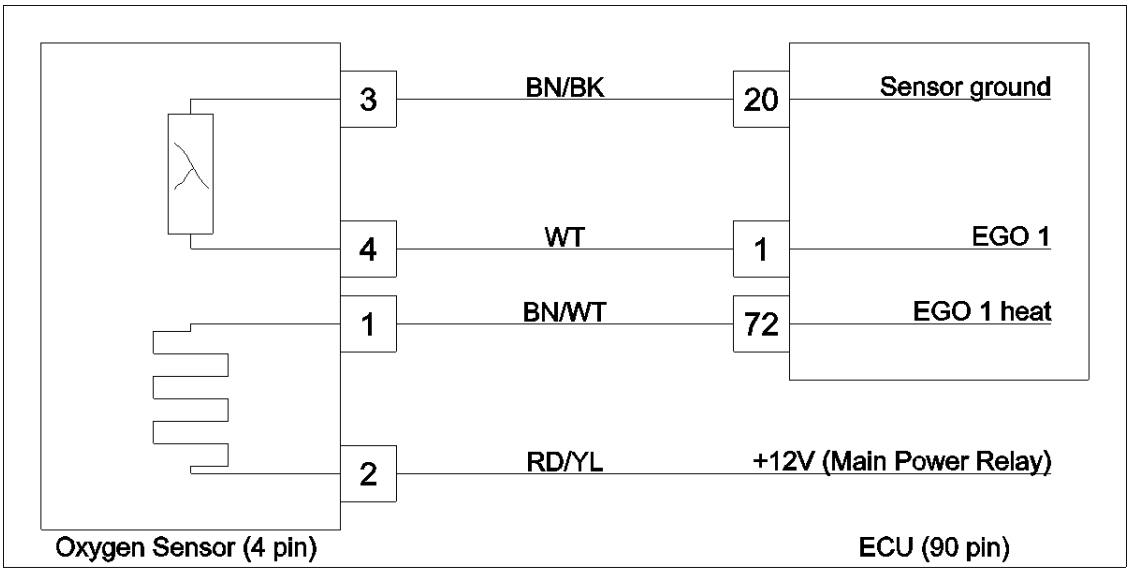
General

The oxygen sensor is used to determine if the fuel flow to the engine is correct by measuring the oxygen content in the exhaust gas. The ECU uses this information to correct the fuel flow with the closed loop multiplier and the adaptive learn. This fault will set if the Closed Loop (CL) multiplier is higher than specified.

For detailed information, please see customer specific DTC settings.

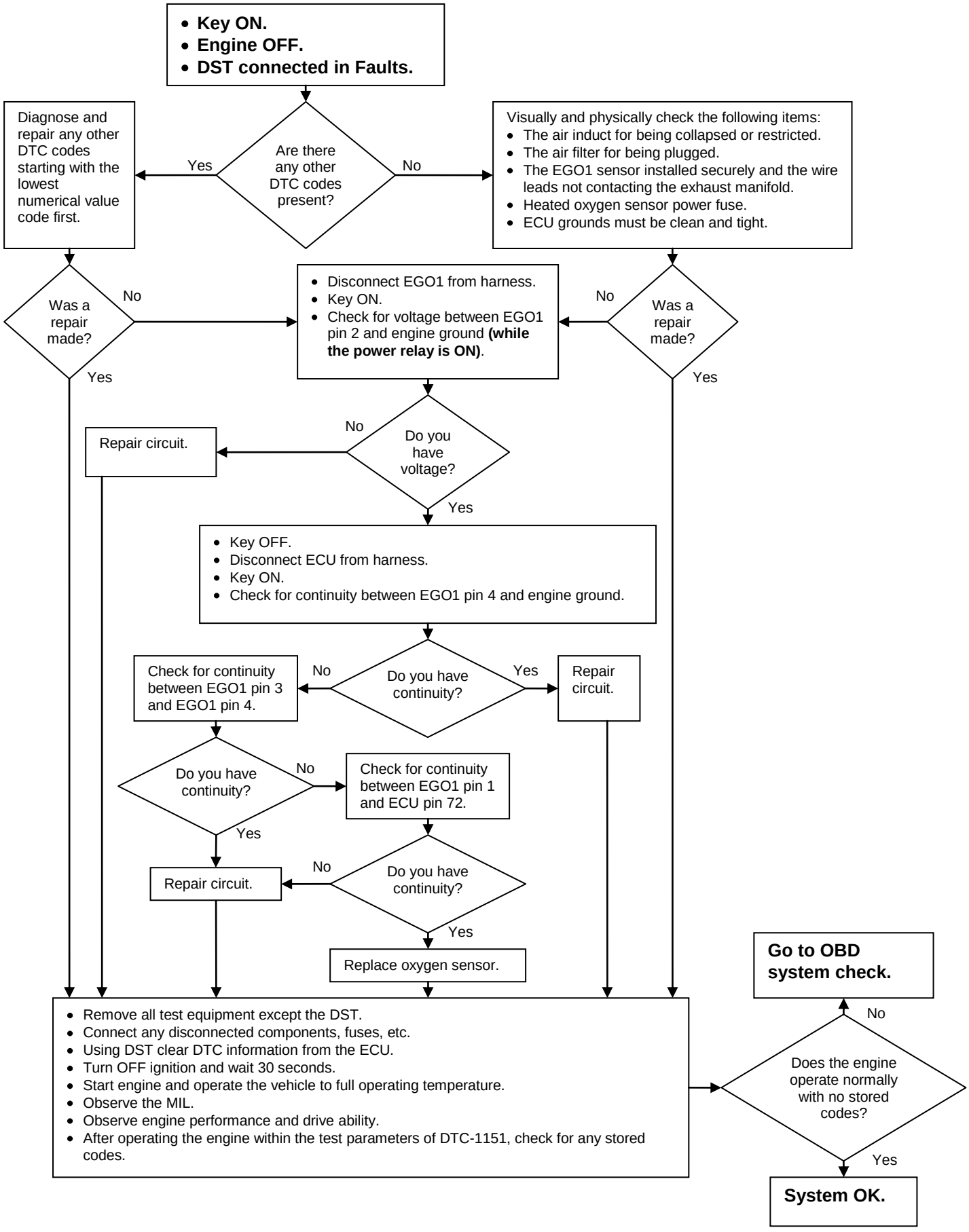
Diagnostic Aid

Oxygen Sensor Wiring:	Heated Oxygen sensor wires may be misrouted and contacting the exhaust manifold.
Vacuum Leaks:	Large vacuum leaks and crankcase leaks can cause a lean exhaust condition especially at low loads.
Fuel Mixer:	The system can be lean due to a faulty fuel mixer.
Fuel Pressure:	Low fuel pressure, a faulty fuel regulator or a contaminated fuel filter can cause the system to run lean.
Exhaust leaks:	If there is an exhaust leak, outside air can be pulled into the exhaust and pass the oxygen sensor causing a false lean condition.
Fuel Quality:	Contaminated or spoiled fuel can cause the fuel system to be lean.
Ground problem:	ECU grounds must be clean, tight and in the proper location.



All information concerning connectors, pin numbers and wire colors can be found in the connector documentation.

DTC 1151 – CL high LPG



All measurements are done at the connector unless otherwise stated.

DTC 1152 – CL low LPG

General

The oxygen sensor is used to determine if the fuel flow to the engine is correct by measuring the oxygen content in the exhaust gas. The ECU uses this information to correct the fuel flow with the closed loop multiplier and the adaptive learn. This fault will set if the Closed Loop (CL) multiplier is lower than specified.

For detailed information, please see customer specific DTC settings.

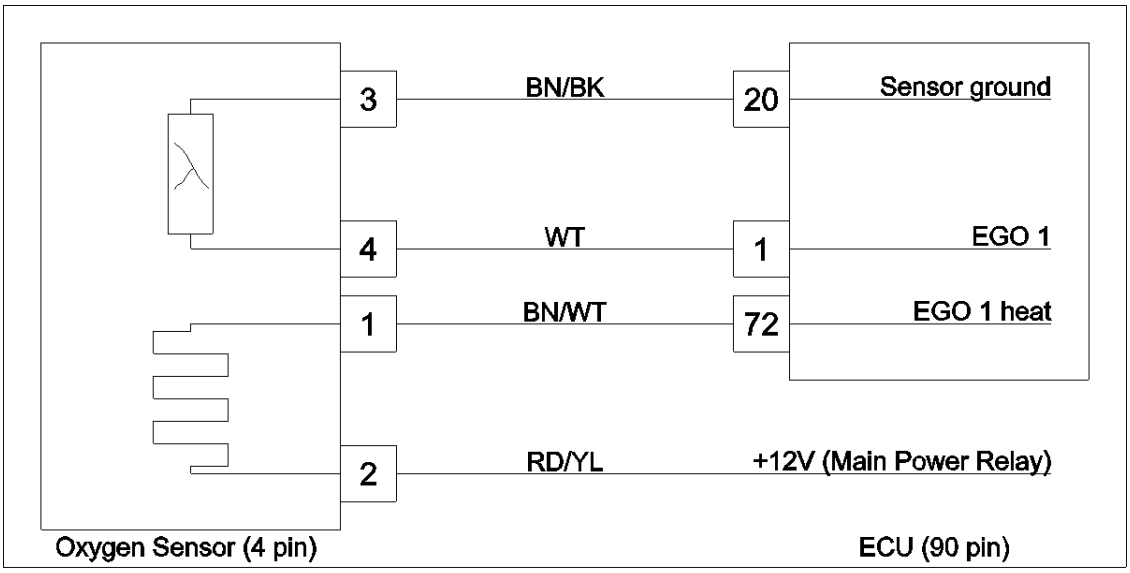
Diagnostic Aid

Fuel Mixer: The system can be rich due to a faulty fuel mixer.

Fuel Pressure: High fuel pressure or a faulty fuel regulator can cause the system to run rich.

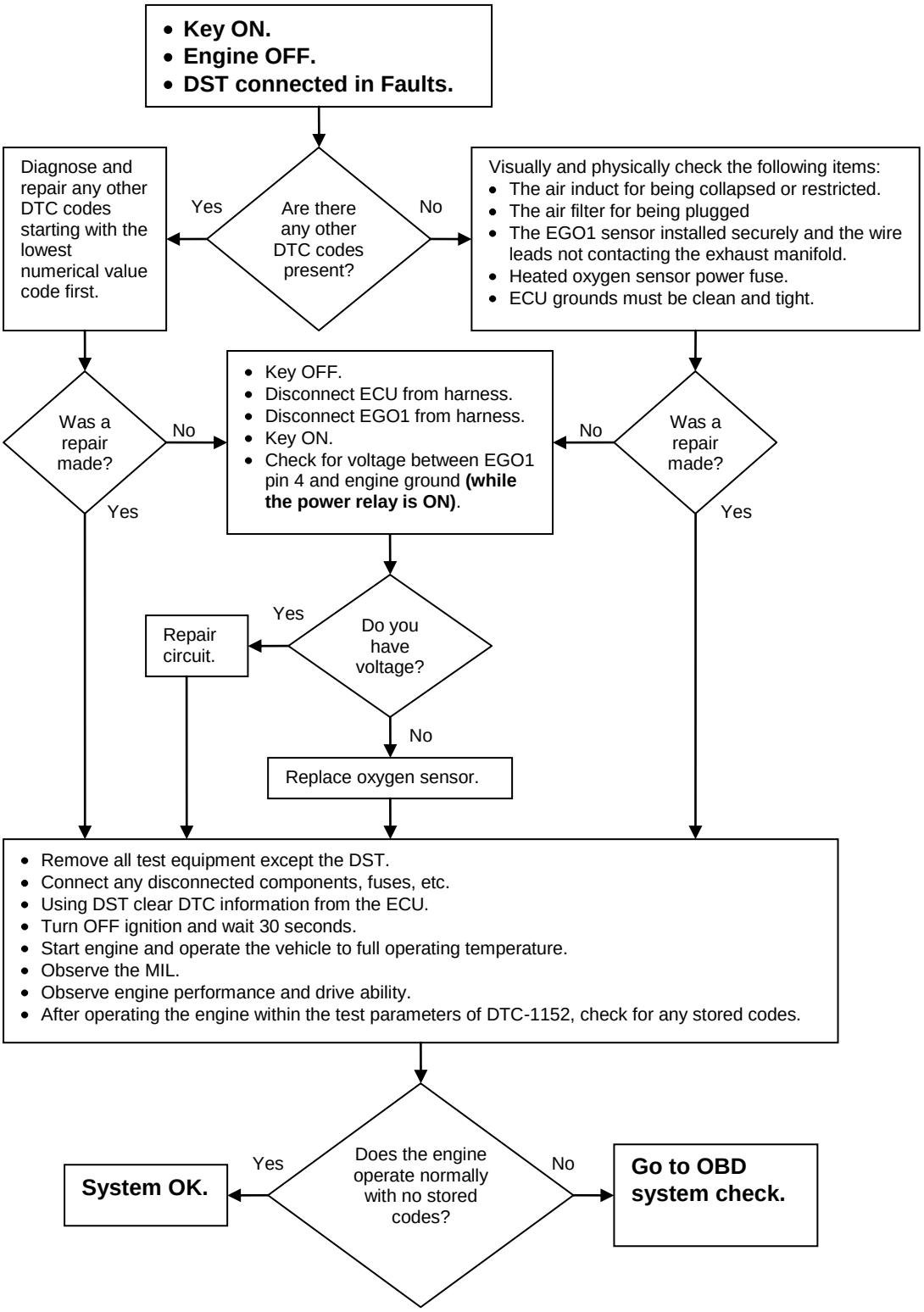
Fuel Quality: A drastic variation in fuel quality (very high butane content) may cause the fuel system to run rich.

Air Filter: A plugged, damaged or modified air filter may cause the system to run rich.



All information concerning connectors, pin numbers and wire colors can be found in the connector documentation.

DTC 1152 – CL low LPG



All measurements are done at the connector unless otherwise stated.

DTC 1153 – CL high NG

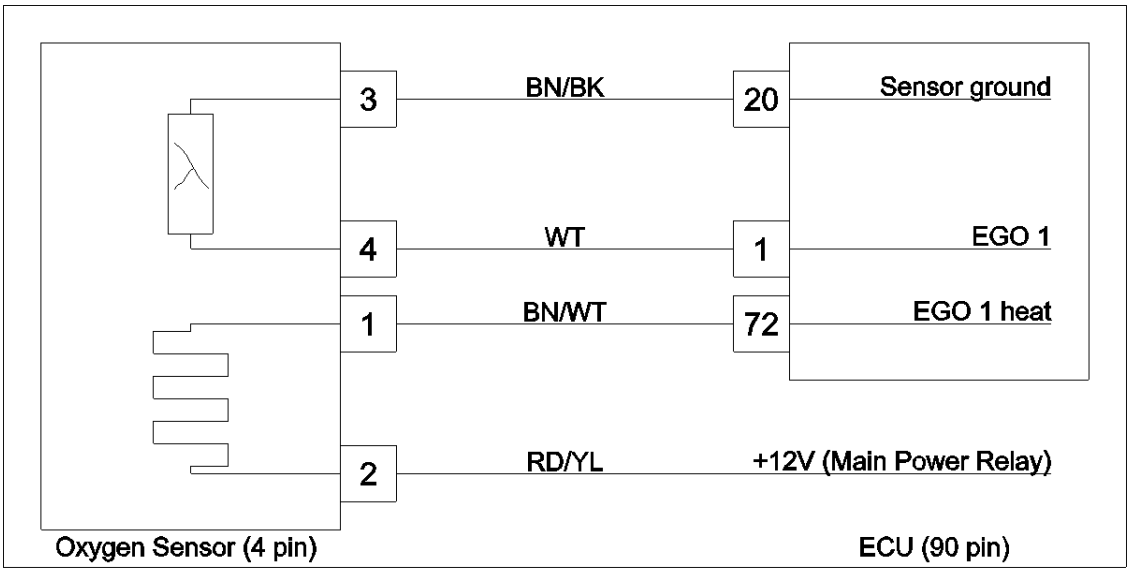
General

The oxygen sensor is used to determine if the fuel flow to the engine is correct by measuring the oxygen content in the exhaust gas. The ECU uses this information to correct the fuel flow with the closed loop multiplier and the adaptive learn. This fault will set if the Closed Loop (CL) multiplier is higher than specified.

For detailed information, please see customer specific DTC settings.

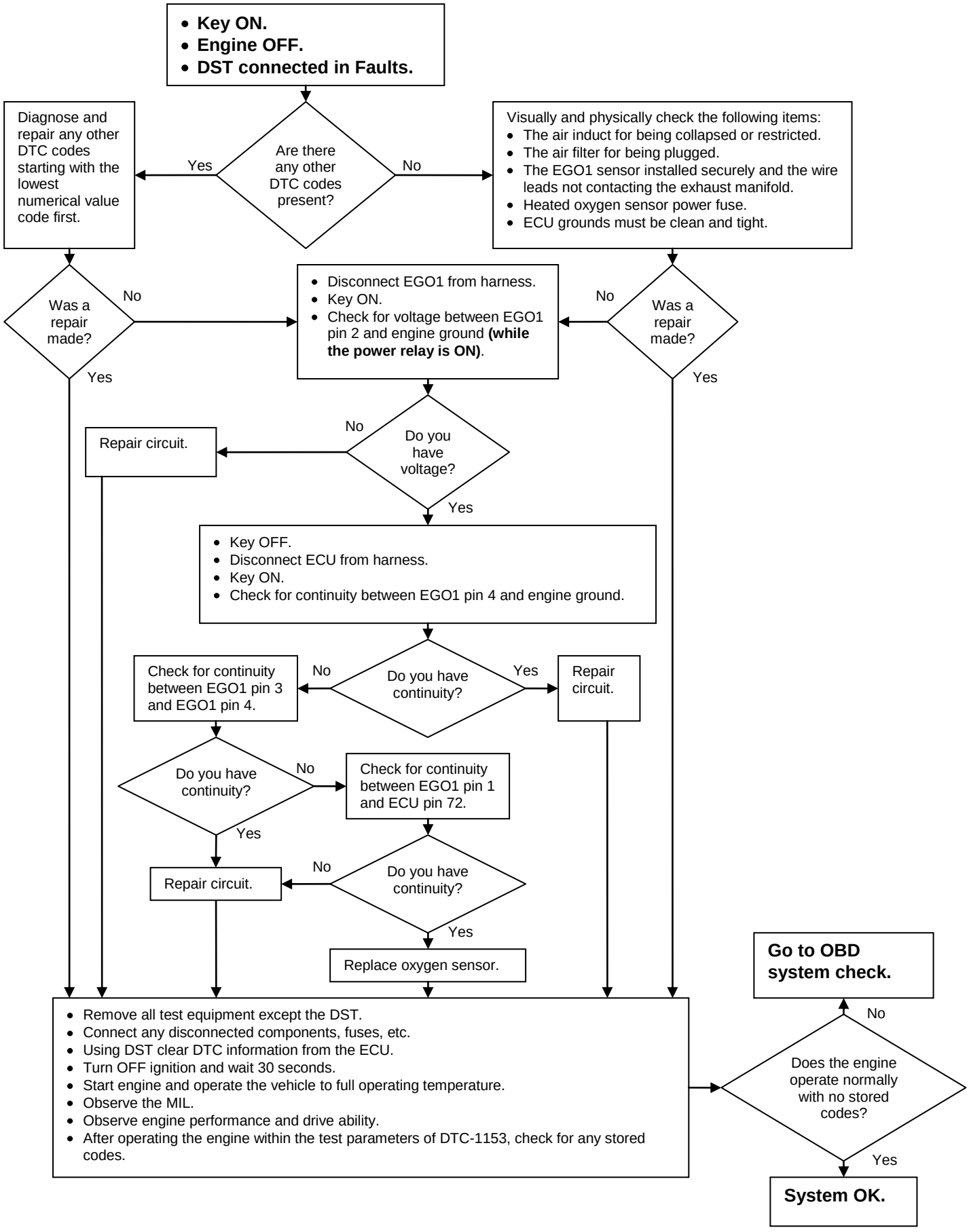
Diagnostic Aid

Oxygen Sensor Wiring:	Heated Oxygen sensor wires may be mis-routed and contacting the exhaust manifold.
Vacuum Leaks:	Large vacuum leaks and crankcase leaks can cause a lean exhaust condition especially at low loads.
Fuel Mixer:	The system can be lean due to a faulty fuel mixer.
Fuel Pressure:	Low fuel pressure, a faulty fuel regulator or a contaminated fuel filter can cause the system to run lean.
Exhaust leaks:	If there is an exhaust leak, outside air can be pulled into the exhaust and pass the oxygen sensor causing a false lean condition.
Fuel Quality:	Contaminated or spoiled fuel can cause the fuel system to be lean.
Ground problem:	ECU grounds must be clean, tight and in the proper location.



All information concerning connectors, pin numbers and wire colors can be found in the connector documentation.

DTC 1153 – CL high NG



All measurements are done at the connector unless otherwise stated.

DTC 1154 – CL low NG

General

The oxygen sensor is used to determine if the fuel flow to the engine is correct by measuring the oxygen content in the exhaust gas. The ECU uses this information to correct the fuel flow with the closed loop multiplier and the adaptive learn. This fault will set if the Closed Loop (CL) multiplier is lower than specified.

For detailed information, please see customer specific DTC settings.

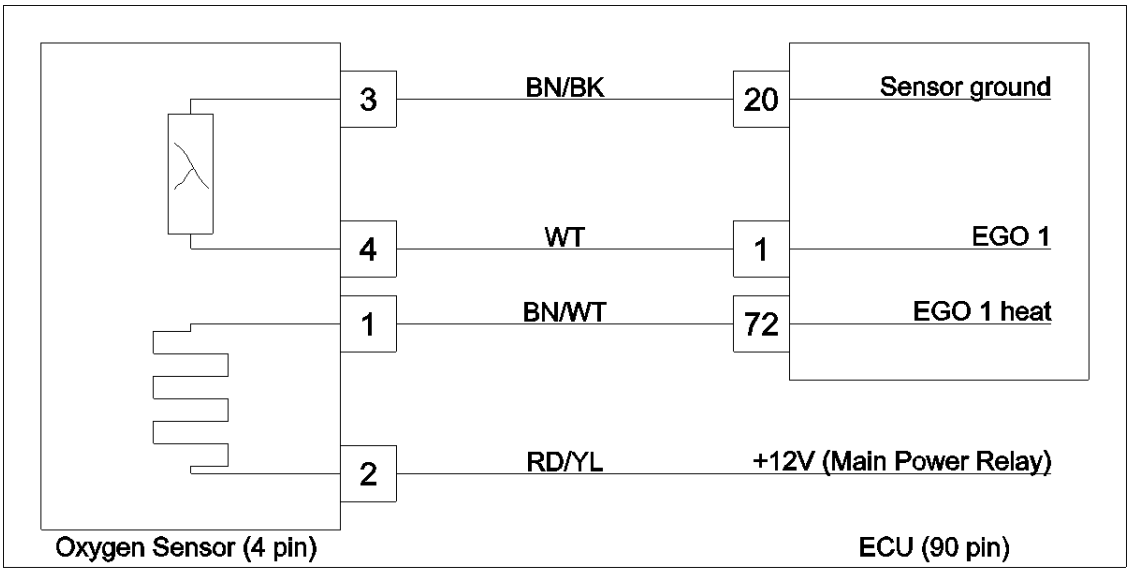
Diagnostic Aid

Fuel Mixer: The system can be rich due to a faulty fuel mixer.

Fuel Pressure: High fuel pressure or a faulty fuel regulator can cause the system to run rich.

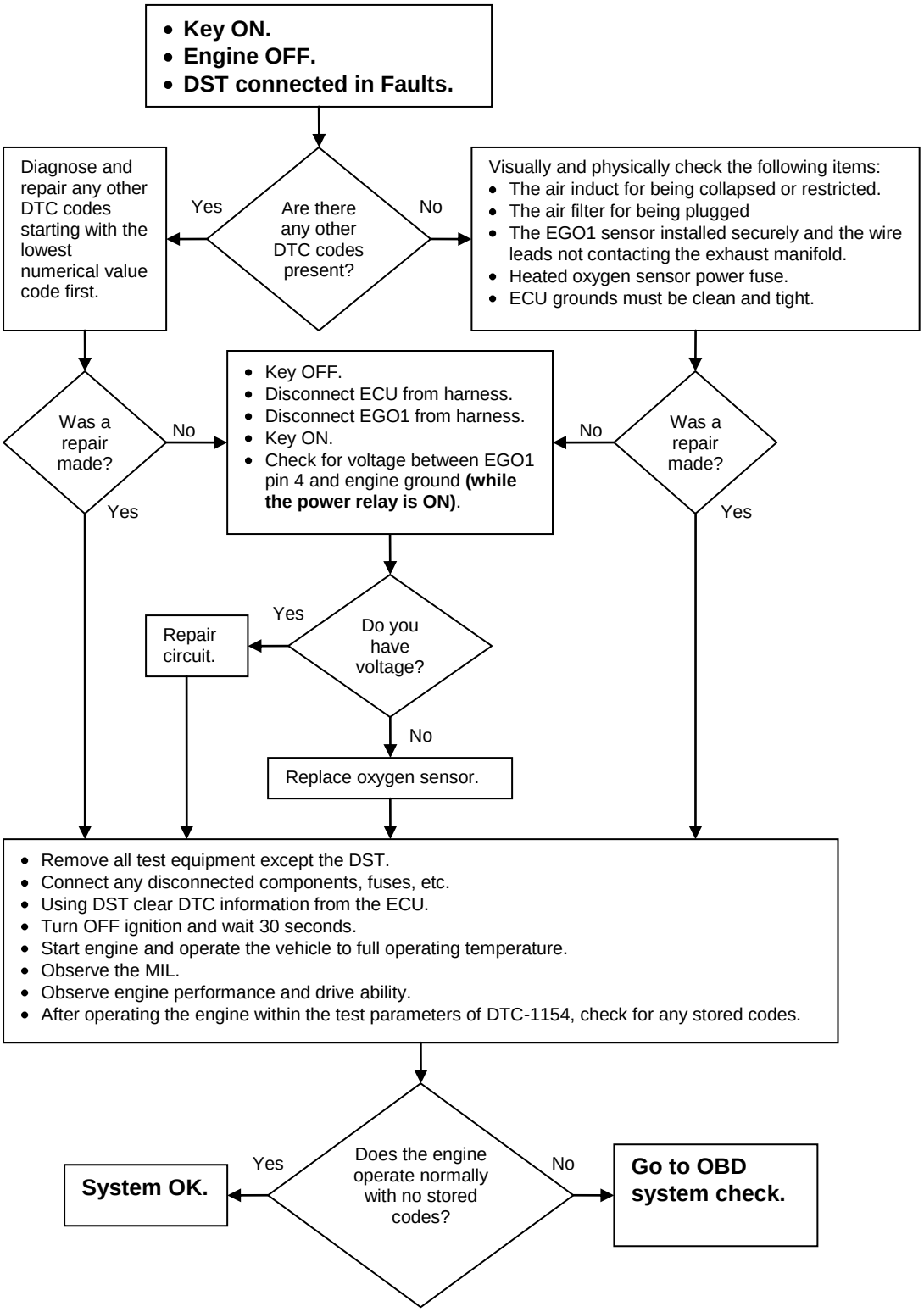
Fuel Quality: A drastic variation in fuel quality (very high butane content) may cause the fuel system to run rich.

Air Filter: A plugged, damaged or modified air filter may cause the system to run rich.



All information concerning connectors, pin numbers and wire colors can be found in the connector documentation.

DTC 1154 – CL low NG



All measurements are done at the connector unless otherwise stated.

DTC 1161 – AL high LPG

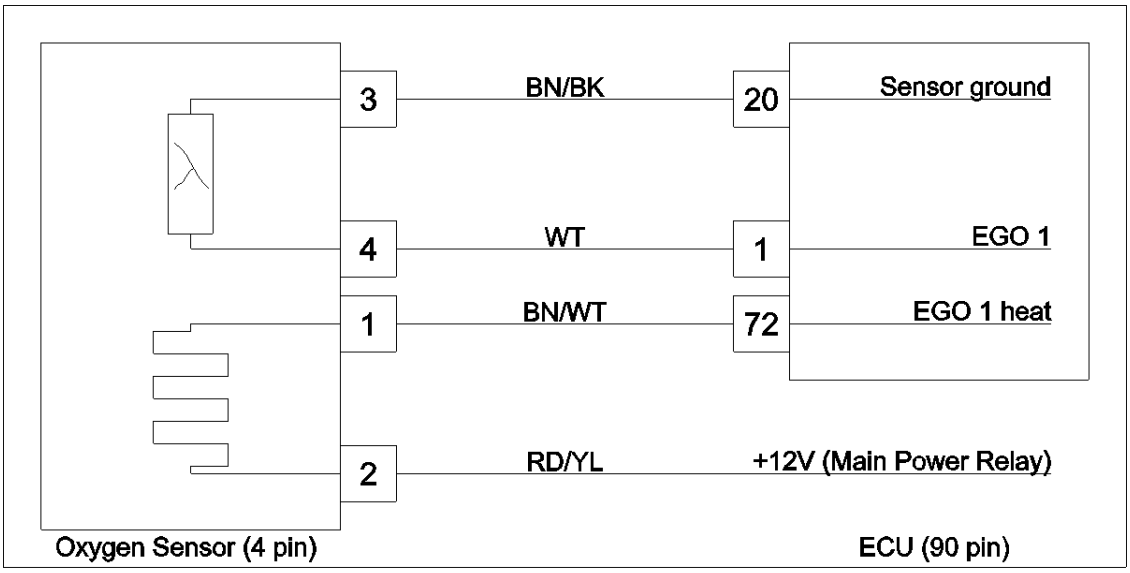
General

The oxygen sensor is used to determine if the fuel flow to the engine is correct by measuring the oxygen content in the exhaust gas. The ECU uses this information to correct the fuel flow with the closed loop multiplier and the adaptive learn. This fault will set if the Adaptive Learn (AL) is higher than specified.

For detailed information, please see customer specific DTC settings.

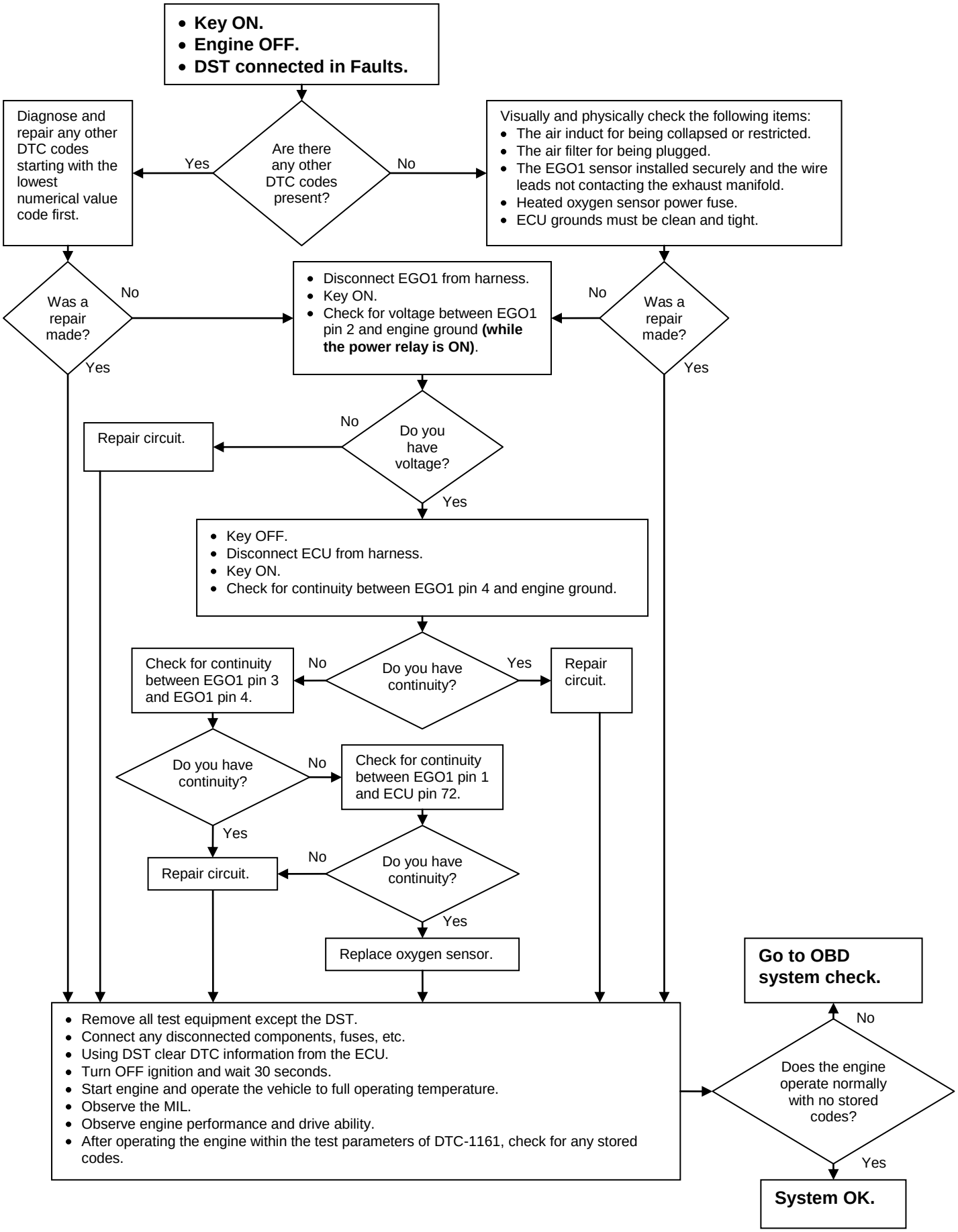
Diagnostic Aid

Oxygen Sensor Wiring:	Heated Oxygen sensor wires may be mis-routed and contacting the exhaust manifold.
Vacuum Leaks:	Large vacuum leaks and crankcase leaks can cause a lean exhaust condition especially at low loads.
Fuel Mixer:	The system can be lean due to a faulty fuel mixer.
Fuel Pressure:	Low fuel pressure, a faulty fuel regulator or a contaminated fuel filter can cause the system to run lean.
Exhaust leaks:	If there is an exhaust leak, outside air can be pulled into the exhaust and pass the oxygen sensor causing a false lean condition.
Fuel Quality:	Contaminated or spoiled fuel can cause the fuel system to be lean.
Ground problem:	ECU grounds must be clean, tight and in the proper location.



All information concerning connectors, pin numbers and wire colors can be found in the connector documentation.

DTC 1161 – AL high LPG



All measurements are done at the connector unless otherwise stated.

DTC 1162 – AL low LPG

General

The oxygen sensor is used to determine if the fuel flow to the engine is correct by measuring the oxygen content in the exhaust gas. The ECU uses this information to correct the fuel flow with the closed loop multiplier and the adaptive learn. This fault will set if the Adaptive Learn (AL) is lower than specified.

For detailed information, please see customer specific DTC settings.

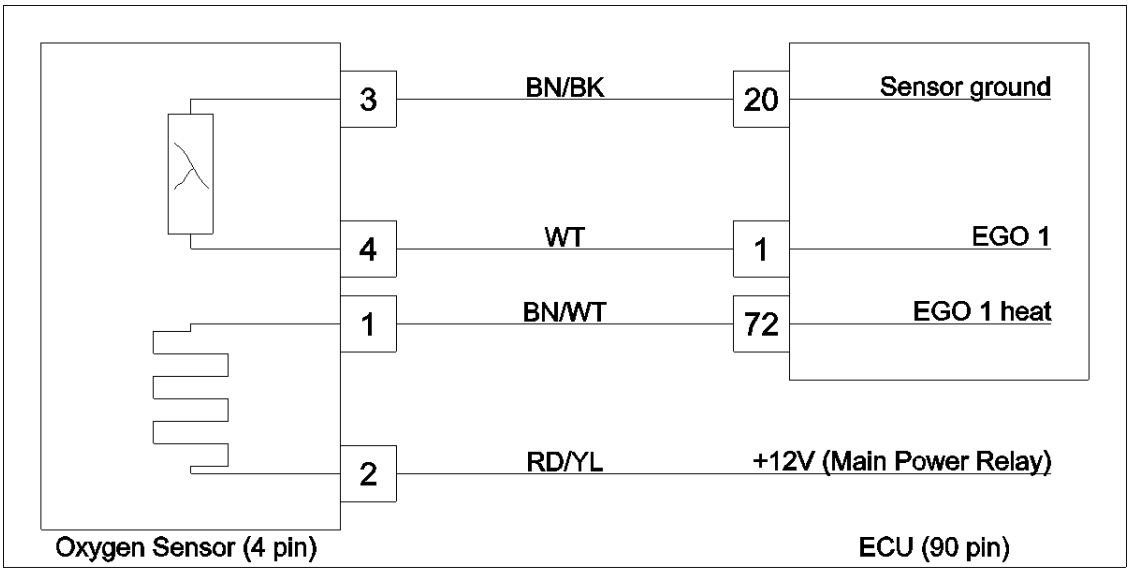
Diagnostic Aid

Fuel Mixer: The system can be rich due to a faulty fuel mixer.

Fuel Pressure: High fuel pressure or a faulty fuel regulator can cause the system to run rich.

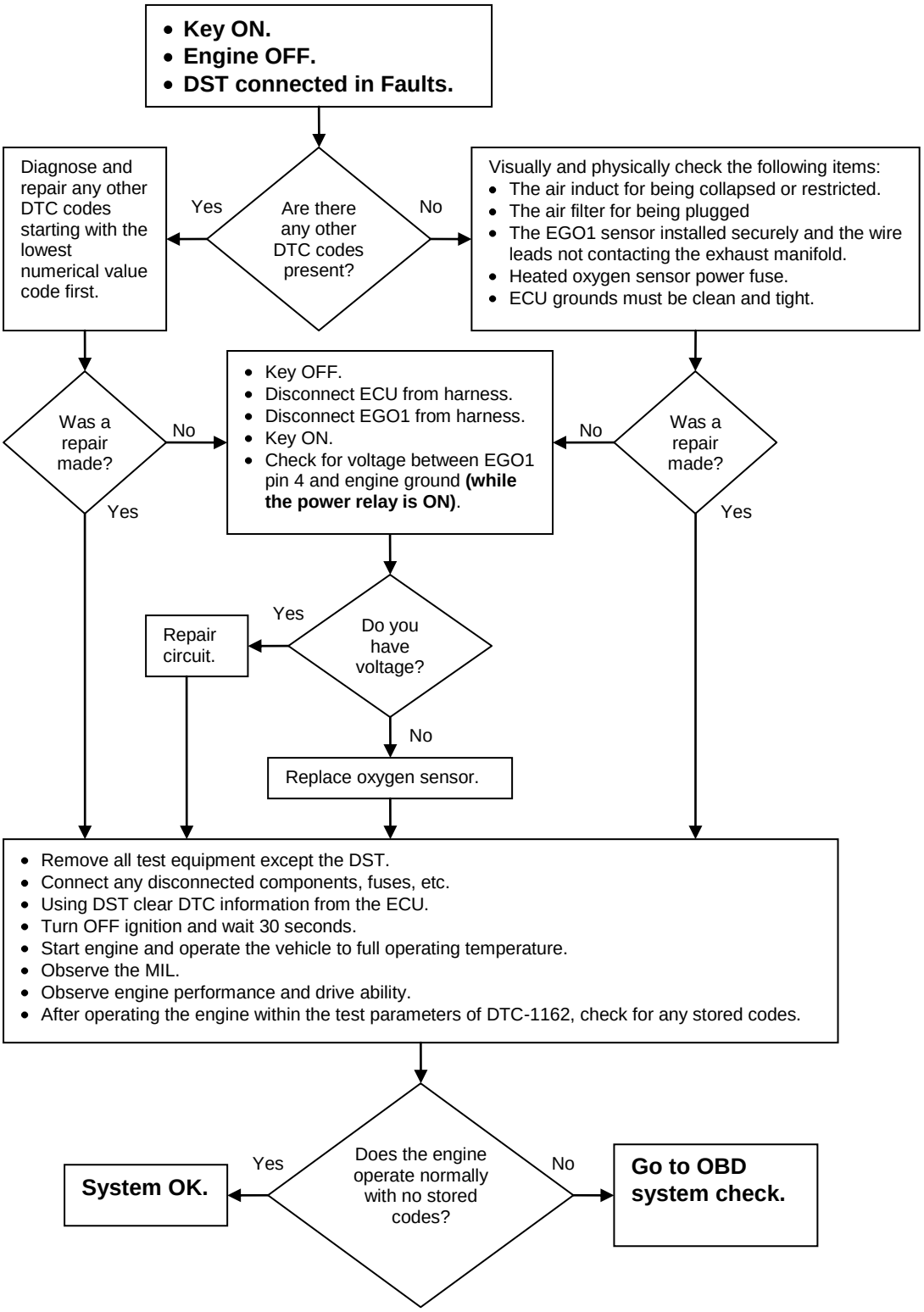
Fuel Quality: A drastic variation in fuel quality (very high butane content) may cause the fuel system to run rich.

Air Filter: A plugged, damaged or modified air filter may cause the system to run rich.



All information concerning connectors, pin numbers and wire colors can be found in the connector documentation.

DTC 1162 – AL low LPG



All measurements are done at the connector unless otherwise stated.

DTC 1163 – AL high NG

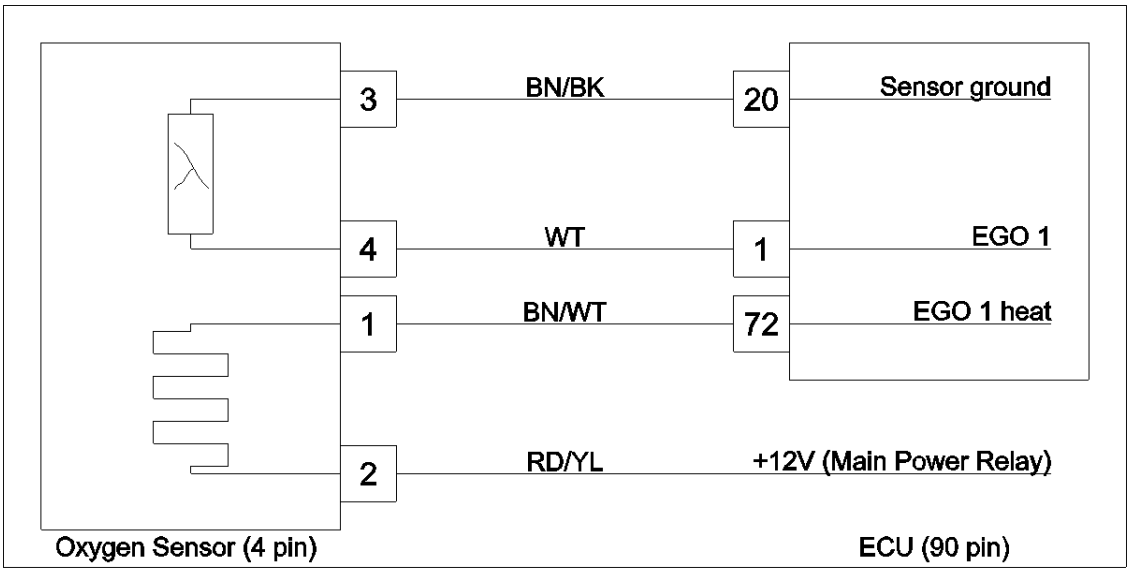
General

The oxygen sensor is used to determine if the fuel flow to the engine is correct by measuring the oxygen content in the exhaust gas. The ECU uses this information to correct the fuel flow with the closed loop multiplier and the adaptive learn. This fault will set if the Adaptive Learn (AL) is higher than specified.

For detailed information, please see customer specific DTC settings.

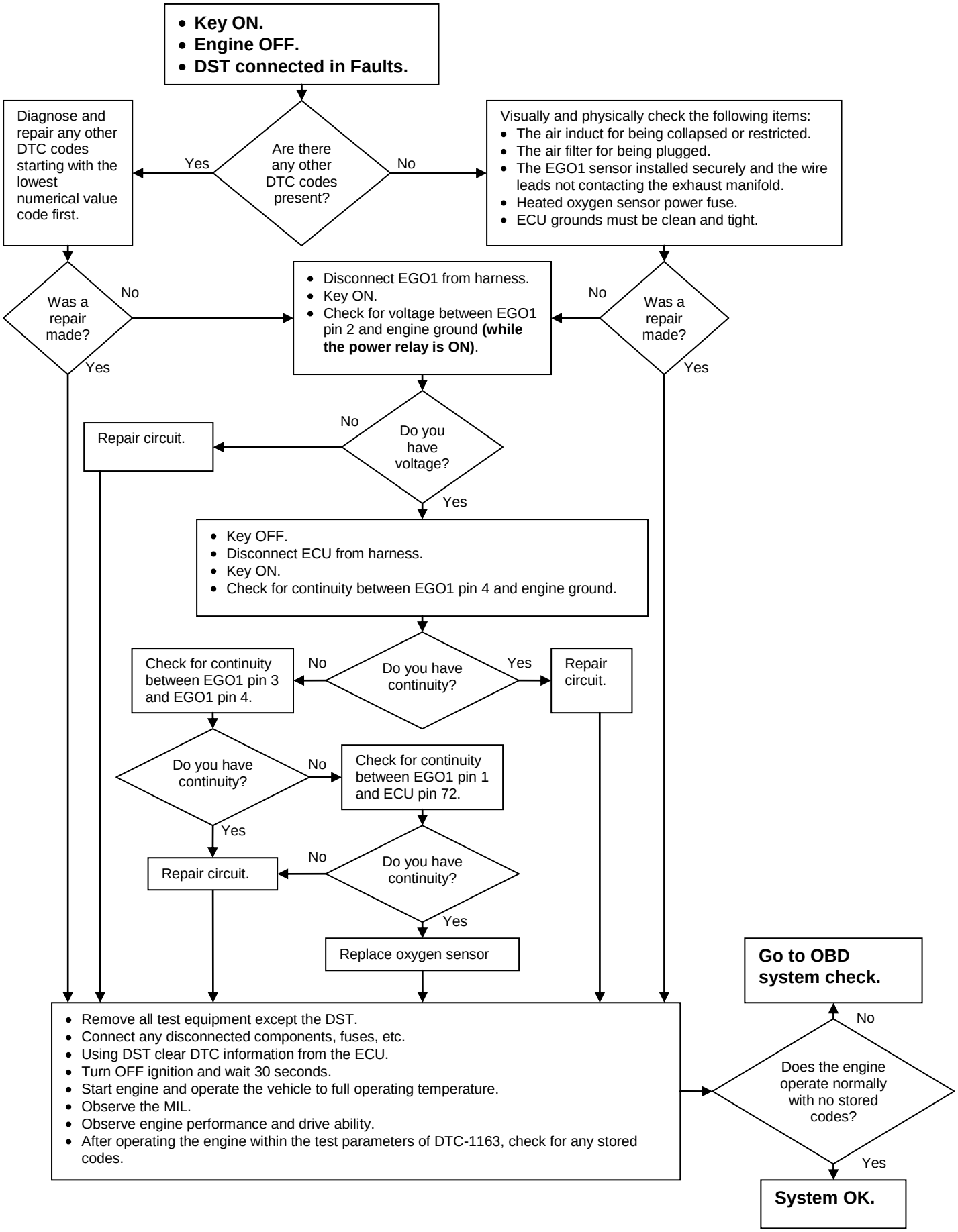
Diagnostic Aid

Oxygen Sensor Wiring:	Heated Oxygen sensor wires may be mis-routed and contacting the exhaust manifold.
Vacuum Leaks:	Large vacuum leaks and crankcase leaks can cause a lean exhaust condition especially at low loads.
Fuel Mixer:	The system can be lean due to a faulty fuel mixer.
Fuel Pressure:	Low fuel pressure, a faulty fuel regulator or a contaminated fuel filter can cause the system to run lean.
Exhaust leaks:	If there is an exhaust leak, outside air can be pulled into the exhaust and pass the oxygen sensor causing a false lean condition.
Fuel Quality:	Contaminated or spoiled fuel can cause the fuel system to be lean.
Ground problem:	ECU grounds must be clean, tight and in the proper location.



All information concerning connectors, pin numbers and wire colors can be found in the connector documentation.

DTC 1163 – AL high NG



All measurements are done at the connector unless otherwise stated.

DTC 1164 – AL low NG

General

The oxygen sensor is used to determine if the fuel flow to the engine is correct by measuring the oxygen content in the exhaust gas. The ECU uses this information to correct the fuel flow with the closed loop multiplier and the adaptive learn. This fault will set if the Adaptive Learn (AL) is lower than specified.

For detailed information, please see customer specific DTC settings.

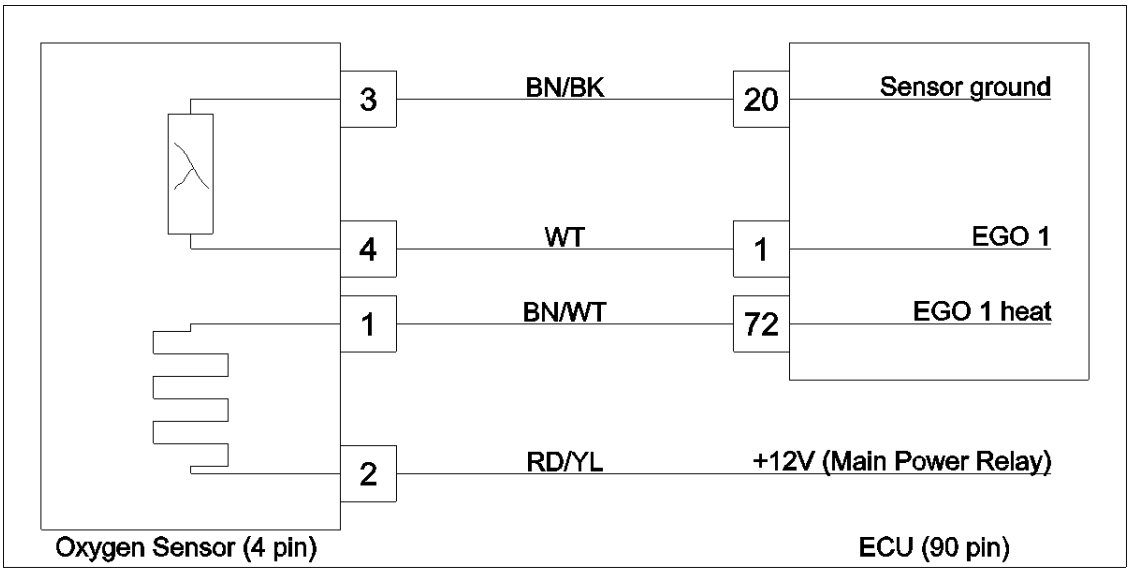
Diagnostic Aid

Fuel Mixer: The system can be rich due to a faulty fuel mixer.

Fuel Pressure: High fuel pressure or a faulty fuel regulator can cause the system to run rich.

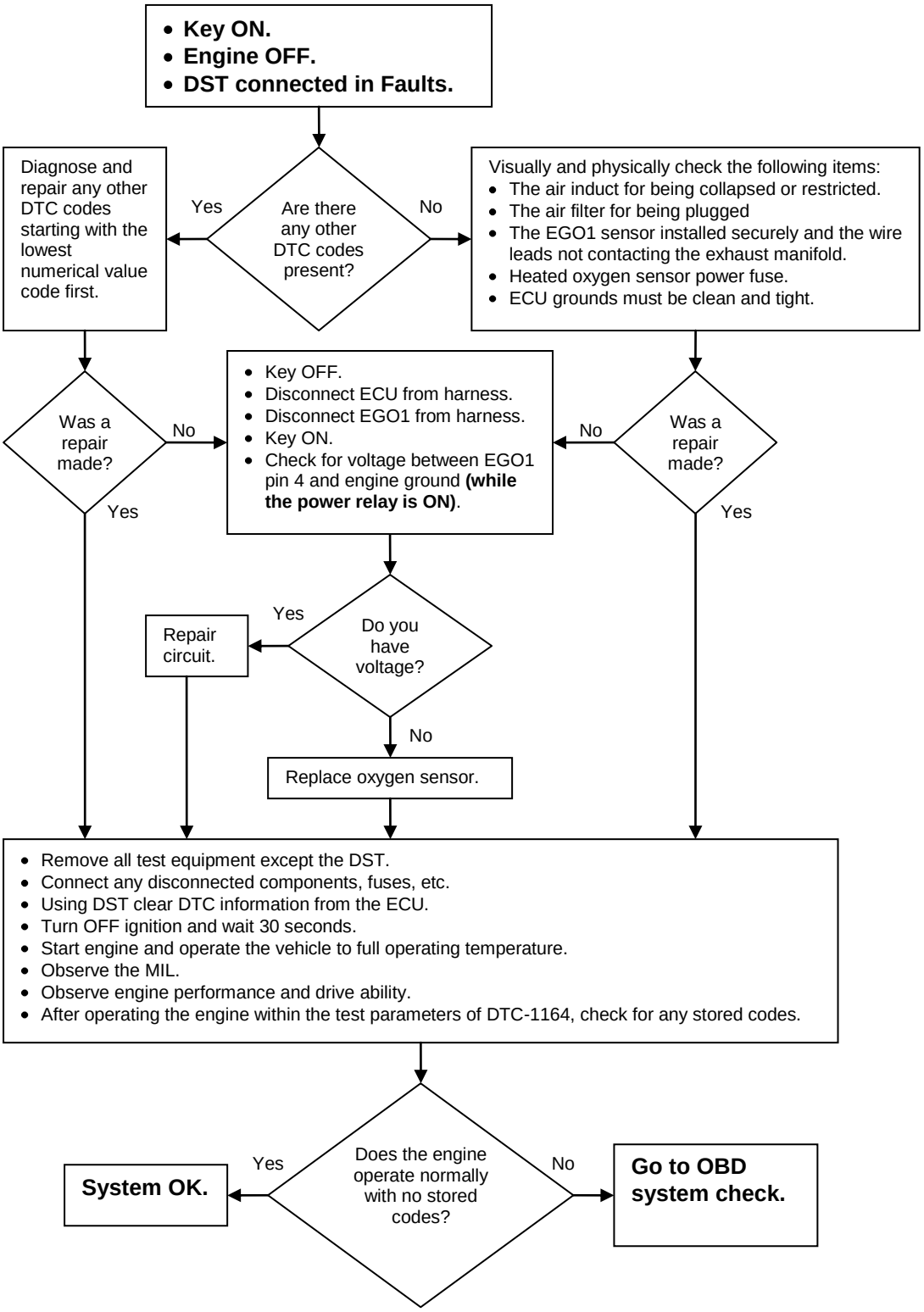
Fuel Quality: A drastic variation in fuel quality (very high butane content) may cause the fuel system to run rich.

Air Filter: A plugged, damaged or modified air filter may cause the system to run rich.



All information concerning connectors, pin numbers and wire colors can be found in the connector documentation.

DTC 1164 – AL low NG



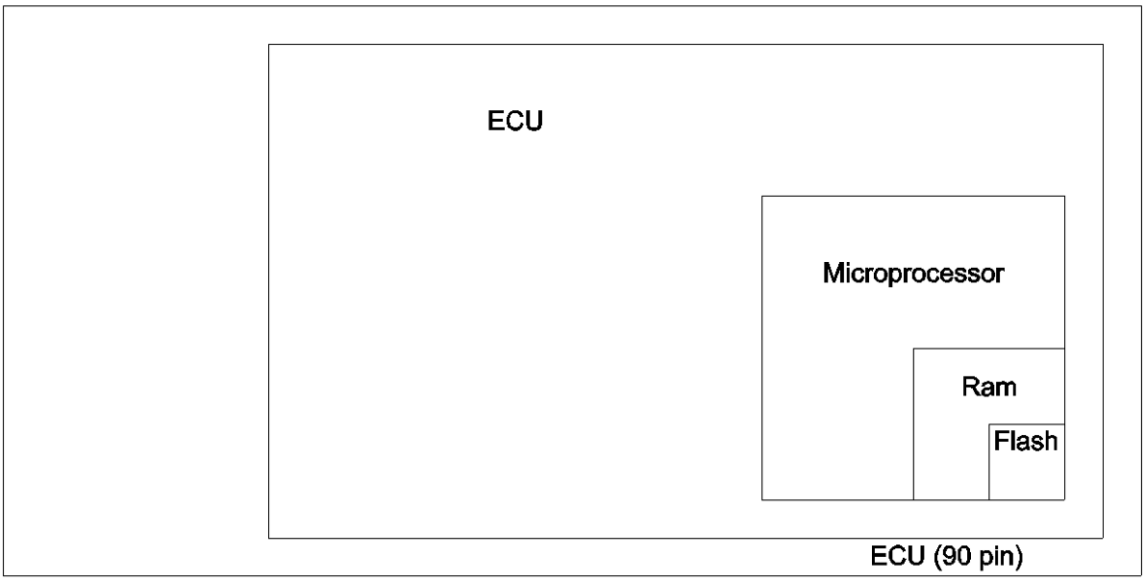
All measurements are done at the connector unless otherwise stated.

DTC 1612 – RTI 1 loss

General

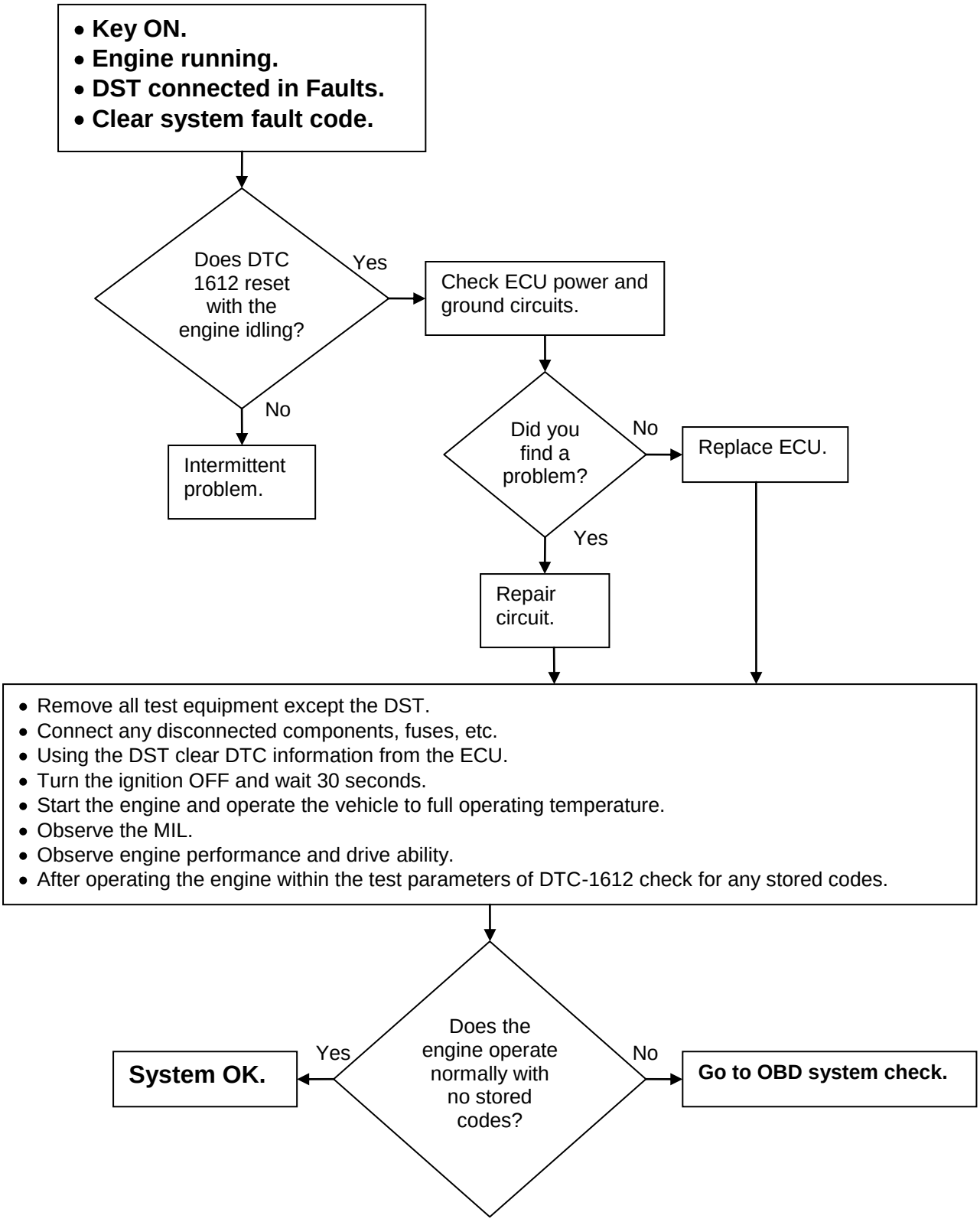
The ECU has several internal checks that must be satisfied each time an instruction is executed. Several different things can happen within the microprocessor that will cause this fault.
The ECU will attempt to reset itself in the event this fault is set.

For detailed information, please see customer specific DTC settings.



All information concerning connectors, pin numbers and wire colors can be found in the connector documentation.

DTC 1612 – RTI 1 loss



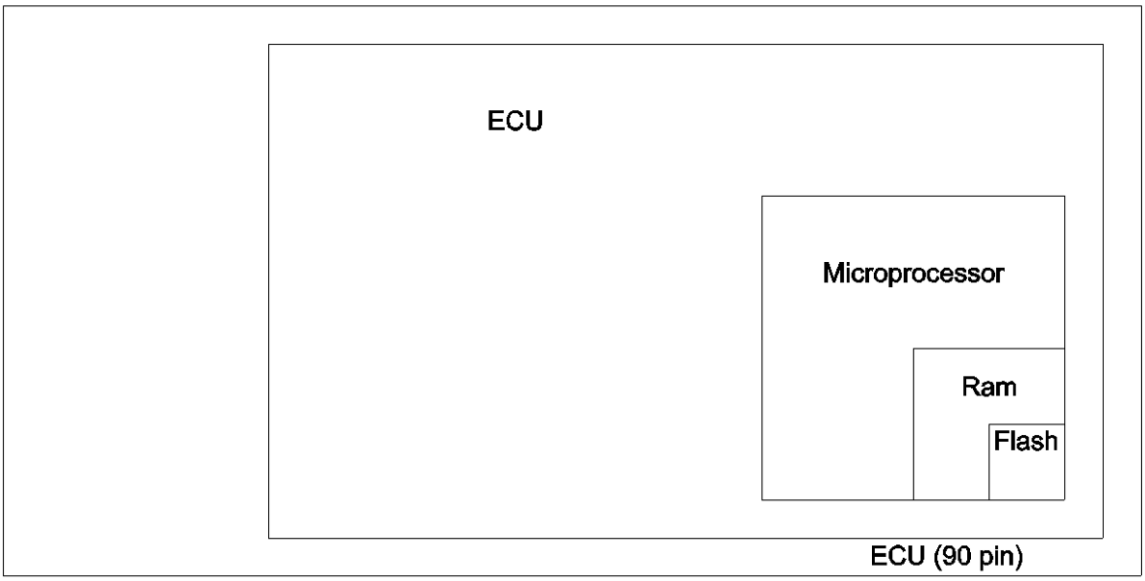
All measurements are done at the connector unless otherwise stated.

DTC 1613 – RTI 2 loss

General

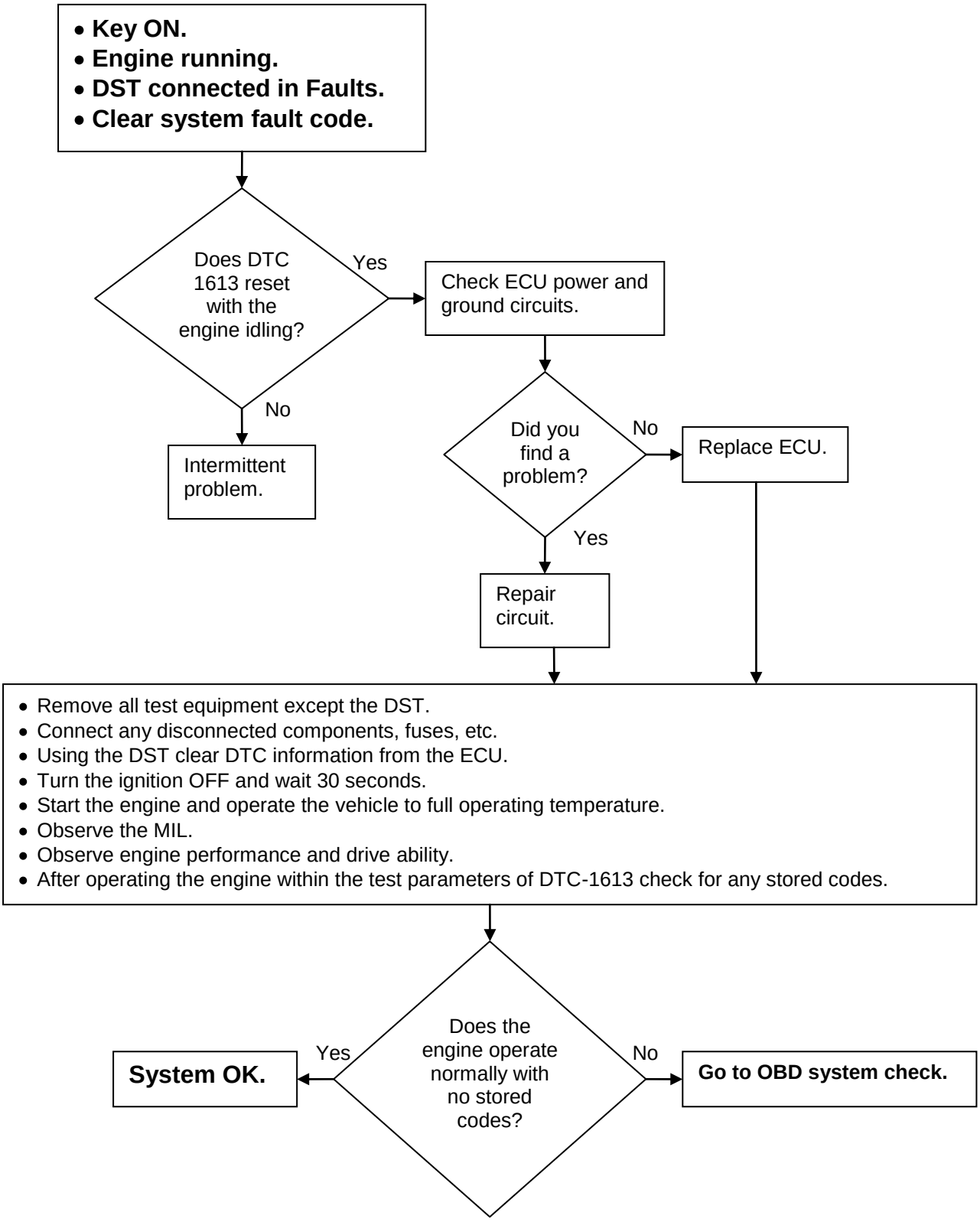
The ECU has several internal checks that must be satisfied each time an instruction is executed. Several different things can happen within the microprocessor that will cause this fault.
The ECU will attempt to reset itself in the event this fault is set.

For detailed information, please see customer specific DTC settings.



All information concerning connectors, pin numbers and wire colors can be found in the connector documentation.

DTC 1613 – RTI 2 loss



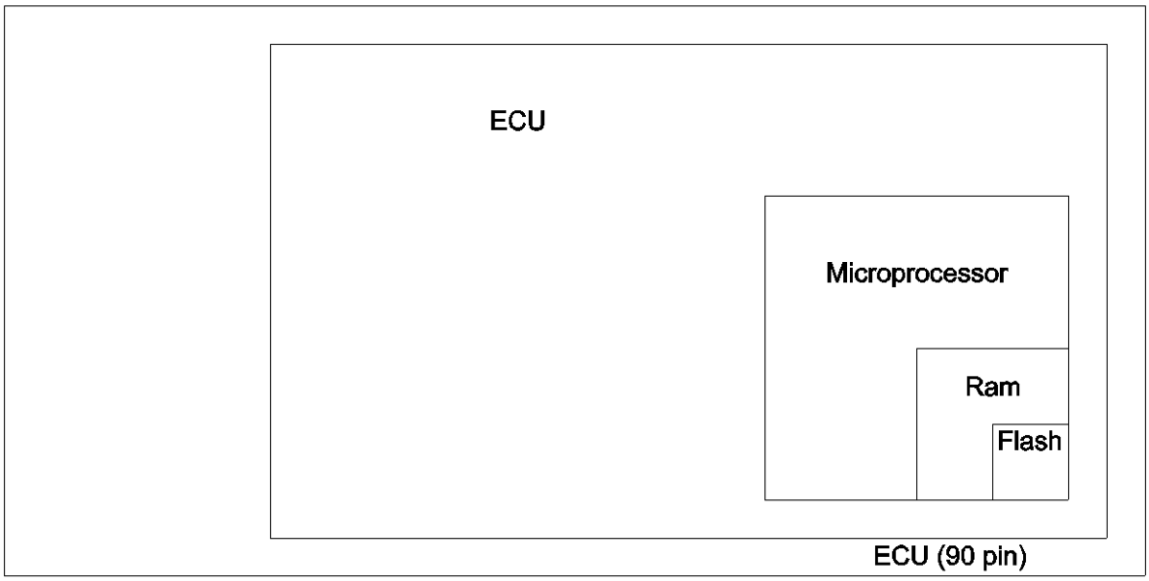
All measurements are done at the connector unless otherwise stated.

DTC 1614 – RTI 3 loss

General

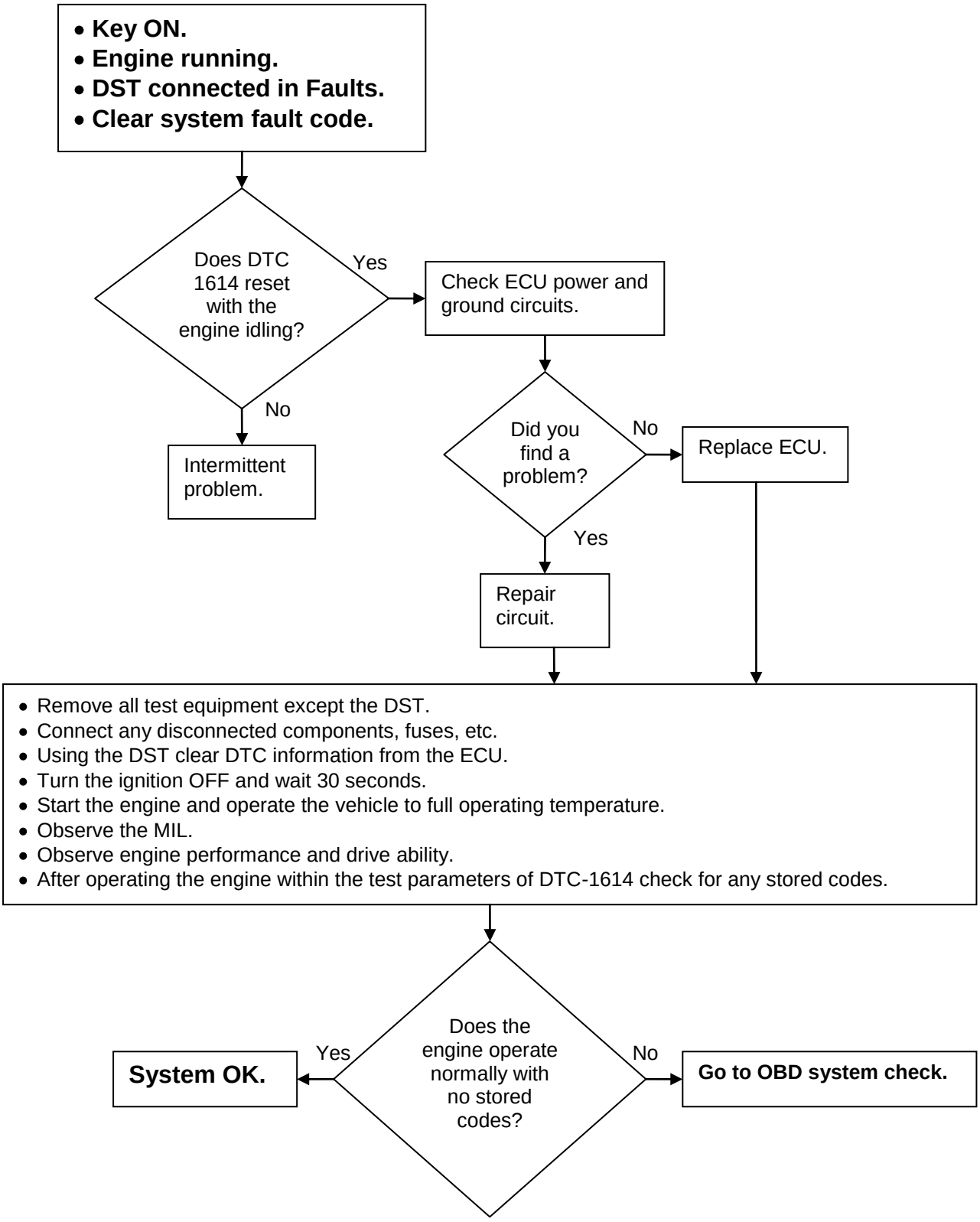
The ECU has several internal checks that must be satisfied each time an instruction is executed. Several different things can happen within the microprocessor that will cause this fault.
The ECU will attempt to reset itself in the event this fault is set.

For detailed information, please see customer specific DTC settings.



All information concerning connectors, pin numbers and wire colors can be found in the connector documentation.

DTC 1614 – RTI 3 loss



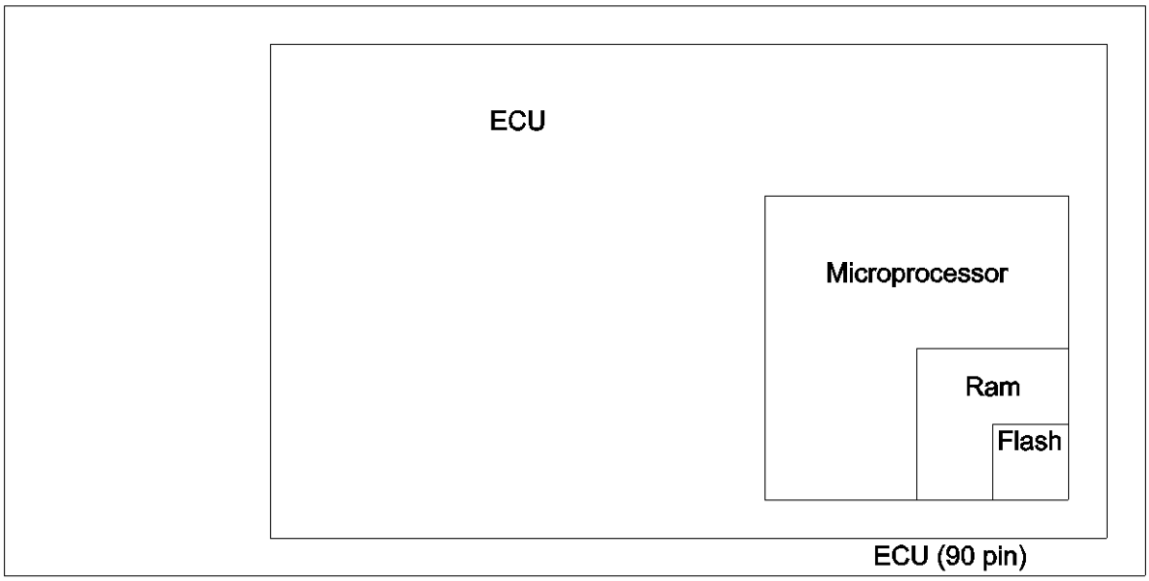
All measurements are done at the connector unless otherwise stated.

DTC 1615 – A/D loss

General

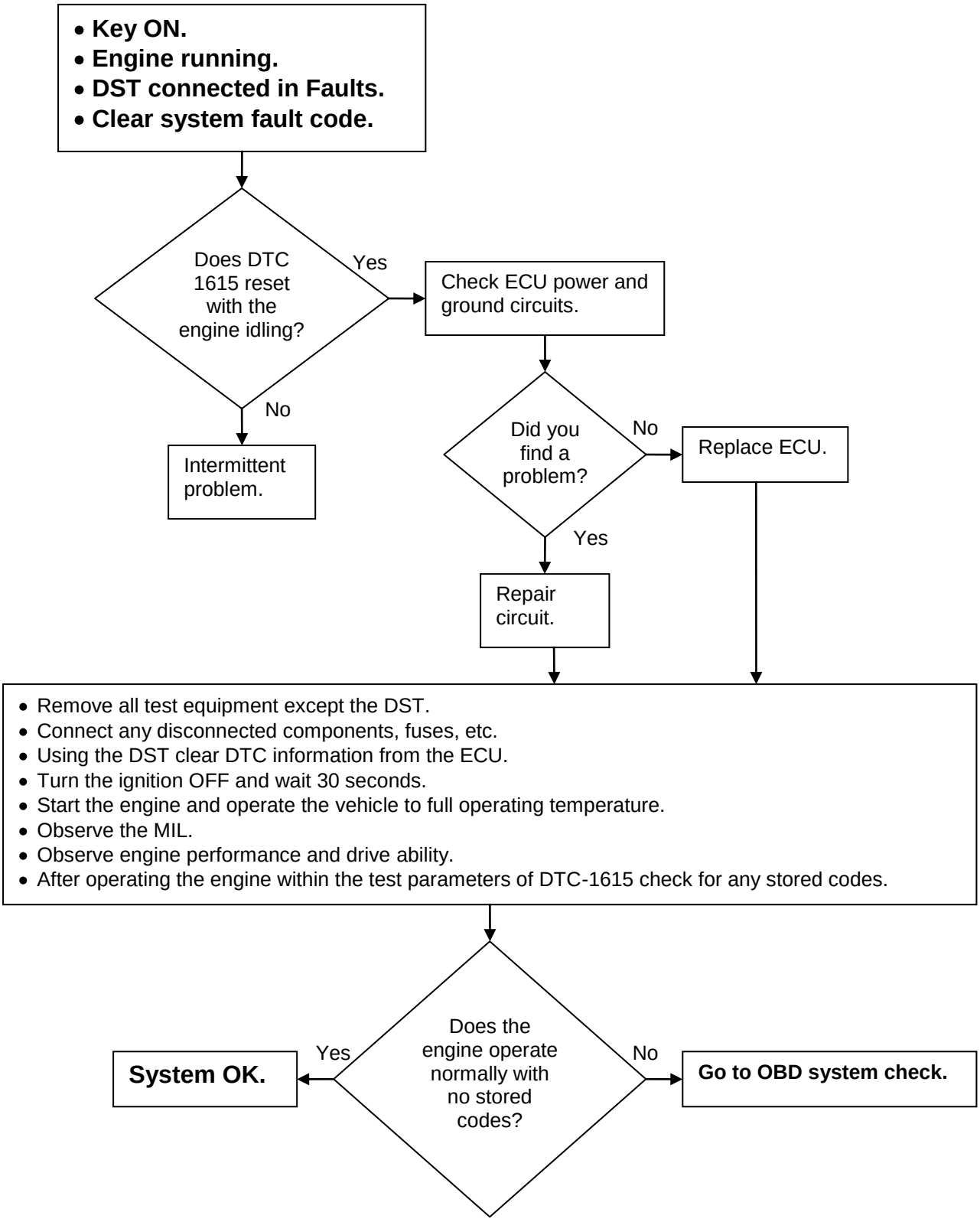
The ECU has several internal checks that must be satisfied each time an instruction is executed. Several different things can happen within the microprocessor that will cause this fault.
The ECU will attempt to reset itself in the event this fault is set.

For detailed information, please see customer specific DTC settings.



All information concerning connectors, pin numbers and wire colors can be found in the connector documentation.

DTC 1615 – A/D loss



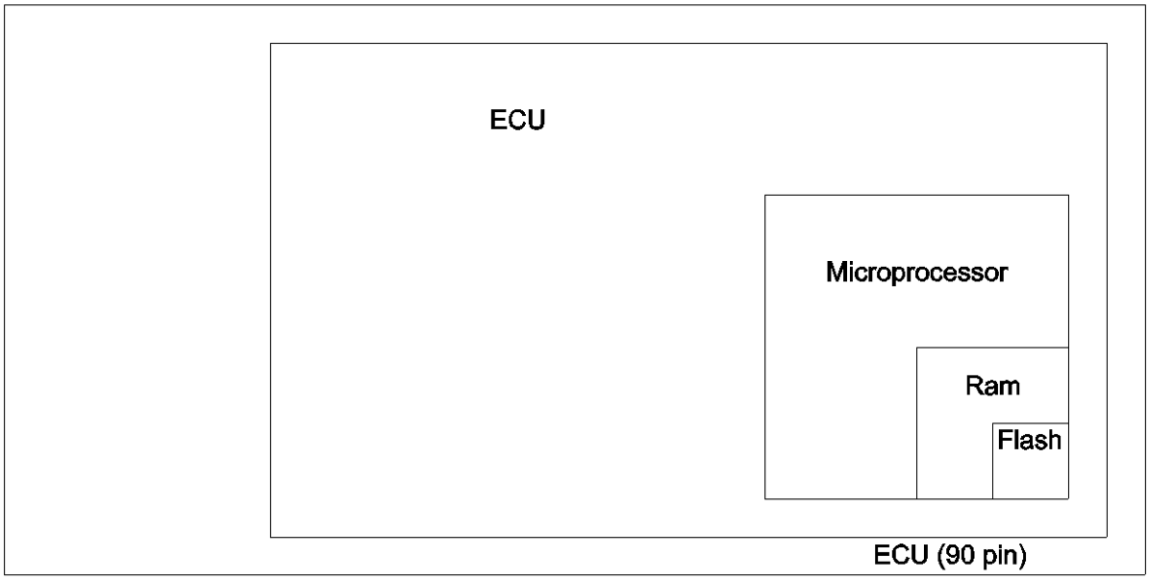
All measurements are done at the connector unless otherwise stated.

DTC 1616 – Invalid interrupt

General

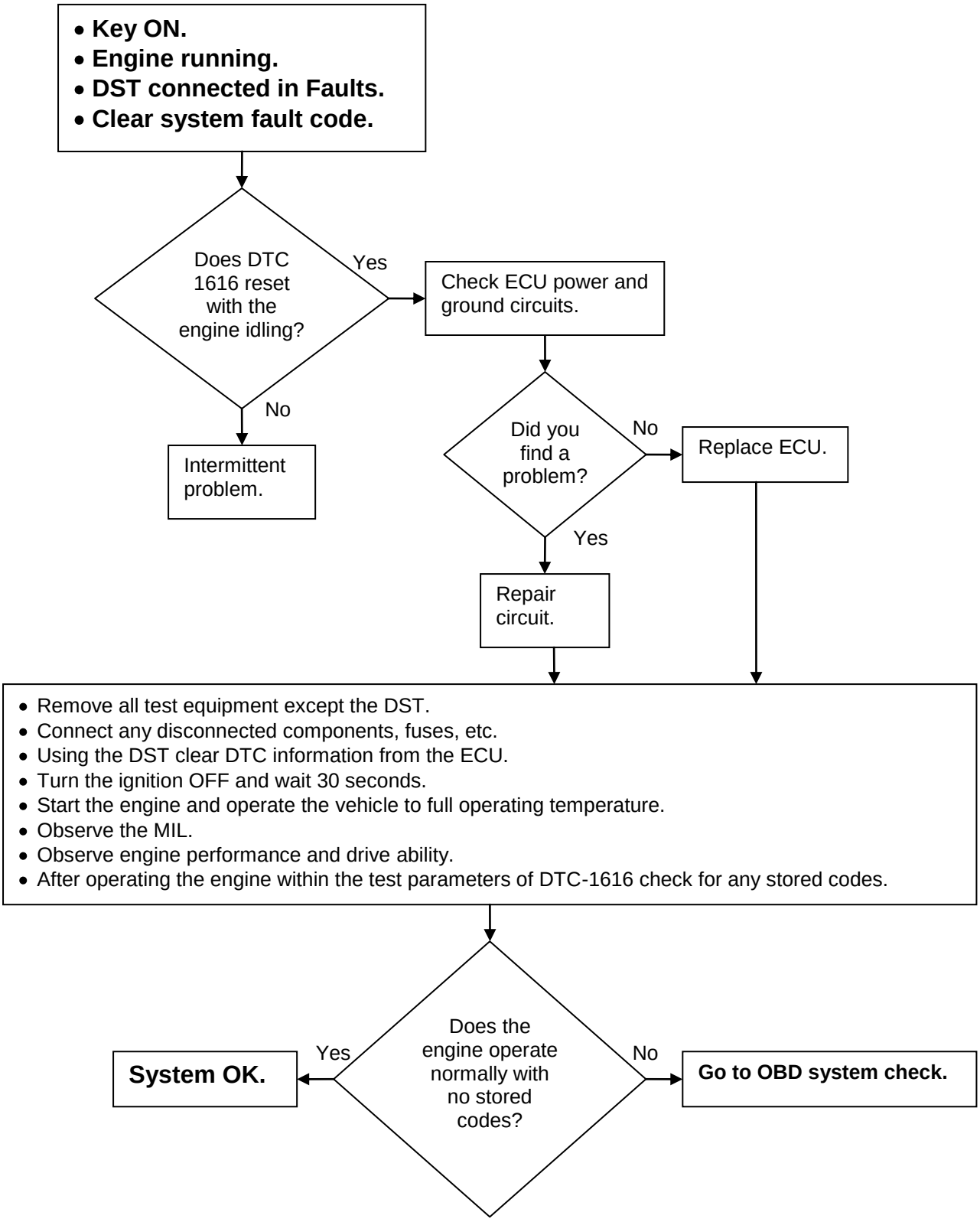
The ECU has several internal checks that must be satisfied each time an instruction is executed. Several different things can happen within the microprocessor that will cause this fault.
The ECU will attempt to reset itself in the event this fault is set.

For detailed information, please see customer specific DTC settings.



All information concerning connectors, pin numbers and wire colors can be found in the connector documentation.

DTC 1616 – Invalid interrupt



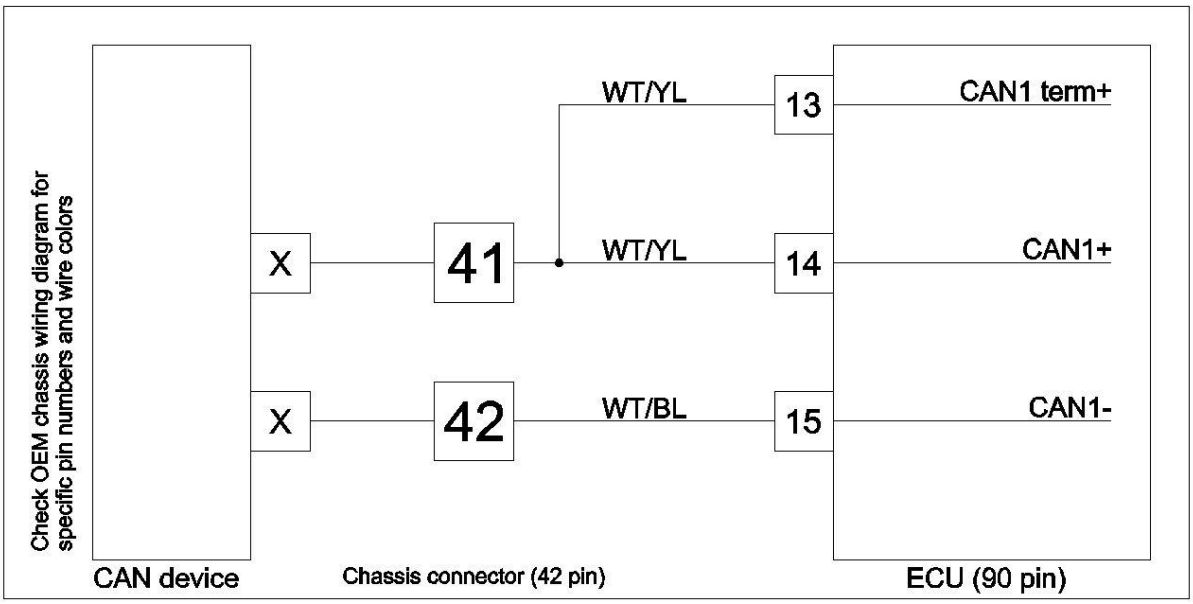
All measurements are done at the connector unless otherwise stated.

DTC 1626 – CAN Tx failure

General

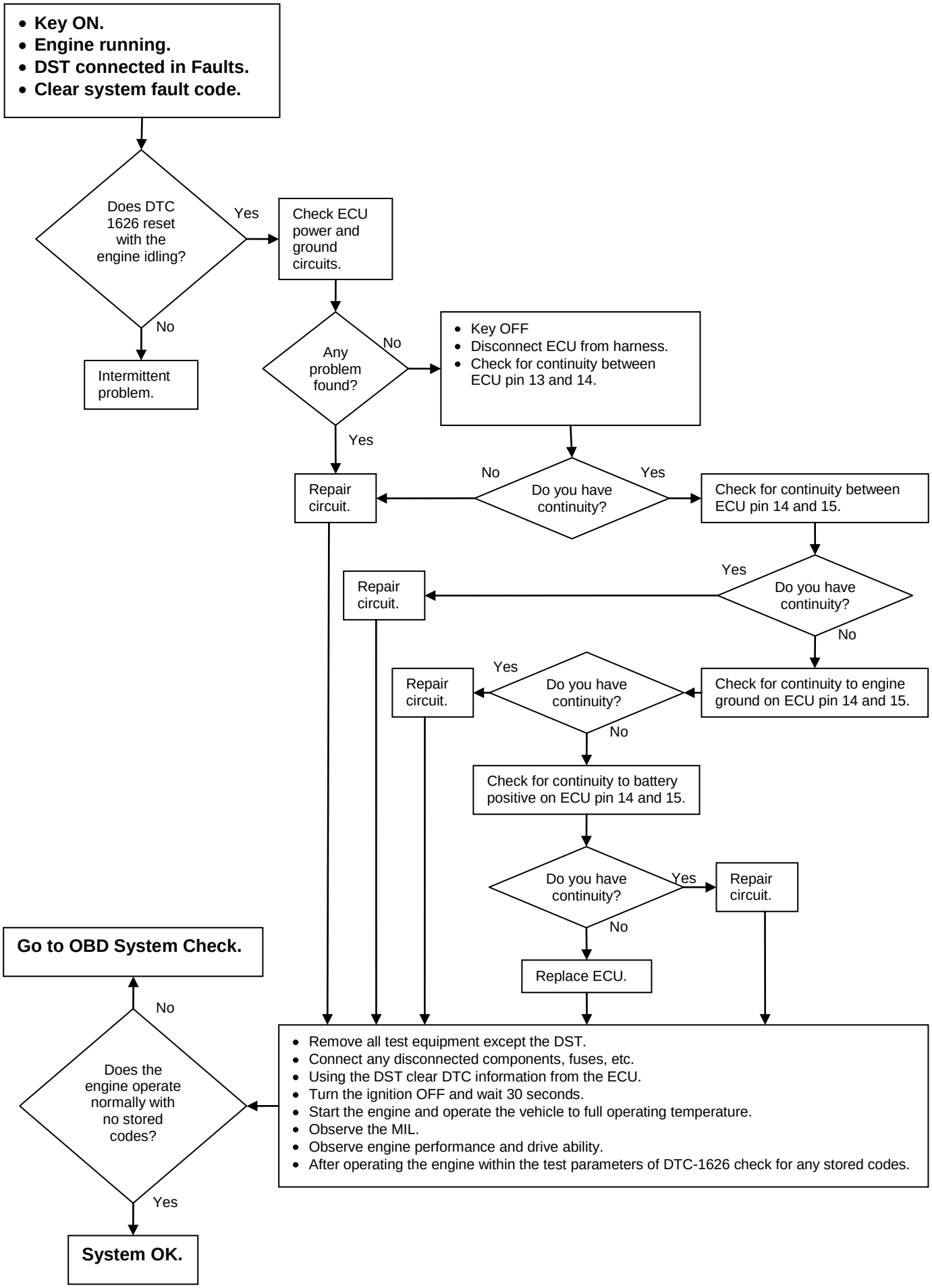
The Controller Area Network (CAN) bus is used by the ECU to communicate with other digital devices used throughout the fuel system. Information is sent over the CAN bus in digital information “packets” that contain information for various control functions. This fault will set if there is a problem with the CAN communication.

For detailed information, please see customer specific DTC settings.



All information concerning connectors, pin numbers and wire colors can be found in the connector documentation.

DTC 1626 – CAN Tx failure



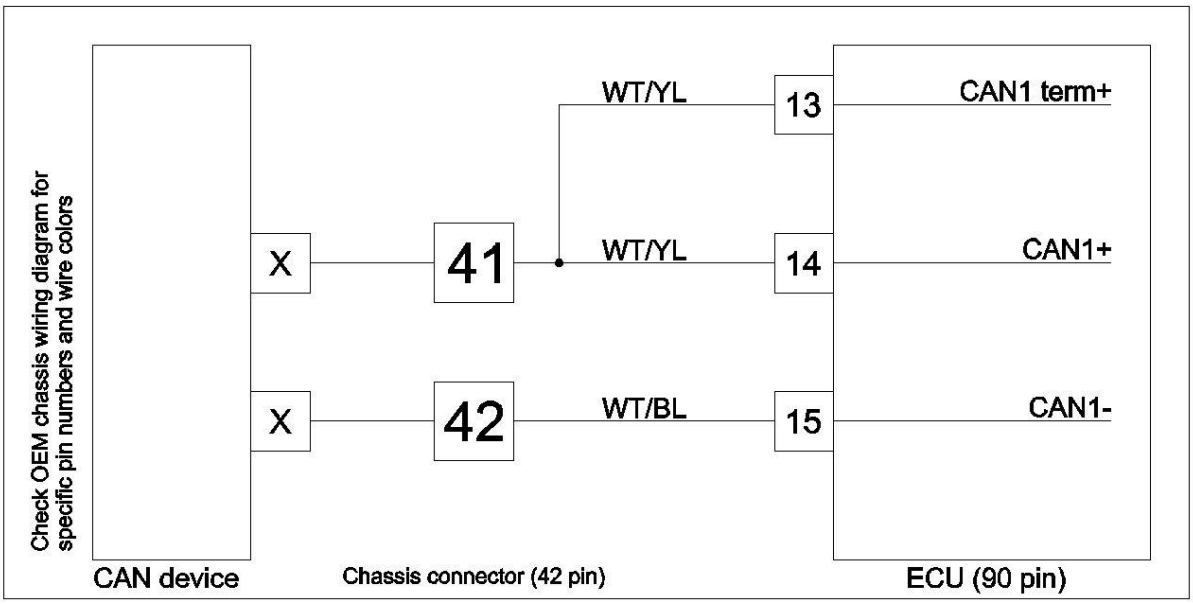
All measurements are done at the connector unless otherwise stated.

DTC 1627 – CAN Rx failure

General

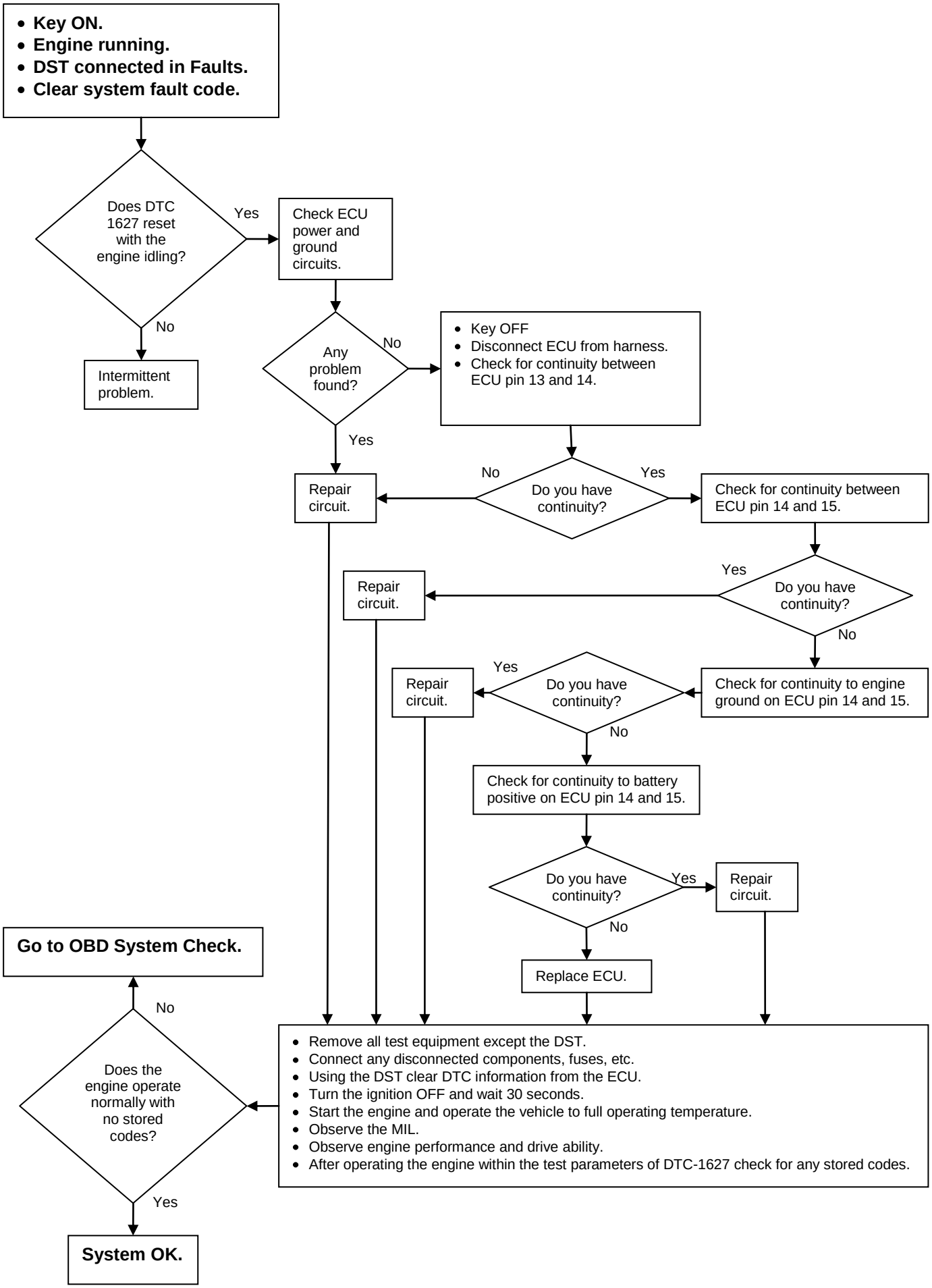
The Controller Area Network (CAN) bus is used by the ECU to communicate with other digital devices used throughout the fuel system. Information is sent over the CAN bus in digital information “packets” that contain information for various control functions. This fault will set if there is a problem with the CAN communication.

For detailed information, please see customer specific DTC settings.



All information concerning connectors, pin numbers and wire colors can be found in the connector documentation.

DTC 1627 – CAN Rx failure



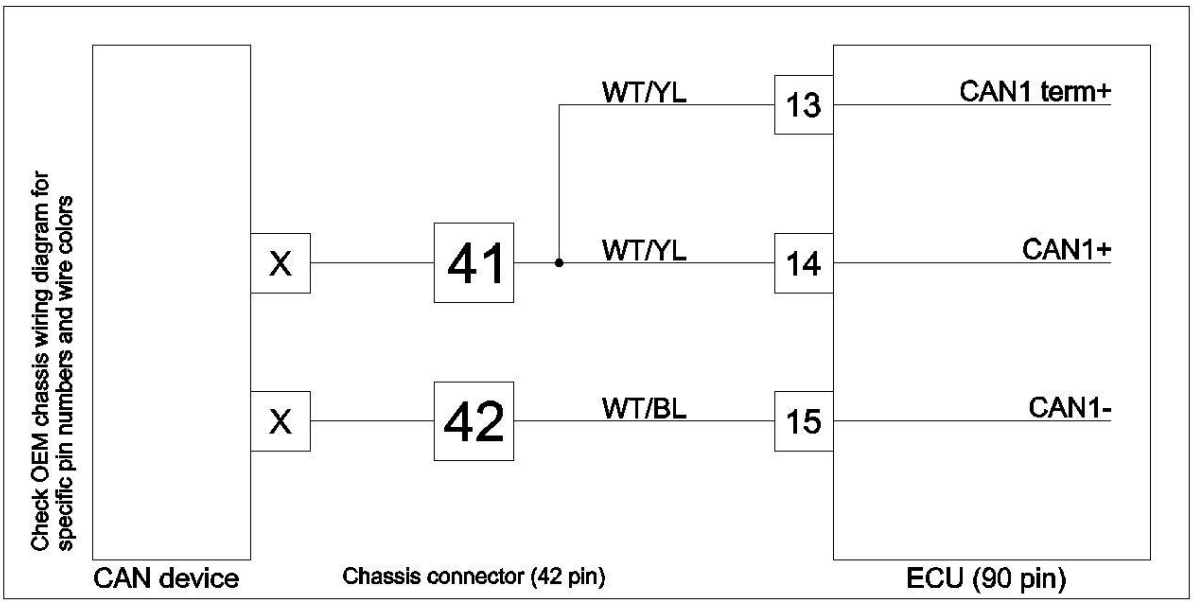
All measurements are done at the connector unless otherwise stated.

DTC 1629 – J1939 TSC1 message receipt lost

General

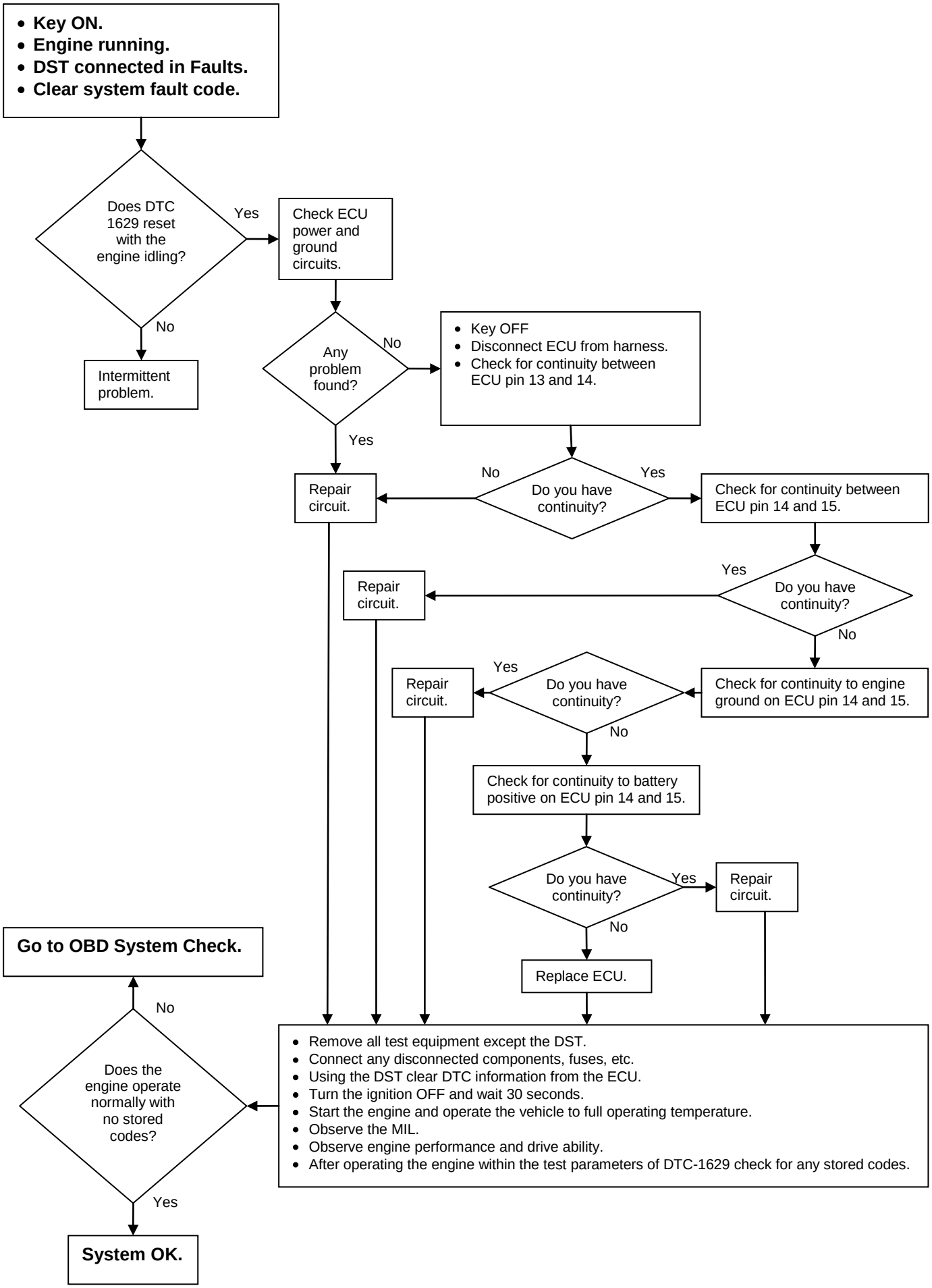
The Controller Area Network (CAN) bus is used by the ECU to communicate with other digital devices used throughout the fuel system. Information is sent over the CAN bus in digital information “packets” that contain information for various control functions. This fault will set if there is a problem with the CAN communication.

For detailed information, please see customer specific DTC settings.



All information concerning connectors, pin numbers and wire colors can be found in the connector documentation.

DTC 1629 – J1939 TSC1 message receipt lost



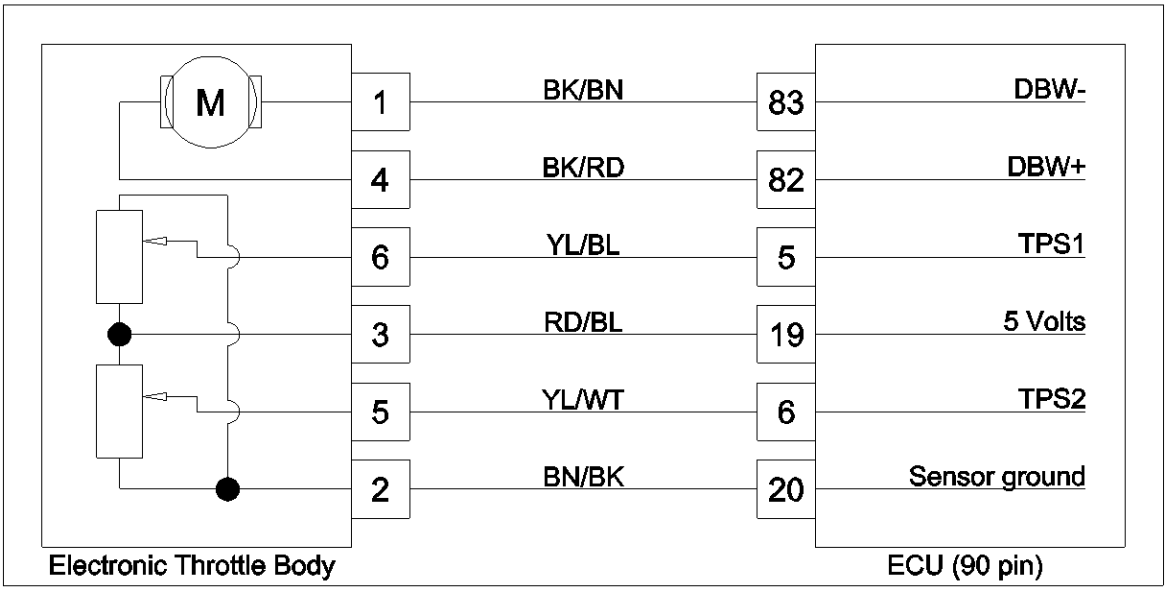
All measurements are done at the connector unless otherwise stated.

DTC 2111 – Unable to reach lower TPS

General

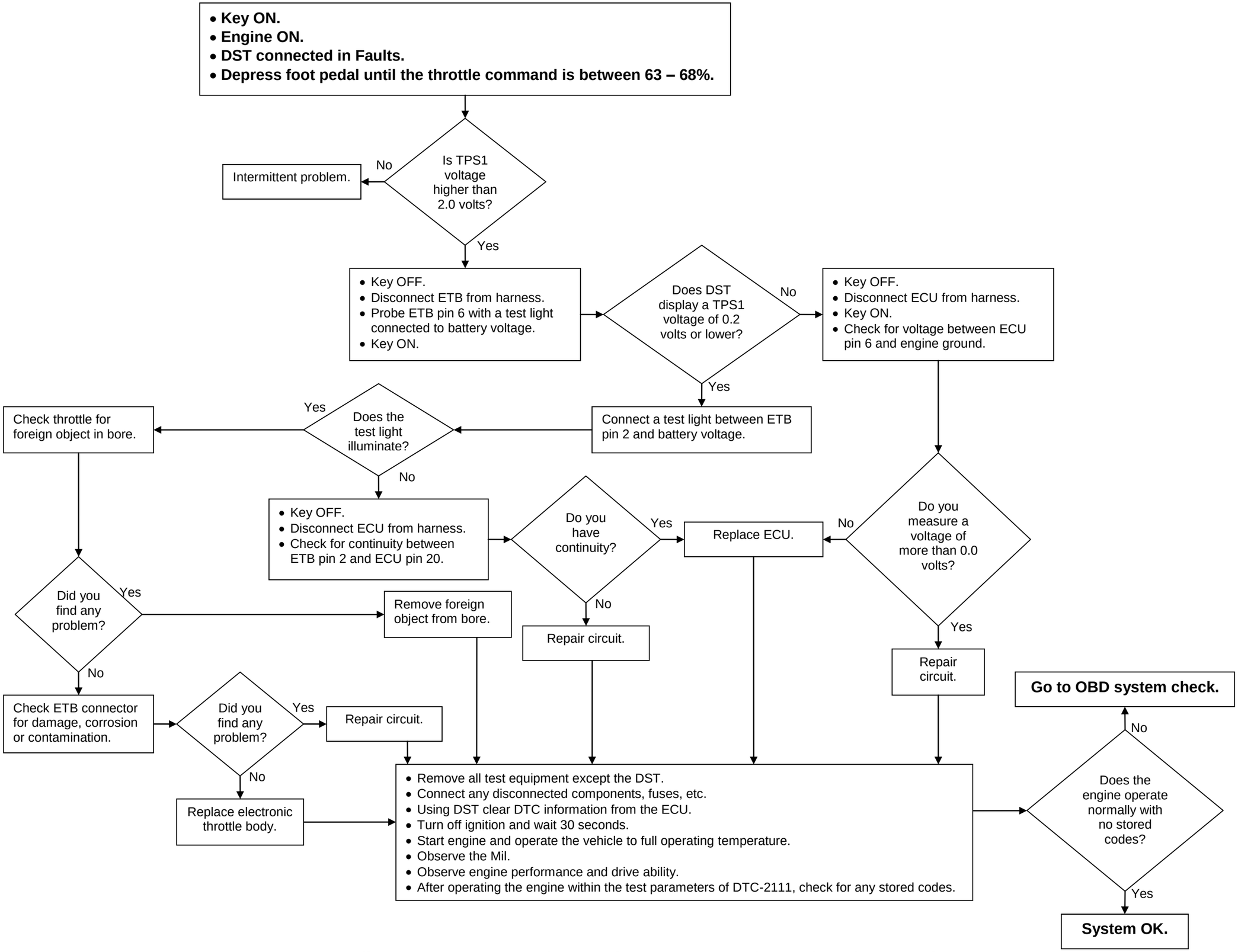
There are two Throttle Position Sensors (TPS) located within the throttle which use variable resistors to determine signal voltage based on throttle plate position. TPS1 will read low voltage when closed and TPS2 will read high voltage when closed. The TPS1 and TPS2 percentages are calculated from these voltages. Although the voltages are different, the calculated values for the throttle position percentages should be very close to the same. The TPS values are used by the ECU to determine if the throttle is opening as commanded. This fault will set if the throttle is unable to close.

For detailed information, please see customer specific DTC settings.



All information concerning connectors, pin numbers and wire colors can be found in the connector documentation.

DTC 2111 – Unable to reach lower TPS



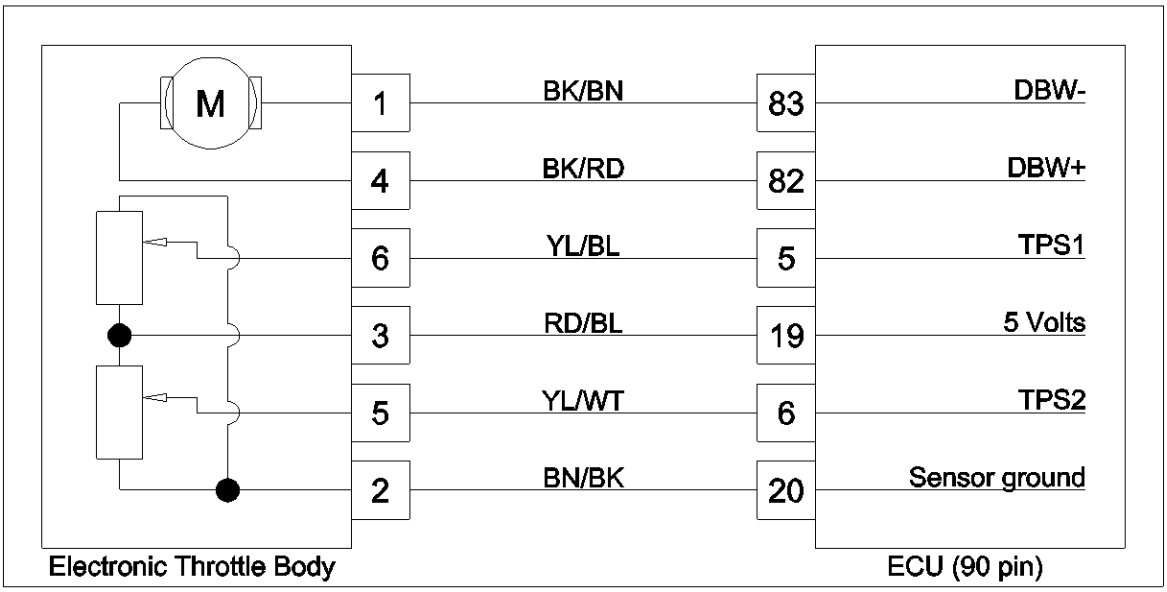
All measurements are done at the connector unless otherwise stated.

DTC 2112 – Unable to reach higher TPS

General

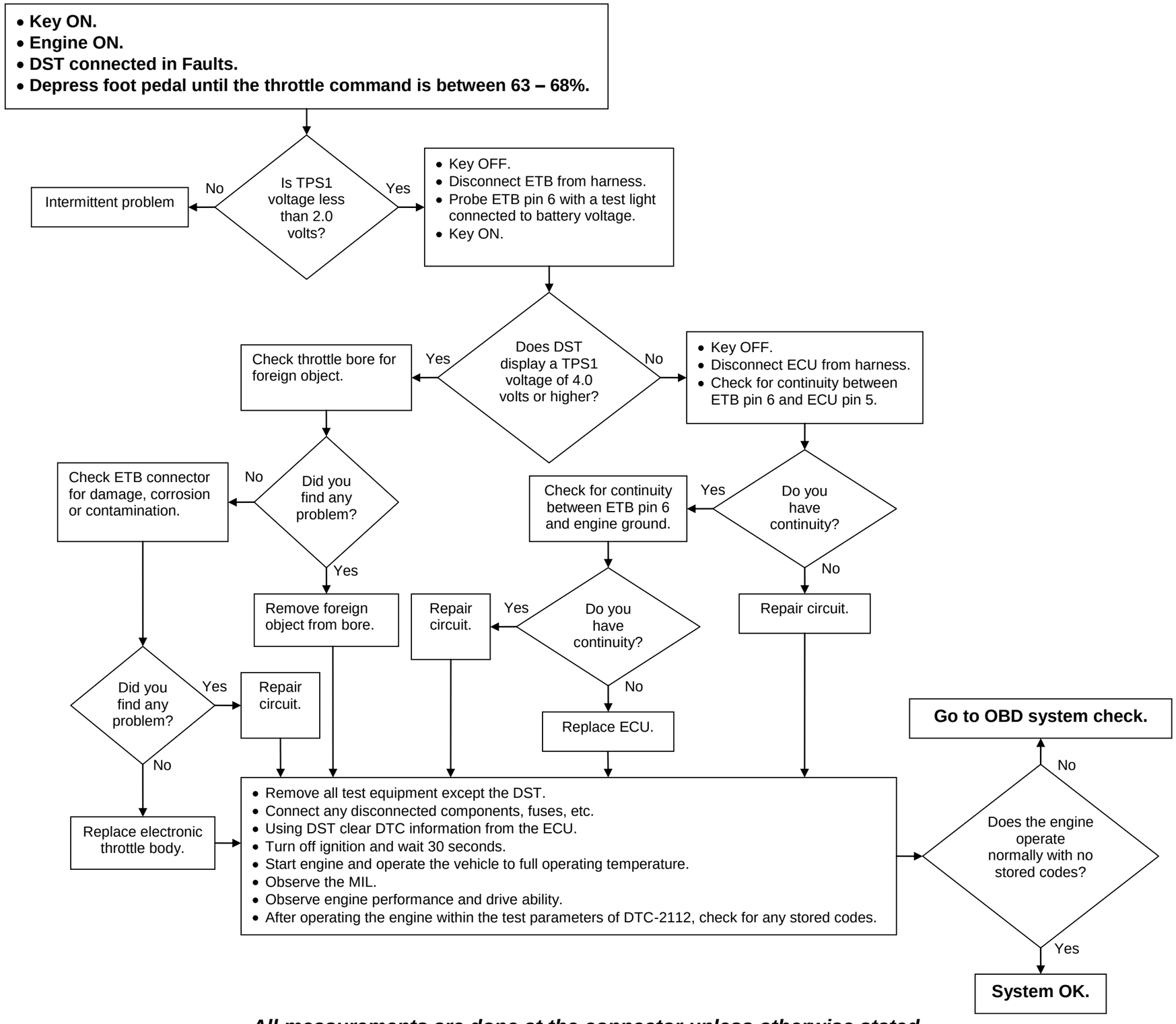
There are two Throttle Position Sensors (TPS) located within the throttle which use variable resistors to determine signal voltage based on throttle plate position. TPS1 will read low voltage when closed and TPS2 will read high voltage when closed. The TPS1 and TPS2 percentages are calculated from these voltages. Although the voltages are different, the calculated values for the throttle position percentages should be very close to the same. The TPS values are used by the ECU to determine if the throttle is opening as commanded. This fault will set if the throttle is unable to open.

For detailed information, please see customer specific DTC settings.



All information concerning connectors, pin numbers and wire colors can be found in the connector documentation.

DTC 2112 – Unable to reach higher TPS



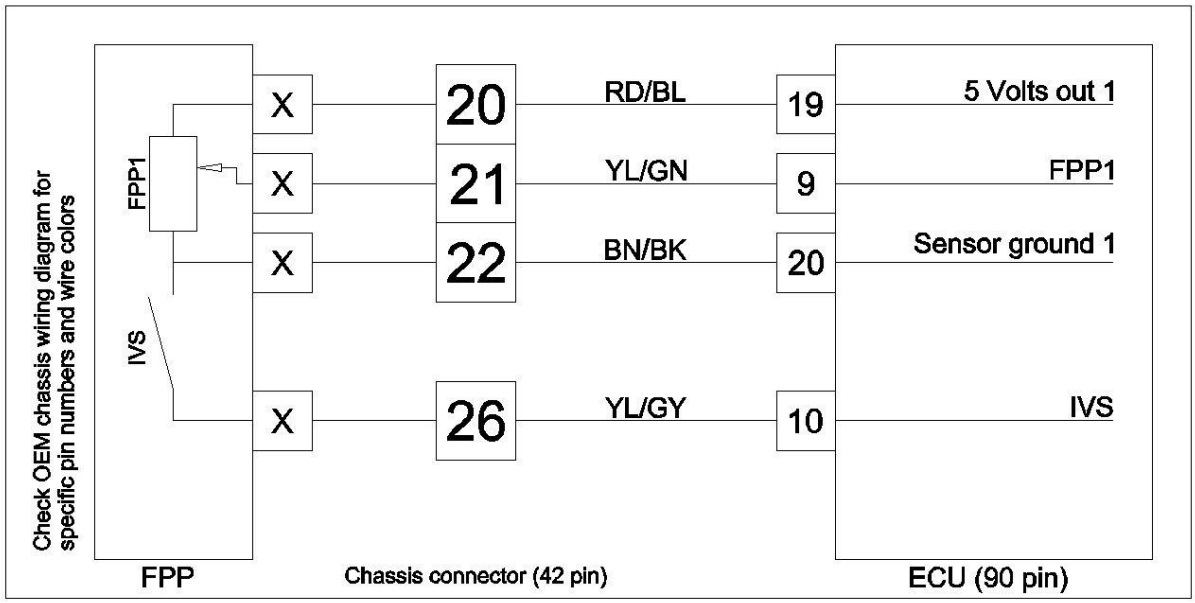
All measurements are done at the connector unless otherwise stated.

DTC 2115 – FPP1 higher than IVS limit

General

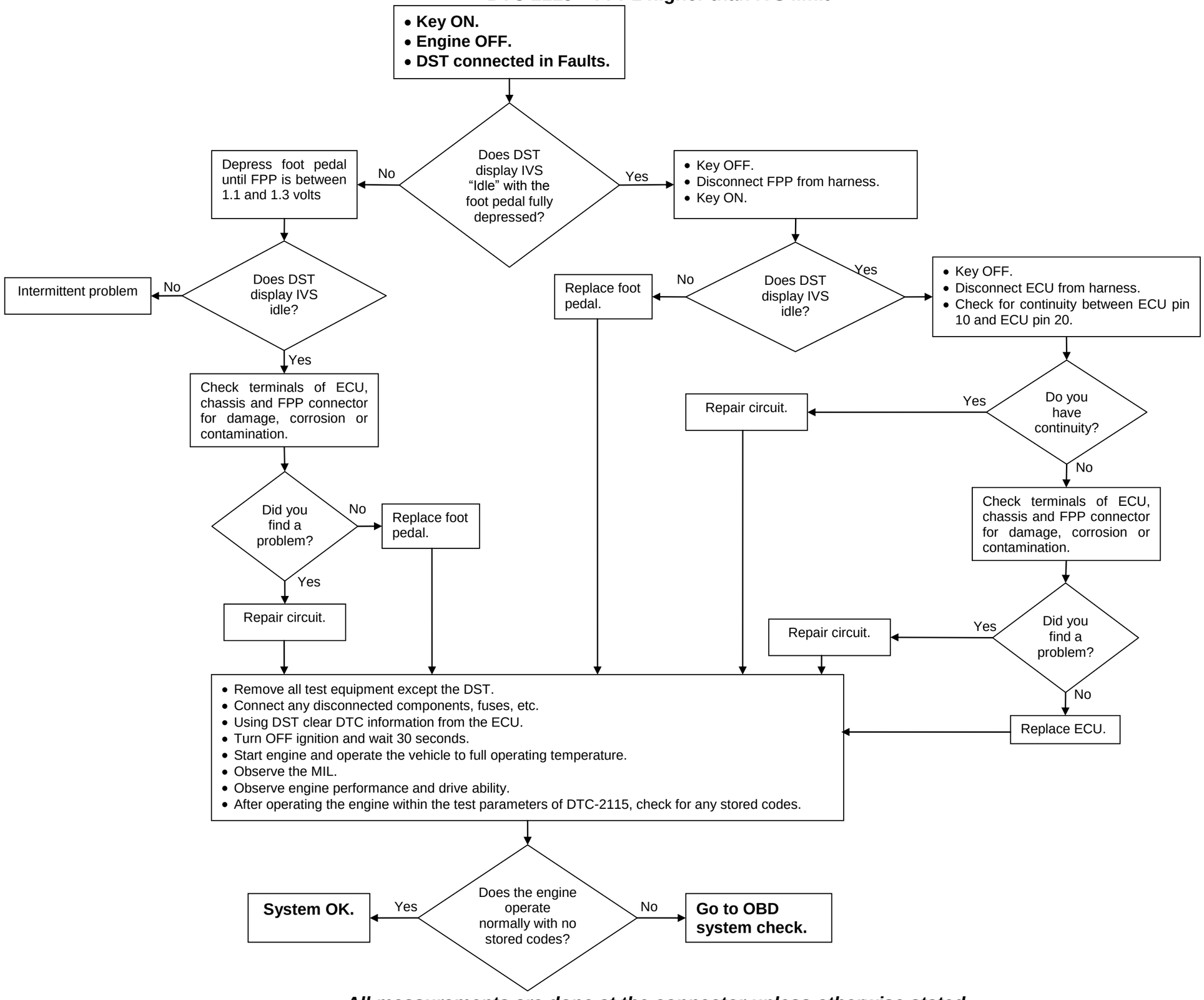
The Foot Pedal Position (FPP) sensor is an electronic device that is coupled to a mechanically driven input which is commanded by the vehicle/engine operator. General sensor configurations consist of a single potentiometer with an Idle Valid Switch (IVS), or two potentiometers. The FPP sensor outputs are proportional to the commanded input. The ECU uses the FPP sensor inputs to control the throttle and adjust the engine's load in order to achieve the requested power. Since the FPP sensor inputs directly affect the engine's power output, redundant sensors are generally used to ensure safe, reliable operation. This fault is only applicable for a single potentiometer sensor with IVS and indicates that FPP1 does not correlate with the IVS. This fault will set if the FPP1 doesn't agree with the IVS.

For detailed information, please see customer specific DTC settings.



All information concerning connectors, pin numbers and wire colors can be found in the connector documentation.

DTC 2115 – FPP1 higher than IVS limit



All measurements are done at the connector unless otherwise stated.

DTC 2121 – FPP1 lower than FPP2

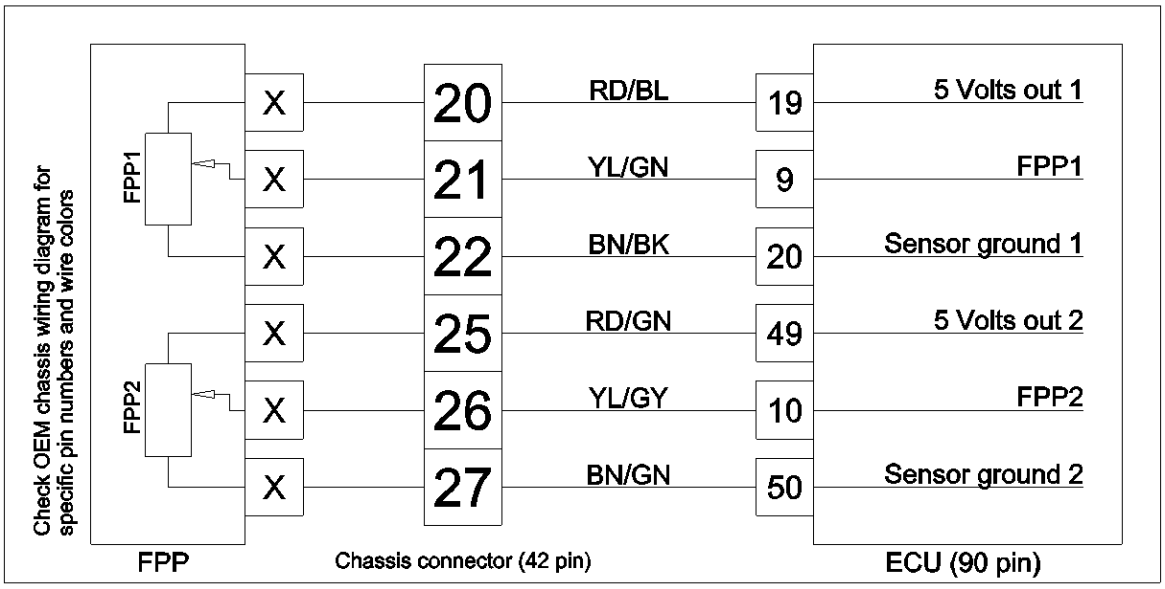
General

There are two Foot Pedal Position (FPP) sensors located within the foot pedal which use variable resistors to determine signal voltage based on foot pedal position. FPP1 and FPP2 will read different voltages from closed to fully open. The FPP1 and FPP2 percentages are calculated from these voltages. Although the voltages are different, the calculated values for the foot pedal position percentages should be very close to the same. This fault will set if the difference between FPP1 and FPP2 is higher than specified.

For detailed information, please see customer specific DTC settings.

Diagnostic Aid

FPP sensors are OEM specific and vary in configuration. The exact wire color and pin numbers for the FPP connection must be verified in the OEM chassis wiring diagram.



All information concerning connectors, pin numbers and wire colors can be found in the connector documentation.

DTC 2121 – FPP1 lower than FPP2

- **Key ON.**
- **Engine OFF.**
- **DST connected in Faults.**

Does DST display a difference between FPP1 and FPP2 voltage higher than specified?

No → Intermittent problem.

Yes →
- Key OFF.
- Disconnect FPP from harness.
- Key ON.

Is the voltage for FPP1 less than 0.2 volts?

Yes → Connect FPP pin FPP1 to FPP pin 5 volts out 1.

No →
- Key OFF.
- Disconnect ECU from harness.
- Key ON.
- Check the voltage between ECU pin 9 and engine ground.

Is the voltage more than 0.0 volts?

No →
- Key OFF.
- Disconnect ECU from harness.
- Check for continuity between FPP pin FPP1 and ECU pin 9.

Do you have continuity?

Yes → Check for continuity between FPP pin sensor ground 1 and ECU pin 20.

Do you have continuity?

Yes → Check terminals of ECU, chassis and FPP connector for damage, corrosion or

Did you find a problem?

Yes → Repair circuit.

No → Replace ECU.

Does DST display TPS1 voltage over 4.00 volts?

Yes → Check terminals of ECU, chassis and FPP connector for damage, corrosion or contamination.

Did you find a problem?

No → Replace foot pedal.

Yes → Repair circuit.

Yes → Repair circuit.

No → Repair circuit.

- Remove all test equipment except the DST.
- Connect any disconnected components, fuses, etc.
- Using DST clear DTC information from the ECU.
- Turn OFF ignition and wait 30 seconds.
- Start engine and operate the vehicle to full operating temperature.
- Observe the MIL.
- Observe engine performance and drive ability.
- After operating the engine within the test parameters of DTC-2121, check for any stored codes.

Does the engine operate normally with no stored codes?

Yes → **System OK.**

No → **Go to OBD system check.**

All measurements are done at the connector unless otherwise stated.

DTC 2122 – FPP1 high voltage

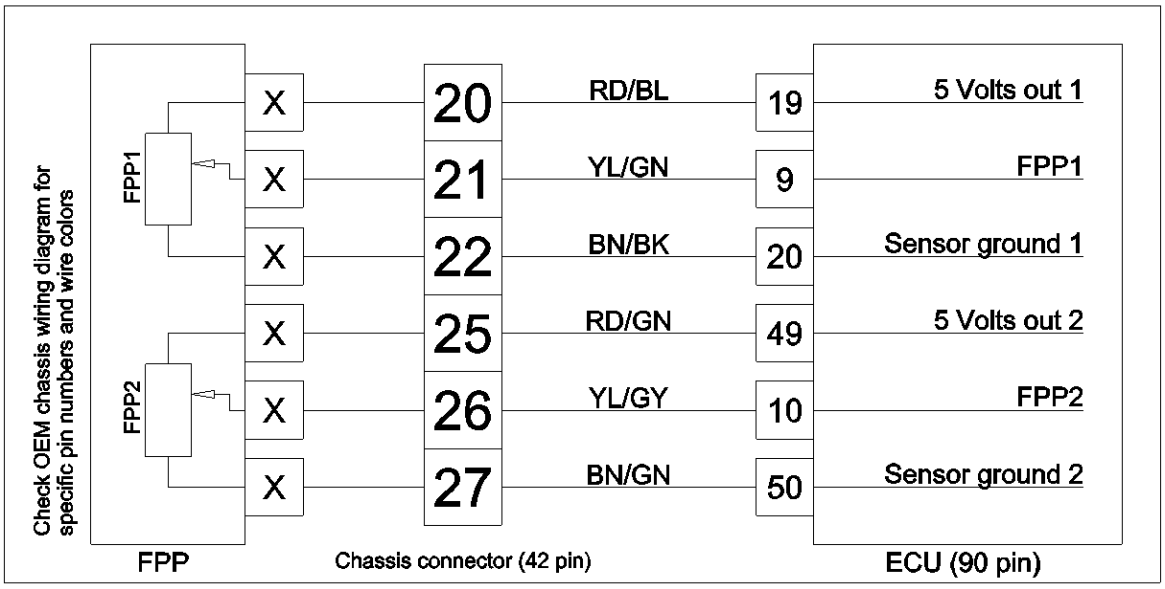
General

There are two Foot Pedal Position (FPP) sensors located within the foot pedal which use variable resistors to determine signal voltage based on foot pedal position. FPP1 and FPP2 will read different voltages from closed to fully open. The FPP1 and FPP2 percentages are calculated from these voltages. Although the voltages are different, the calculated values for the foot pedal position percentages should be very close to the same. This fault will set if the sensor voltage is too high.

For detailed information, please see customer specific DTC settings.

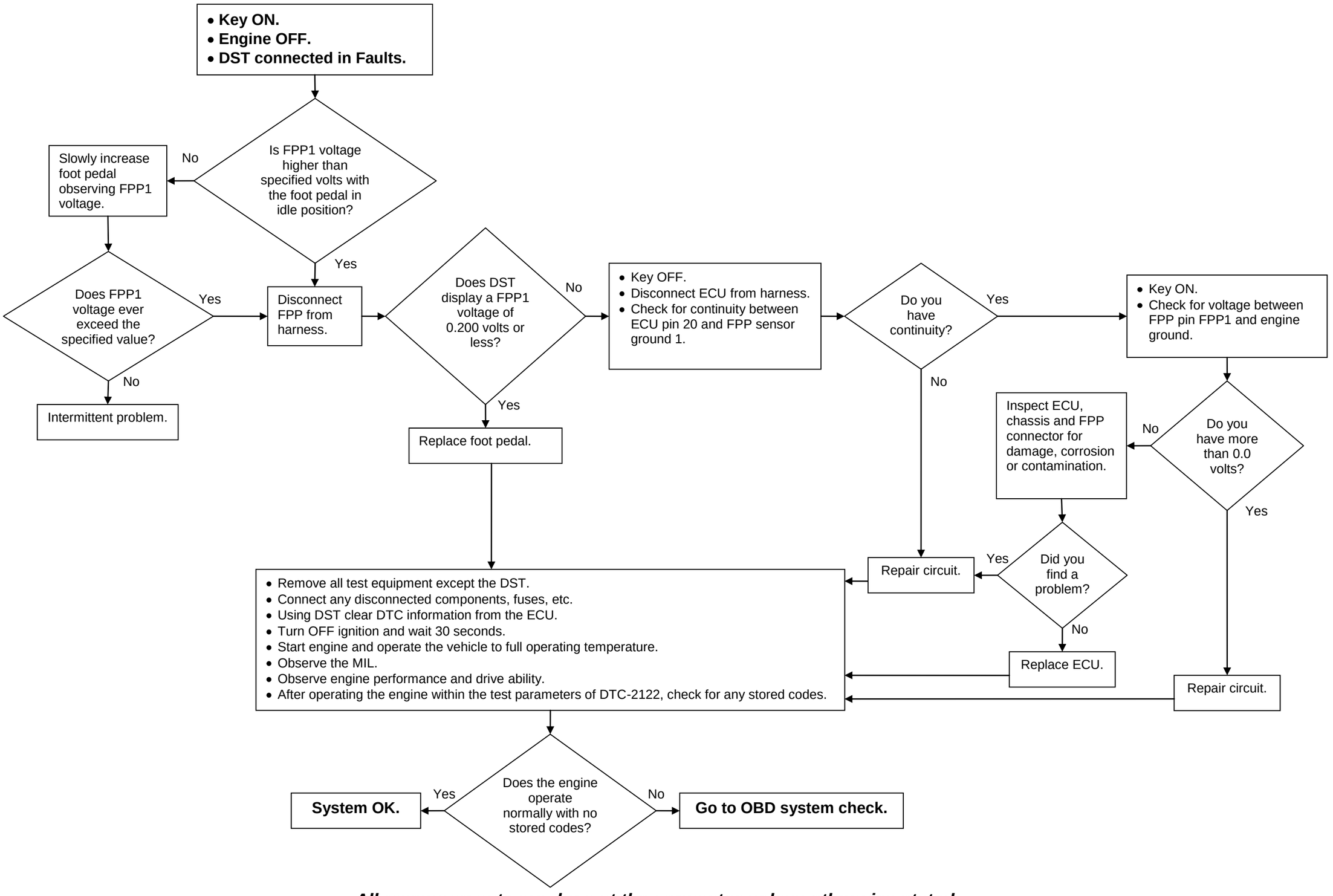
Diagnostic Aid

FPP sensors are OEM specific and vary in configuration. The exact wire color and pin numbers for the FPP connection must be verified in the OEM chassis wiring diagram.



All information concerning connectors, pin numbers and wire colors can be found in the connector documentation.

DTC 2122 – FPP1 high voltage



All measurements are done at the connector unless otherwise stated.

DTC 2123 – FPP1 low voltage

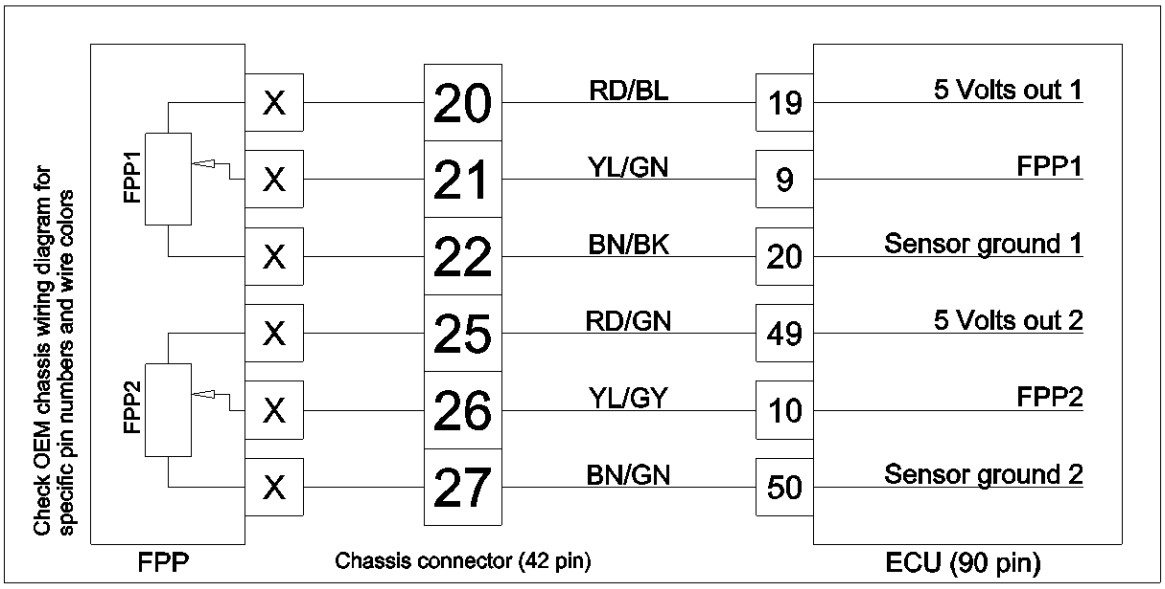
General

There are two Foot Pedal Position (FPP) sensors located within the foot pedal which use variable resistors to determine signal voltage based on foot pedal position. FPP1 and FPP2 will read different voltages from closed to fully open. The FPP1 and FPP2 percentages are calculated from these voltages. Although the voltages are different, the calculated values for the foot pedal position percentages should be very close to the same. This fault will set if the sensor voltage is too low.

For detailed information, please see customer specific DTC settings.

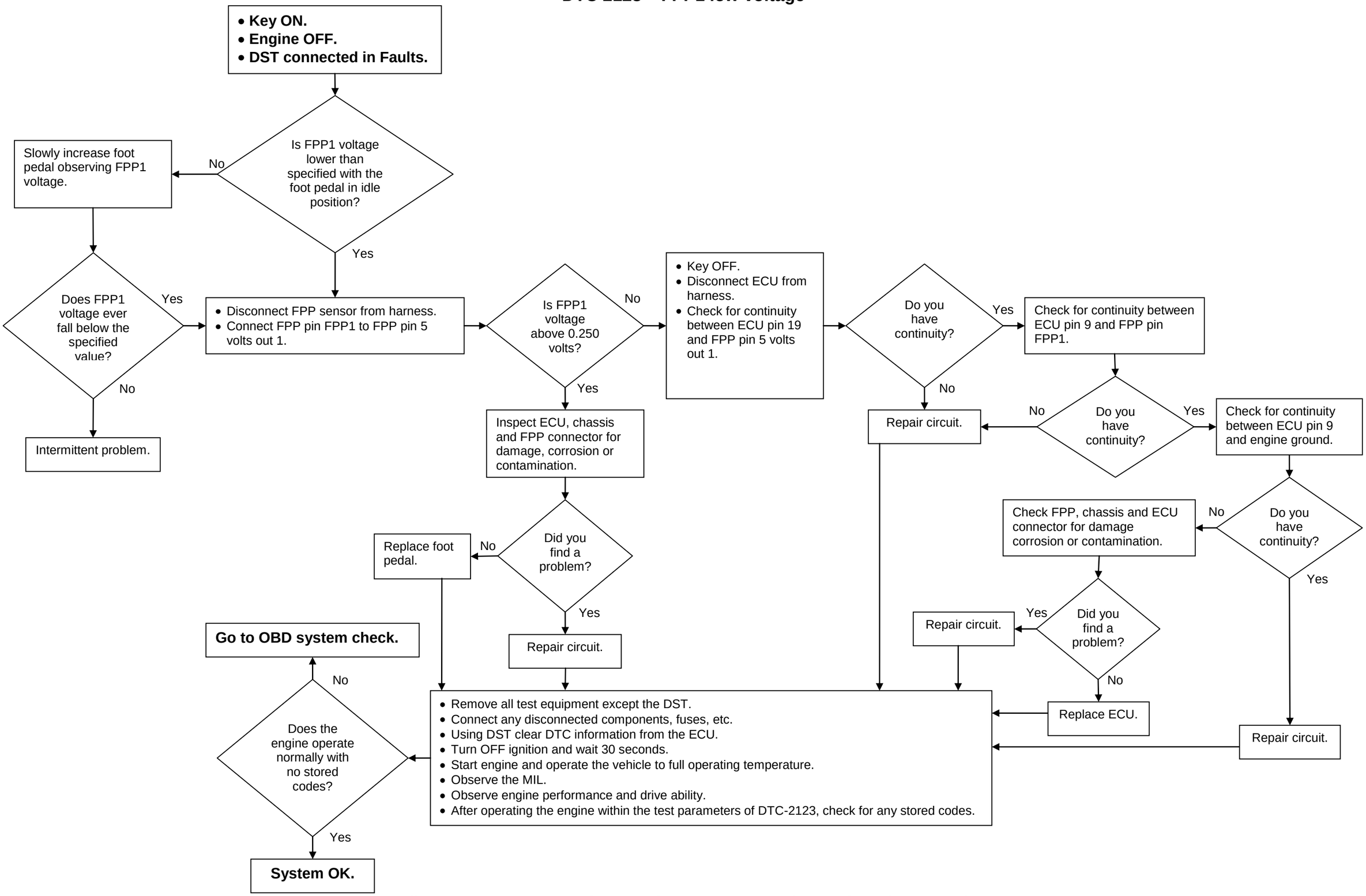
Diagnostic Aid

FPP sensors are OEM specific and vary in configuration. The exact wire color and pin numbers for the FPP connection must be verified in the OEM chassis wiring schematic.



All information concerning connectors, pin numbers and wire colors can be found in the connector documentation.

DTC 2123 – FPP1 low voltage



All measurements are done at the connector unless otherwise stated.

DTC 2126 – FPP1 higher than FPP2

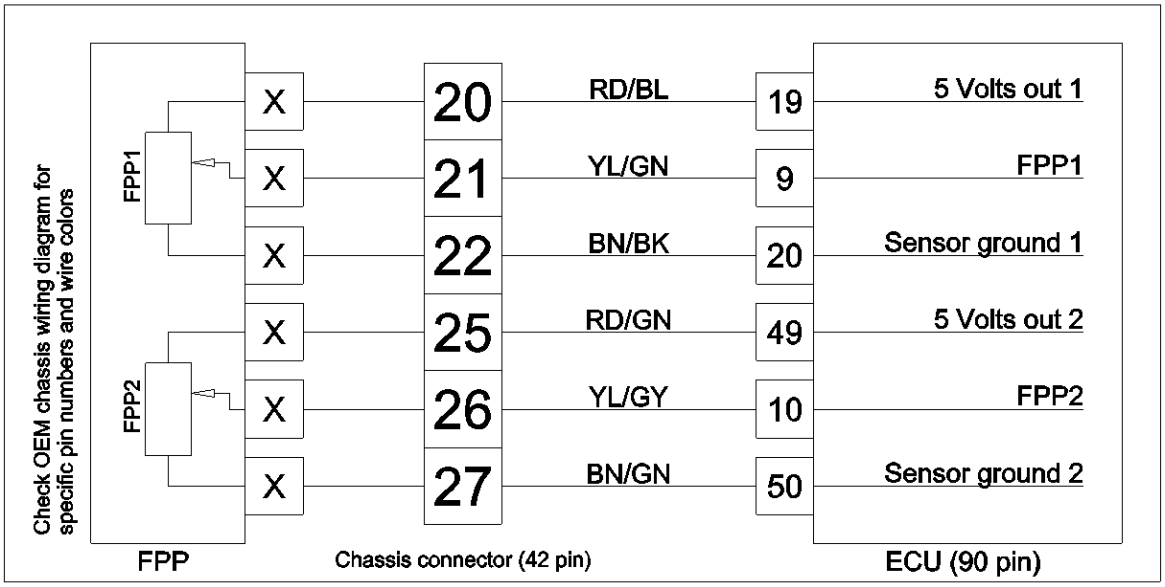
General

There are two Foot Pedal Position (FPP) sensors located within the foot pedal which use variable resistors to determine signal voltage based on foot pedal position. FPP1 and FPP2 will read different voltages from closed to fully open. The FPP1 and FPP2 percentages are calculated from these voltages. Although the voltages are different, the calculated values for the foot pedal position percentages should be very close to the same. This fault will set if the difference between FPP1 and FPP2 is higher than specified.

For detailed information, please see customer specific DTC settings.

Diagnostic Aid

FPP sensors are OEM specific and vary in configuration. The exact wire color and pin numbers for the FPP connection must be verified in the OEM chassis wiring diagram.



All information concerning connectors, pin numbers and wire colors can be found in the connector documentation.

DTC 2126 – FPP1 higher than FPP2

- **Key ON.**
- **Engine OFF.**
- **DST connected in Faults.**

Does DST display a difference between FPP1 and FPP2 voltage lower than specified?

No → Intermittent problem.

Yes →
- Key OFF.
- Disconnect FPP from harness.
- Key ON.

Is the voltage for FPP1 less than 0.2 volts?

Yes → Connect FPP pin FPP1 to FPP pin 5 volts out 1.

No →
- Key OFF.
- Disconnect ECU from harness.
- Key ON.
- Check the voltage between ECU pin 9 and engine ground.

Is the voltage more than 0.0 volts?

No →
- Key OFF.
- Disconnect ECU from harness.
- Check for continuity between FPP pin FPP1 and ECU pin 9.

Do you have continuity?

Yes → Check for continuity between FPP pin sensor ground 1 and ECU pin 20.

Do you have continuity?

Yes → Check terminals of ECU, chassis and FPP connector for damage, corrosion or

Did you find a problem?

Yes → Repair circuit.

No → Replace ECU.

Does DST display TPS1 voltage over 4.00 volts?

No →
- Key OFF.
- Disconnect ECU from harness.
- Check for continuity between FPP pin FPP1 and ECU pin 9.

Do you have continuity?

No → Repair circuit.

Yes → Check terminals of ECU, ChC and FPP connector for damage, corrosion or contamination.

Did you find a problem?

No → Replace foot pedal.

Yes → Repair circuit.

Repair circuit.

- Remove all test equipment except the DST.
- Connect any disconnected components, fuses, etc.
- Using DST clear DTC information from the ECU.
- Turn OFF ignition and wait 30 seconds.
- Start engine and operate the vehicle to full operating temperature.
- Observe the MIL.
- Observe engine performance and drive ability.
- After operating the engine within the test parameters of DTC-2126, check for any stored codes.

Does the engine operate normally with no stored codes?

Yes → **System OK.**

No → **Go to OBD system check.**

All measurements are done at the connector unless otherwise stated.

DTC 2127 – FPP2 low voltage

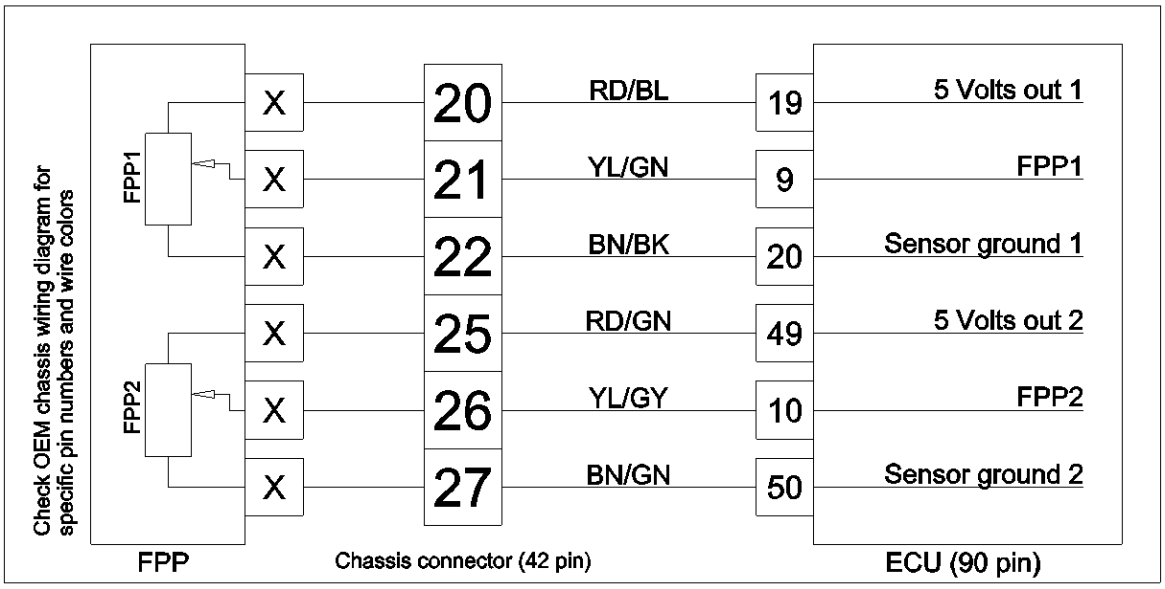
General

There are two Foot Pedal Position (FPP) sensors located within the foot pedal which use variable resistors to determine signal voltage based on foot pedal position. FPP1 and FPP2 will read different voltages from closed to fully open. The FPP1 and FPP2 percentages are calculated from these voltages. Although the voltages are different, the calculated values for the foot pedal position percentages should be very close to the same. This fault will set if the sensor voltage is too low.

For detailed information, please see customer specific DTC settings.

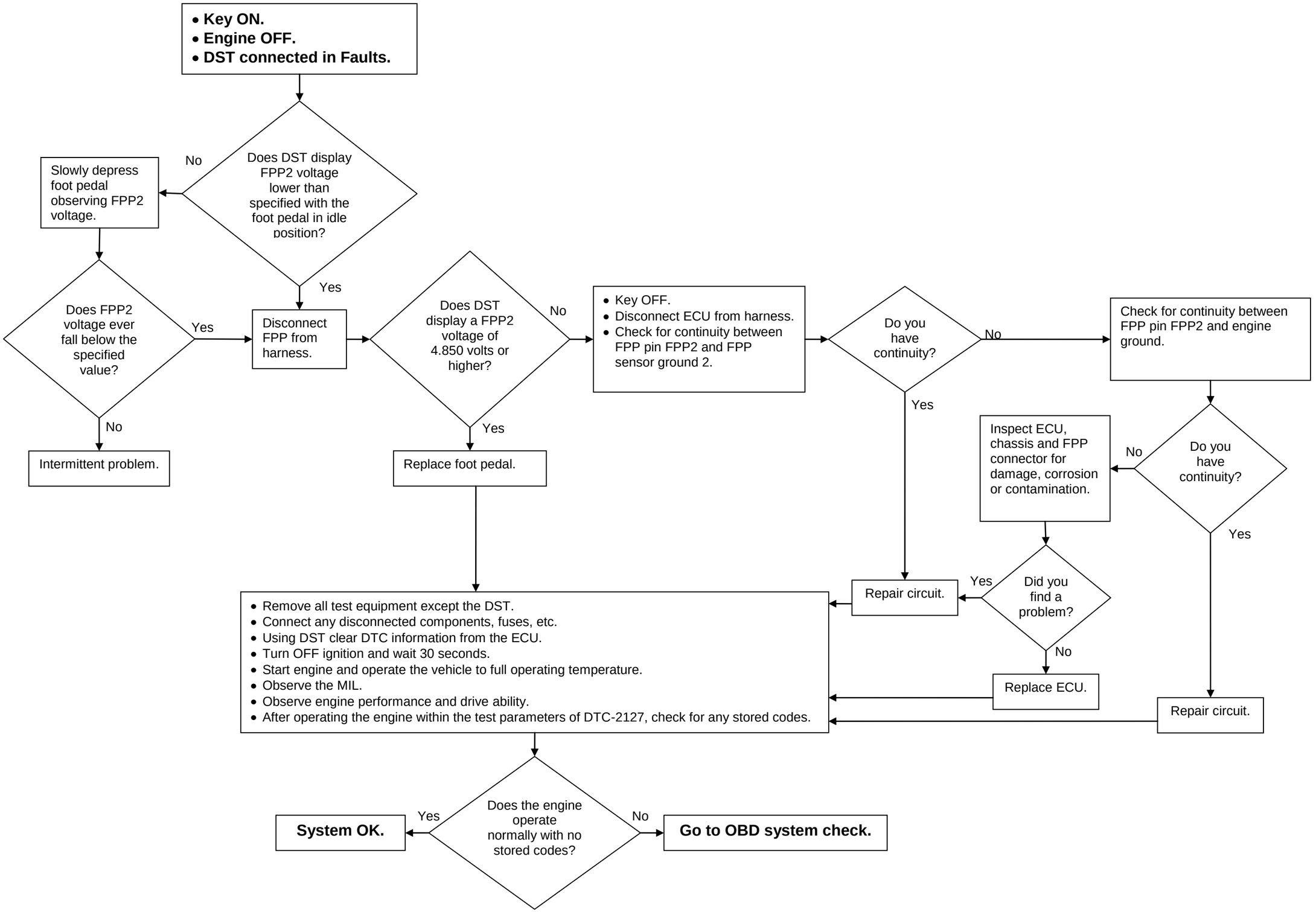
Diagnostic Aid

FPP sensors are OEM specific and vary in configuration. The exact wire color and pin numbers for the FPP connection must be verified in the OEM chassis wiring schematic.



All information concerning connectors, pin numbers and wire colors can be found in the connector documentation.

DTC 2127 – FPP2 low voltage



All measurements are done at the connector unless otherwise stated.

DTC 2128 – FPP2 high voltage

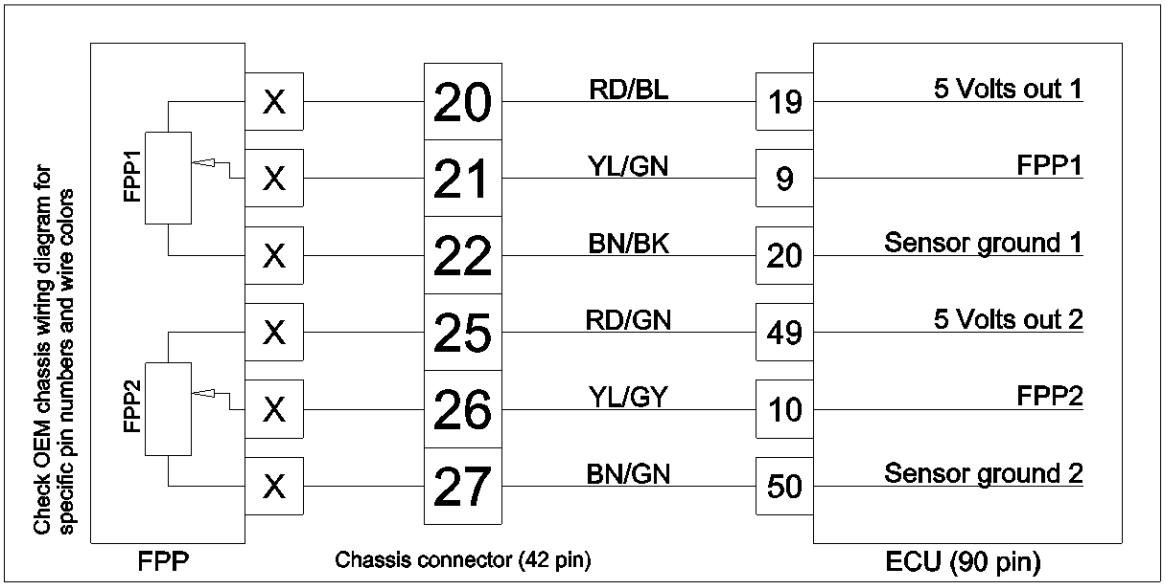
General

There are two Foot Pedal Position (FPP) sensors located within the foot pedal which use variable resistors to determine signal voltage based on foot pedal position. FPP1 and FPP2 will read different voltages from closed to fully open. The FPP1 and FPP2 percentages are calculated from these voltages. Although the voltages are different, the calculated values for the foot pedal position percentages should be very close to the same. This fault will set if the sensor voltage is too high.

For detailed information, please see customer specific DTC settings.

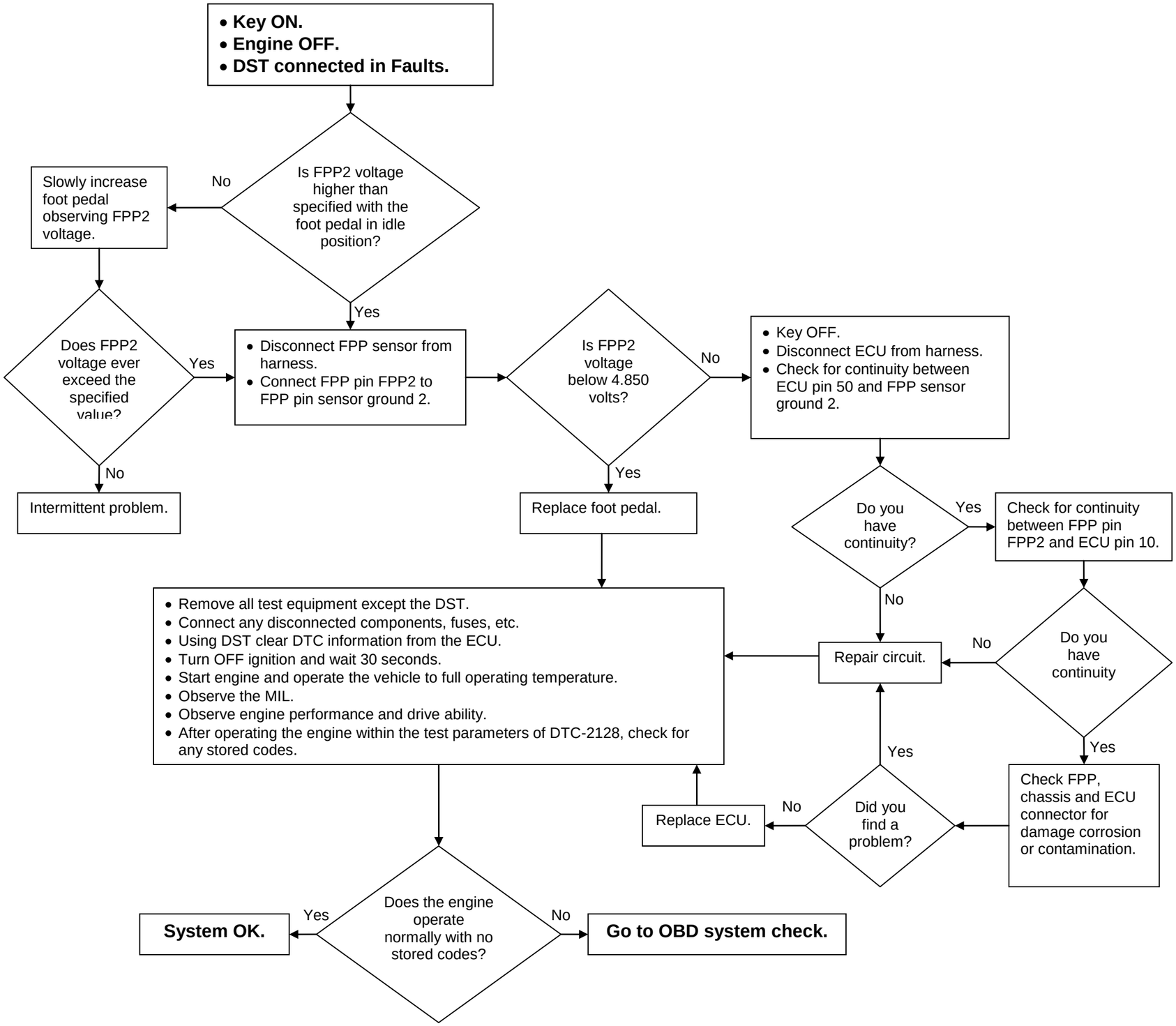
Diagnostic Aid

FPP sensors are OEM specific and vary in configuration. The exact wire color and pin numbers for the FPP connection must be verified in the OEM chassis wiring schematic.



All information concerning connectors, pin numbers and wire colors can be found in the connector documentation.

DTC 2128 – FPP2 high voltage



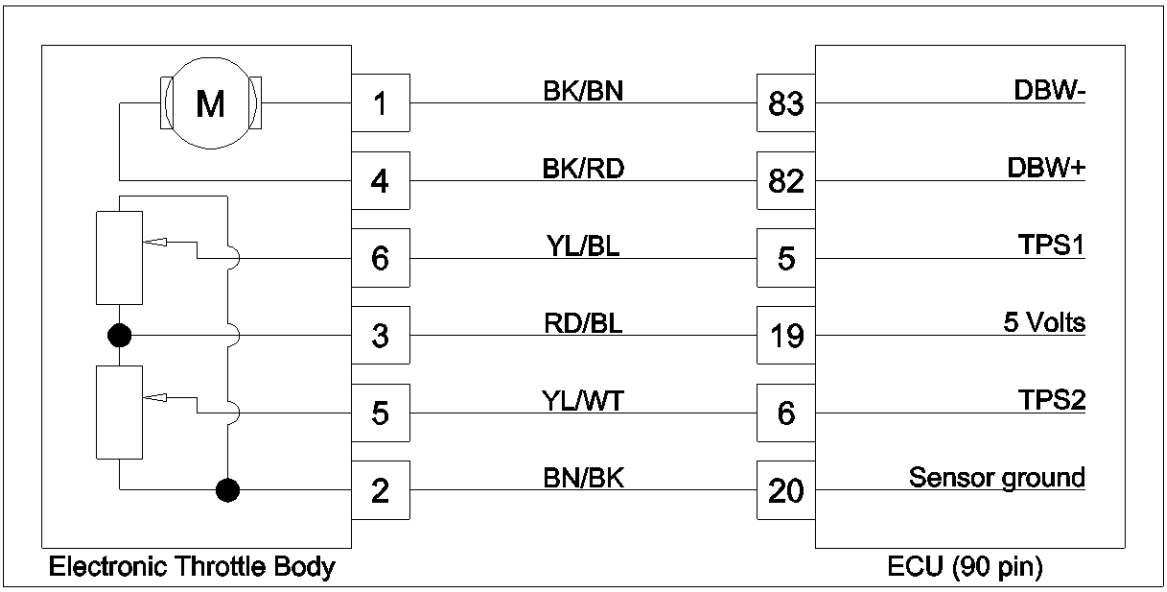
All measurements are done at the connector unless otherwise stated.

DTC 2135 – TPS1/2 simultaneous voltages out of range

General

There are two Throttle Position Sensors (TPS) located within the throttle which use variable resistors to determine signal voltage based on throttle plate position. TPS1 will read low voltage when closed and TPS2 will read high voltage when closed. The TPS1 and TPS2 percentages are calculated from these voltages. Although the voltages are different, the calculated values for the throttle position percentages should be very close to the same. The TPS values are used by the ECU to determine if the throttle is opening as commanded. This fault will set if TPS1 doesn't agree with TPS2.

For detailed information, please see customer specific DTC settings.



All information concerning connectors, pin numbers and wire colors can be found in the connector documentation.

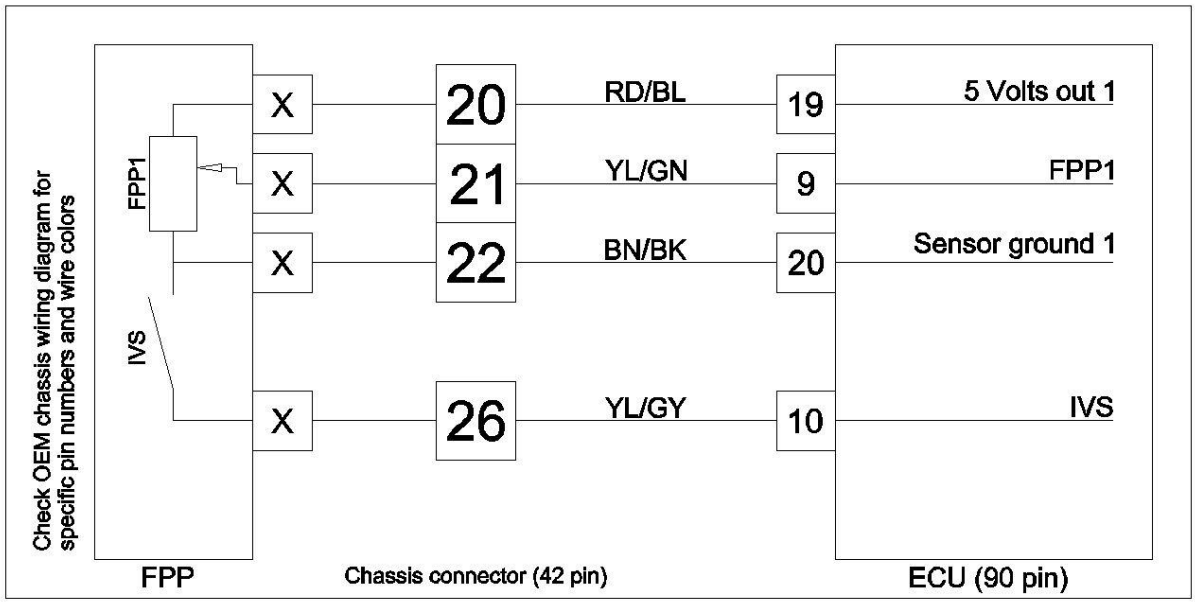
For troubleshooting, refer to DTC 122, 123, 222 and 223 procedures.

DTC 2139 – FPP1 lower than IVS limit

General

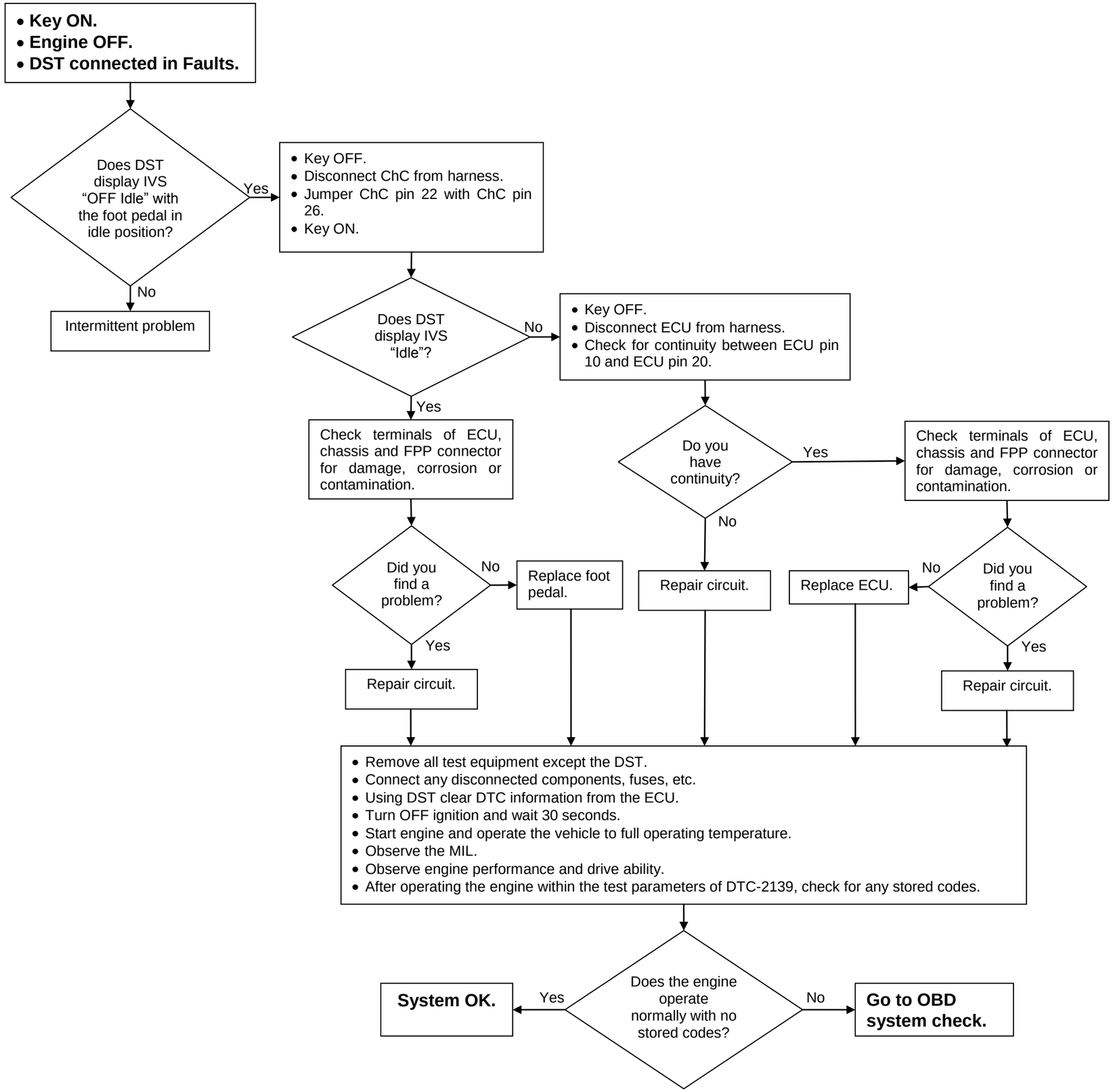
The Foot Pedal Position (FPP) sensor is an electronic device that is coupled to a mechanically driven input which is commanded by the vehicle/engine operator. General sensor configurations consist of a single potentiometer with an Idle Valid Switch (IVS), or two potentiometers. The FPP sensor outputs are proportional to the commanded input. The ECU uses the FPP sensor inputs to control the throttle and adjust the engine's load in order to achieve the requested power. Since the FPP sensor inputs directly affect the engine's power output, redundant sensors are generally used to ensure safe, reliable operation. This fault is only applicable for a single potentiometer sensor with IVS and indicates that FPP1 does not correlate with the IVS. This fault will set if the FPP1 doesn't agree with the IVS.

For detailed information, please see customer specific DTC settings.



All information concerning connectors, pin numbers and wire colors can be found in the connector documentation.

DTC 2139 – FPP1 lower than IVS limit



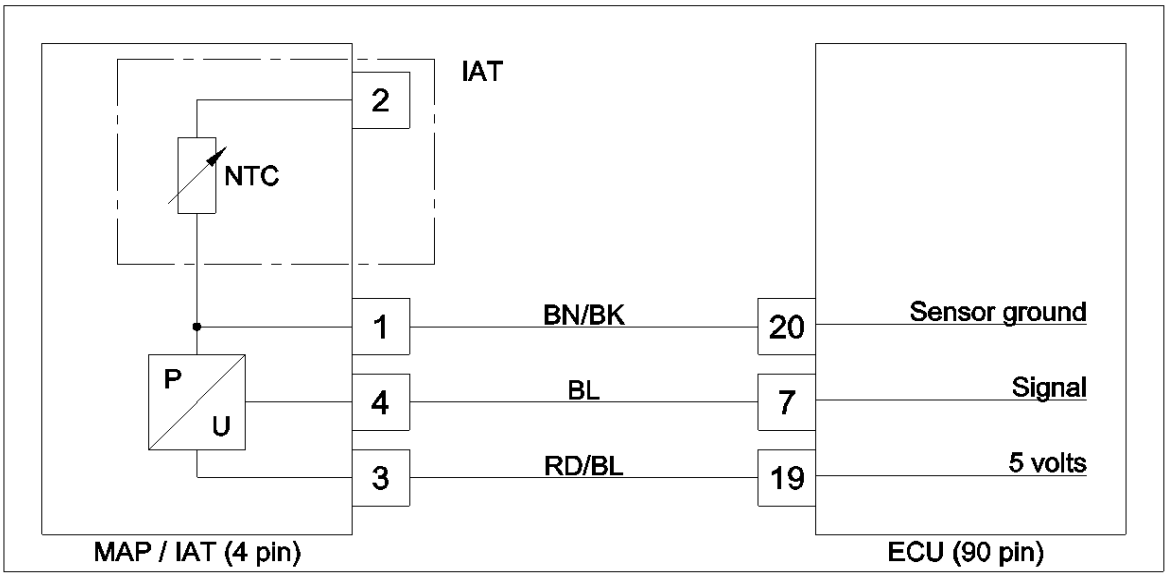
All measurements are done at the connector unless otherwise stated.

DTC 2229 – BP high pressure

General

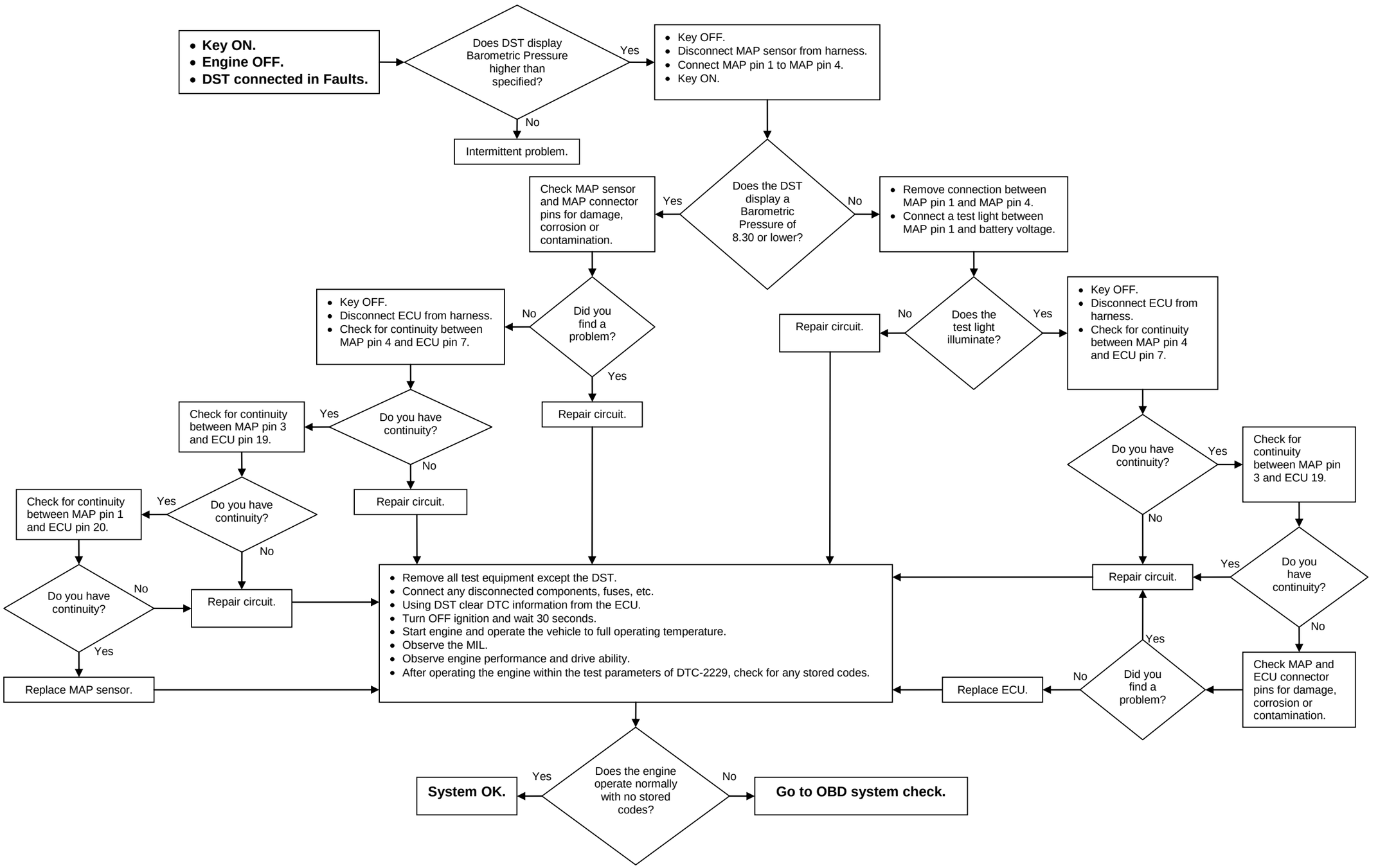
The Barometric Pressure (BP) is estimated from the MAP sensor. The barometric pressure value is used for fuel and airflow calculations. This fault will set if the BP is too high.

For detailed information, please see customer specific DTC settings.



All information concerning connectors, pin numbers and wire colors can be found in the connector documentation.

DTC 2229 – BP high pressure



All measurements are done at the connector unless otherwise stated.

Connector documentation

In this part of the manual contains the connector information. all the connectors used on this engine are explained with a description, location, pinning table and figure.

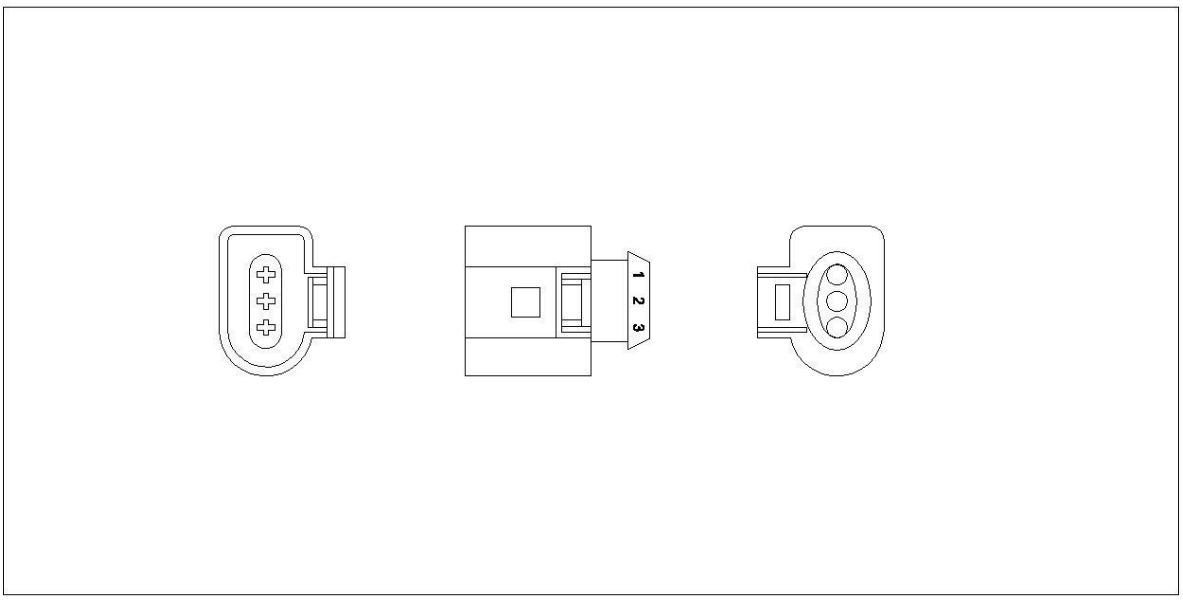
Camshaft position sensor (CAM)

The camshaft position (CAM) sensor connector is a 3 pin plug which is located on cold side of the engine next to the intake manifold. See picture 2 connector location on cold engine side for detailed location.

The CAM+ and CAM- wires are a so called twisted pair, this is to protect the CAM signal from electronic noise.

Always keep the connector and wires away from ignition coils, spark plugs and spark plug leads.

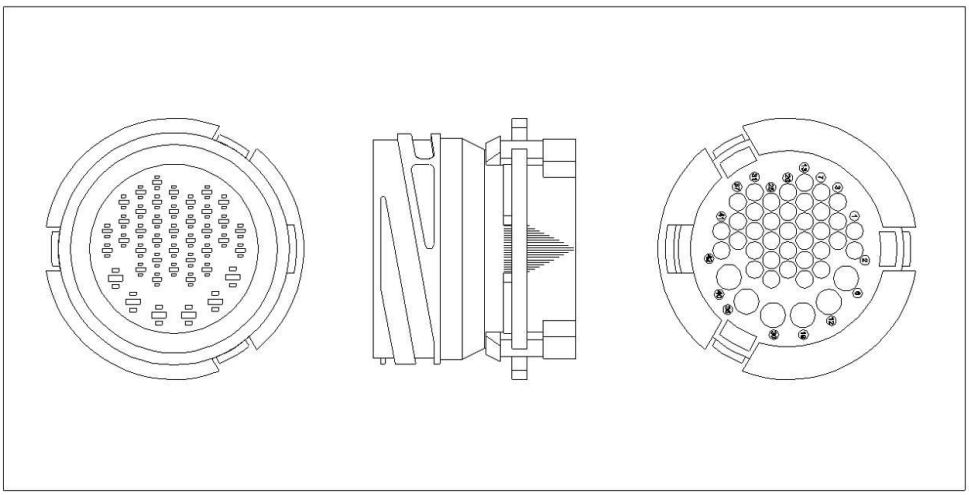
Pin nr:	Function:	Color:	Thickness:
1	5 volts	red / blue	0.75
2	CAM+	blue / red	0.75
3	CAM-	white / red	0.75



Chassis connector (ChC)

The chassis (ChC) connector is used for connecting the engine to the rest of the vehicle. It is a 42 pin plug which has 6 high power pins and 36 low power pins. The connector is located on the back side of the engine above the flywheel housing. See picture 1 connector location on back side of engine for detailed location.

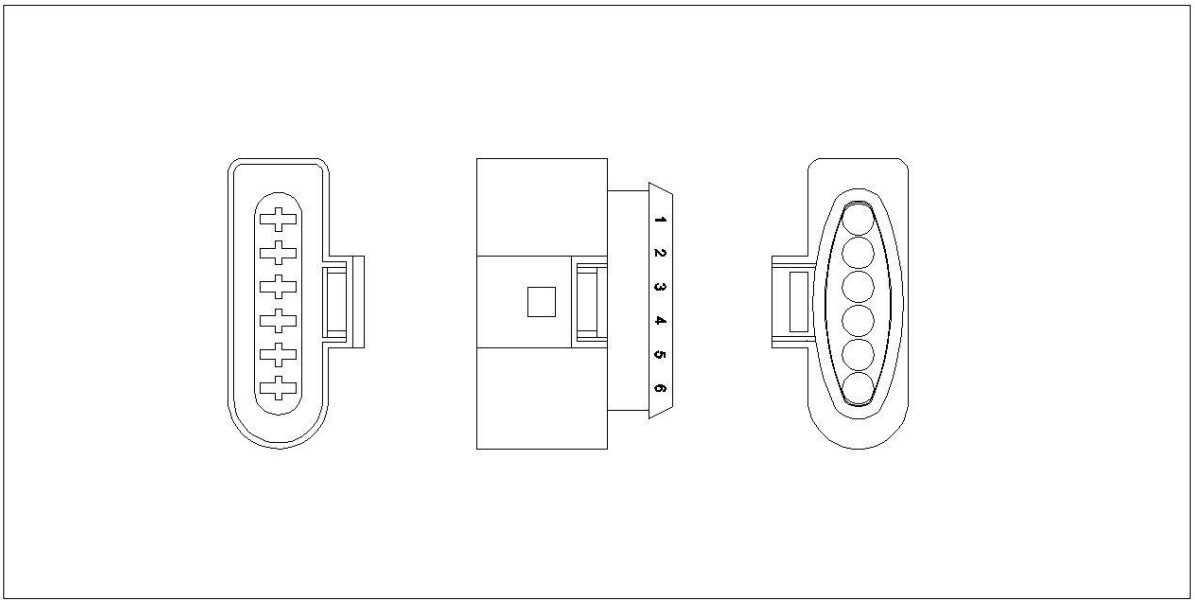
Pin nr:	Function:	Color:	Thickness:
1	diagnostic PC Tx	white / green	0.75
2	diagnostic PC Rx	white / red	0.75
5	buzzer	green / white	0.75
6	battery voltage (+30)	red	1.5
7	malfunction indicator lamp (MIL)	green / red	0.75
8	generator Lamp	blue / green	0.75
9	tacho signal	green	0.75
10	oil pressure switch	green / black	0.75
12	battery voltage (+30)	red	1.5
13	gauge out 1	violet / black	0.75
14	switched voltage (+15)	red / black	0.75
15	governor 1	violet	0.75
16	governor 2	violet / red	0.75
17	gauge out 2	violet / blue	0.75
18	starter signal output	violet / brown	0.75
19	starter motor signal	red / white	1.5
20	5 volt sensor supply 1	red / blue	0.75
21	foot pedal position sensor 1	yellow / green	0.75
22	sensor ground 1	brown / black	0.75
23	auxiliary digital output 1	blue / white	0.75
24	auxiliary digital output 2	green / blue	0.75
25	5 volt sensor supply 2	red / green	0.75
26	foot pedal position sensor 2	yellow / green	0.75
27	sensor ground 2	brown / green	0.75
28	lock off solenoid	green / yellow	0.75
30	voltage main power relay	red / yellow	1.5
31	auxiliary digital input 2	violet / yellow	0.75
32	auxiliary digital input 3	violet / gray	0.75
33	auxiliary digital output 4	blue / gray	0.75
34	auxiliary digital output 5	green / gray	0.75
35	main power relay coil	yellow / brown	0.75
36	main ground	brown	1.5
39	auxiliary digital input 1	yellow / violet	0.75
40	main ground	brown	1.5
41	CAN1 +	white / yellow	0.5
42	CAN1 -	white / blue	0.5



Coil pack (CP)

The coil pack (CP) connector is a 6 pin plug which is located on the ignition coil pack on the cold side of the engine. See picture 1 connector location on back side of engine for detailed location.

Pin nr:	Function:	Color:	Thickness:
1	chassis ground	brown	1.5
2	ignition cylinder 1	gray / black	0.75
3	ignition cylinder 2	gray / green	0.75
4	ignition cylinder 3	gray / blue	0.75
5	ignition cylinder 4	gray / red	0.75
6	voltage main power relay	red / yellow	1.5



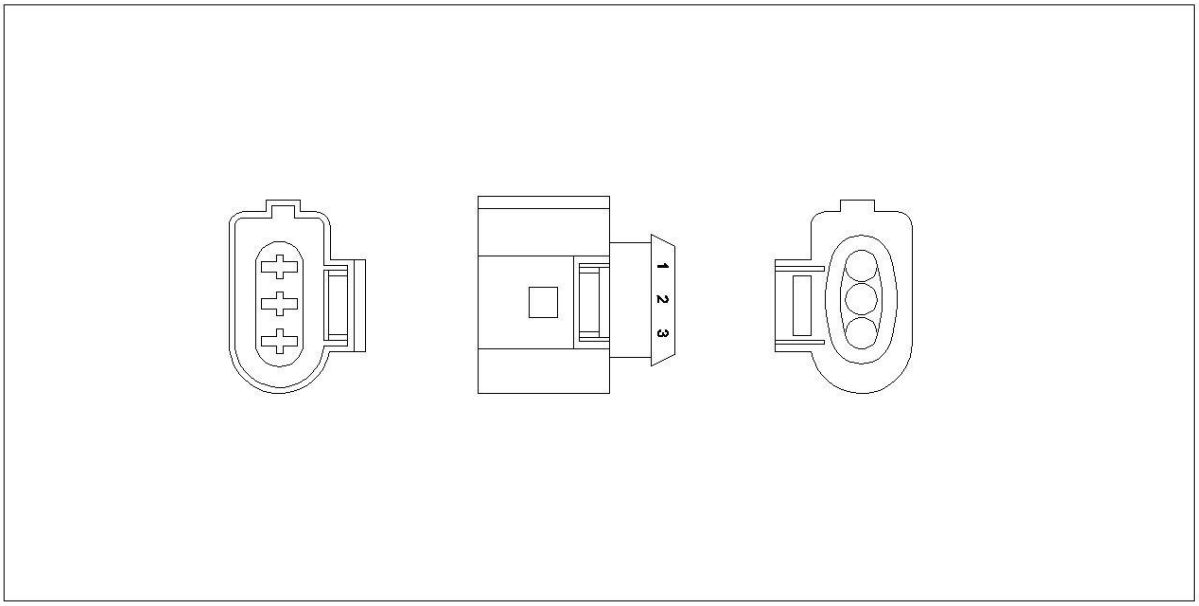
Crankshaft position sensor (CRK)

The crankshaft position (CRK) sensor connector is a 3 pin plug which is located on the cold side of the engine above the oil filter. See picture 2 connector location on cold engine side for detailed location.

The CRK+ and CRK- wires are a so called shielded wire. This is to protect the CRK signal from electronic noise.

Always keep the connector and wires away from ignition coils, spark plugs and spark plug leads.

Pin nr:	Function:	Color:	Thickness:
1	shield	black	0.35
2	CRK -	blue	0.35
3	CRK +	gray	0.35

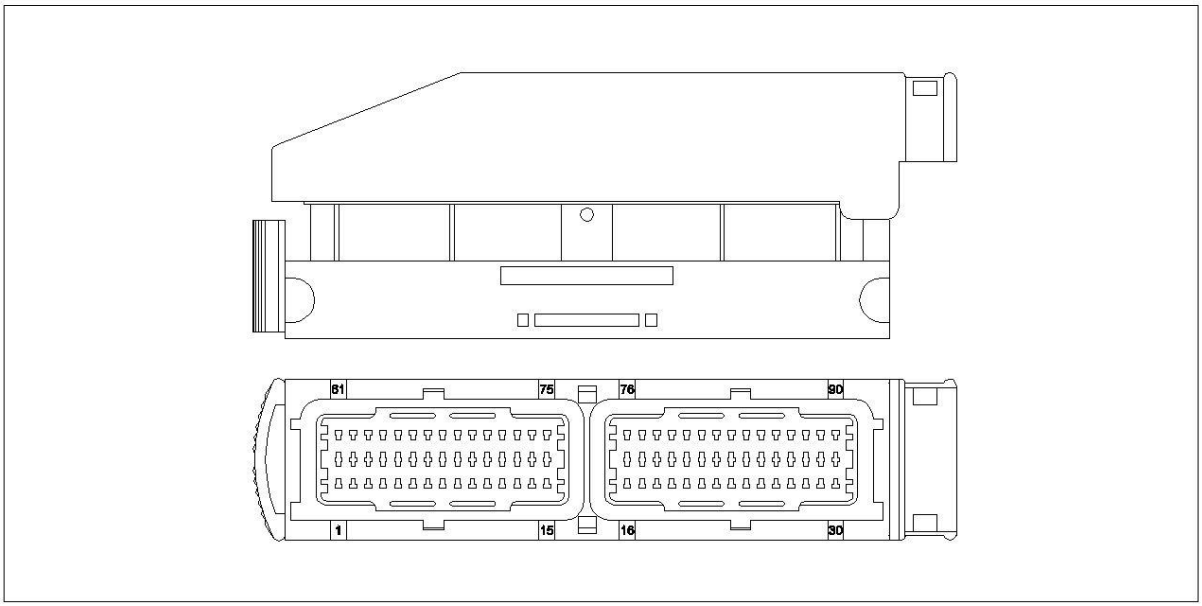


Electronic control unit (ECU)

The electronic control unit (ECU) connector is a 90 pin plug which is located on the cold side of the engine underneath the generator. See picture 2 connector location on cold engine side for detailed location.

Pin nr:	Function:	Color:	Thickness:
1	lambda 1 sensor signal (pre cat)	white	0.75
2	lambda 2 sensor signal (post cat)	white / yellow	0.75
5	throttle position sensor 1	yellow / blue	0.75
6	throttle position sensor 2	yellow / white	0.75
7	manifold absolute pressure signal	blue	0.75
9	foot pedal position sensor 1	yellow / green	0.75
10	foot pedal position sensor 2	yellow / gray	0.75
13	CAN1 termination +	white / yellow	0.75
14	CAN1 +	white / yellow	0.5
15	CAN1 -	white / blue	0.5
19	5 volts out 1	red / blue	0.75
20	sensor ground 1	brown / black	0.75
21	CRK +	gray	0.75
22	CRK -	blue	0.75
23	camshaft position +	blue / red	0.75
24	camshaft position -	white / red	0.75
27	knock sensor 1 +	yellow / green	0.75
28	knock sensor 1 -	blue / green	0.75
29	knock sensor 2 +	yellow / violet	0.75
30	knock sensor 2 -	blue / violet	0.75
31	ignition cylinder 1	gray / black	0.75
32	ignition cylinder 4	gray / red	0.75
33	ignition cylinder 3	gray / blue	0.75
34	ignition cylinder 2	gray / green	0.75
39	intake air temperature signal	blue / brown	0.75
40	engine coolant temperature signal	white / blue	0.75
42	auxiliary digital input 1	yellow / violet	0.75
43	auxiliary digital input 2	violet / yellow	0.75
44	auxiliary digital input 3	violet / gray	0.75
45	switched voltage (+15)	red / black	0.75
49	5 volts out 2	red / green	0.75
50	sensor ground 2	brown / green	0.75
51	governor 1	violet	0.75
52	governor 2	violet / red	0.75
53	oil pressure switch	green / black	0.75
55	PC Tx	white / green	0.75
56	PC Rx	white / red	0.75
58	tacho signal	green	0.75
60	battery voltage (+30)	red	1.5
61	gauge out 1	violet / black	0.75
62	gauge out 2	violet / blue	0.75
69	main ground	brown	1.5
70	starter signal	violet / brown	0.75
71	main power relay coil (-)	yellow / brown	0.75
72	lambda 1 sensor heater (-) (pre cat)	brown / white	0.75
73	lambda 2 sensor heater (-) (post cat)	brown / gray	0.75
75	lock off valve signal (-)	green / yellow	0.75
76	buzzer	green / white	0.75
77	auxiliary digital output 5 (-)	green / gray	0.75
79	battery voltage (+30)	red	1.5
80	malfunction indicator lamp (MIL)	green / red	0.75
81	main ground	brown	1.5
82	drive by wire +	black / red	0.75

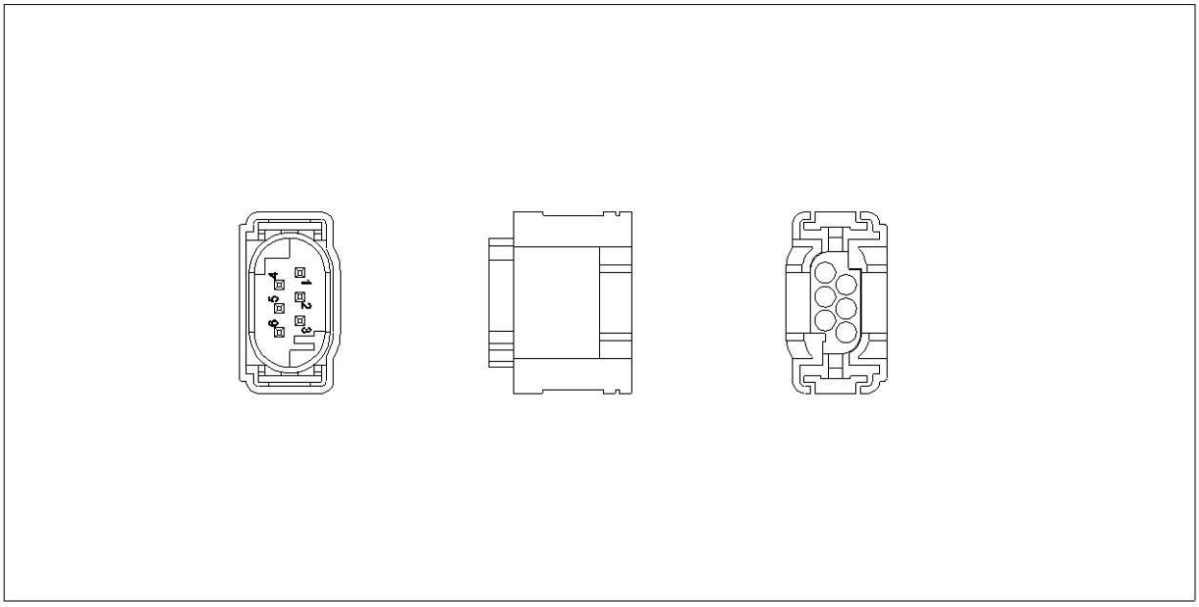
Pin nr:	Function:	Color:	Thickness:
83	drive by wire -	black / brown	0.75
86	fuel control valve signal (-)	white / brown	0.75
87	auxiliary digital output 1 (-)	blue / white	0.75
88	auxiliary digital output 2 (-)	green / blue	0.75
89	auxiliary digital output 4 (-)	blue / gray	0.75



Electronic throttle body (ETB)

The electronic throttle body (ETB) connector is a 6 pin plug which is located on the cold side of the engine on the throttle body. See picture 2 connector location on cold engine side for detailed location.

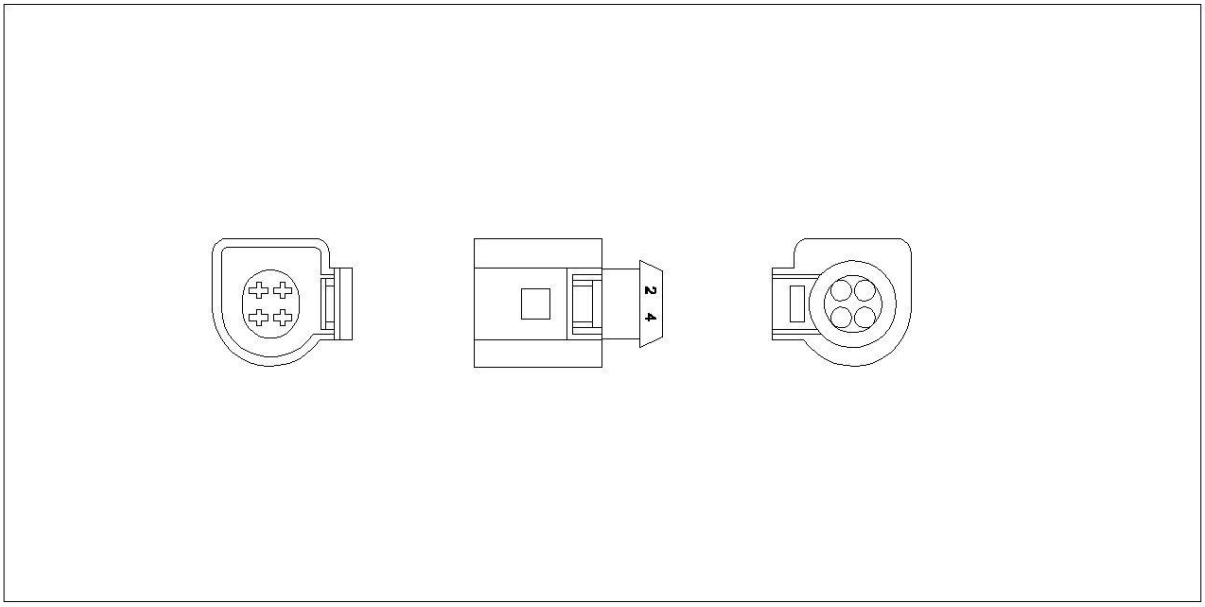
Pin nr:	Function:	Color:	Thickness:
1	drive by wire -	black / brown	0.75
2	sensor ground	brown / black	0.75
3	5 volts	red / blue	0.75
4	drive by wire +	black / red	0.75
5	throttle position sensor 2	yellow / white	0.75
6	throttle position sensor 1	yellow / blue	0.75



Engine coolant temperature sensor (ECT)

The engine coolant temperature (ECT) sensor connector is a 4 pin plug (only 2 are used) which is located on the back side of the engine above the flywheel housing. See picture 1 connector location on back side of engine for detailed location.

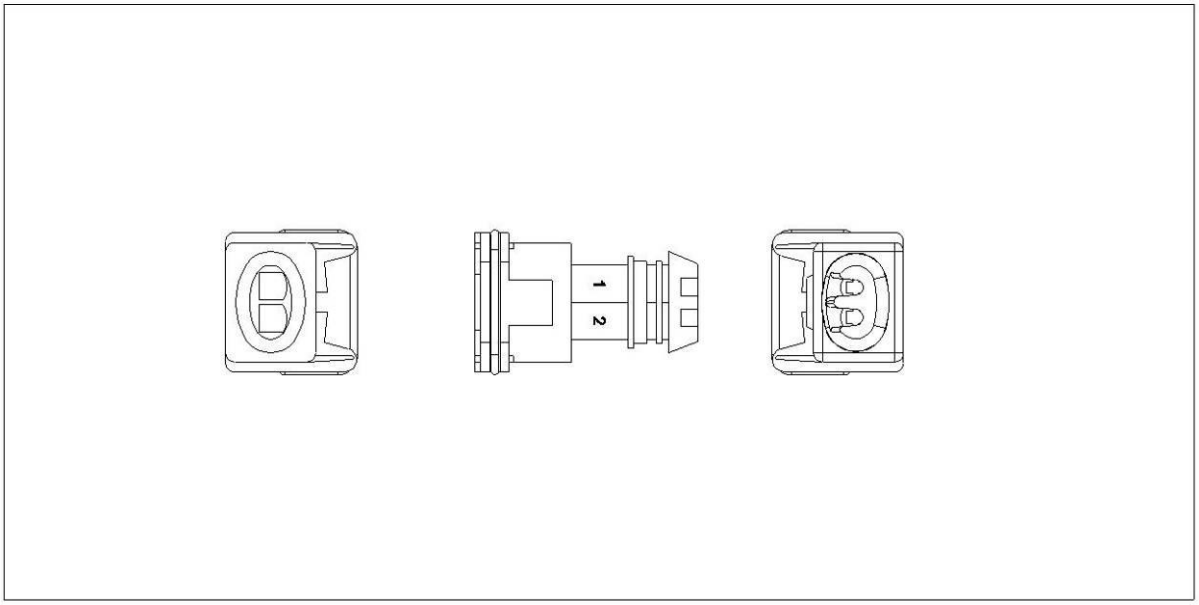
Pin nr:	Function:	Color:	Thickness:
1	- -	- -	- -
2	- -	- -	- -
3	engine coolant temperature signal	white / blue	0.75
4	sensor ground	brown / black	0.75



Fuel control valve (FCV)

The fuel control valve (FCV) connector is a 2 pin plug which is located on the cold side of the engine near the throttle body. See picture 2 connector location on cold engine side for detailed location.

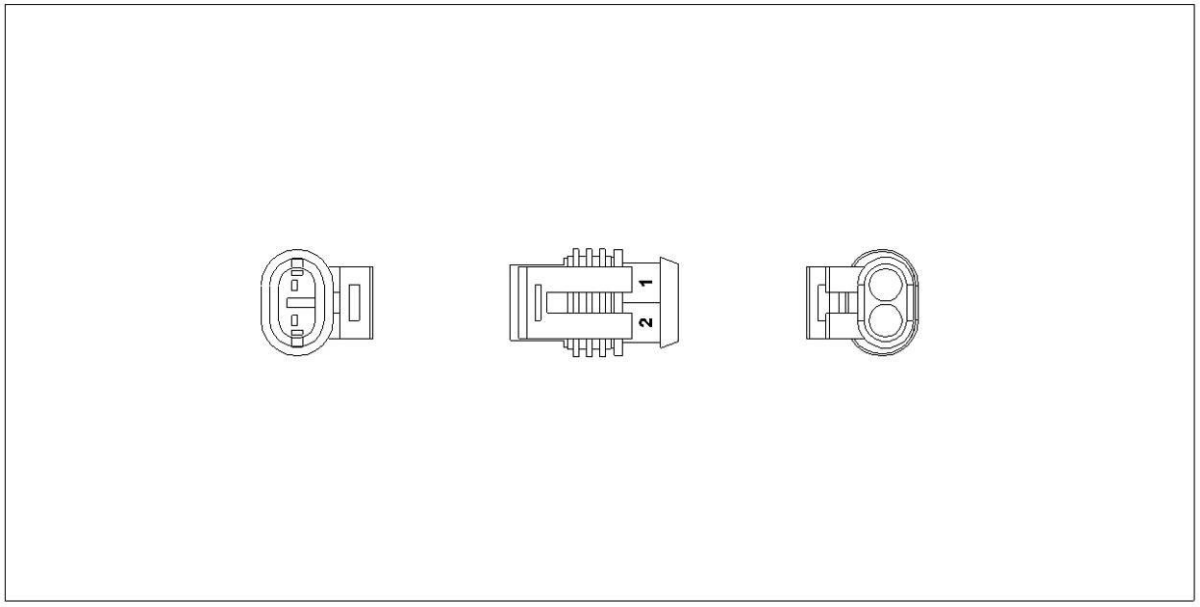
Pin nr:	Function:	Color:	Thickness:
1	voltage main power relay (+)	red / yellow	0.75
2	fuel control valve signal (-)	white / brown	0.75



Gaseous fuel lock off valve (LO)

The gaseous fuel lock off valve (LO) connector is a 2 pin plug which is located on the top of the engine next to the vaporizer. See picture 3 connector location on hot / top side of engine for detailed location.

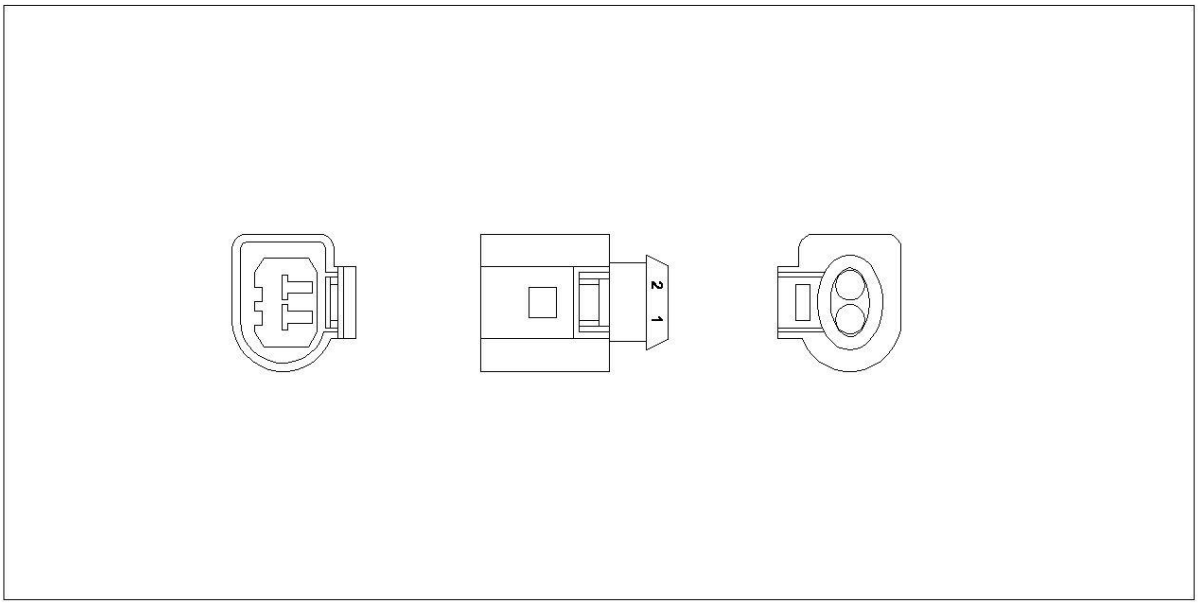
Pin nr:	Function:	Color:	Thickness:
1	lock off valve signal (-)	green / yellow	0.75
2	switched voltage (+15)	red / black	0.75



Generator (GEN)

The generator (GEN) connector is a 2 pin plug which is located on the cold side of the engine on the back of the generator. See picture 2 connector location on cold engine side for detailed location.

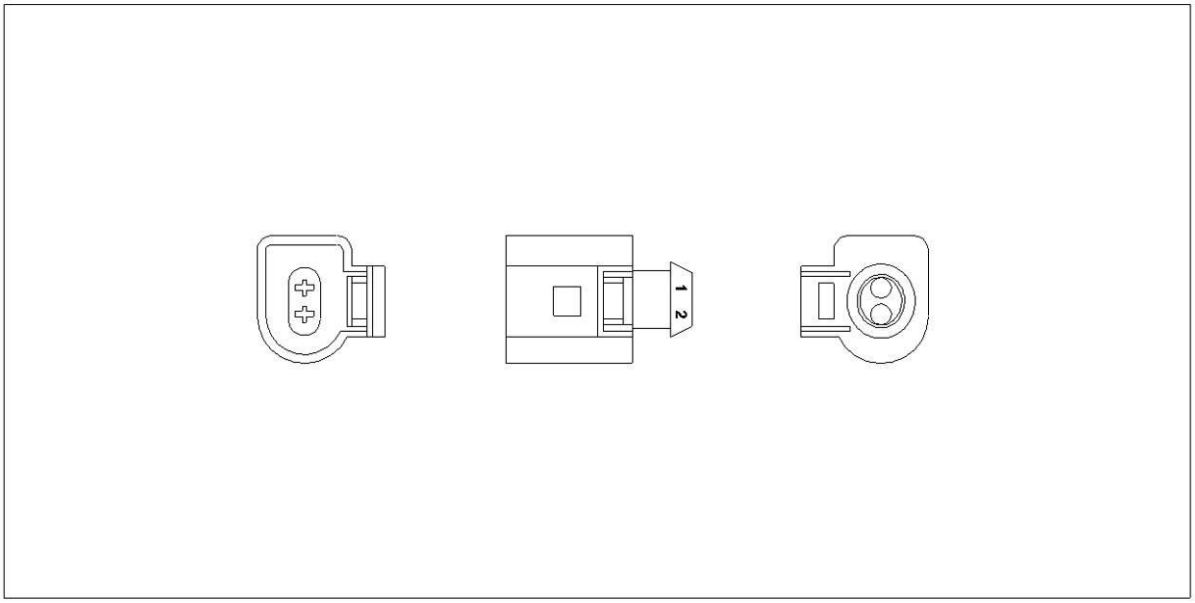
Pin nr:	Function:	Color:	Thickness:
1	gen lamp	red / black	0.75
2	- -	- -	- -



Knock sensor 1 (KNK1)

The knock sensor 1 (KNK1) connector is a 2 pin plug which is located on the cold side of the engine underneath the intake manifold. See picture 2 connector location on cold engine side for detailed information.

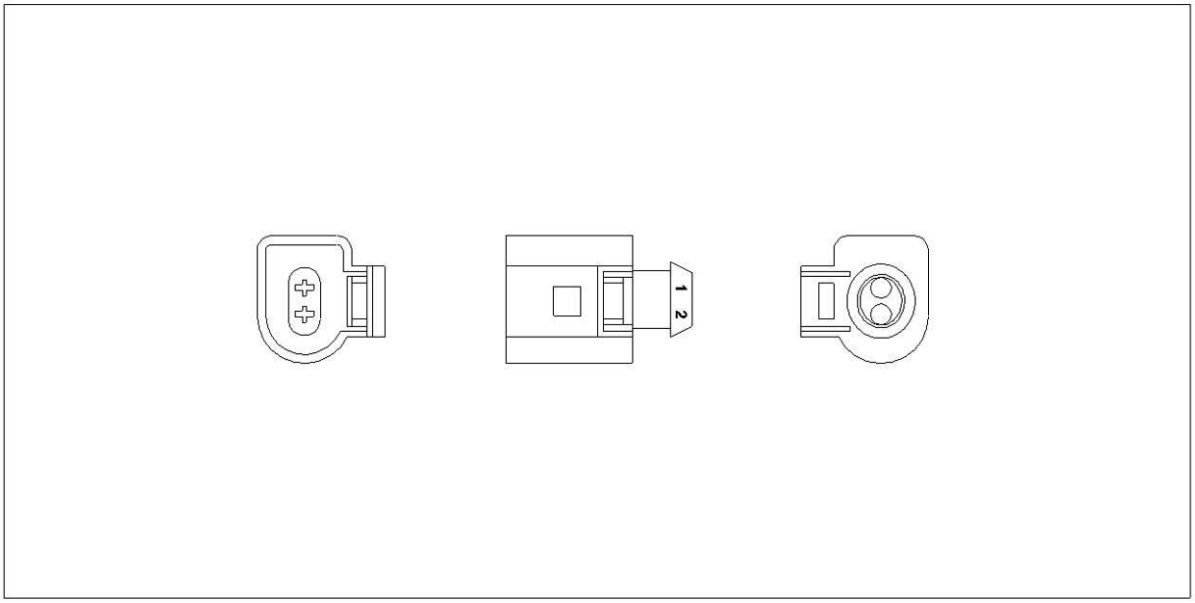
Pin nr:	Function:	Color:	Thickness:
1	knock sensor 1 +	yellow / green	0.75
2	knock sensor 1 -	blue / green	0.75



Knock sensor 2 (KNK2)

The knock sensor 2 (KNK2) connector is a 2 pin plug which is located on the cold side of the engine underneath the intake manifold. See picture 2 connector location on cold engine side for detailed information.

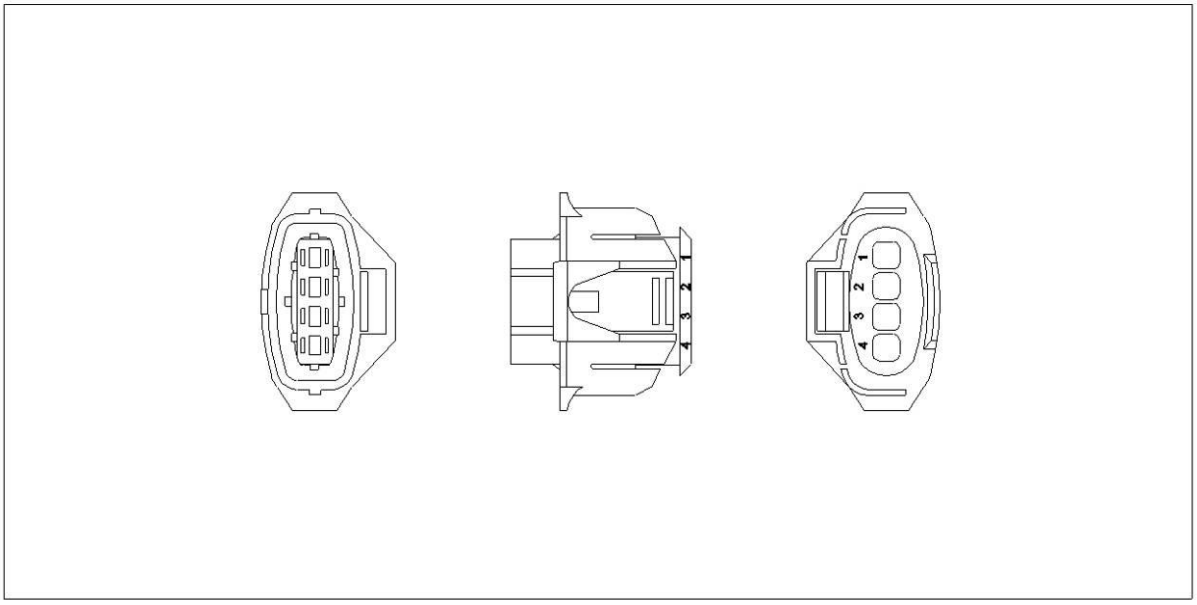
Pin nr:	Function:	Color:	Thickness:
1	knock sensor 2 +	yellow / violet	0.75
2	knock sensor 2 -	blue / violet	0.75



Manifold absolute pressure- and intake air temperature sensor (MAP / IAT)

The manifold absolute pressure and intake air temperature (MAP / IAT) sensor connector is a 4 pin plug which is located on the cold side of the engine on the intake manifold. See picture 1 connector location on back side of engine for detailed location.

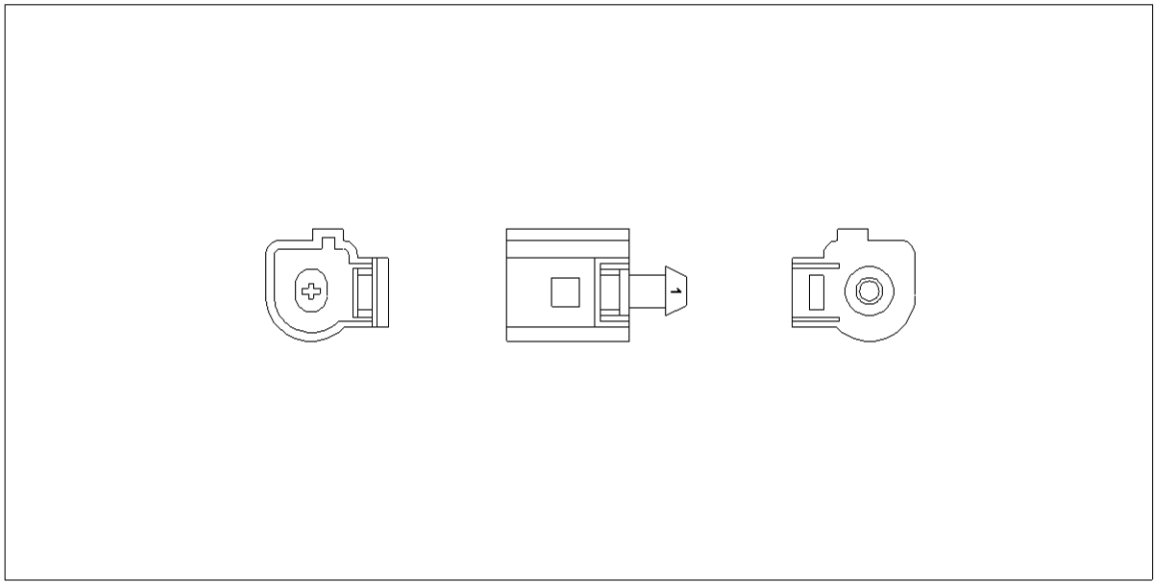
Pin nr:	Function:	Color:	Thickness:
1	sensor ground 1	brown / black	0.75
2	intake air temperature signal	blue / brown	0.75
3	5 volts out 1	red / blue	0.75
4	manifold absolute pressure signal	blue	0.75



Oil pressure sensor connector (OILP)

The OIL Pressure (OILP) sensor connector is a 1 pin plug which is located on the cold side of the engine underneath the coil pack. See picture 2 connector location on cold engine side for detailed location.

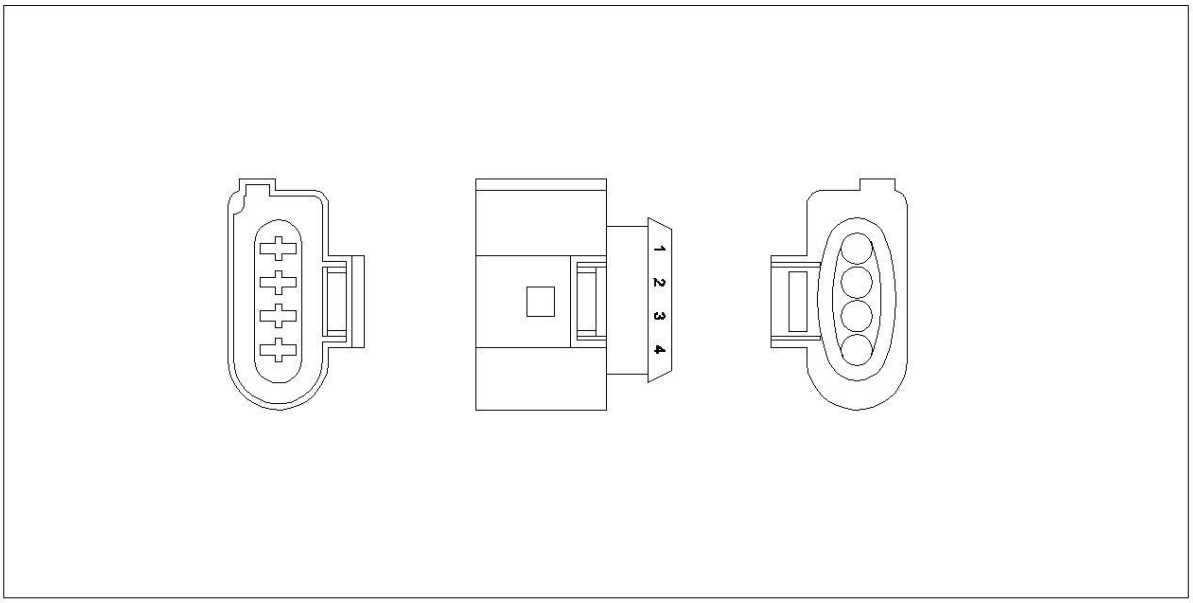
Pin nr:	Function:	Color:	Thickness:
1	oil pressure sensor	green / black	0.75



Post-catalyst oxygen sensor connector (EGO2)

The post-cat oxygen sensor (EGO2) connector is a 4 pin plug which is located on the hot side of the engine behind the catalyst. See picture 2 connector location on cold engine side for detailed information.

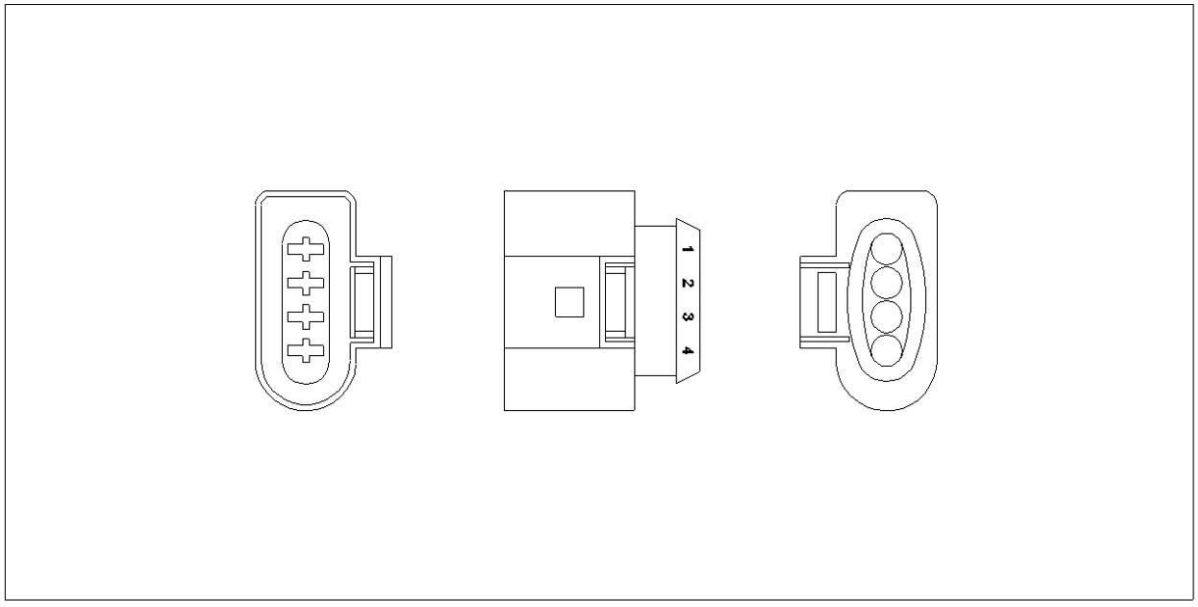
Pin nr:	Function:	Color:	Thickness:
1	lambda sensor heater 2	brown / gray	0.75
2	voltage main power relay	red / yellow	0.75
3	sensor ground 1	brown / black	0.75
4	lambda sensor signal 2	white / yellow	0.75



Pre- catalyst oxygen sensor connector (EGO1)

The pre-cat oxygen sensor (EGO1) connector is a 4 pin plug which is located on the back side of the engine underneath the ChC connector. See picture 1 connector location on back side of engine for detailed location.

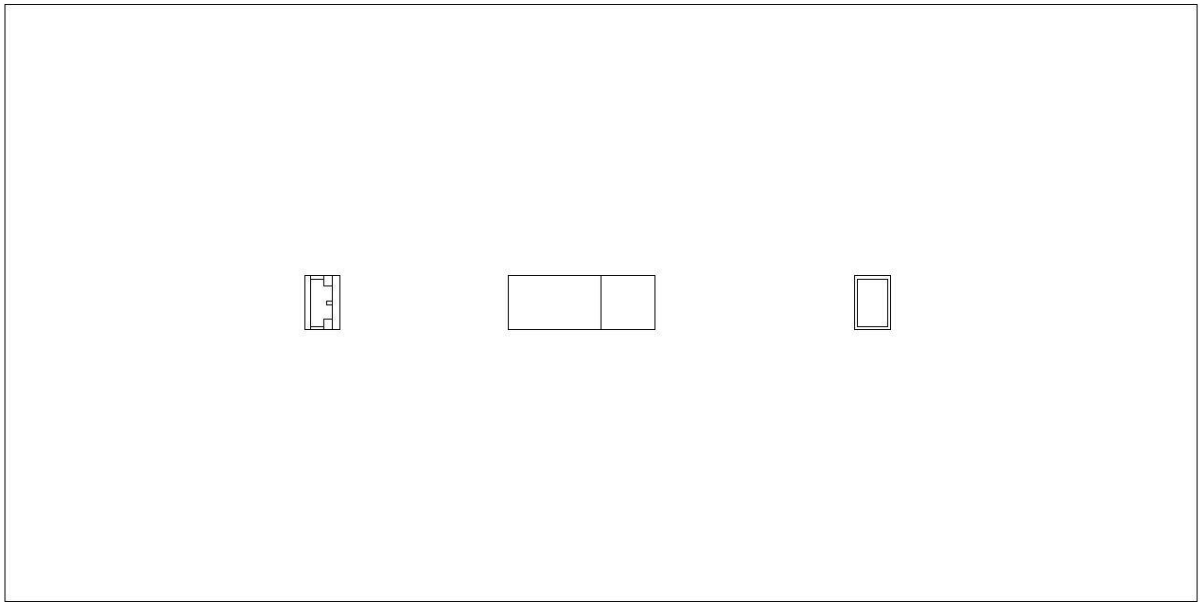
Pin nr:	Function:	Color:	Thickness:
1	lambda sensor heater 1	brown / white	0.75
2	voltage main power relay	red / yellow	0.75
3	sensor ground 1	brown / black	0.75
4	lambda sensor signal 1	white	0.75



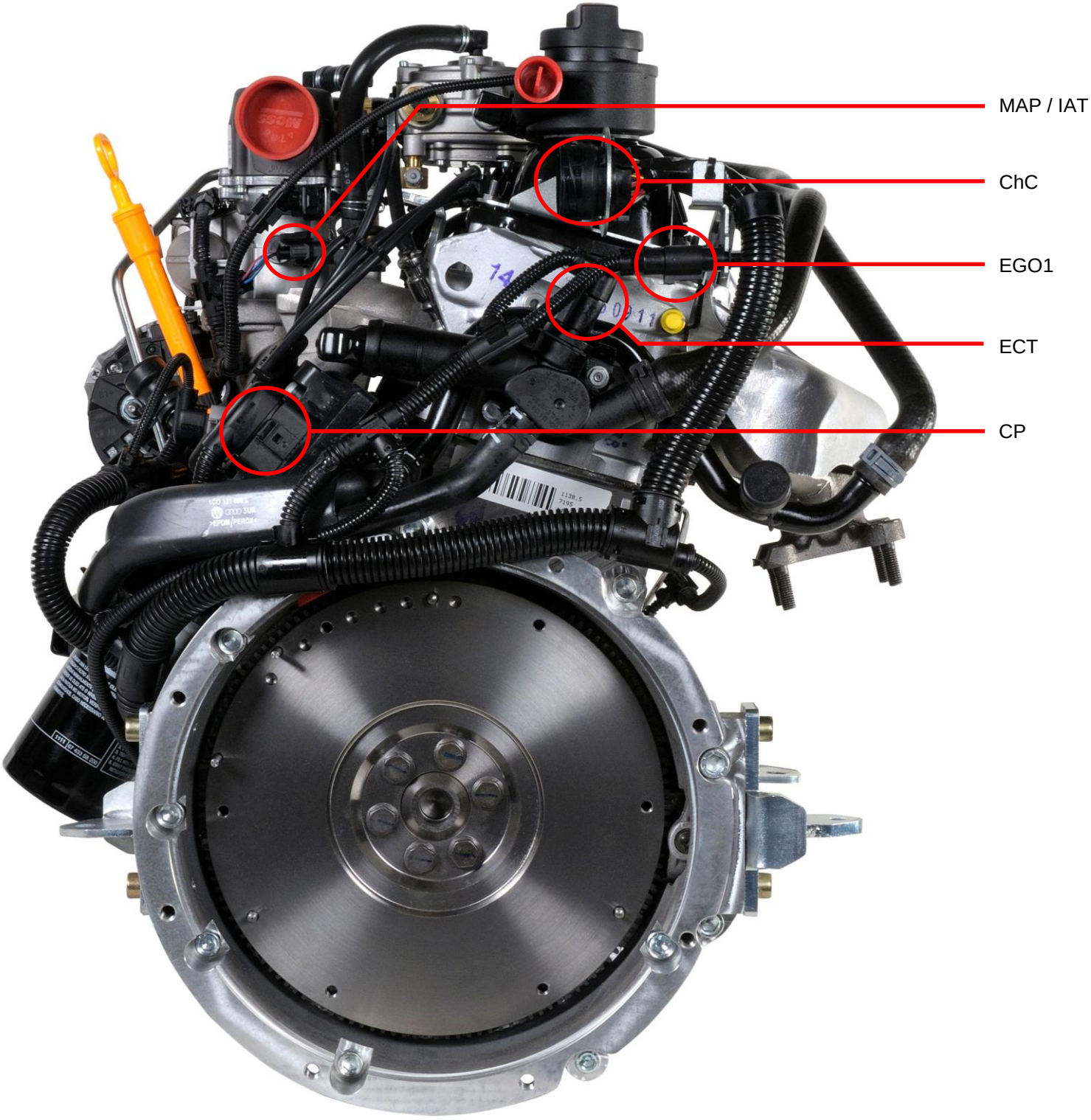
Starter motor connector (STM)

The starter motor (STM) connecter is a 1 pin plug which is located on the hot side of the engine underneath the exhaust manifold. See picture 3 connector location on hot / top side of engine for detailed location.

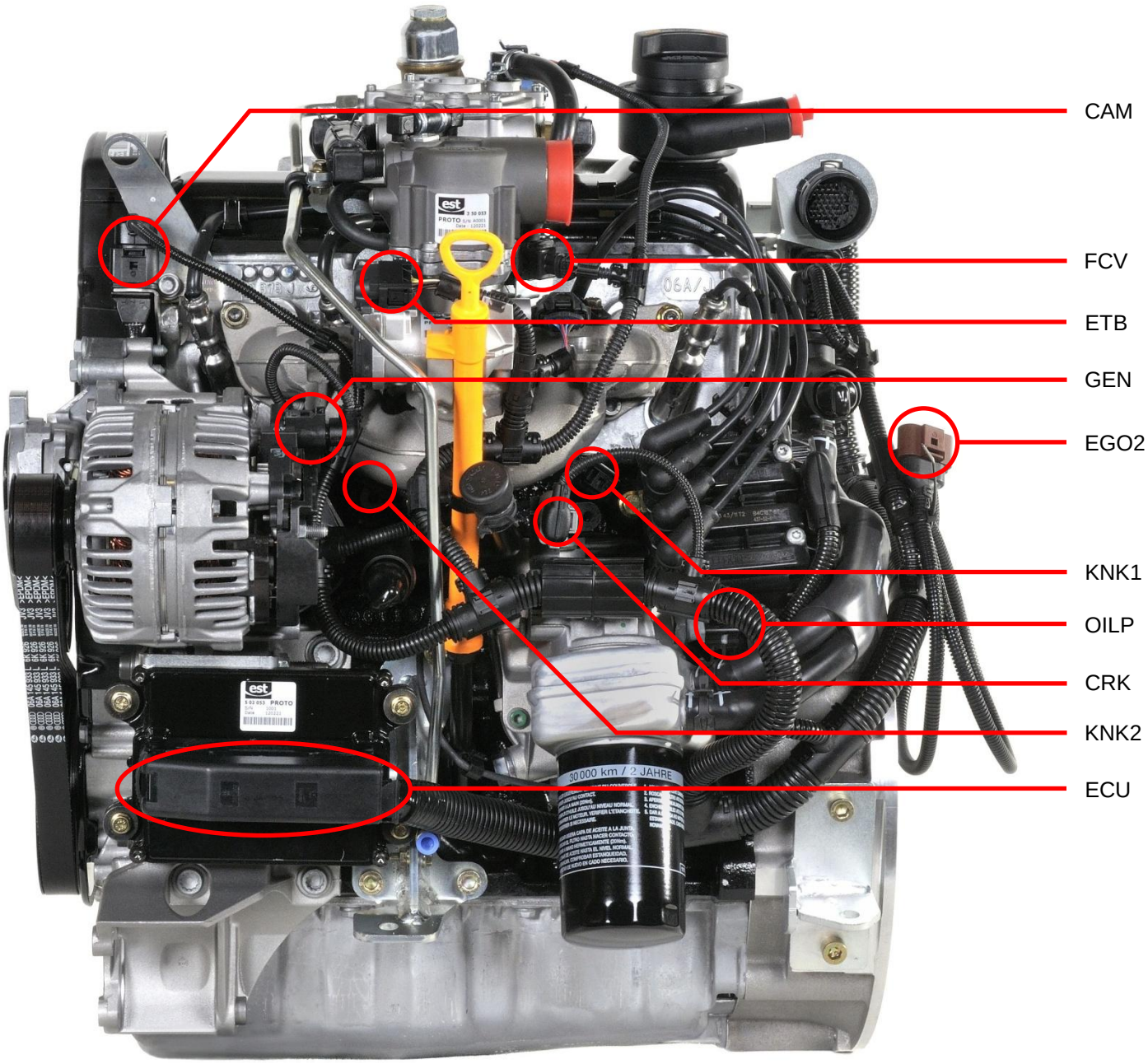
Pin nr:	Function:	Color:	Thickness:
1	starter motor signal	red / white	1.5



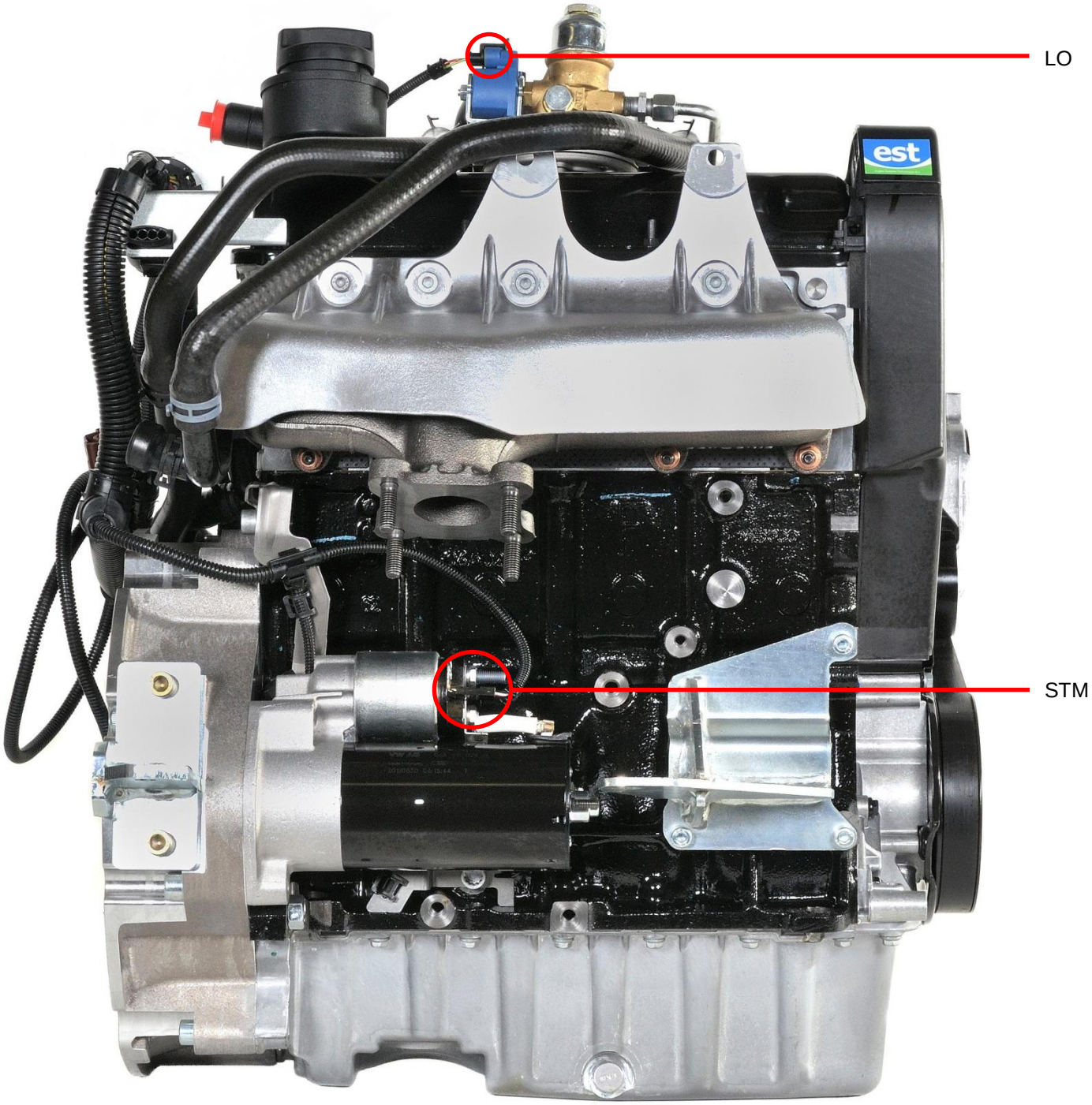
picture 1 connector location on back side of engine



picture 2 connector location on cold engine side



picture 3 connector location on hot / top side of engine



Diagnostic trouble code SPN and FMI reference

DTC	DTC description	SPN	FMI
16	Never crank synced at start	636	8
107	MAP low voltage	106	4
108	MAP high pressure	106	16
111	IAT higher than expected 1	105	15
112	IAT low voltage	105	4
113	IAT high voltage	105	3
116	ECT higher than expected 1	110	15
117	ECT low voltage	110	4
118	ECT high voltage	110	3
121	TPS1 lower than TPS2	51	1
122	TPS1 low voltage	51	4
123	TPS1 high voltage	51	3
127	IAT higher than expected 2	105	0
129	BP low pressure	108	1
134	EGO open / lazy pre – cat 1	3217	5
154	EGO open / lazy post cat 1	3227	5
217	ECT higher than expected 2	110	0
219	Max govern speed override	515	15
221	TPS1 higher than TPS2	51	0
222	TPS2 low voltage	3673	4
223	TPS2 high voltage	3673	3
326	Knock 1 excessive signal	731	2
327	Knock 1 sensor open	731	4
331	Knock 2 excessive signal	520197	2
332	Knock 2 sensor open	520197	4
336	Crank sync noise	636	2
337	Crank loss	636	4
341	Cam sync noise	723	2
342	Cam loss	723	4
359	Fuel run-out longer than expected	632	31
524	Oil pressure low	100	1
562	System voltage low	168	17
563	System voltage high	168	15
601	Flash checksum invalid	628	13
604	RAM failure	630	12
606	COP failure	629	31
642	5VE1 low voltage	1079	4
643	5VE1 high voltage	1079	3
1111	Fuel rev limit	515	16
1112	Spark rev limit	515	0
1121	FPP1/2 Simultaneous voltages out of range	91	31
1151	CL high LPG	4236	0
1152	CL low LPG	4236	1
1153	CL high NG	4236	0
1154	CL low NG	4236	1
1161	AL high LPG	4237	0
1162	AL low LPG	4237	1
1163	AL high NG	4237	0
1164	AL low NG	4237	1
1612	RTI 1 loss	629	31
1613	RTI 2 loss	629	31
1614	RTI 3 loss	629	31
1615	A/D loss	629	31
1616	Invalid Interrupt	629	31
1626	Can Tx failure	639	12
1627	Can Rx failure	639	12
1629	J1939 TSC1 message receipt lost	695	9
2111	Unable to reach lower TPS	51	7
2112	Unable to reach higher TPS	51	7
2115	FPP1 higher than IVS limit	91	0
2121	FPP1 lower than TPS2	91	18
2122	FPP1 high voltage	91	3
2123	FPP1 low voltage	91	4
2126	FPP1 higher than FPP2	91	16
2127	FPP2 low voltage	29	4
2128	FPP2 high voltage	29	3
2135	TPS1/2 simultaneous voltages out-of-range	51	31
2139	FPP1 lower than IVS limit	91	1
2229	BP high pressure	108	0

Electric diagram

